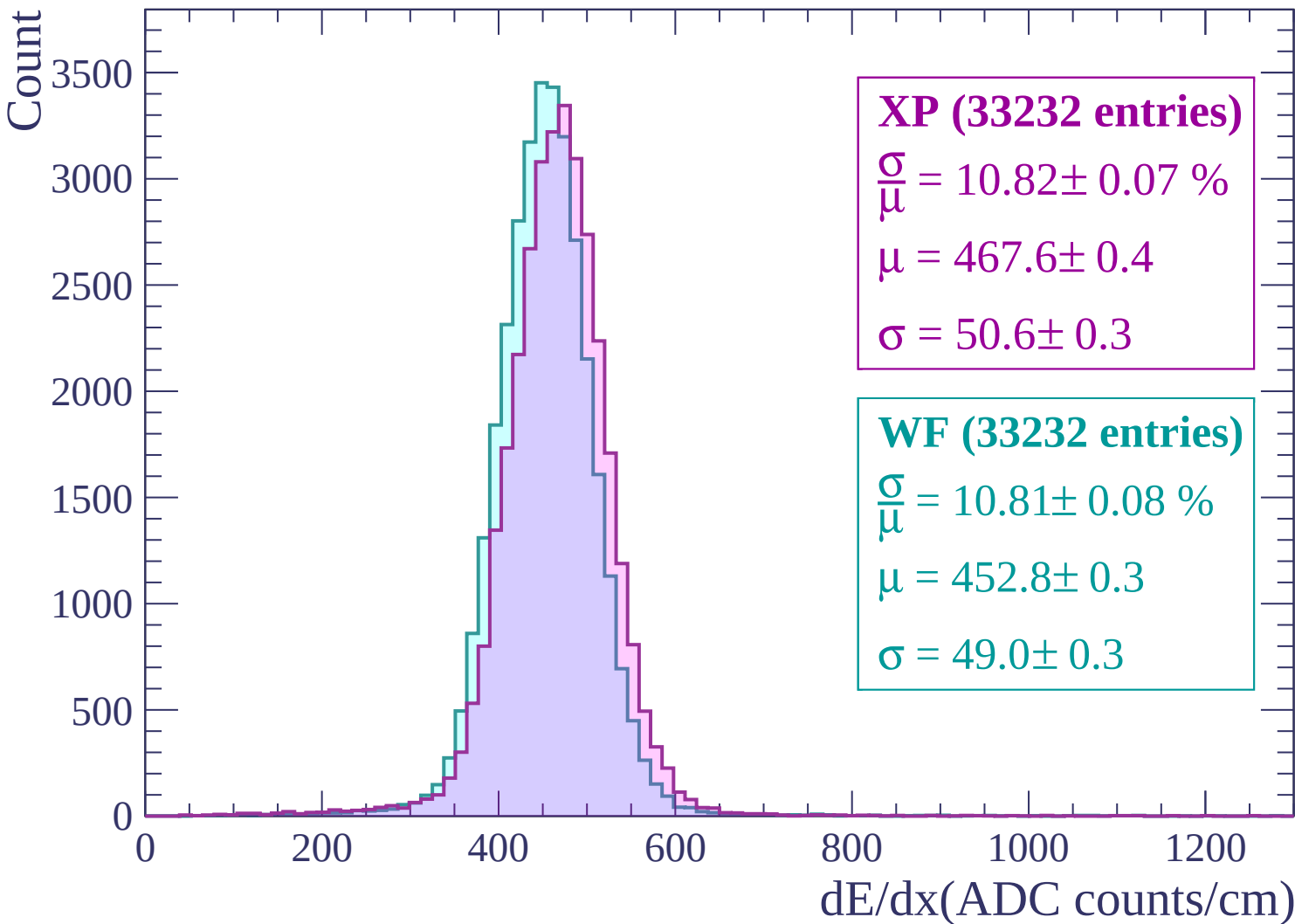
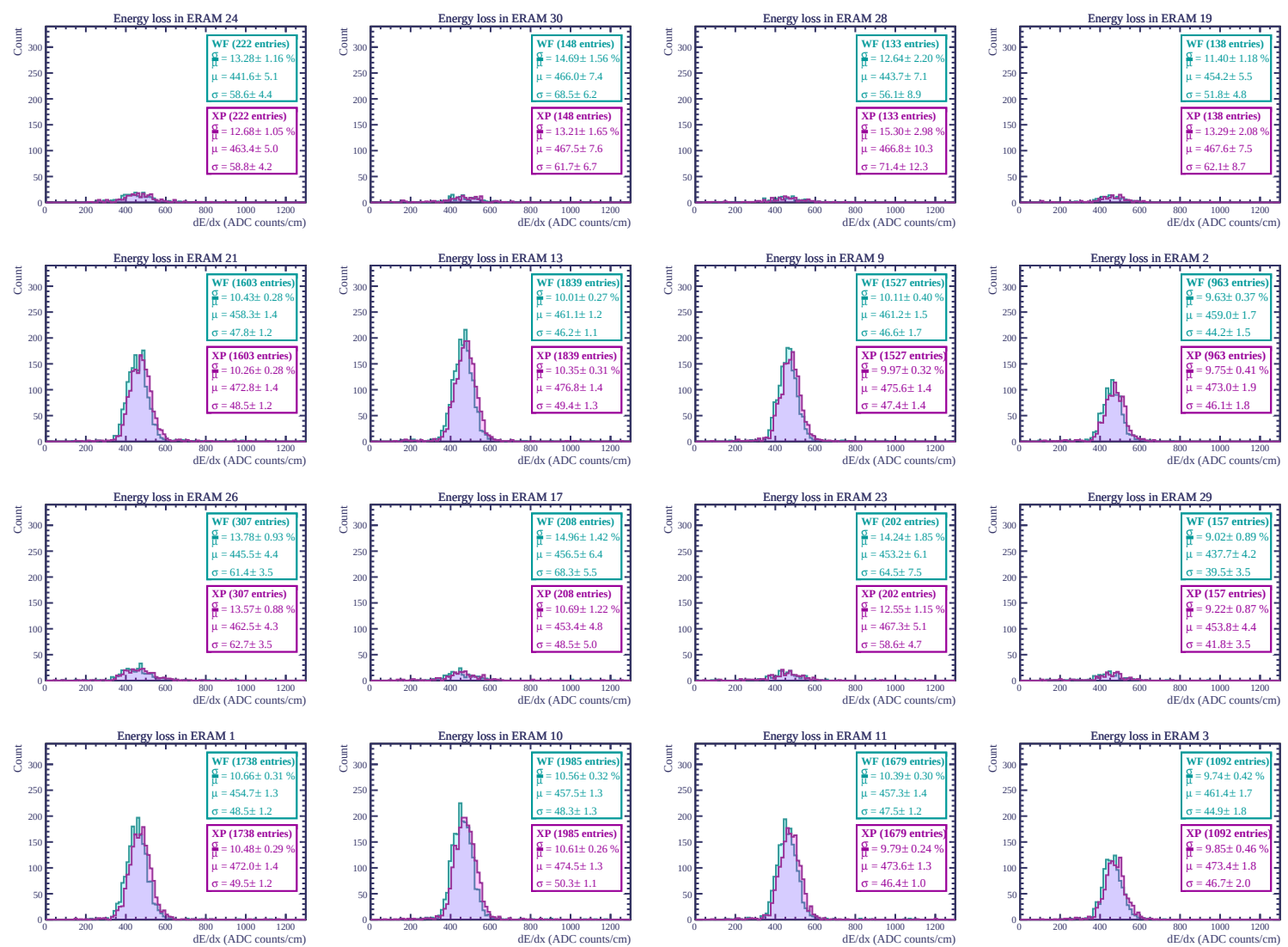
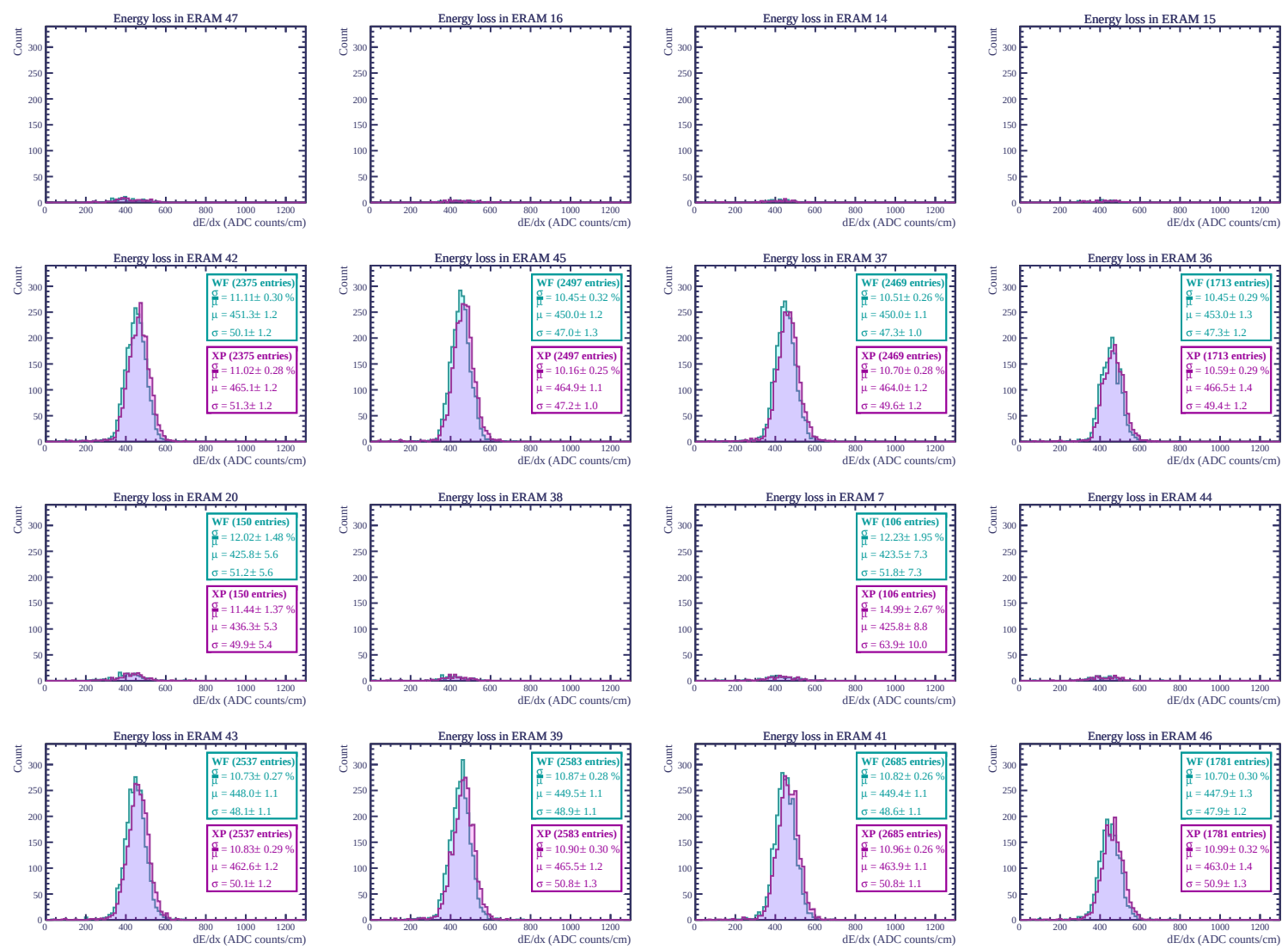
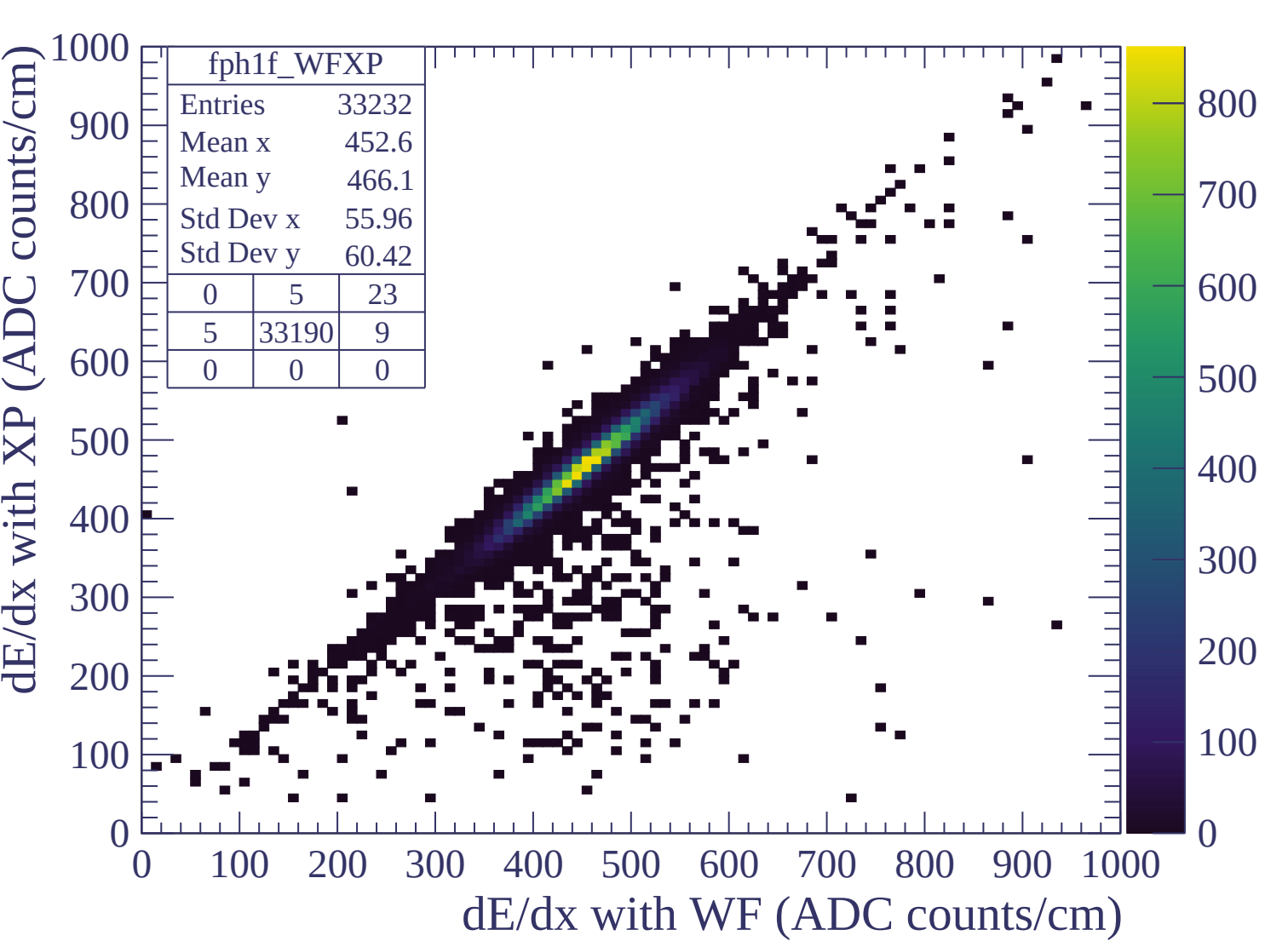
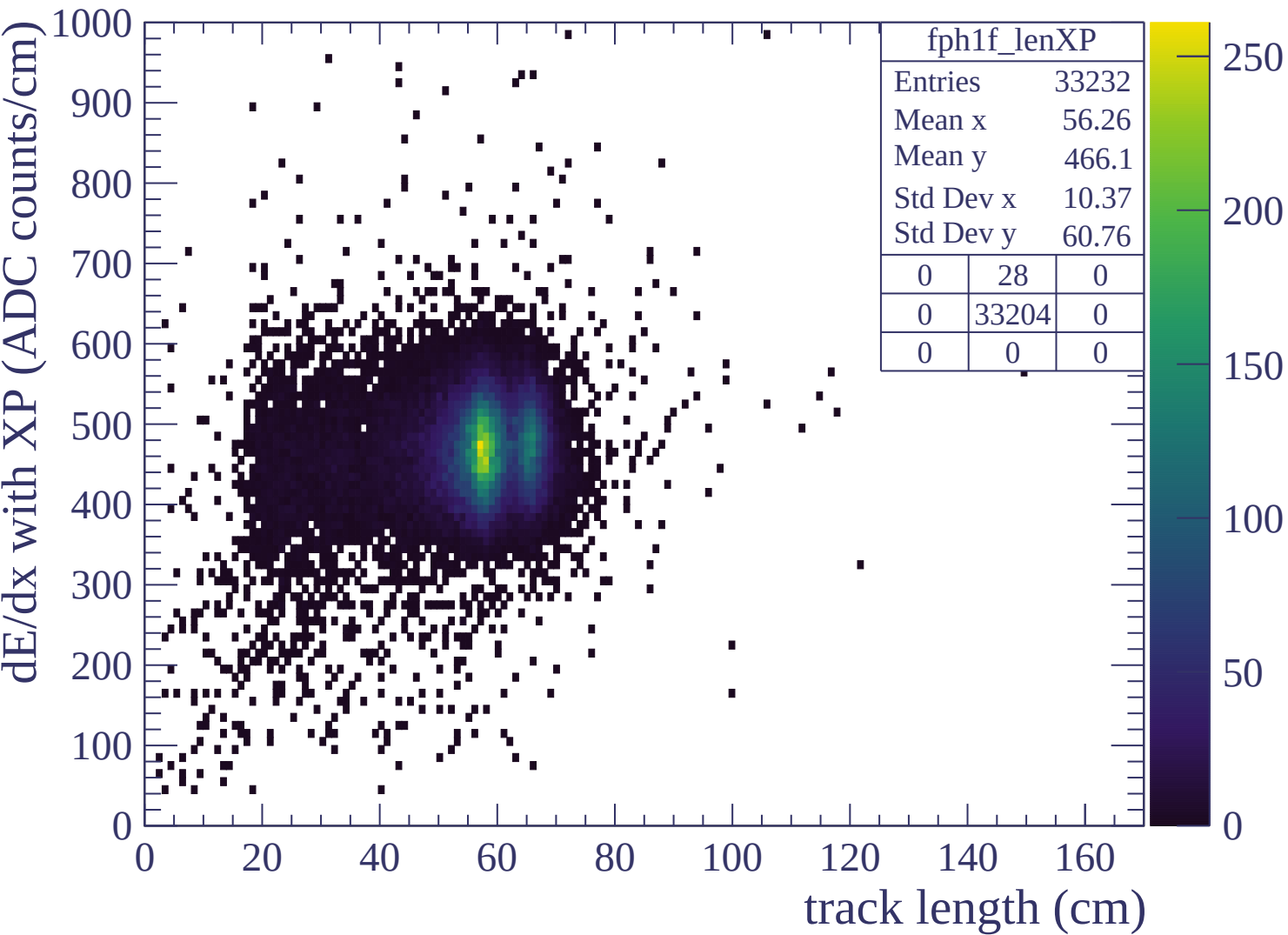


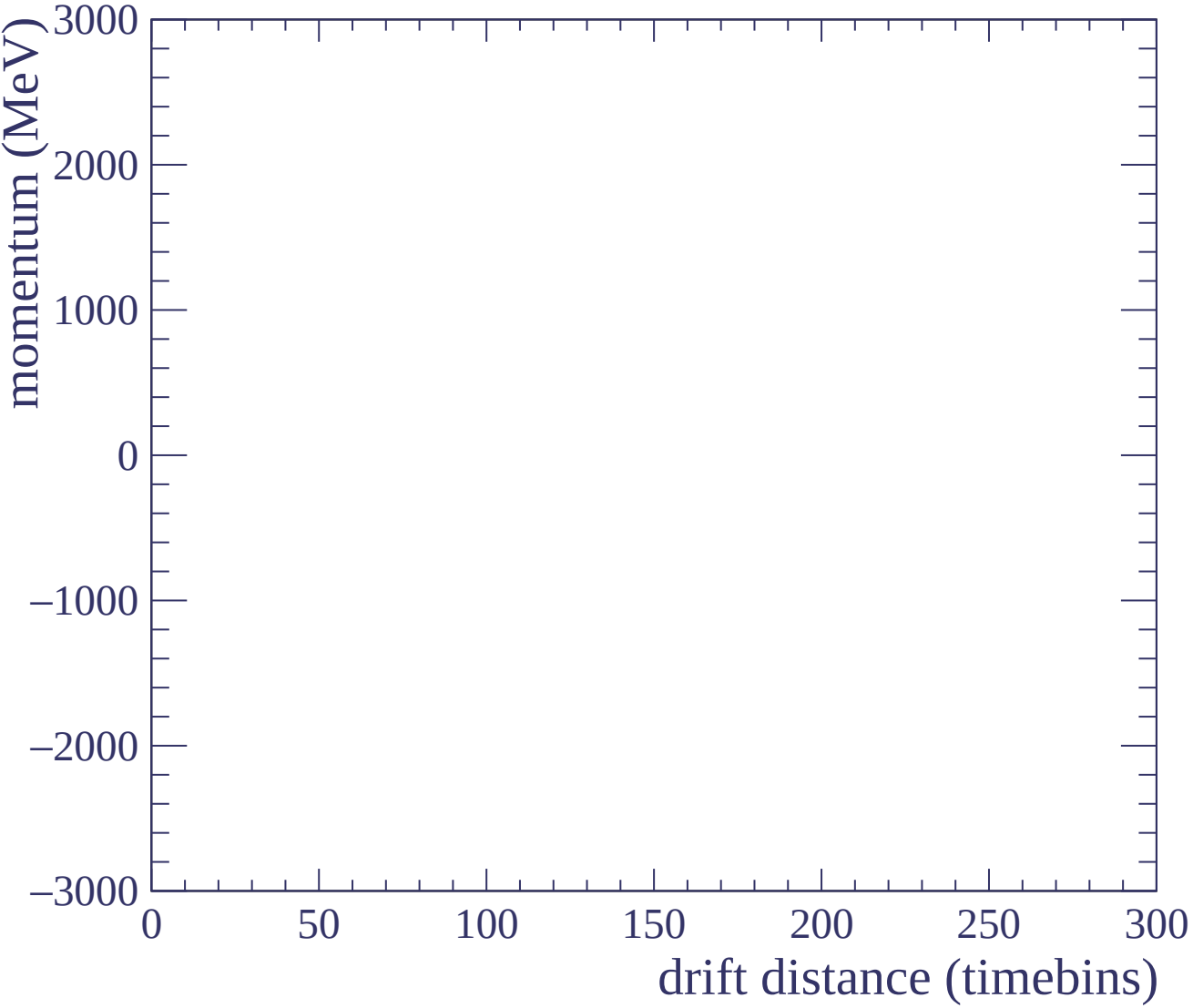


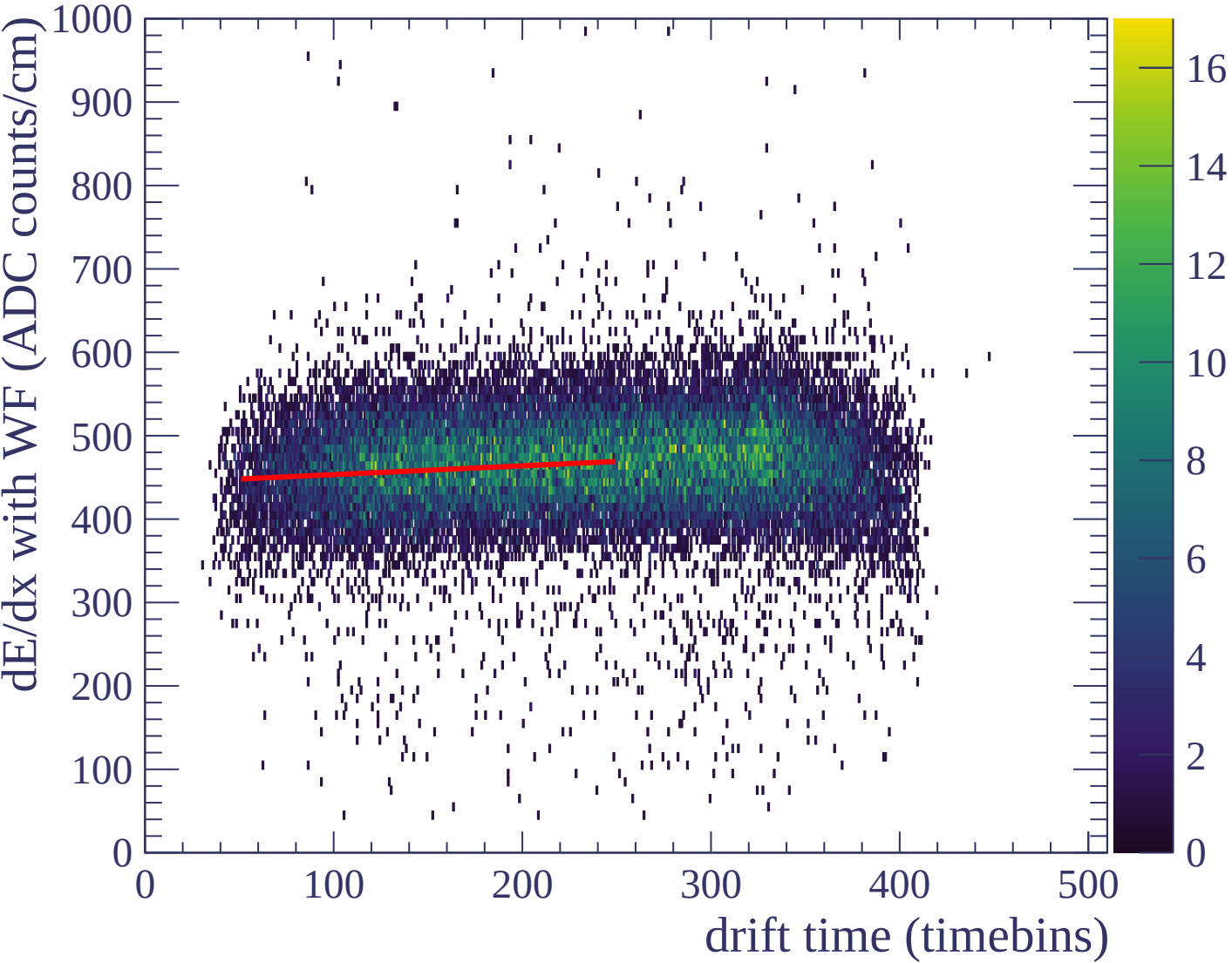


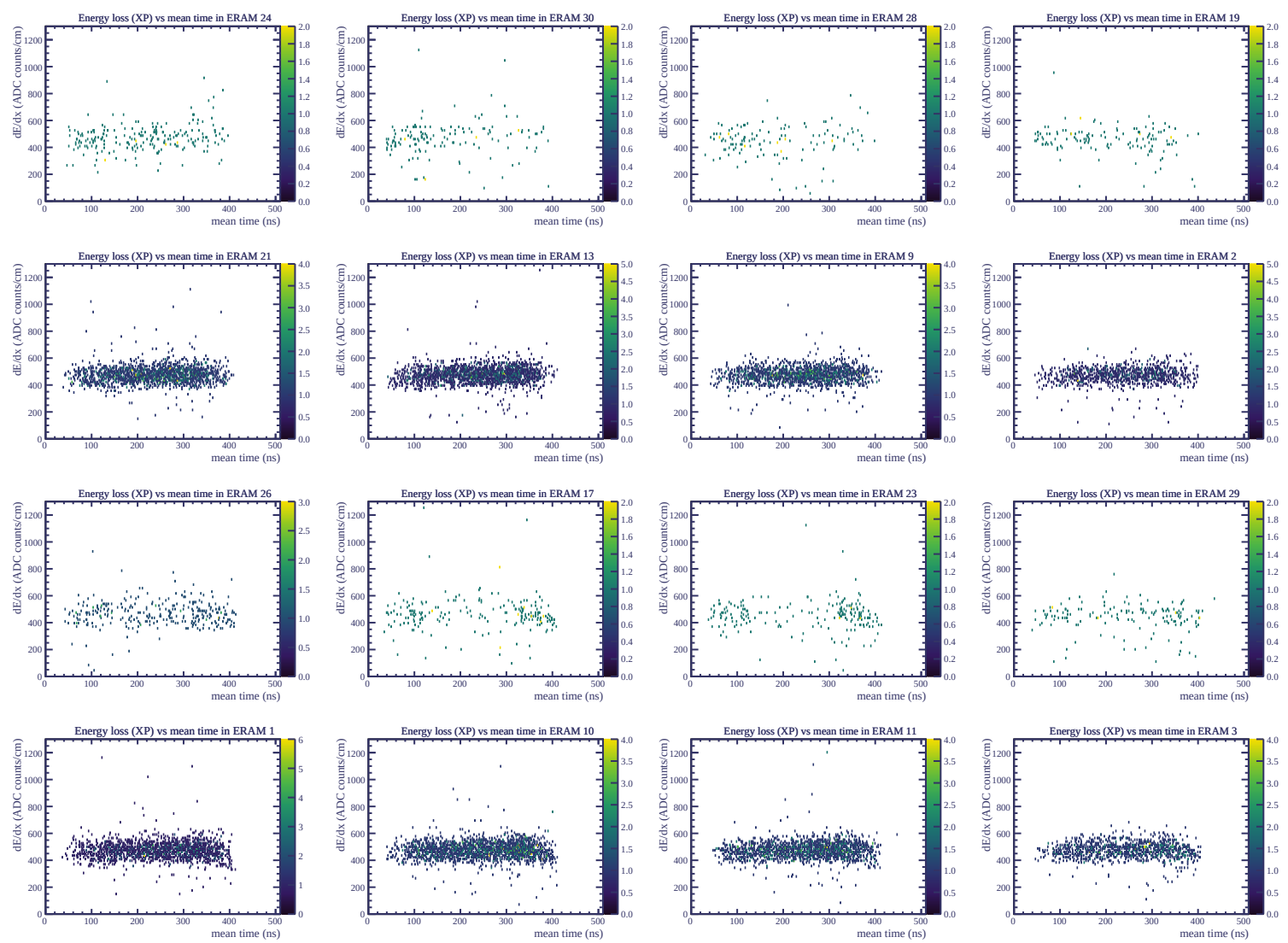


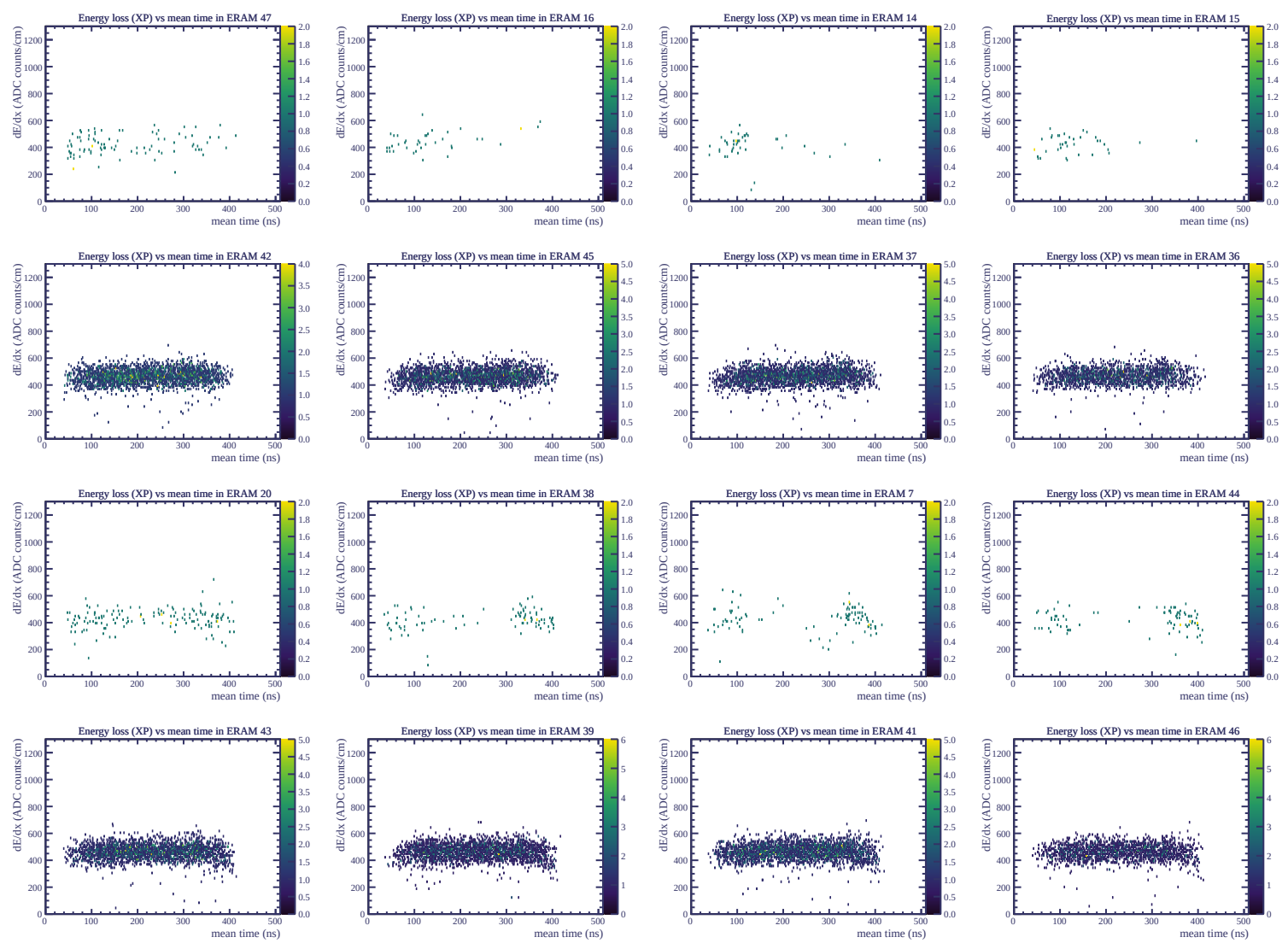


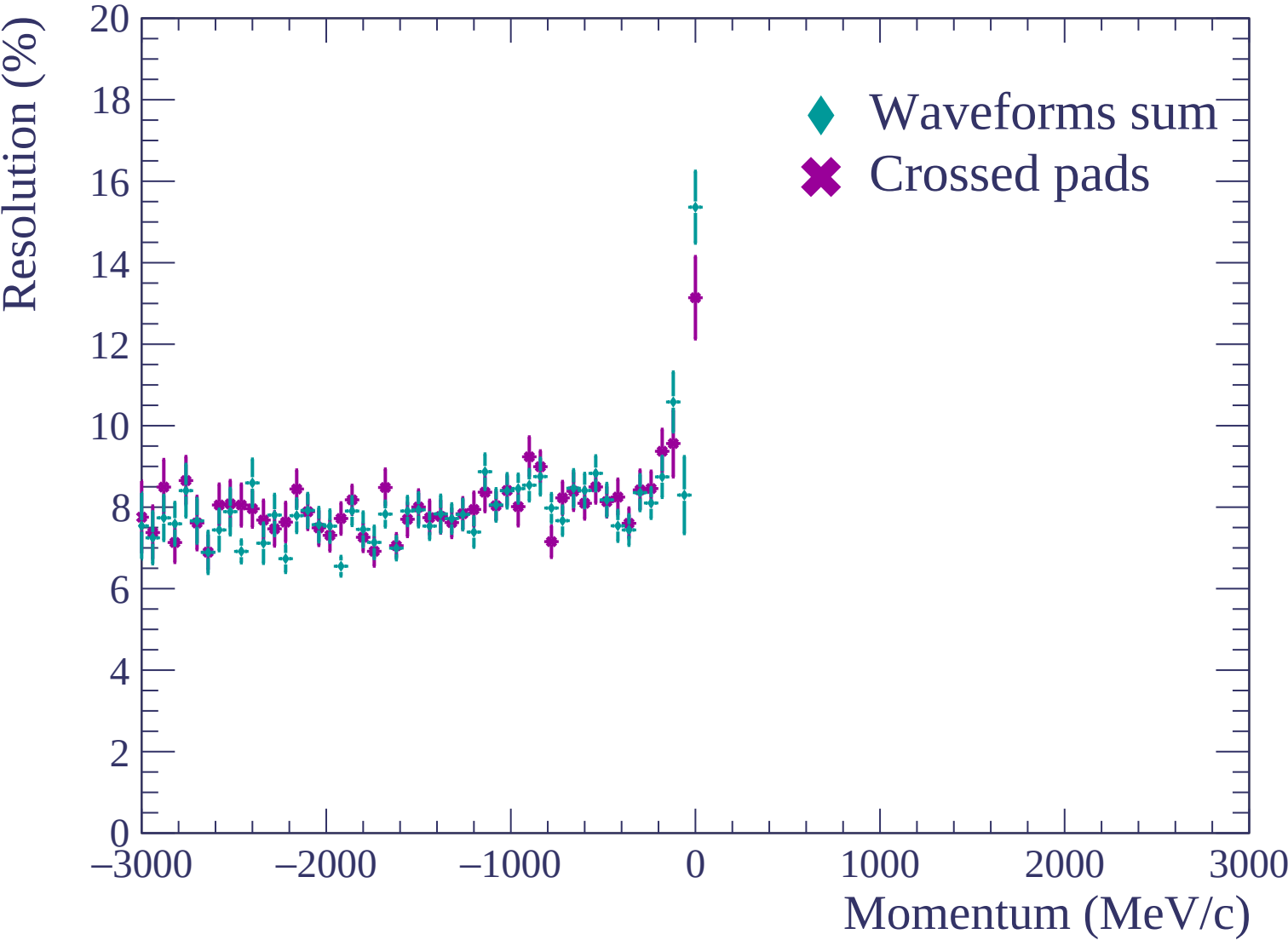


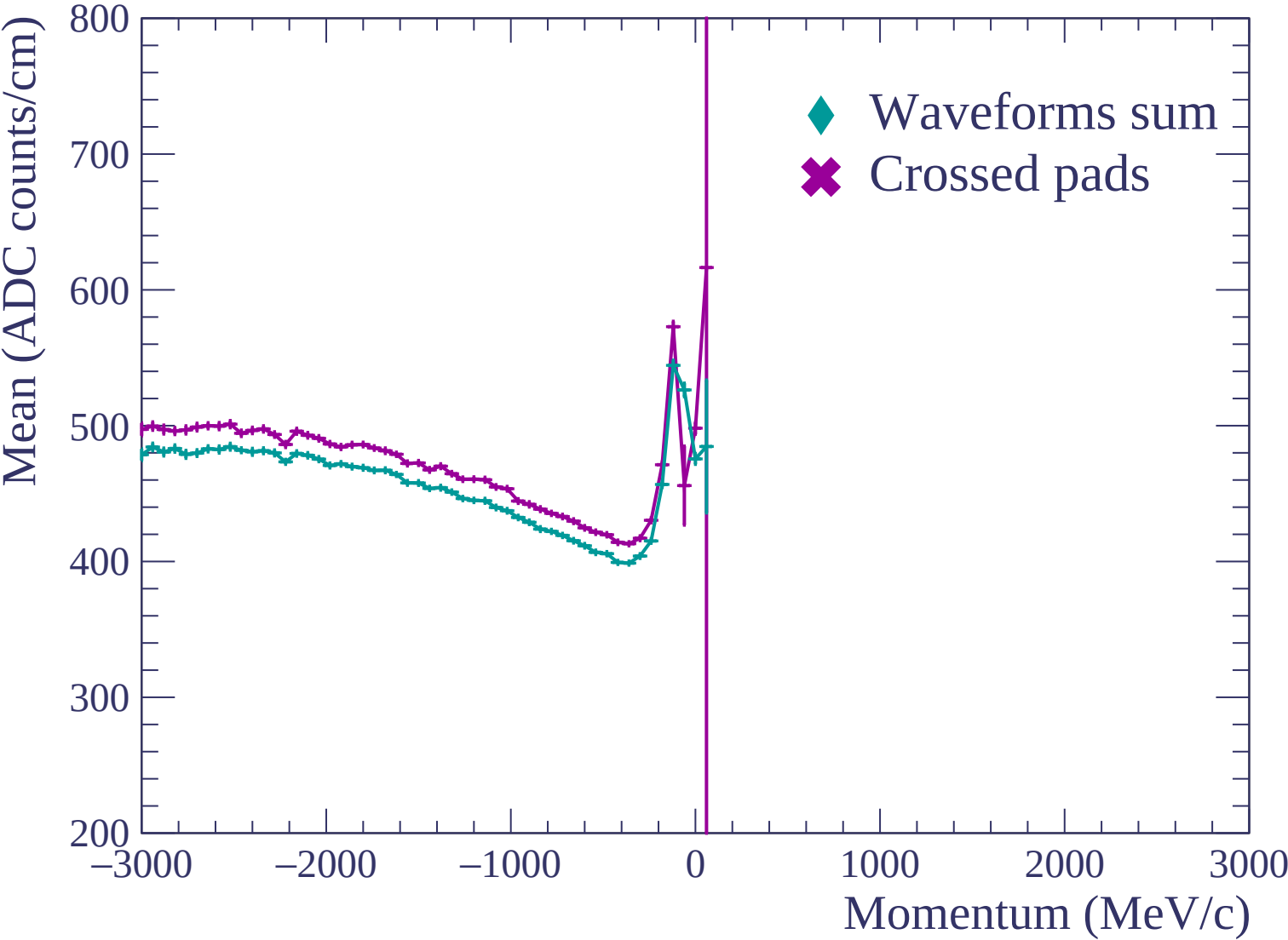


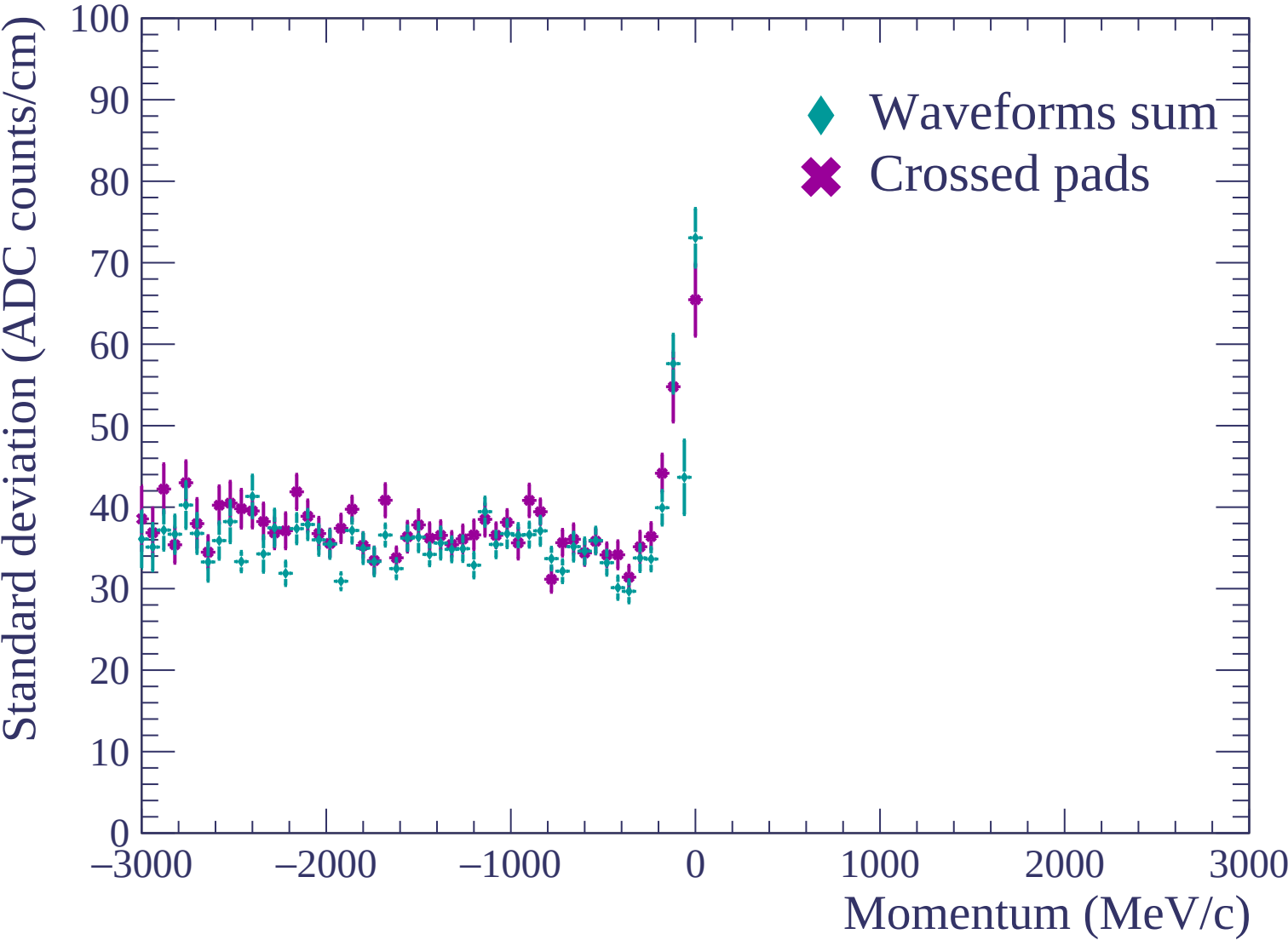


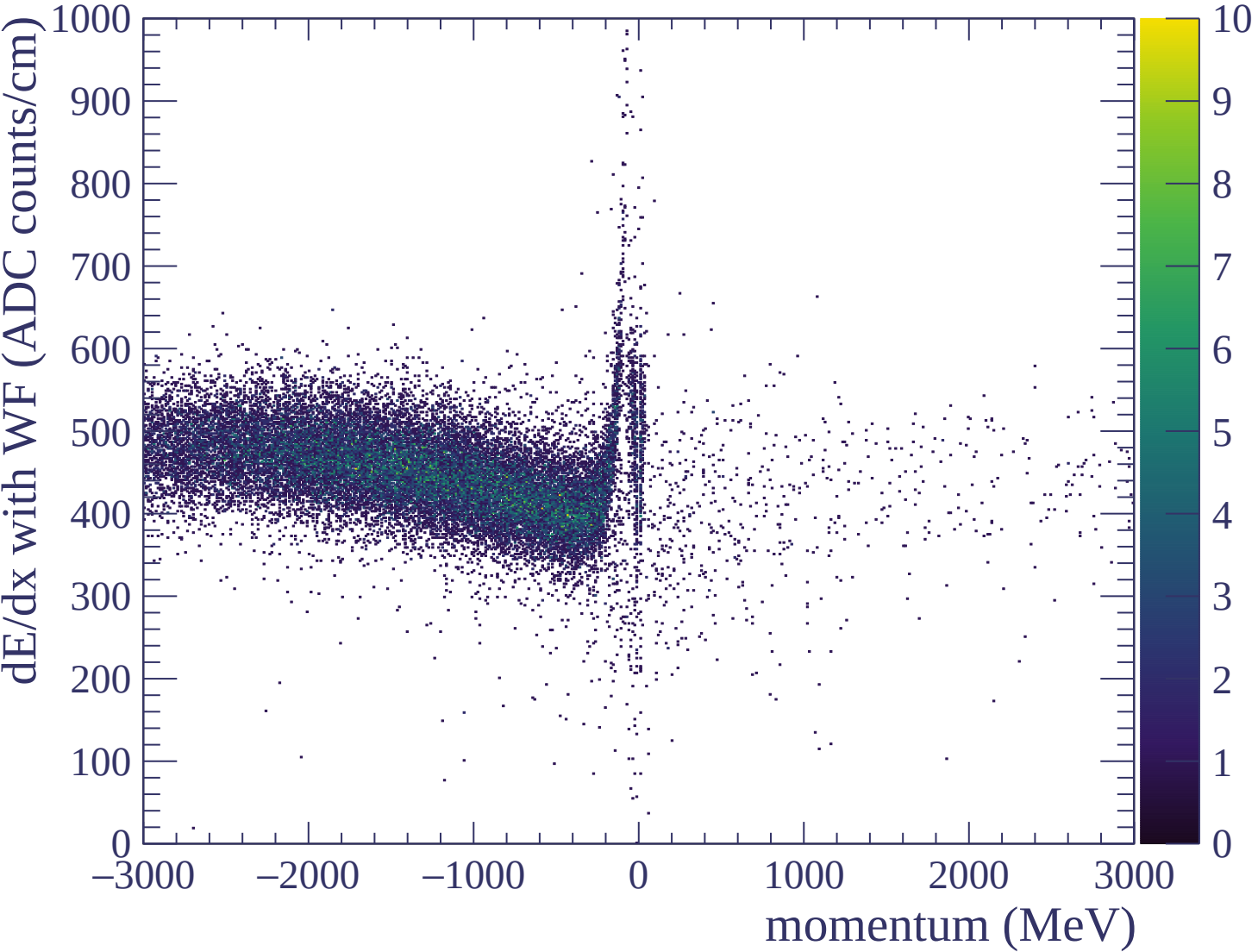


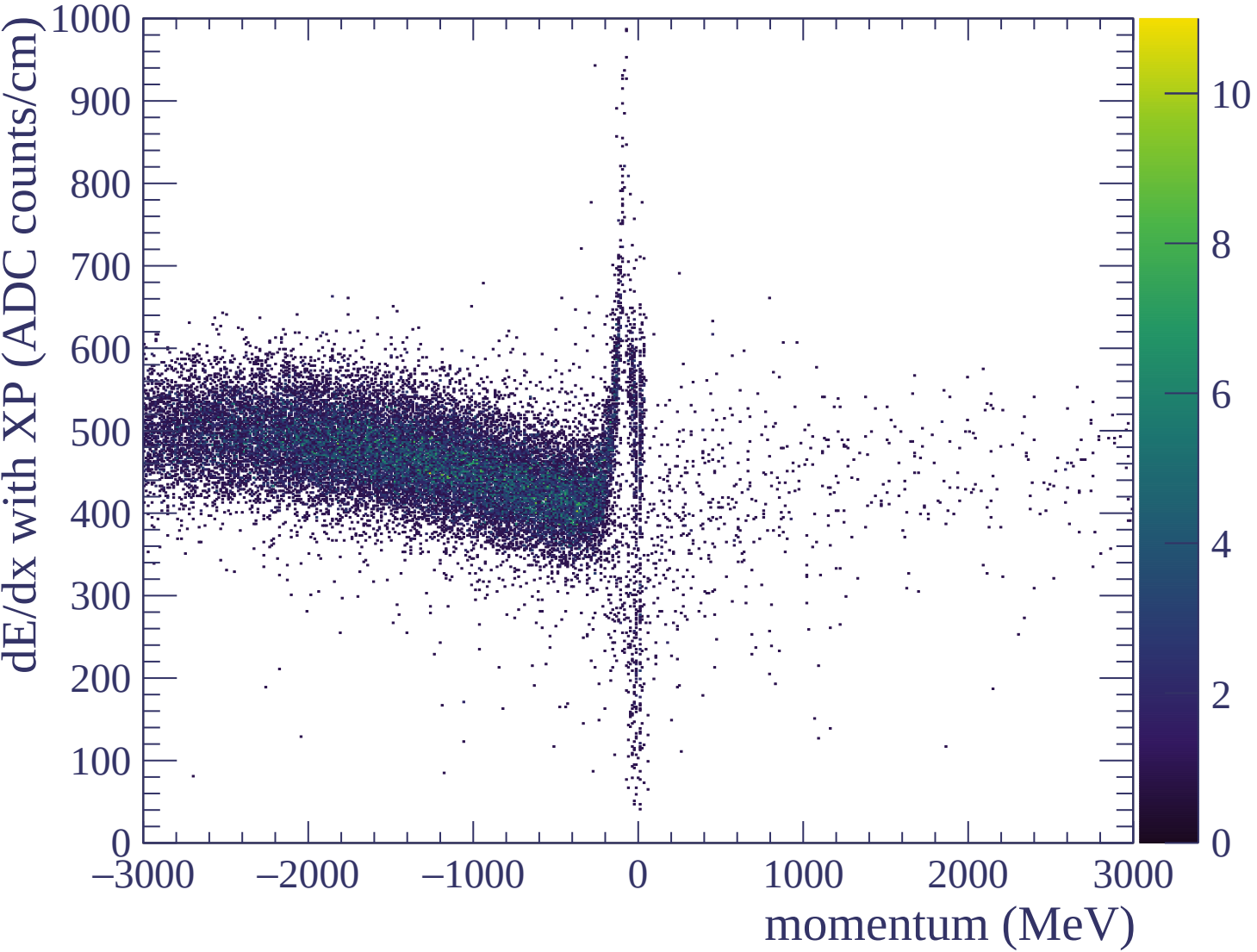


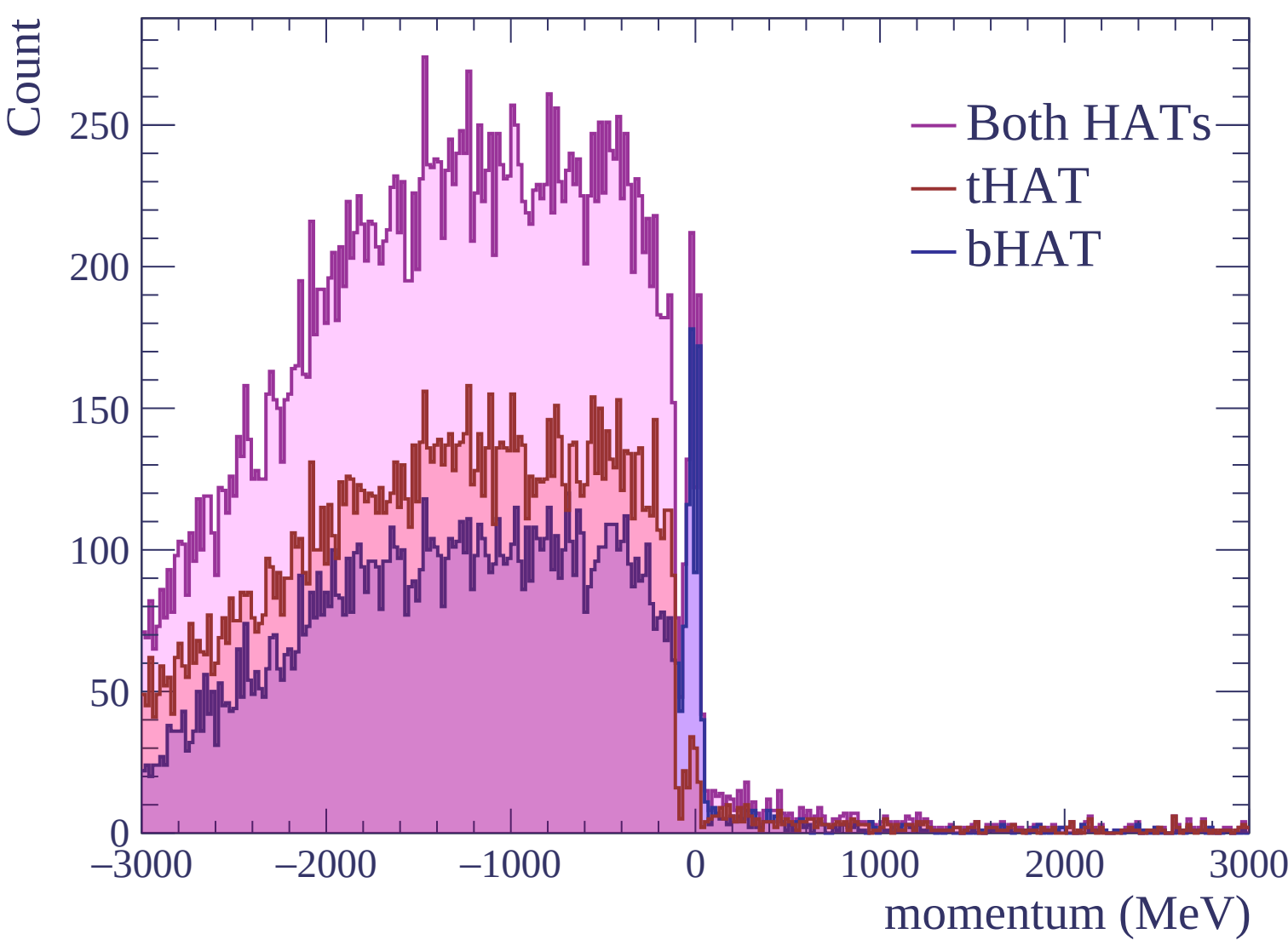


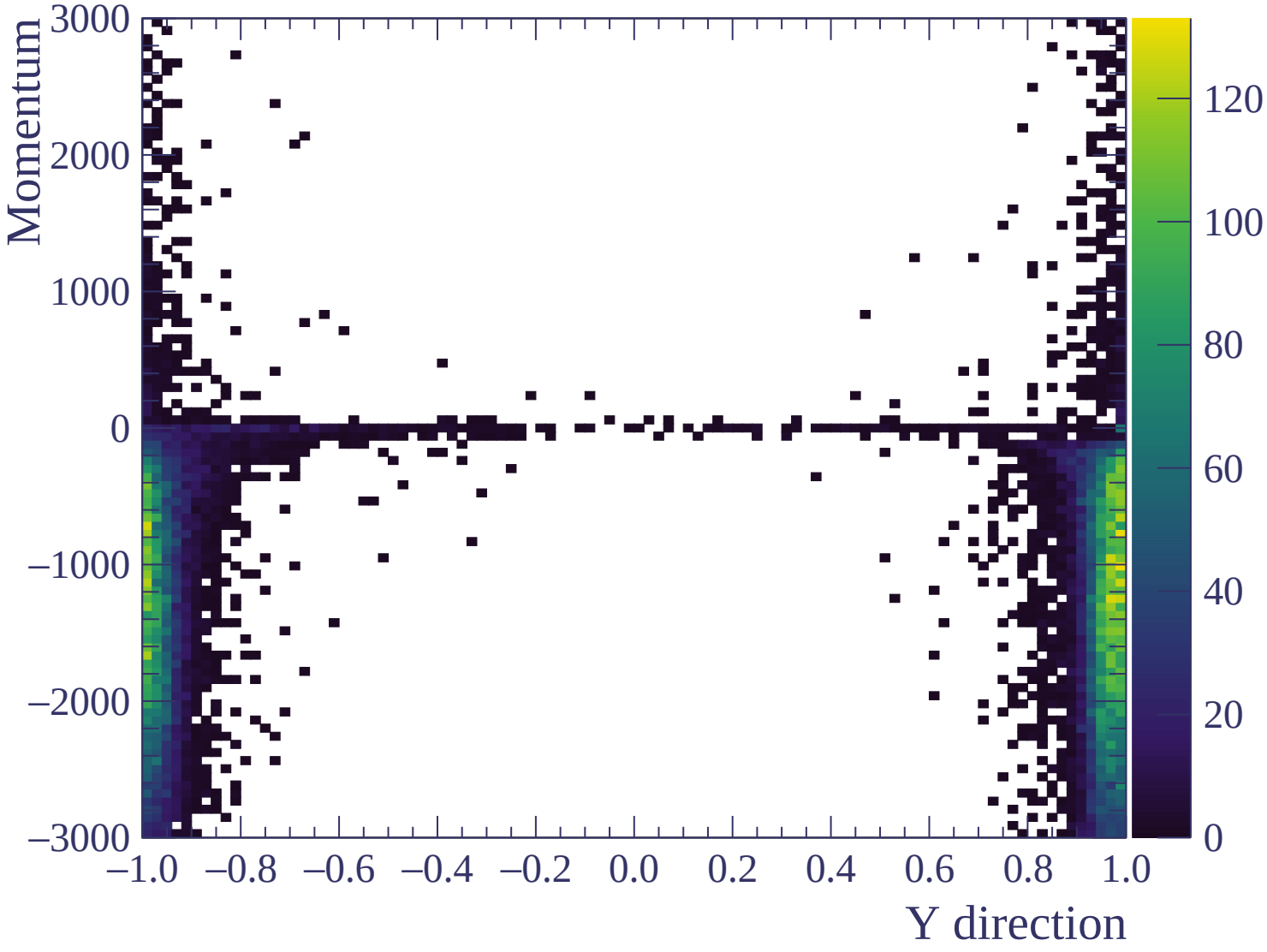


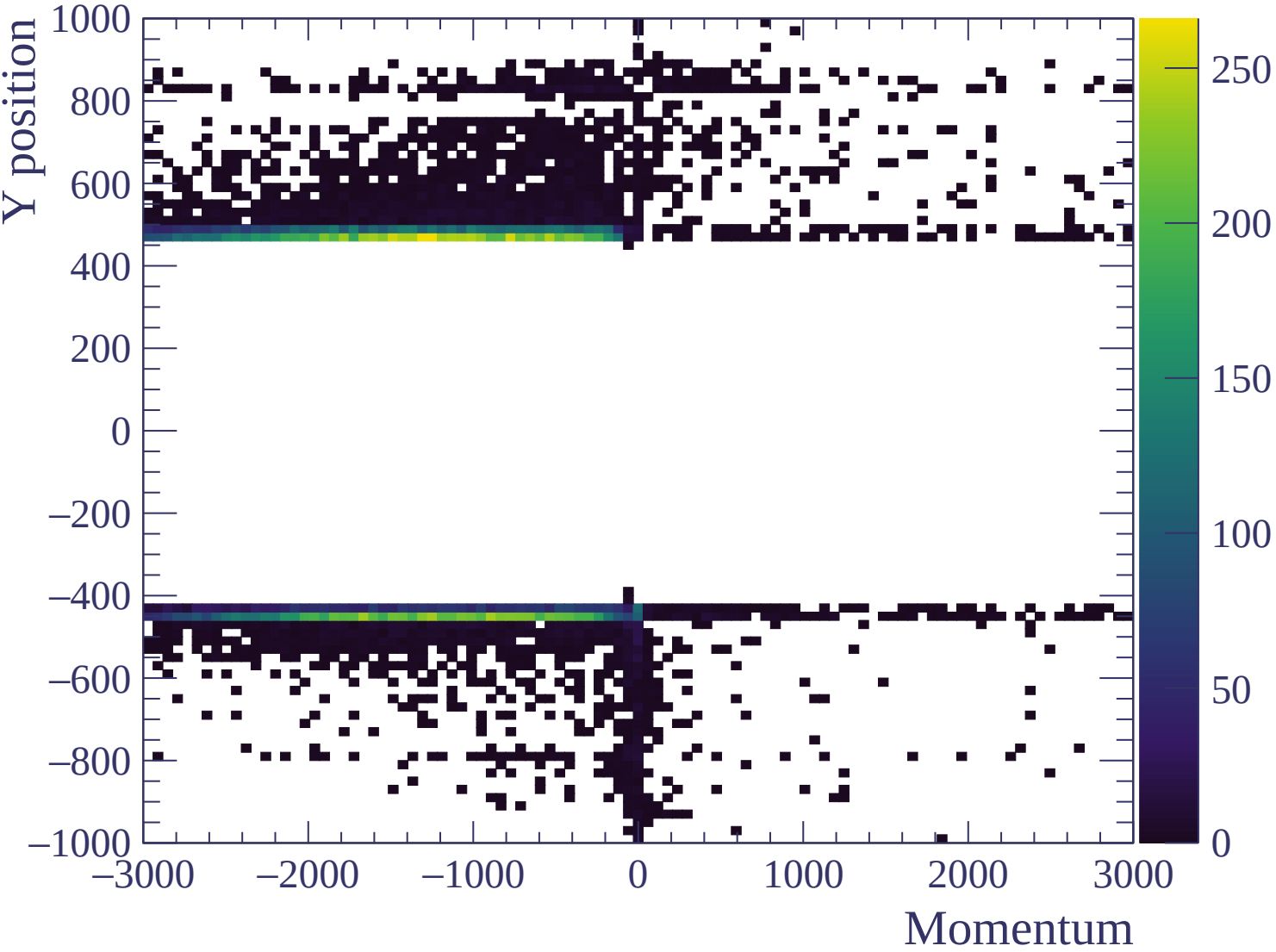


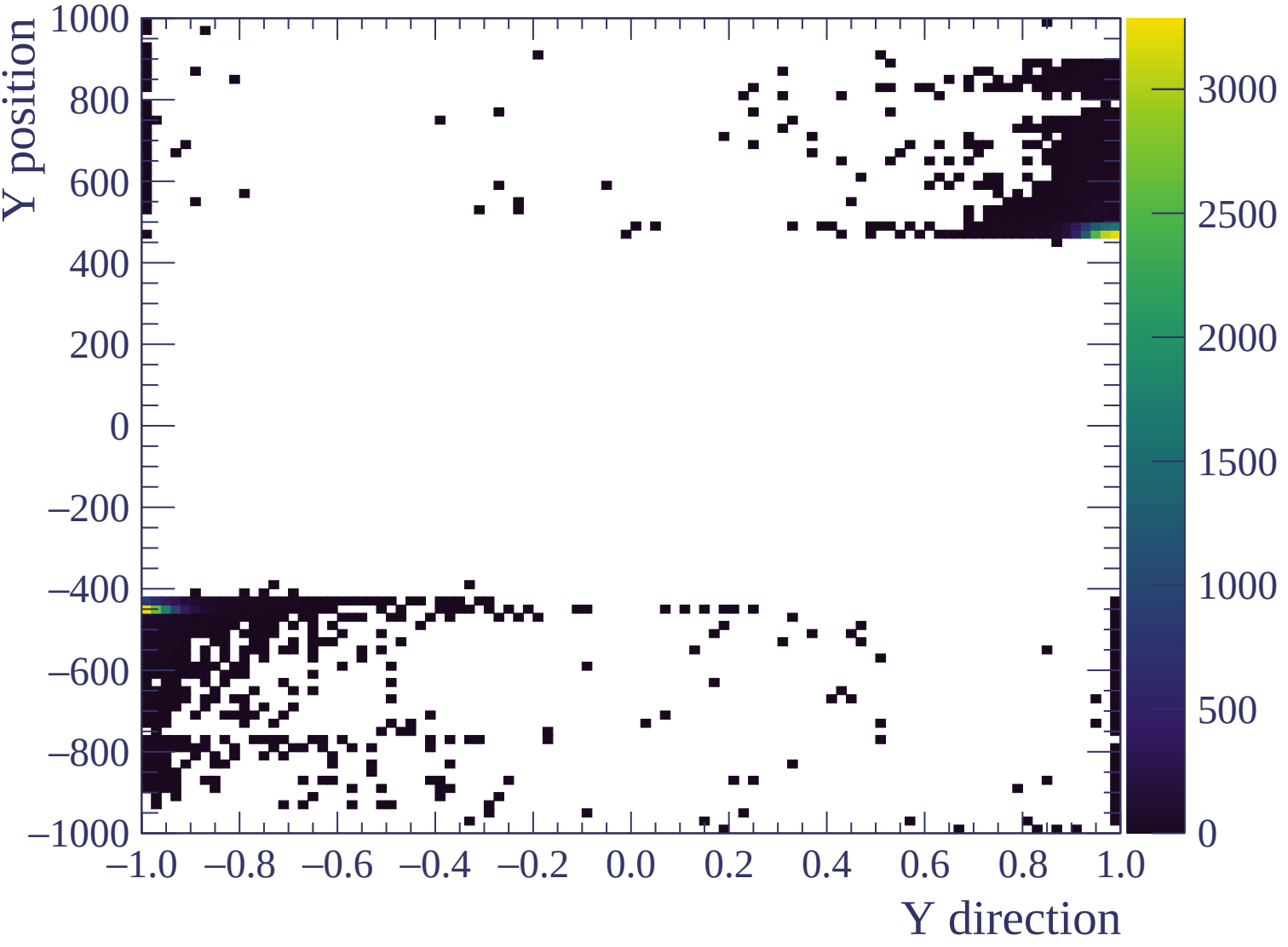


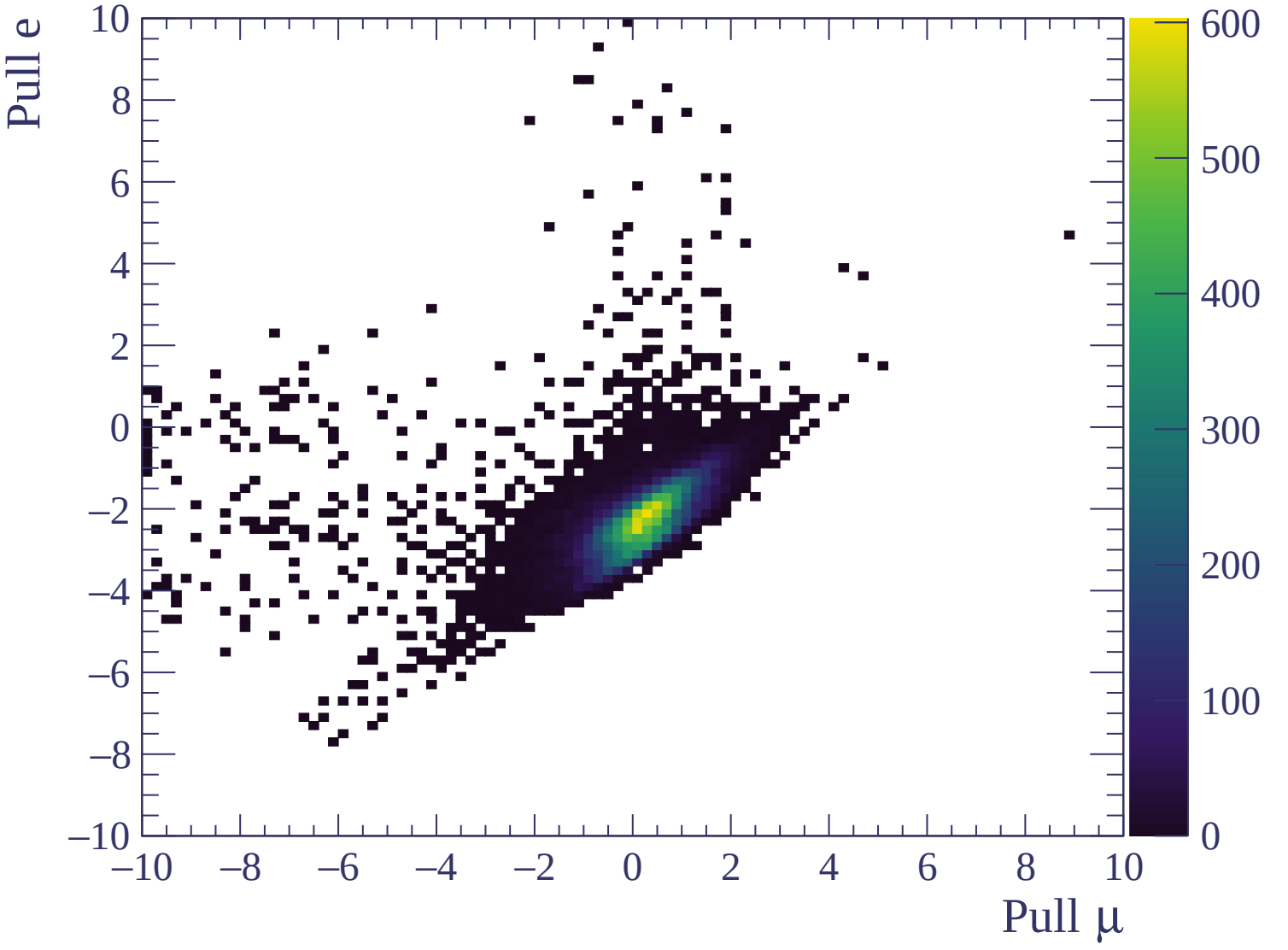






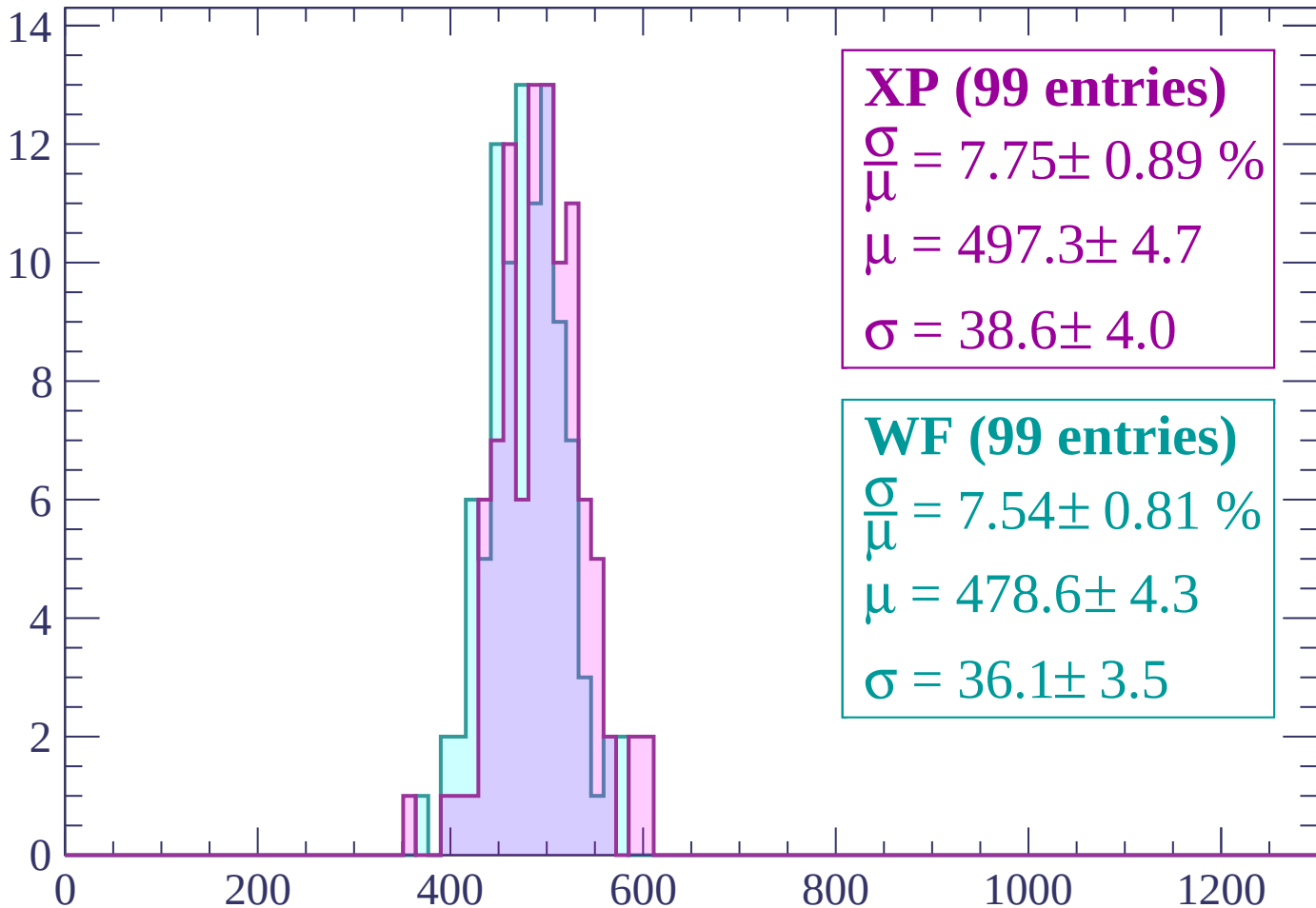






Energy loss | -3000 < p < -2940

Count



XP (99 entries)

$$\frac{\sigma}{\mu} = 7.75 \pm 0.89 \%$$

$$\mu = 497.3 \pm 4.7$$

$$\sigma = 38.6 \pm 4.0$$

WF (99 entries)

$$\frac{\sigma}{\mu} = 7.54 \pm 0.81 \%$$

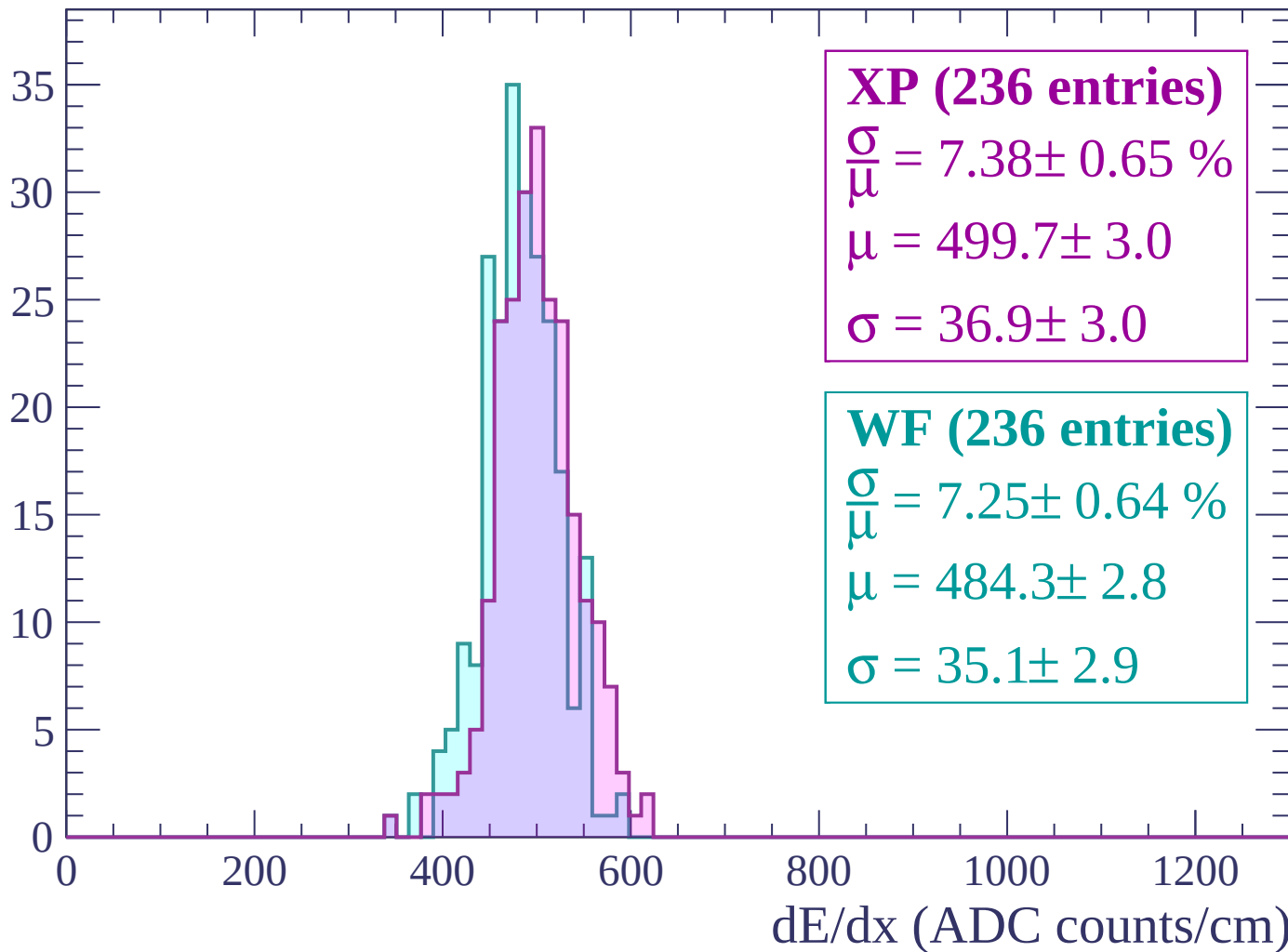
$$\mu = 478.6 \pm 4.3$$

$$\sigma = 36.1 \pm 3.5$$

dE/dx (ADC counts/cm)

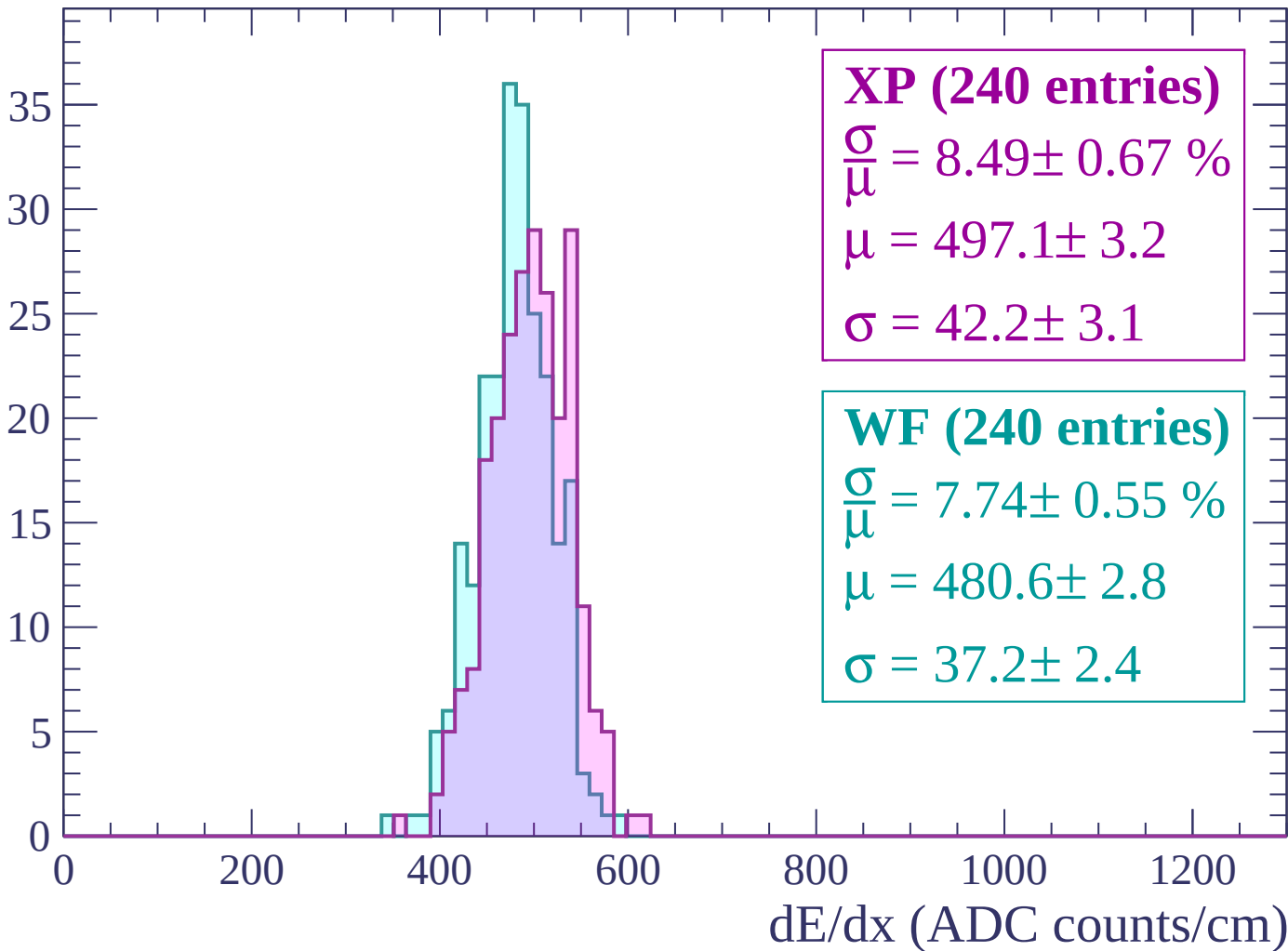
Energy loss | $-2940 < p < -2880$

Count



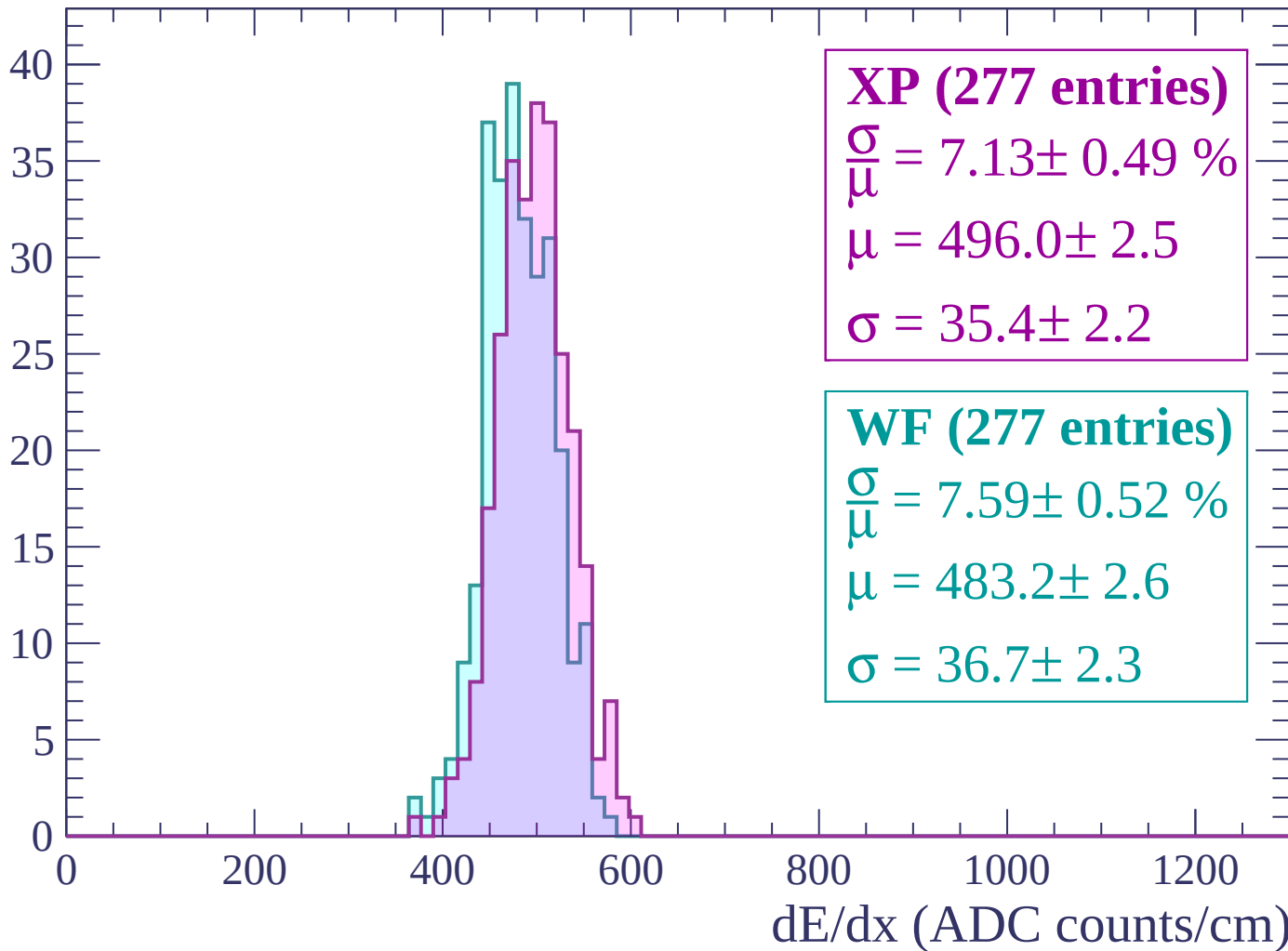
Energy loss | $-2880 < p < -2820$

Count



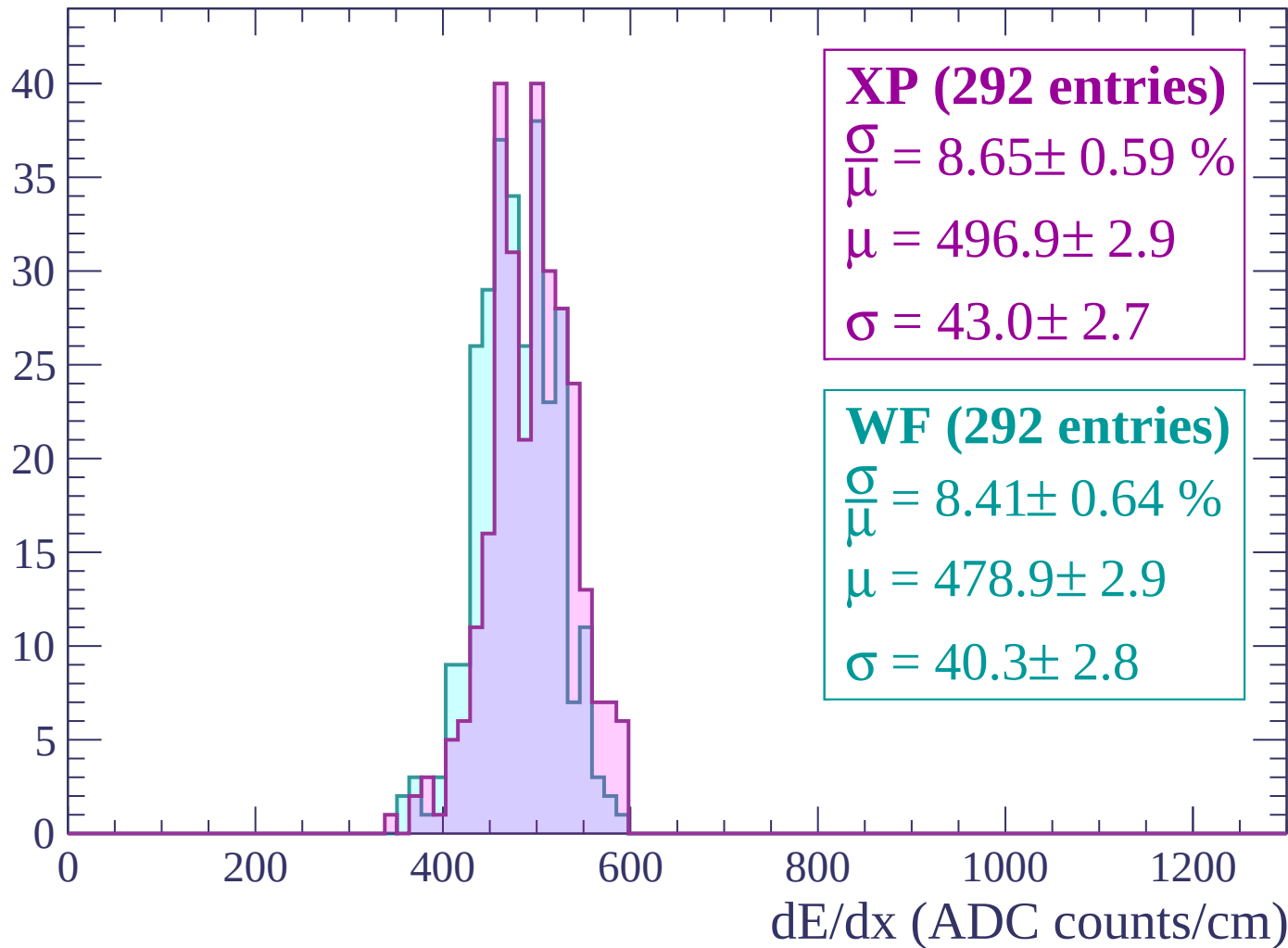
Energy loss | $-2820 < p < -2760$

Count



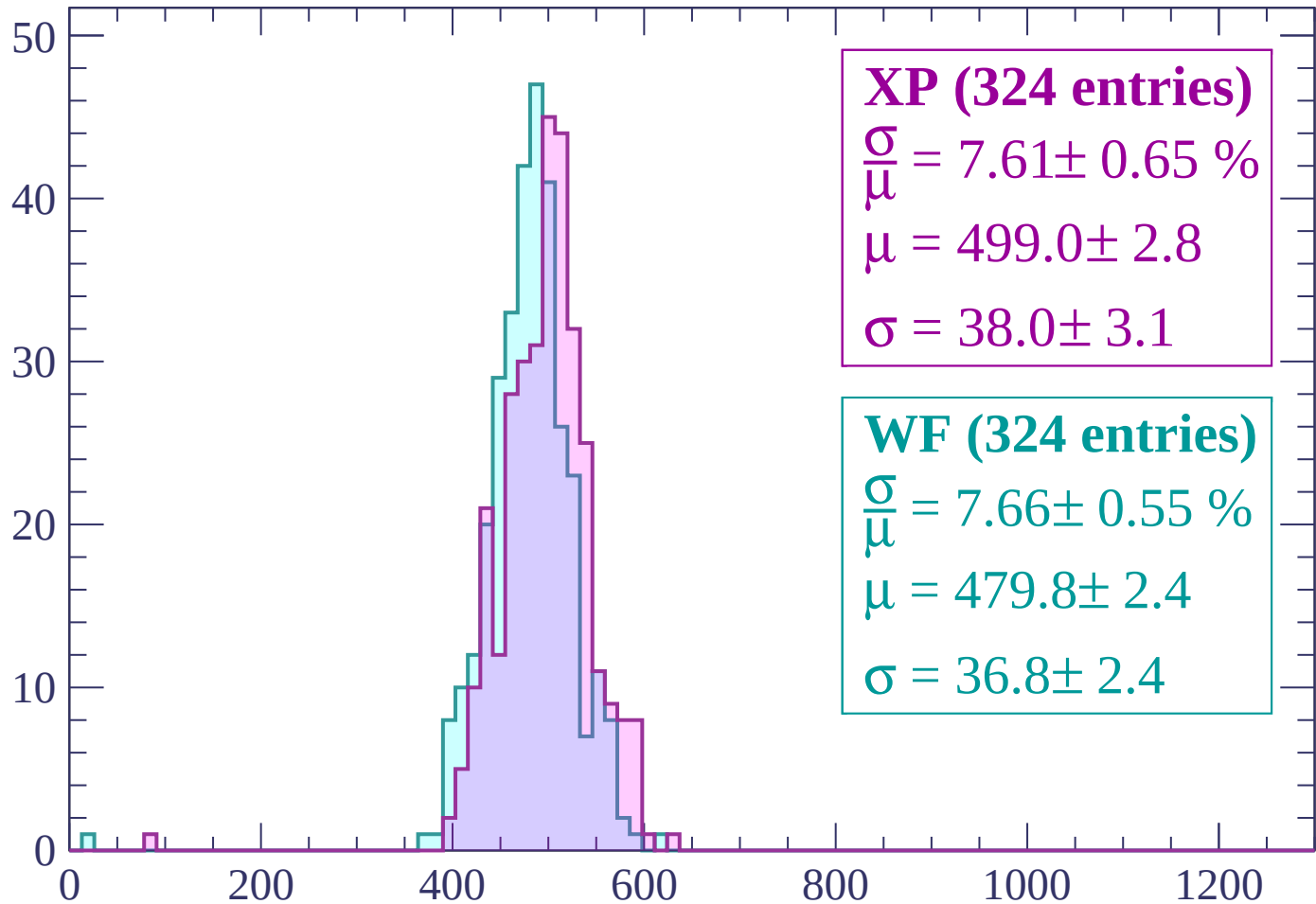
Energy loss | -2760 < p < -2700

Count



Energy loss | -2700 < p < -2640

Count



XP (324 entries)

$\frac{\sigma}{\mu} = 7.61 \pm 0.65 \%$

$\mu = 499.0 \pm 2.8$

$\sigma = 38.0 \pm 3.1$

WF (324 entries)

$\frac{\sigma}{\mu} = 7.66 \pm 0.55 \%$

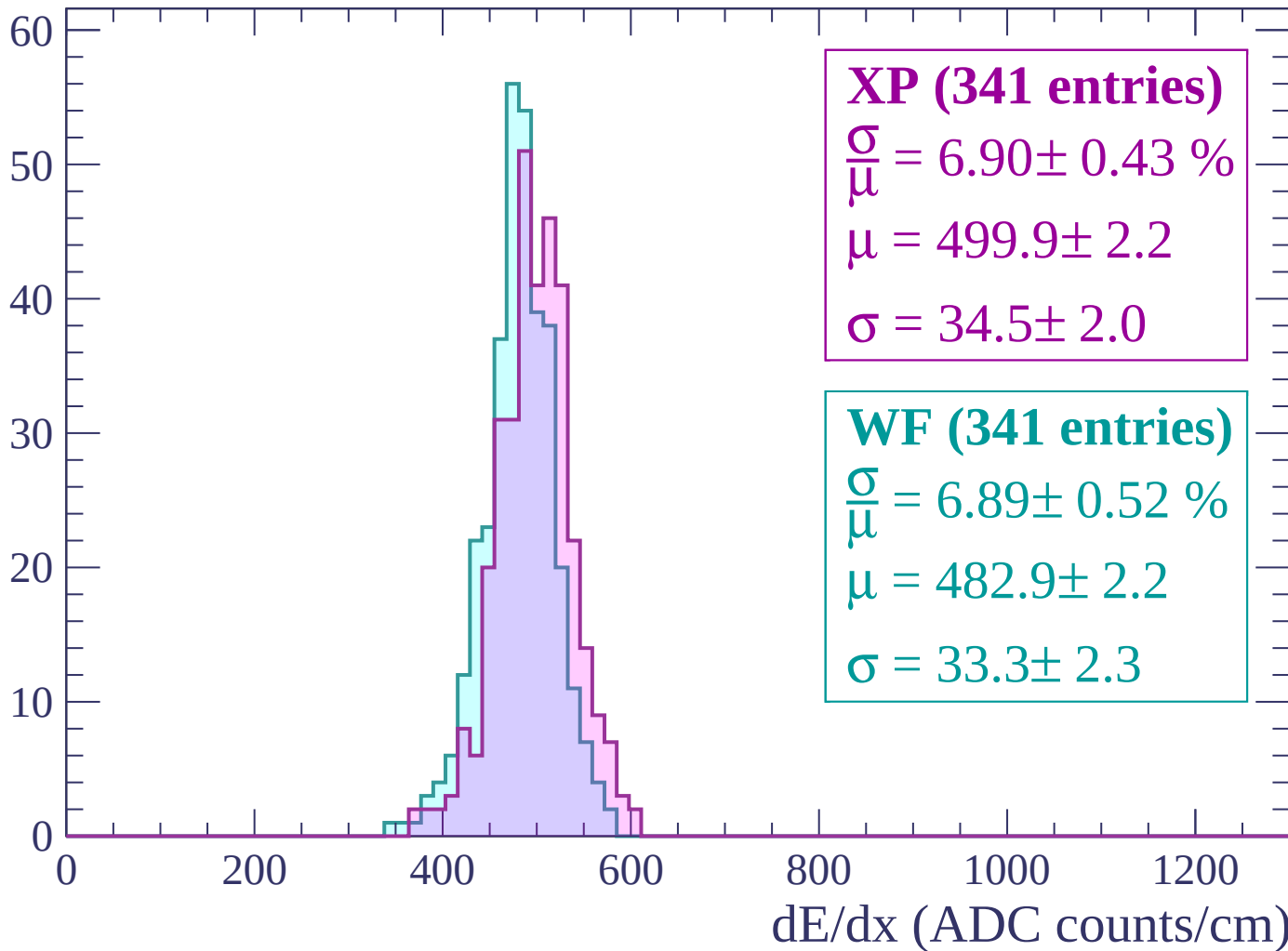
$\mu = 479.8 \pm 2.4$

$\sigma = 36.8 \pm 2.4$

dE/dx (ADC counts/cm)

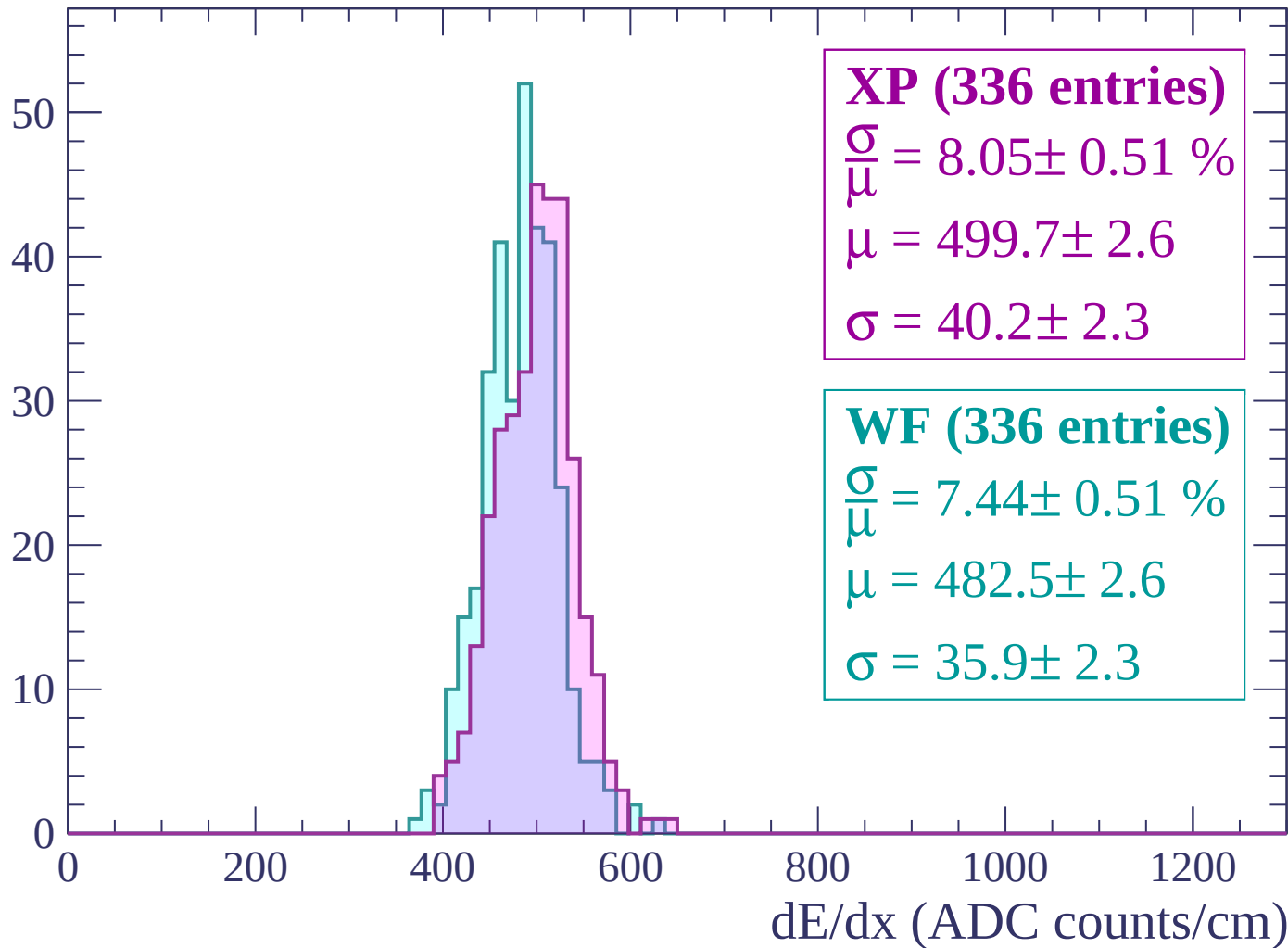
Energy loss | $-2640 < p < -2580$

Count



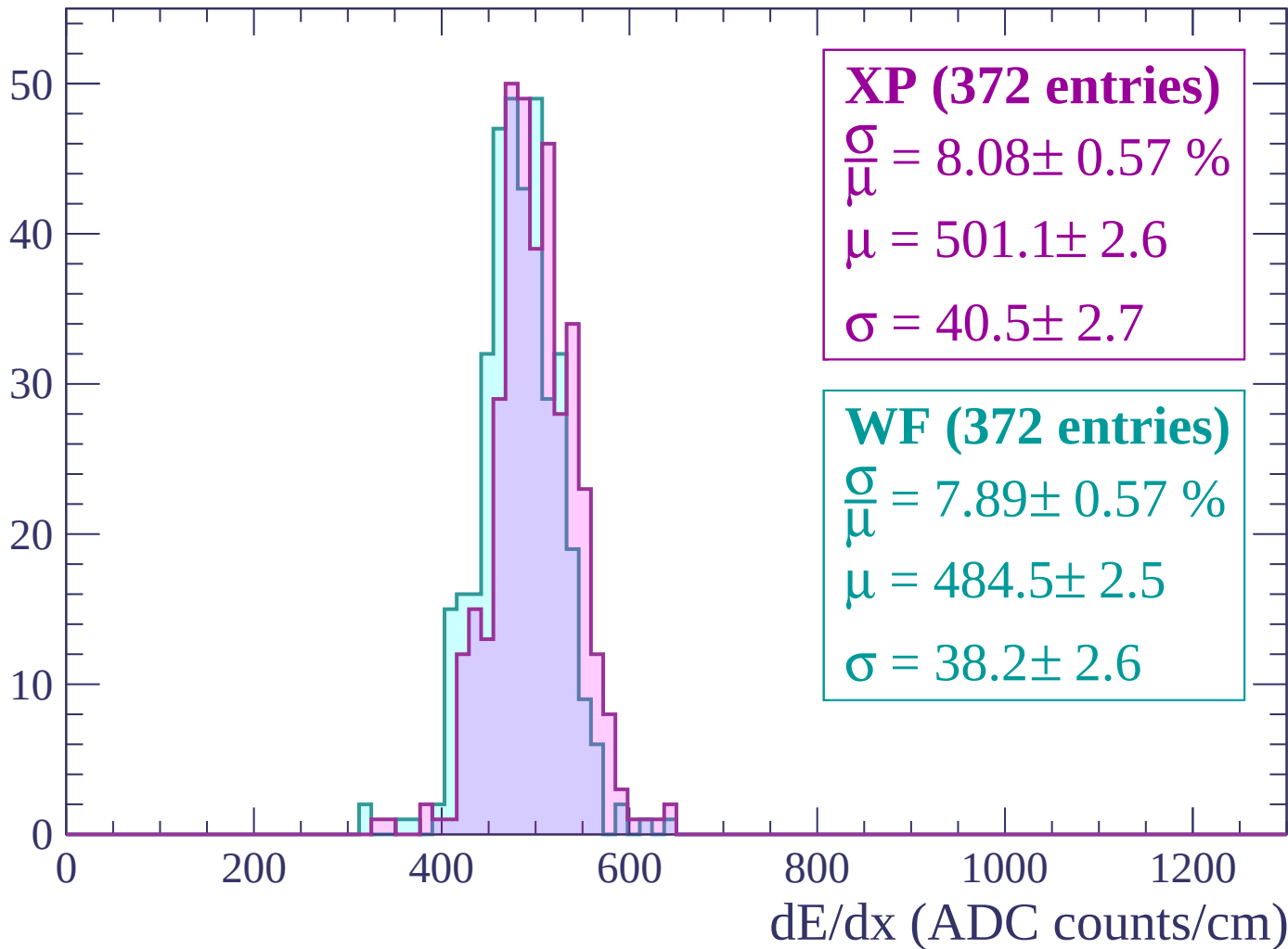
Energy loss | $-2580 < p < -2520$

Count



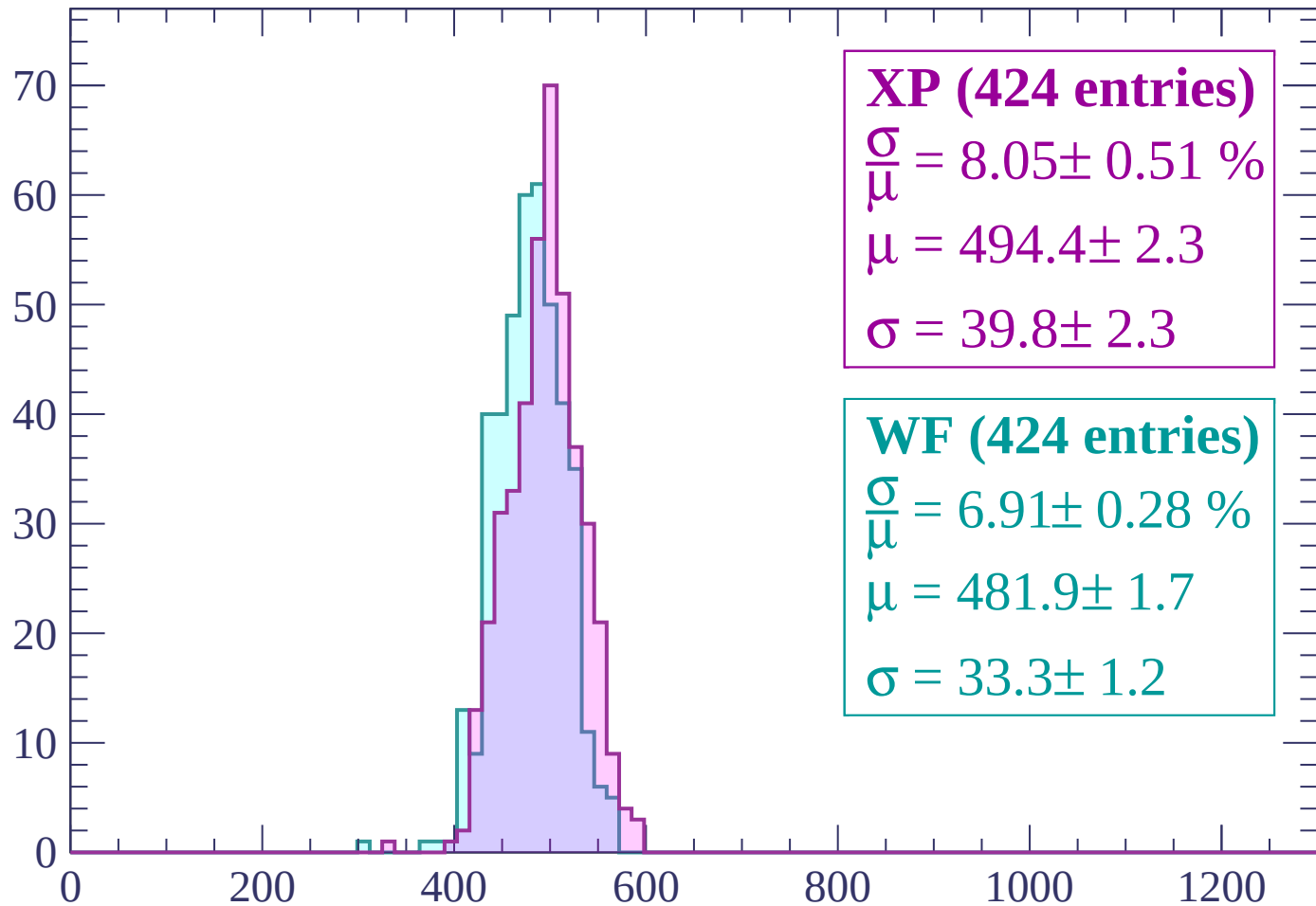
Energy loss | $-2520 < p < -2460$

Count



Energy loss | $-2460 < p < -2400$

Count



XP (424 entries)

$\frac{\sigma}{\mu} = 8.05 \pm 0.51 \%$

$\mu = 494.4 \pm 2.3$

$\sigma = 39.8 \pm 2.3$

WF (424 entries)

$\frac{\sigma}{\mu} = 6.91 \pm 0.28 \%$

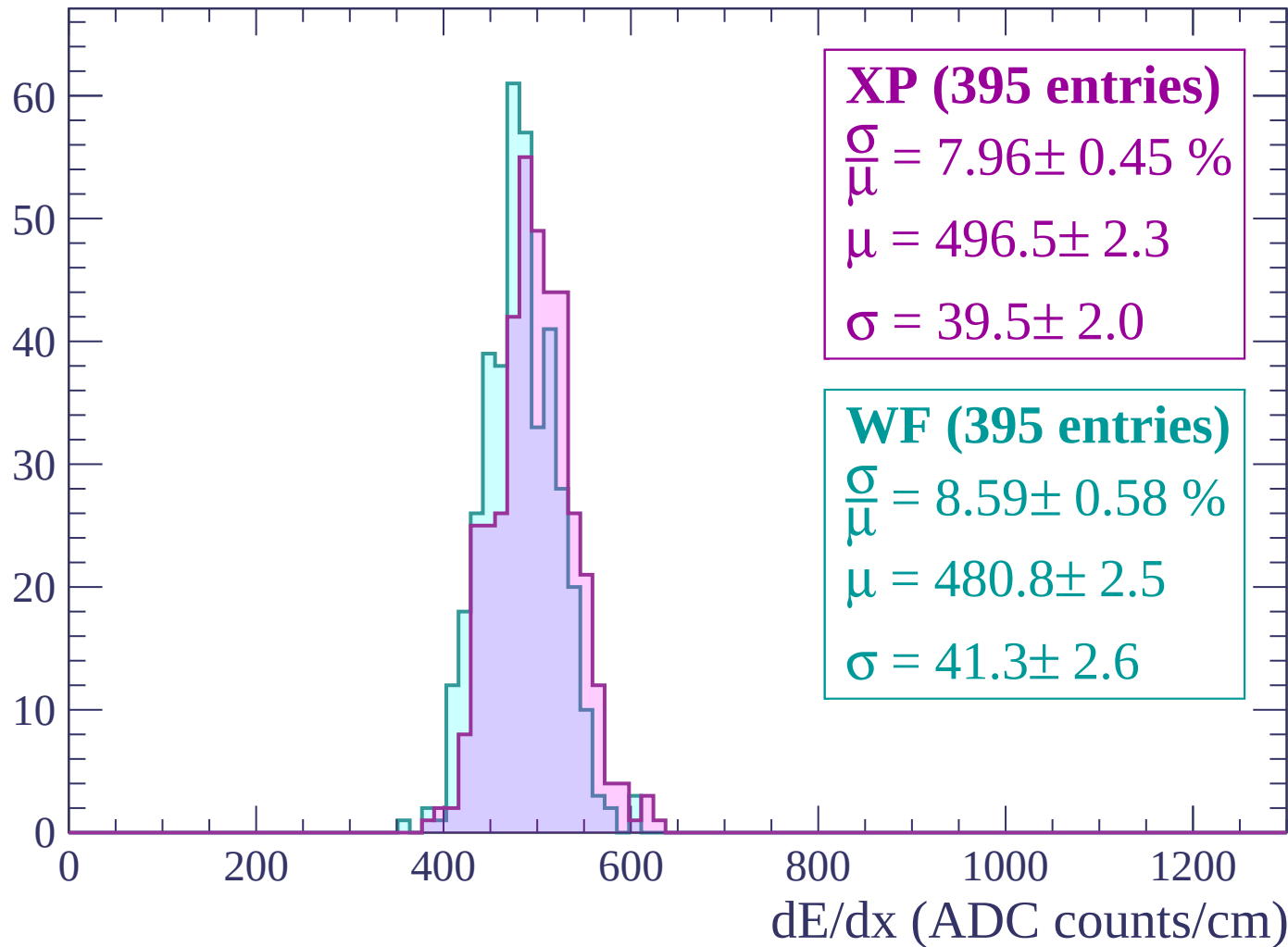
$\mu = 481.9 \pm 1.7$

$\sigma = 33.3 \pm 1.2$

dE/dx (ADC counts/cm)

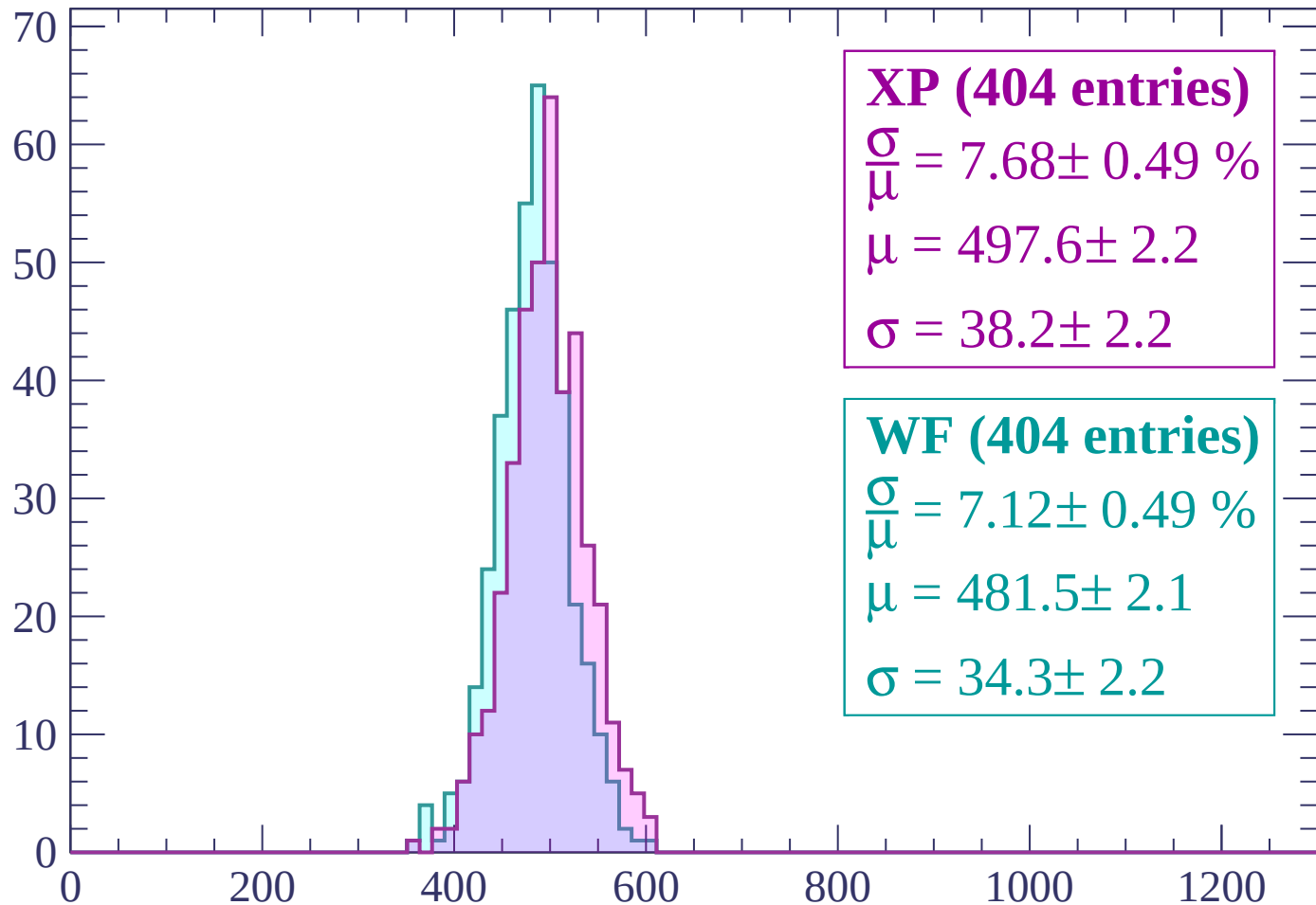
Energy loss | $-2400 < p < -2340$

Count



Energy loss | $-2340 < p < -2280$

Count



XP (404 entries)

$$\frac{\sigma}{\mu} = 7.68 \pm 0.49 \%$$

$$\mu = 497.6 \pm 2.2$$

$$\sigma = 38.2 \pm 2.2$$

WF (404 entries)

$$\frac{\sigma}{\mu} = 7.12 \pm 0.49 \%$$

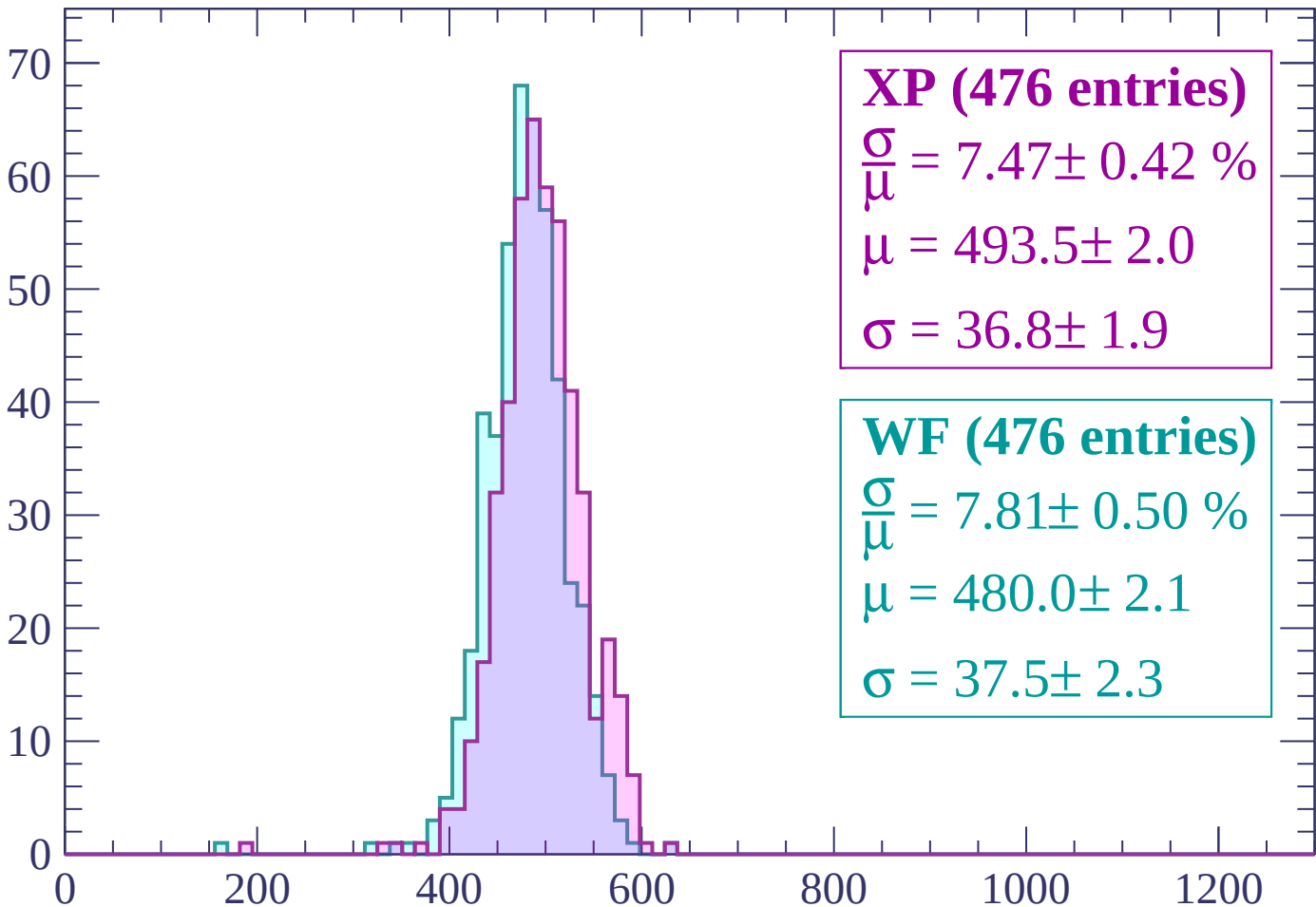
$$\mu = 481.5 \pm 2.1$$

$$\sigma = 34.3 \pm 2.2$$

dE/dx (ADC counts/cm)

Energy loss | $-2280 < p < -2220$

Count



XP (476 entries)

$$\frac{\sigma}{\mu} = 7.47 \pm 0.42 \%$$

$$\mu = 493.5 \pm 2.0$$

$$\sigma = 36.8 \pm 1.9$$

WF (476 entries)

$$\frac{\sigma}{\mu} = 7.81 \pm 0.50 \%$$

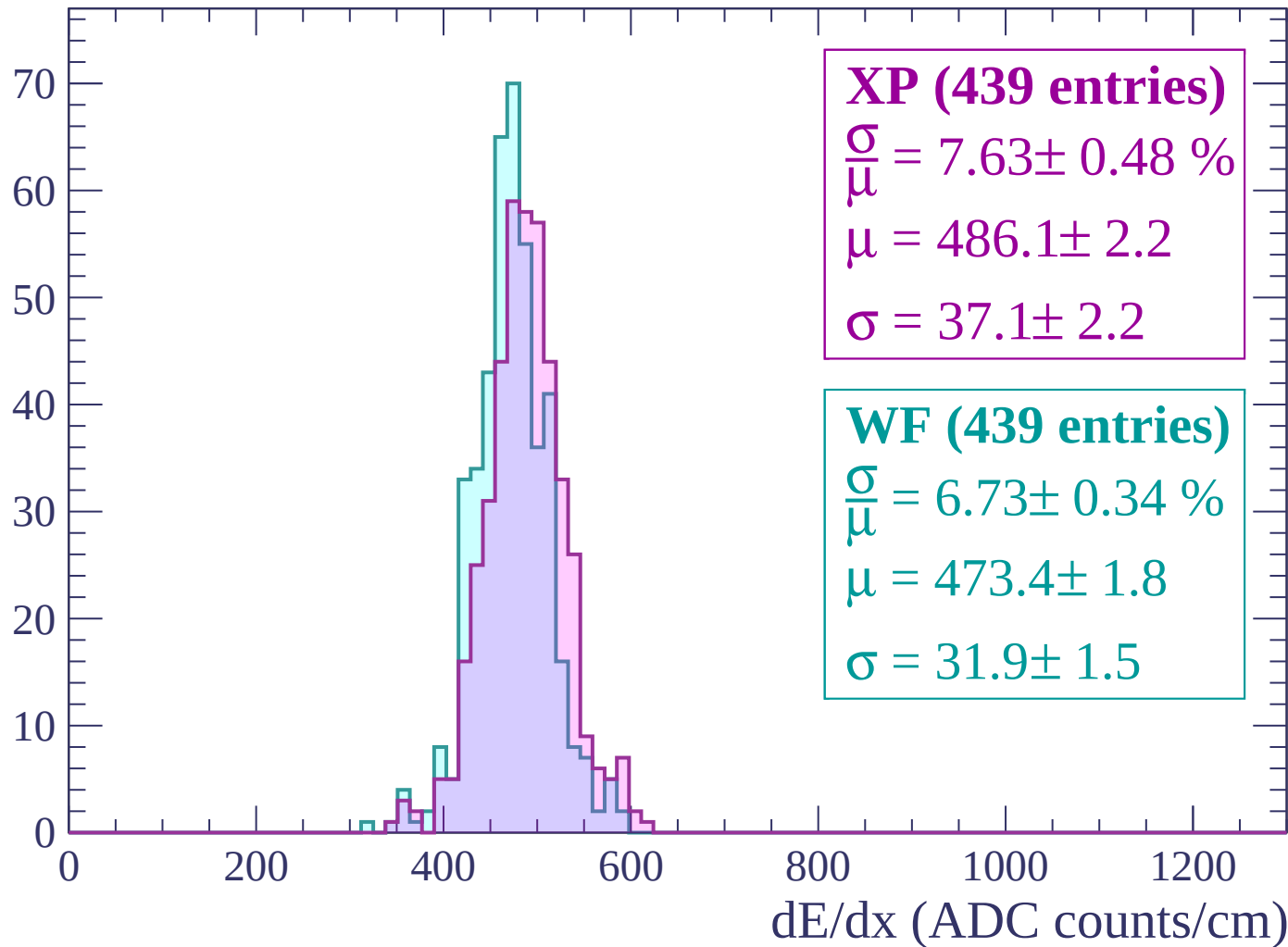
$$\mu = 480.0 \pm 2.1$$

$$\sigma = 37.5 \pm 2.3$$

dE/dx (ADC counts/cm)

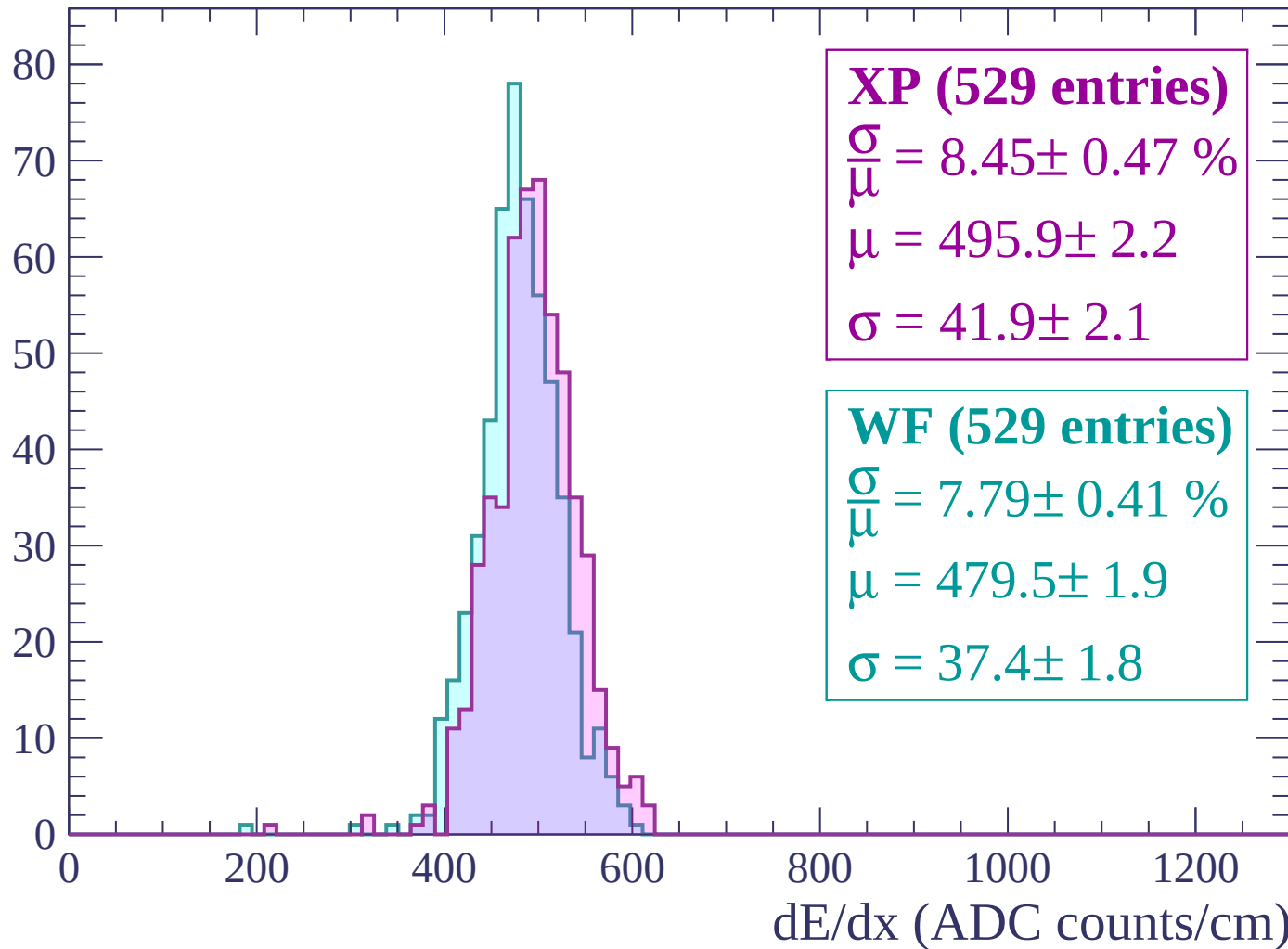
Energy loss | -2220 < p < -2160

Count



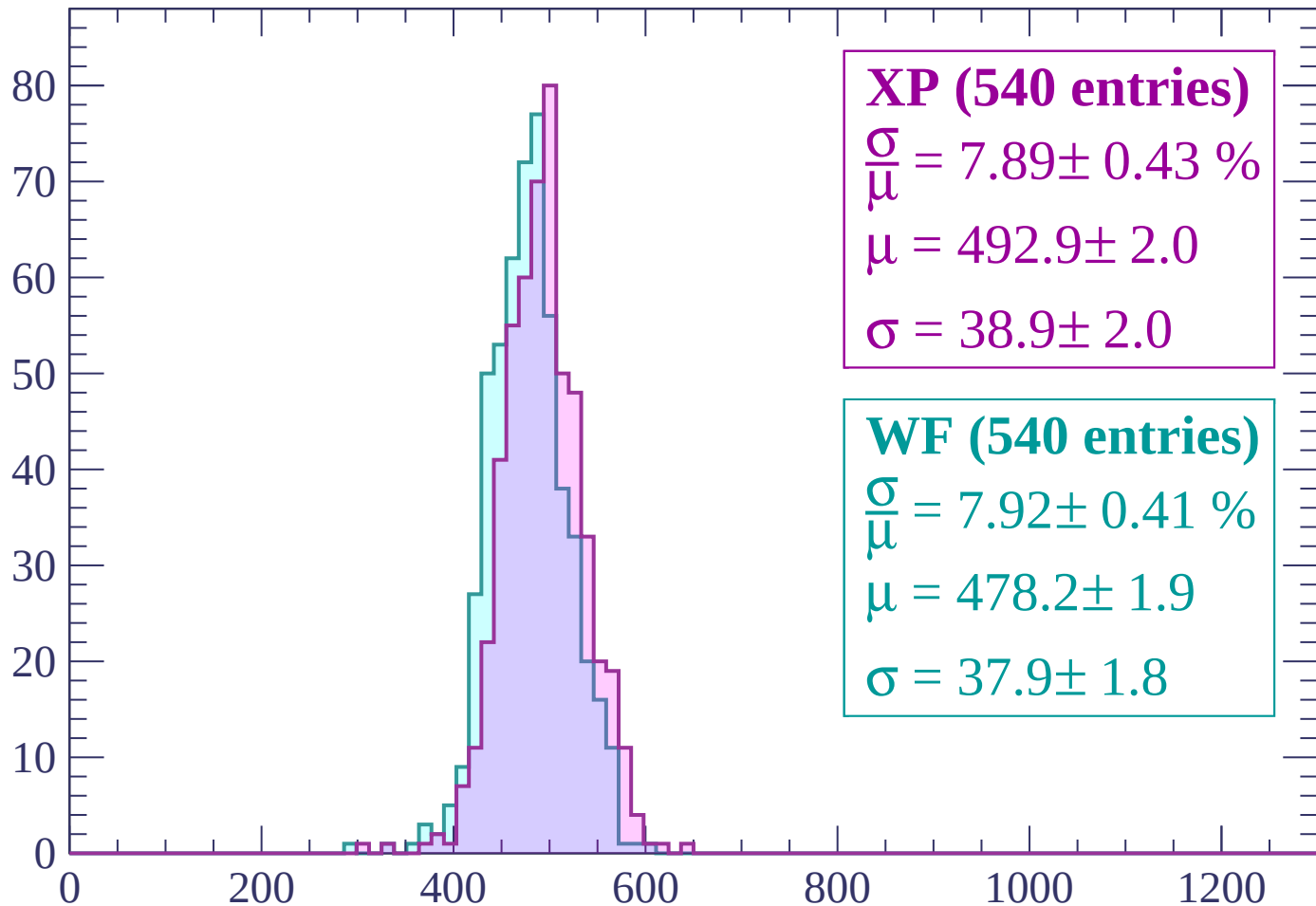
Energy loss | -2160 < p < -2100

Count



Energy loss | $-2100 < p < -2040$

Count



XP (540 entries)

$\frac{\sigma}{\mu} = 7.89 \pm 0.43 \%$

$\mu = 492.9 \pm 2.0$

$\sigma = 38.9 \pm 2.0$

WF (540 entries)

$\frac{\sigma}{\mu} = 7.92 \pm 0.41 \%$

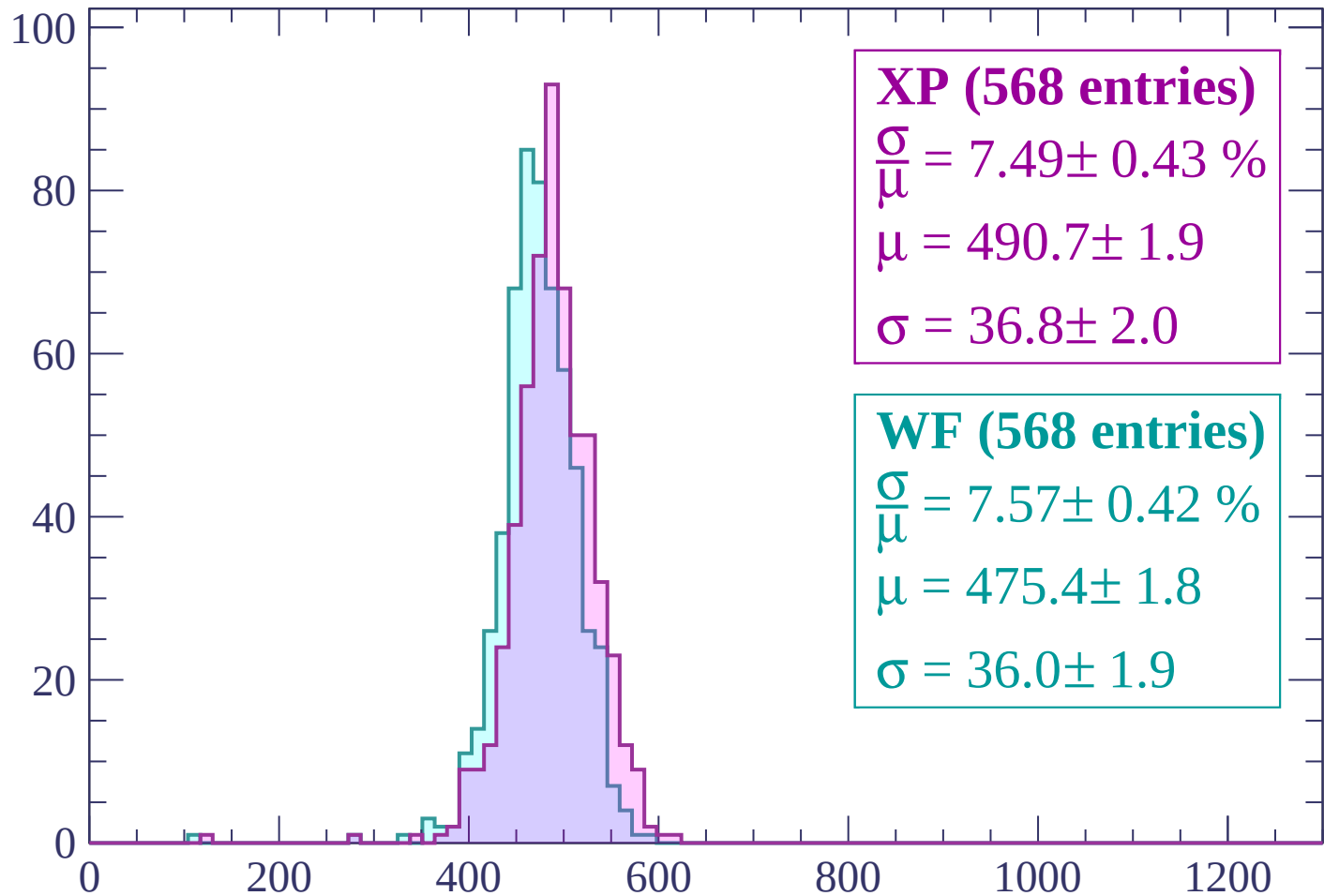
$\mu = 478.2 \pm 1.9$

$\sigma = 37.9 \pm 1.8$

dE/dx (ADC counts/cm)

Energy loss | -2040 < p < -1980

Count



XP (568 entries)

$\frac{\sigma}{\mu} = 7.49 \pm 0.43 \%$

$\mu = 490.7 \pm 1.9$

$\sigma = 36.8 \pm 2.0$

WF (568 entries)

$\frac{\sigma}{\mu} = 7.57 \pm 0.42 \%$

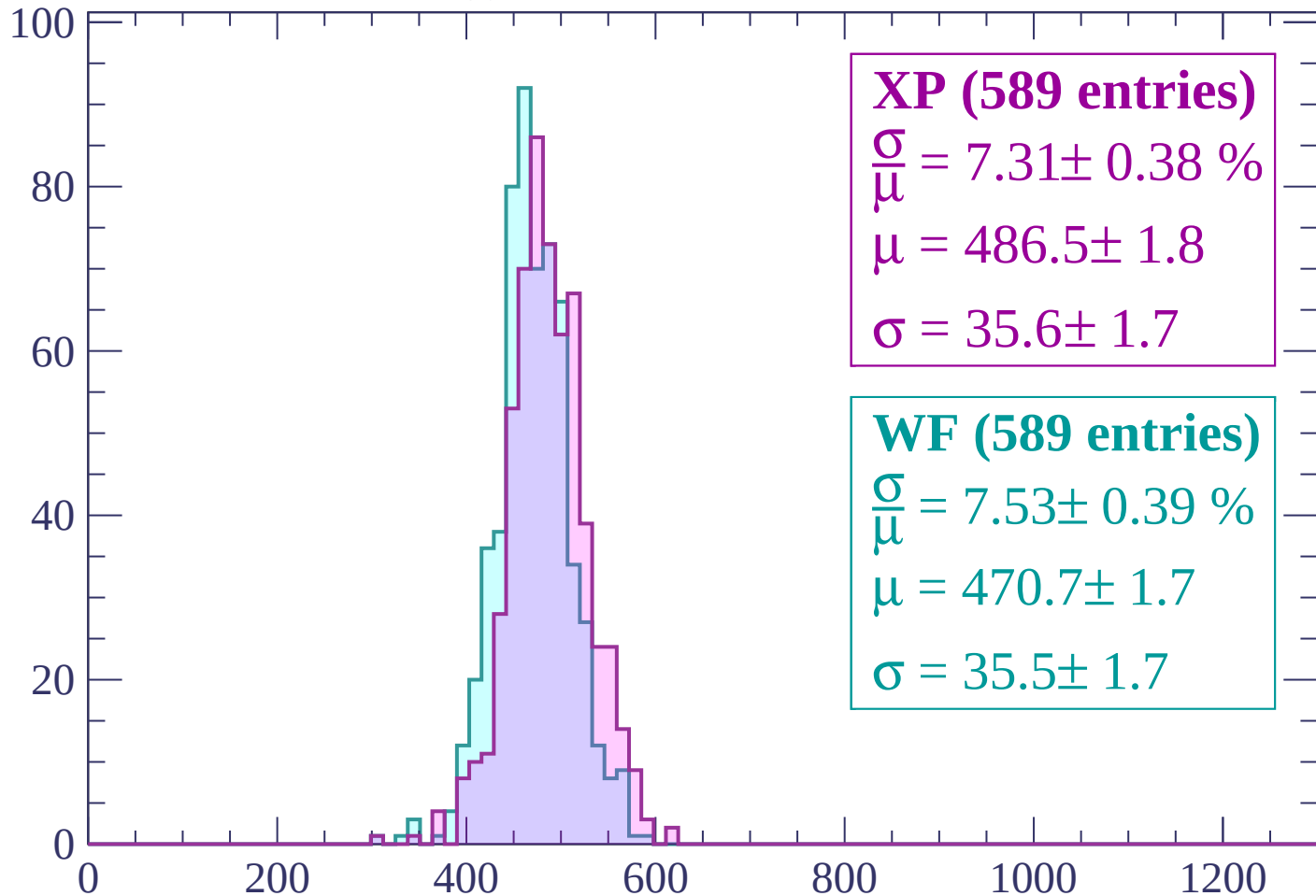
$\mu = 475.4 \pm 1.8$

$\sigma = 36.0 \pm 1.9$

dE/dx (ADC counts/cm)

Energy loss | -1980 < p < -1920

Count



XP (589 entries)

$\frac{\sigma}{\mu} = 7.31 \pm 0.38 \%$

$\mu = 486.5 \pm 1.8$

$\sigma = 35.6 \pm 1.7$

WF (589 entries)

$\frac{\sigma}{\mu} = 7.53 \pm 0.39 \%$

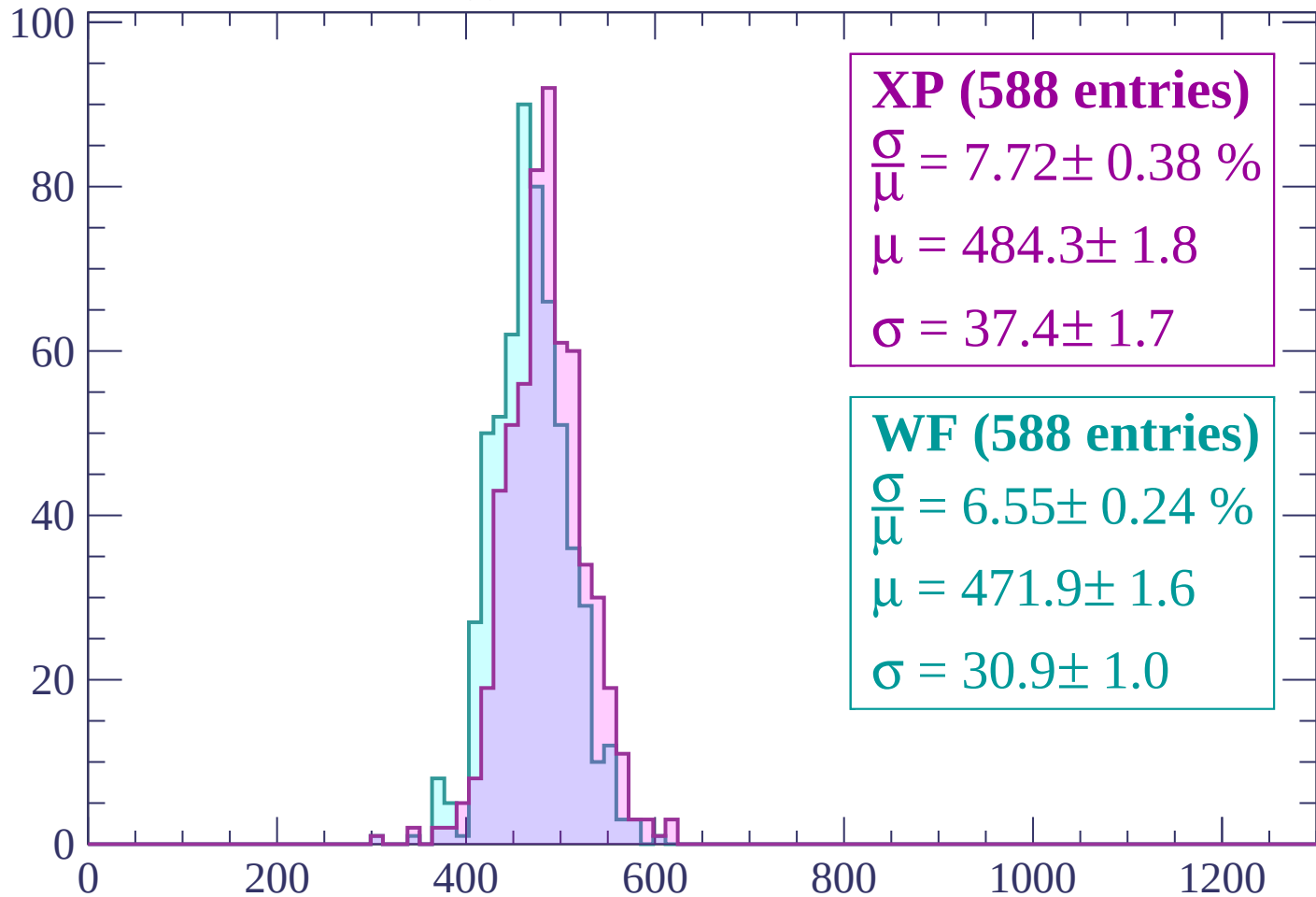
$\mu = 470.7 \pm 1.7$

$\sigma = 35.5 \pm 1.7$

dE/dx (ADC counts/cm)

Energy loss | $-1920 < p < -1860$

Count



XP (588 entries)

$$\frac{\sigma}{\mu} = 7.72 \pm 0.38 \%$$

$$\mu = 484.3 \pm 1.8$$

$$\sigma = 37.4 \pm 1.7$$

WF (588 entries)

$$\frac{\sigma}{\mu} = 6.55 \pm 0.24 \%$$

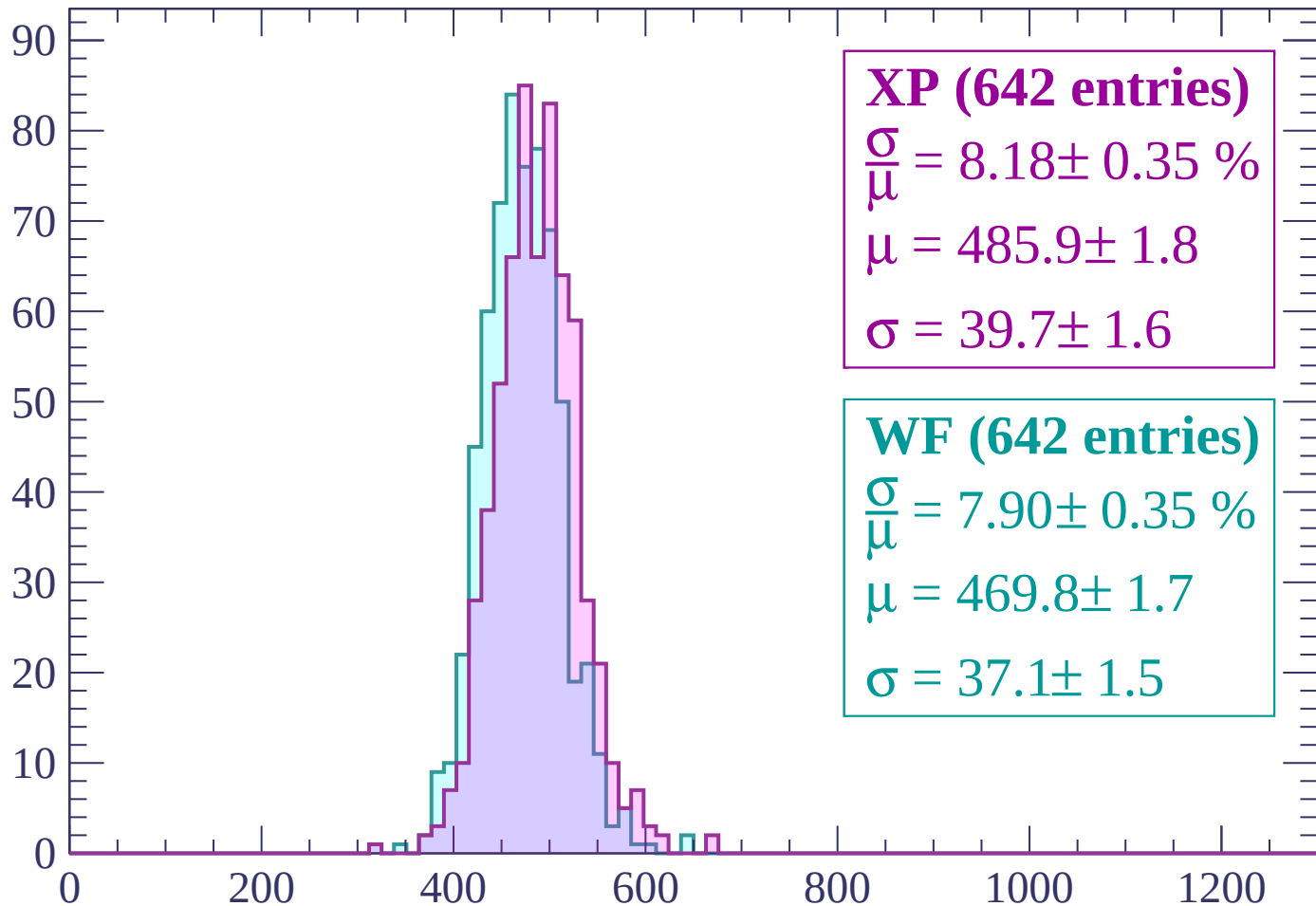
$$\mu = 471.9 \pm 1.6$$

$$\sigma = 30.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-1860 < p < -1800$

Count



XP (642 entries)

$$\frac{\sigma}{\mu} = 8.18 \pm 0.35 \%$$

$$\mu = 485.9 \pm 1.8$$

$$\sigma = 39.7 \pm 1.6$$

WF (642 entries)

$$\frac{\sigma}{\mu} = 7.90 \pm 0.35 \%$$

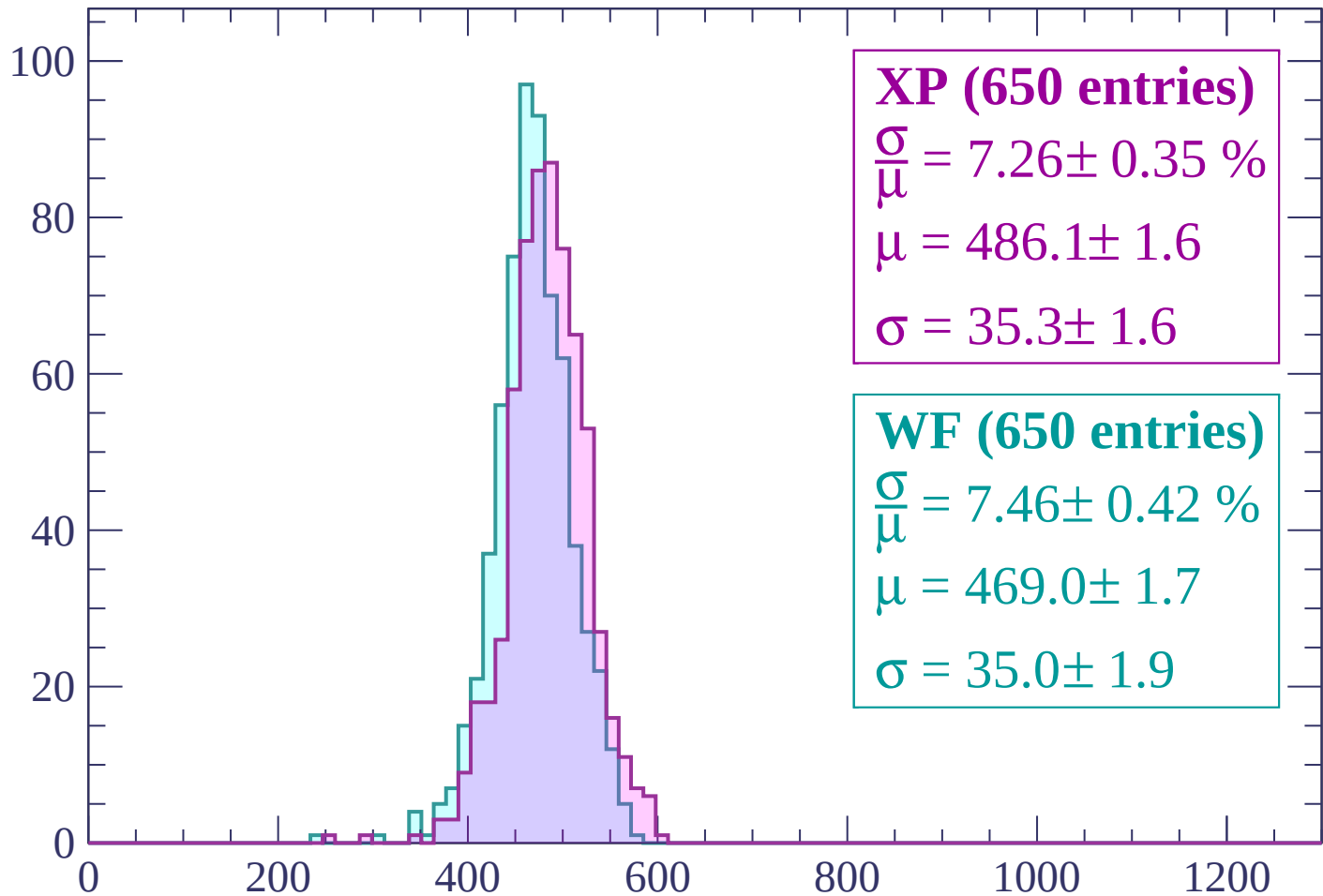
$$\mu = 469.8 \pm 1.7$$

$$\sigma = 37.1 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1800 < p < -1740$

Count



XP (650 entries)

$$\frac{\sigma}{\mu} = 7.26 \pm 0.35 \%$$

$$\mu = 486.1 \pm 1.6$$

$$\sigma = 35.3 \pm 1.6$$

WF (650 entries)

$$\frac{\sigma}{\mu} = 7.46 \pm 0.42 \%$$

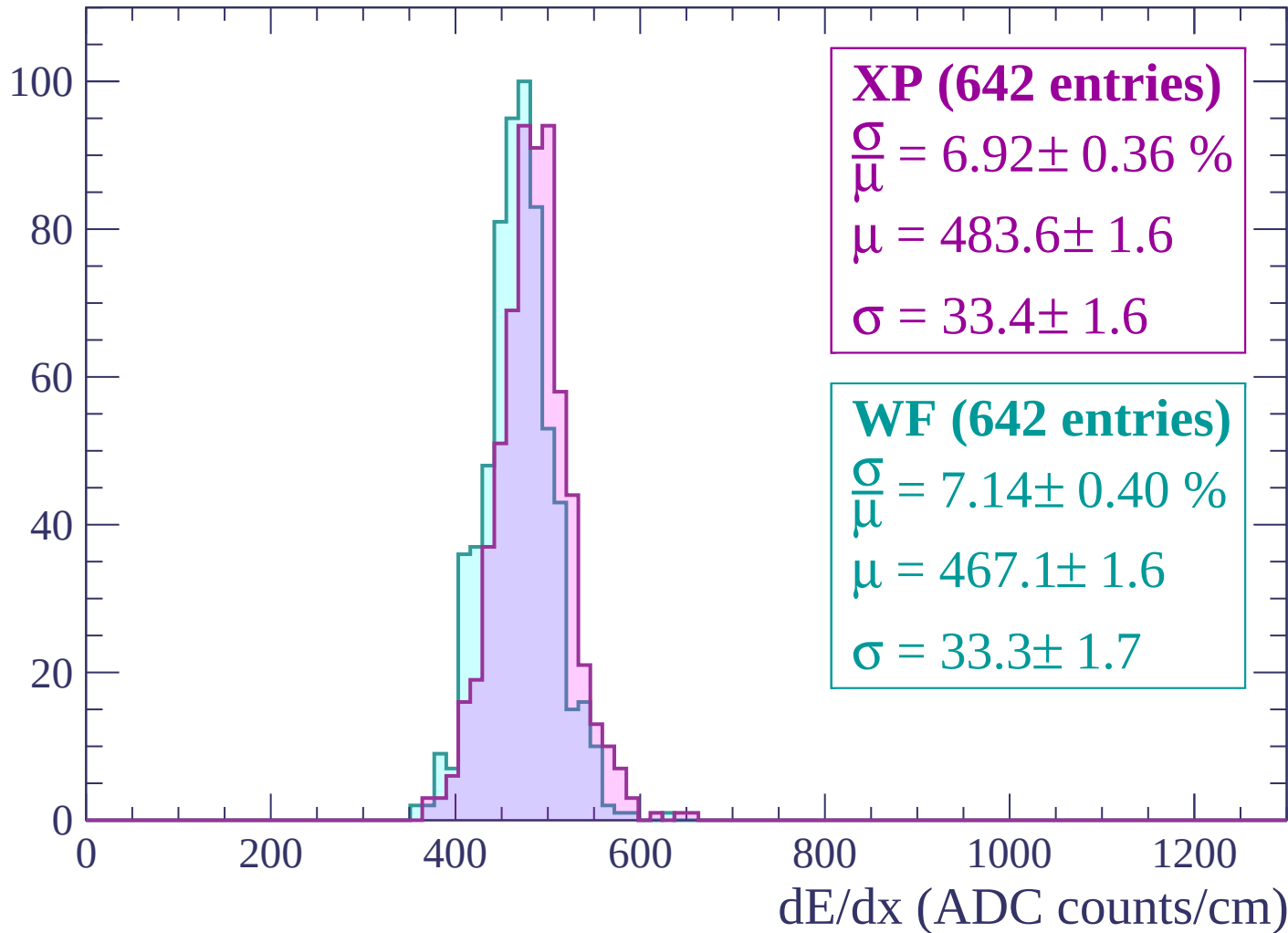
$$\mu = 469.0 \pm 1.7$$

$$\sigma = 35.0 \pm 1.9$$

dE/dx (ADC counts/cm)

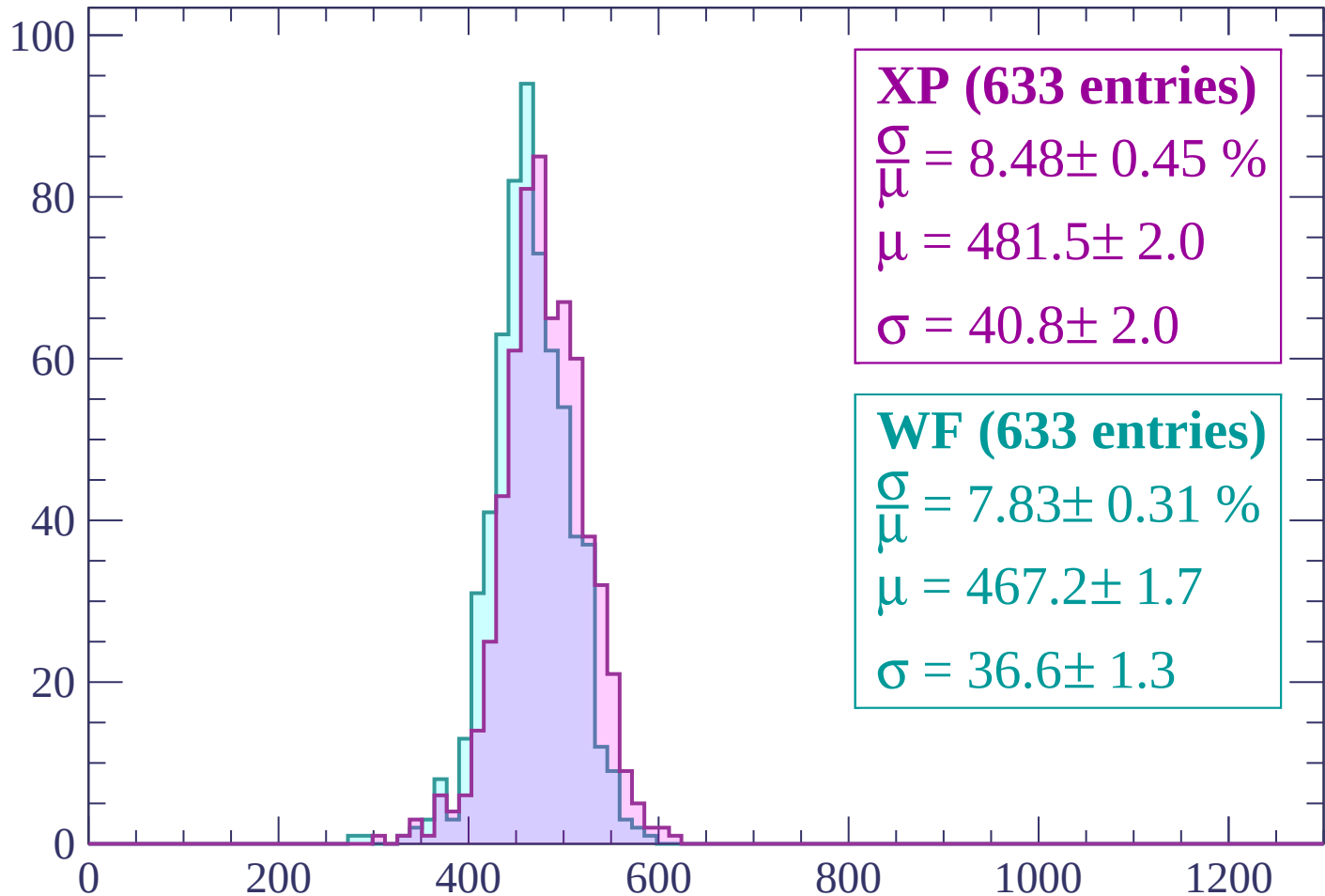
Energy loss | $-1740 < p < -1680$

Count



Energy loss | $-1680 < p < -1620$

Count



XP (633 entries)

$\frac{\sigma}{\mu} = 8.48 \pm 0.45 \%$

$\mu = 481.5 \pm 2.0$

$\sigma = 40.8 \pm 2.0$

WF (633 entries)

$\frac{\sigma}{\mu} = 7.83 \pm 0.31 \%$

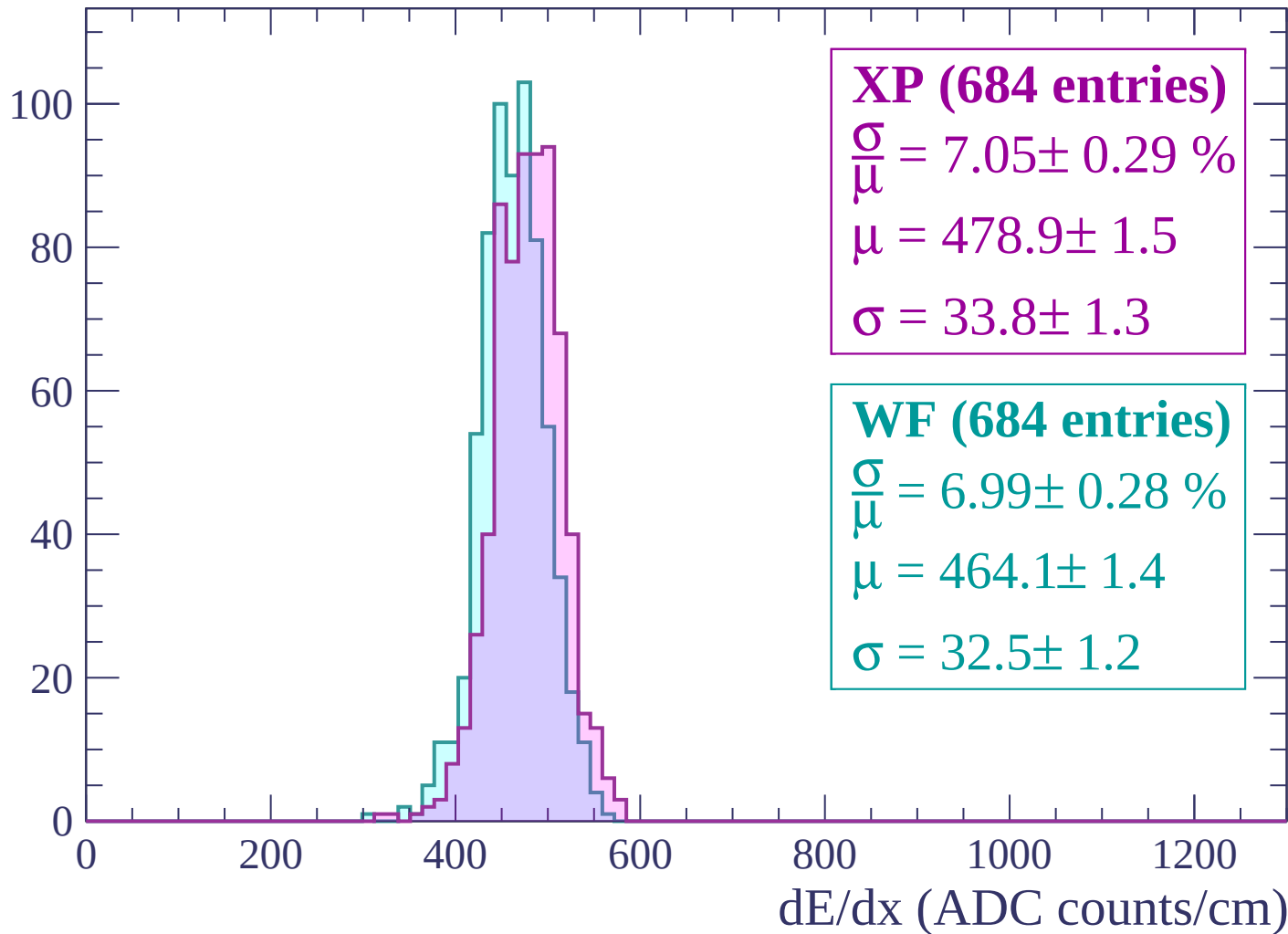
$\mu = 467.2 \pm 1.7$

$\sigma = 36.6 \pm 1.3$

dE/dx (ADC counts/cm)

Energy loss | $-1620 < p < -1560$

Count



Energy loss | -1560 < p < -1500

Count

100

80

60

40

20

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (611 entries)

$\frac{\sigma}{\mu} = 7.70 \pm 0.42 \%$

$\mu = 472.2 \pm 1.8$

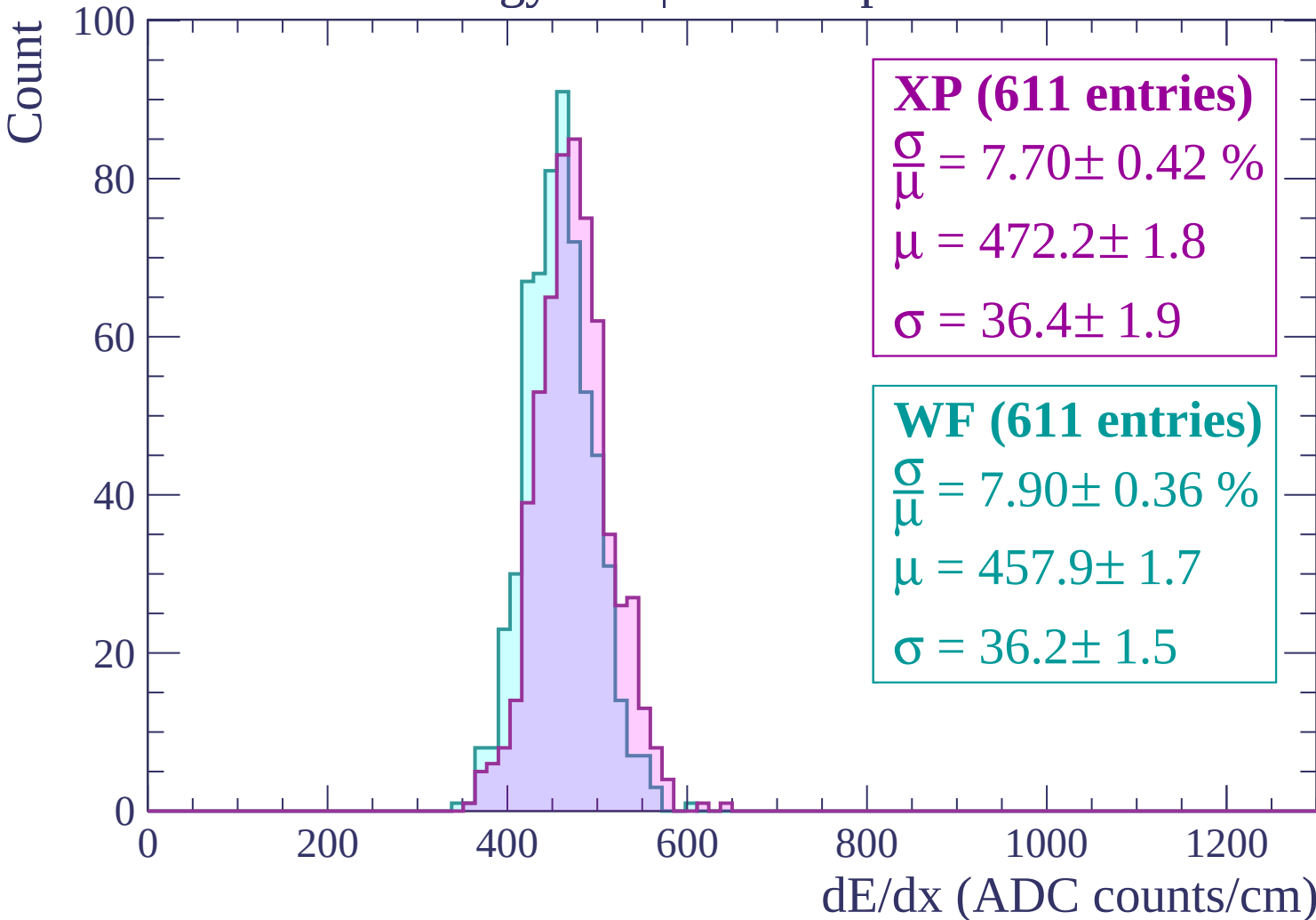
$\sigma = 36.4 \pm 1.9$

WF (611 entries)

$\frac{\sigma}{\mu} = 7.90 \pm 0.36 \%$

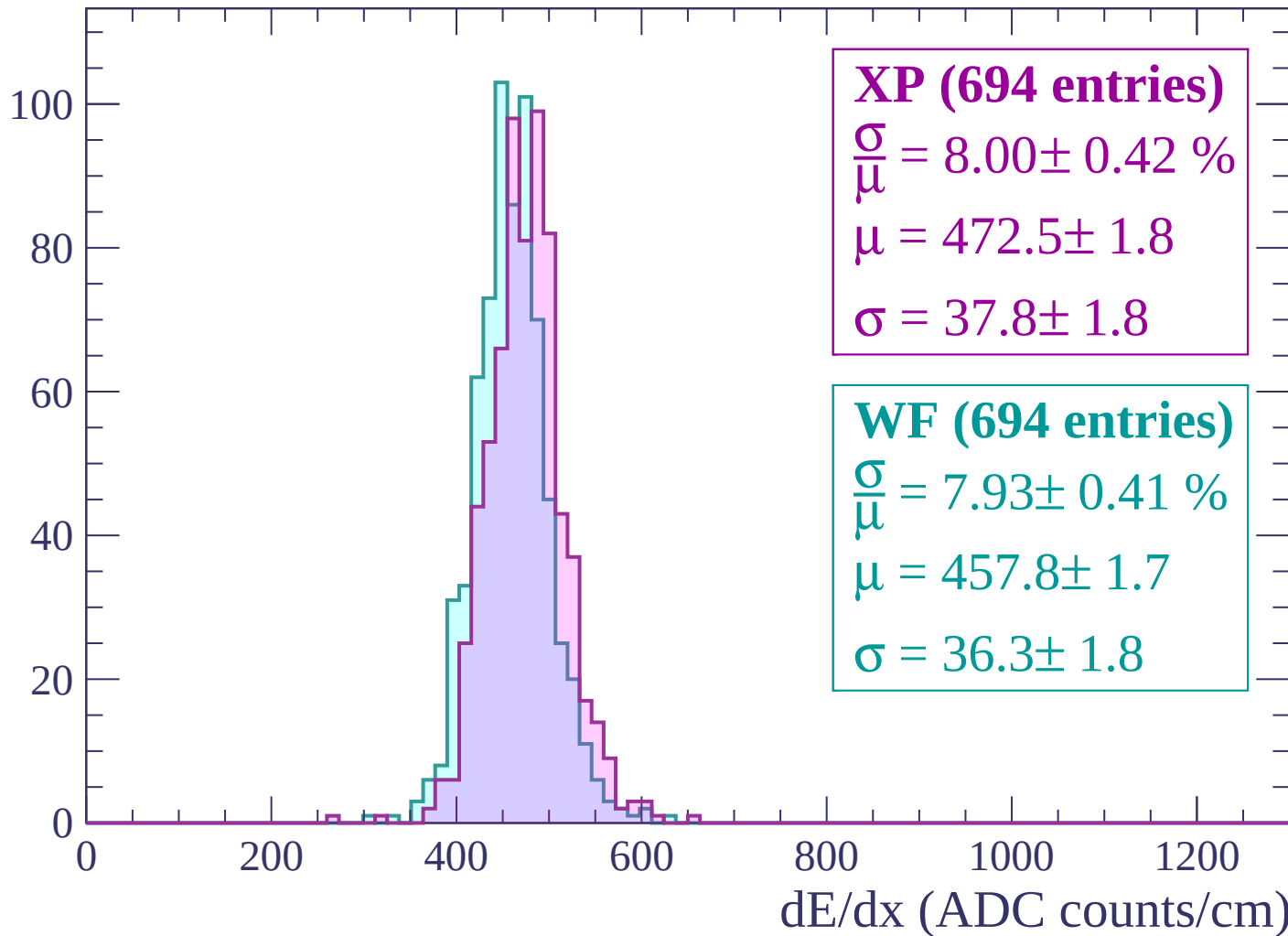
$\mu = 457.9 \pm 1.7$

$\sigma = 36.2 \pm 1.5$



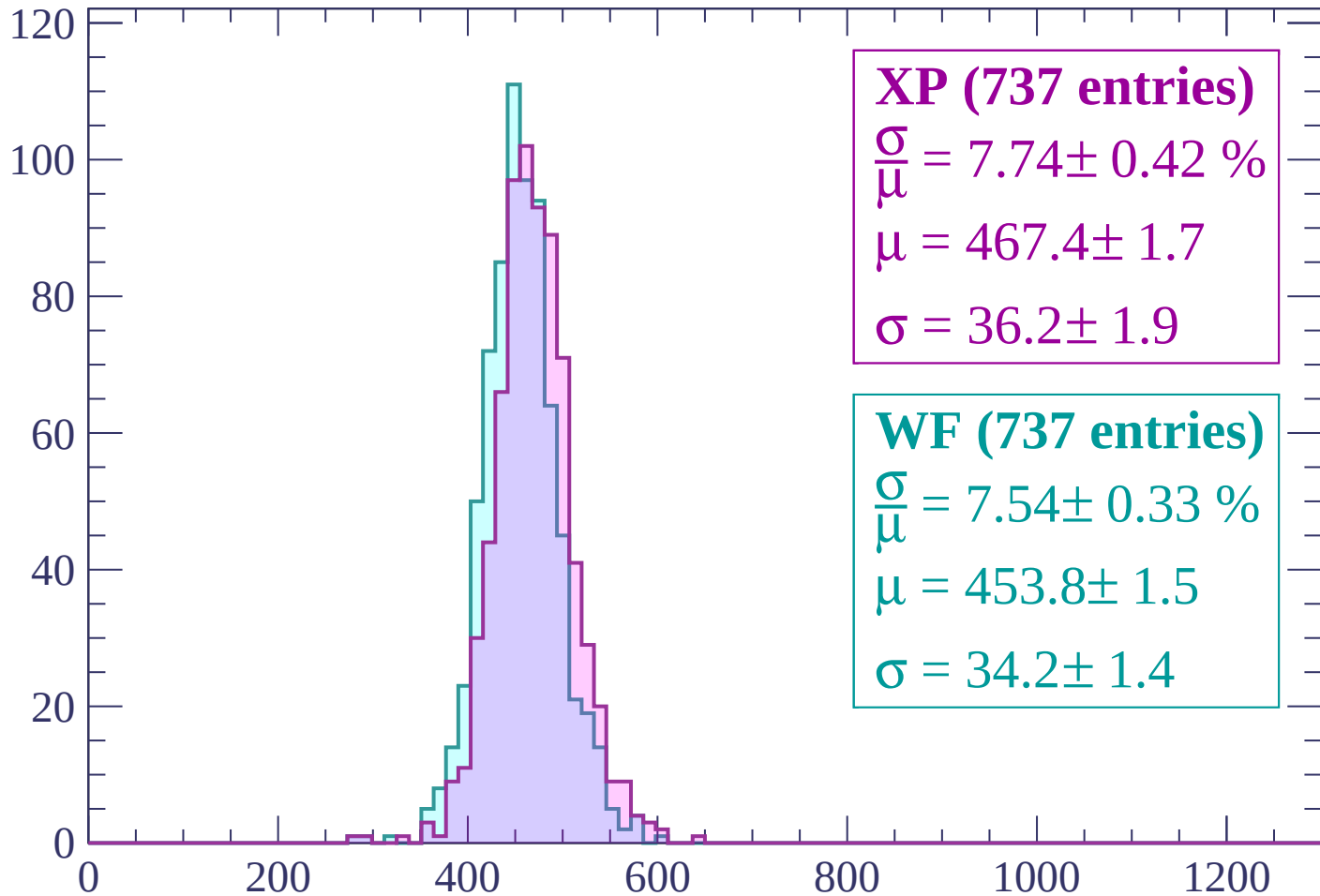
Energy loss | $-1500 < p < -1440$

Count



Energy loss | $-1440 < p < -1380$

Count



XP (737 entries)

$$\frac{\sigma}{\mu} = 7.74 \pm 0.42 \%$$

$$\mu = 467.4 \pm 1.7$$

$$\sigma = 36.2 \pm 1.9$$

WF (737 entries)

$$\frac{\sigma}{\mu} = 7.54 \pm 0.33 \%$$

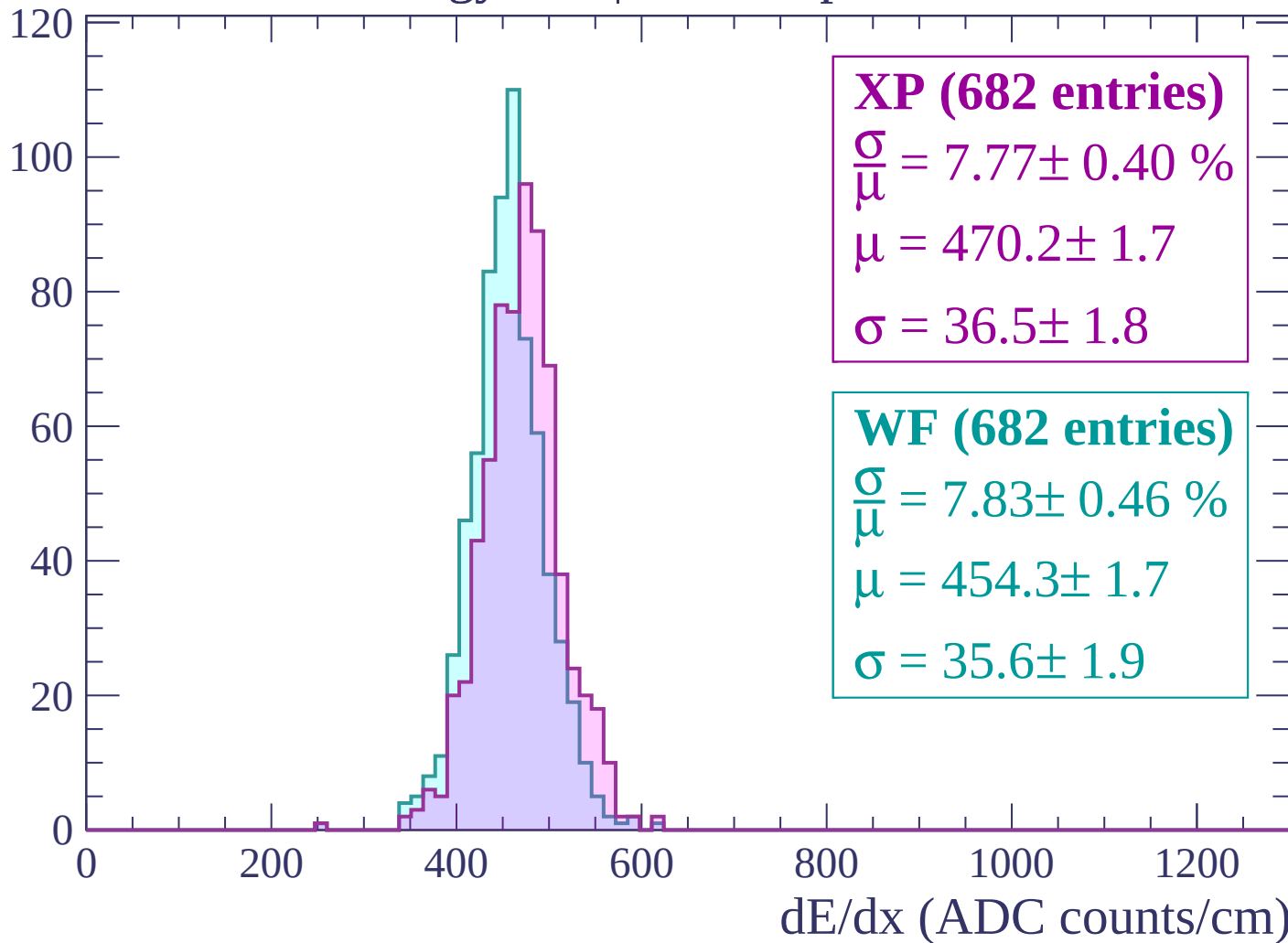
$$\mu = 453.8 \pm 1.5$$

$$\sigma = 34.2 \pm 1.4$$

dE/dx (ADC counts/cm)

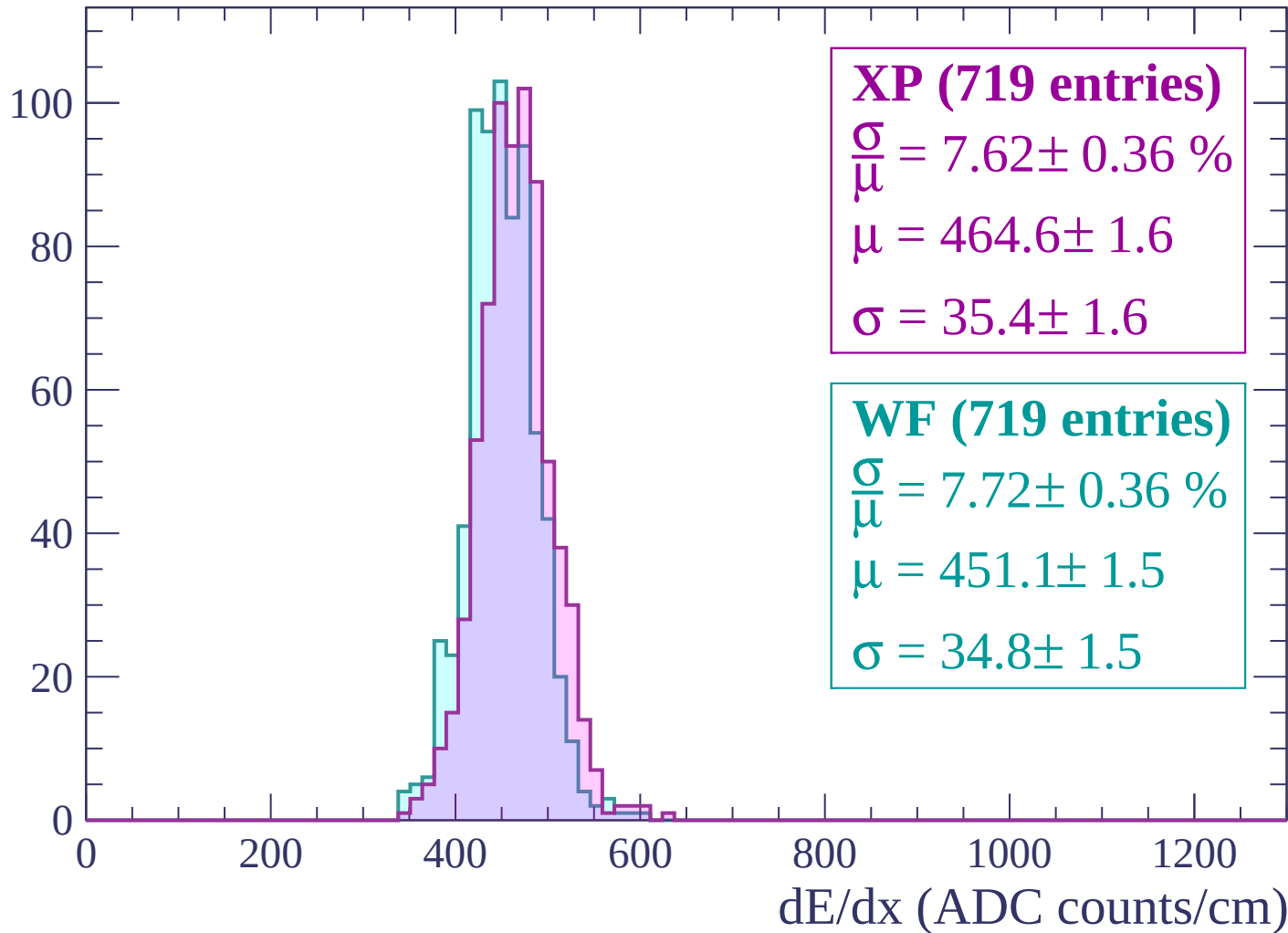
Energy loss | $-1380 < p < -1320$

Count



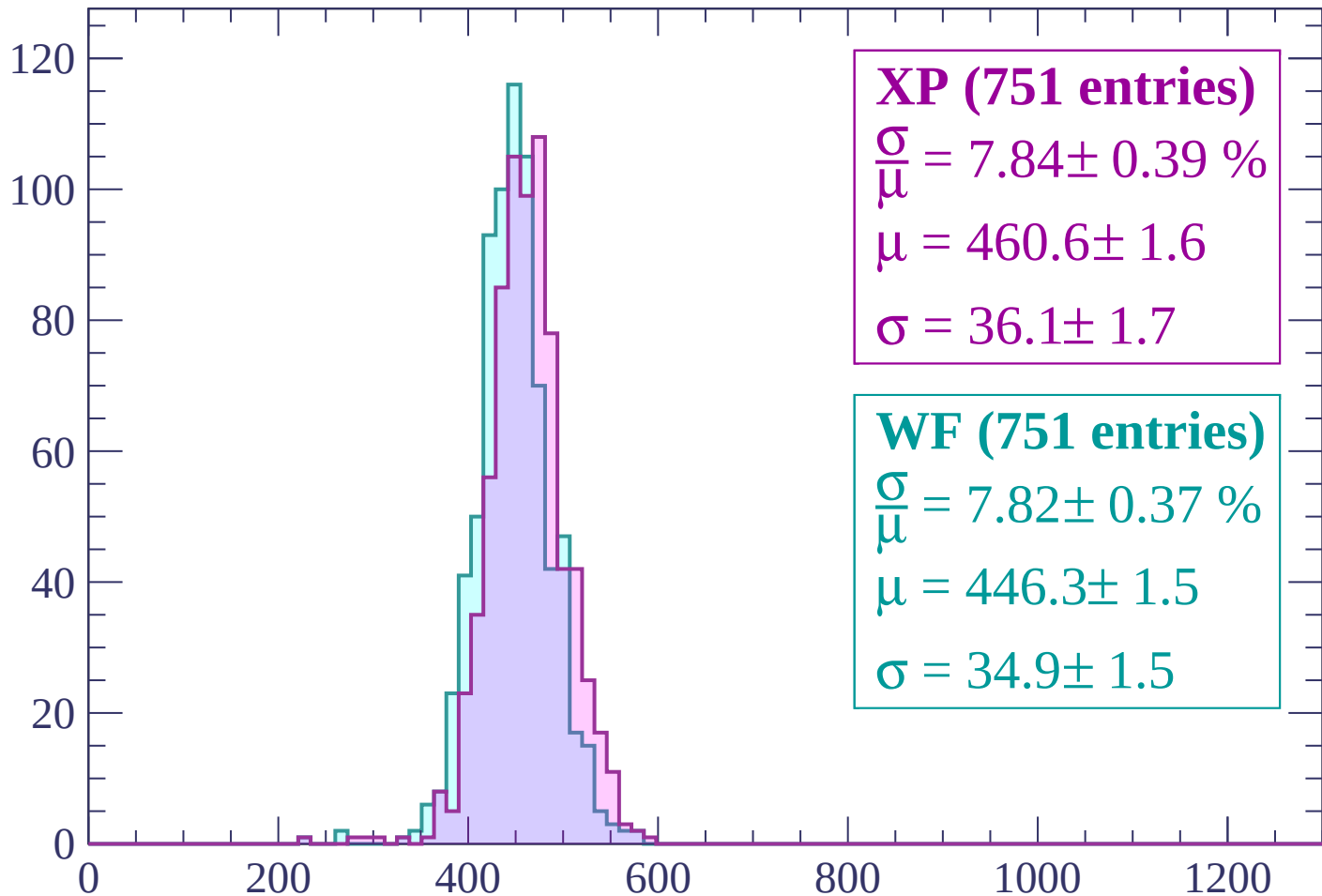
Energy loss | -1320 < p < -1260

Count



Energy loss | -1260 < p < -1200

Count



XP (751 entries)

$\frac{\sigma}{\mu} = 7.84 \pm 0.39 \%$

$\mu = 460.6 \pm 1.6$

$\sigma = 36.1 \pm 1.7$

WF (751 entries)

$\frac{\sigma}{\mu} = 7.82 \pm 0.37 \%$

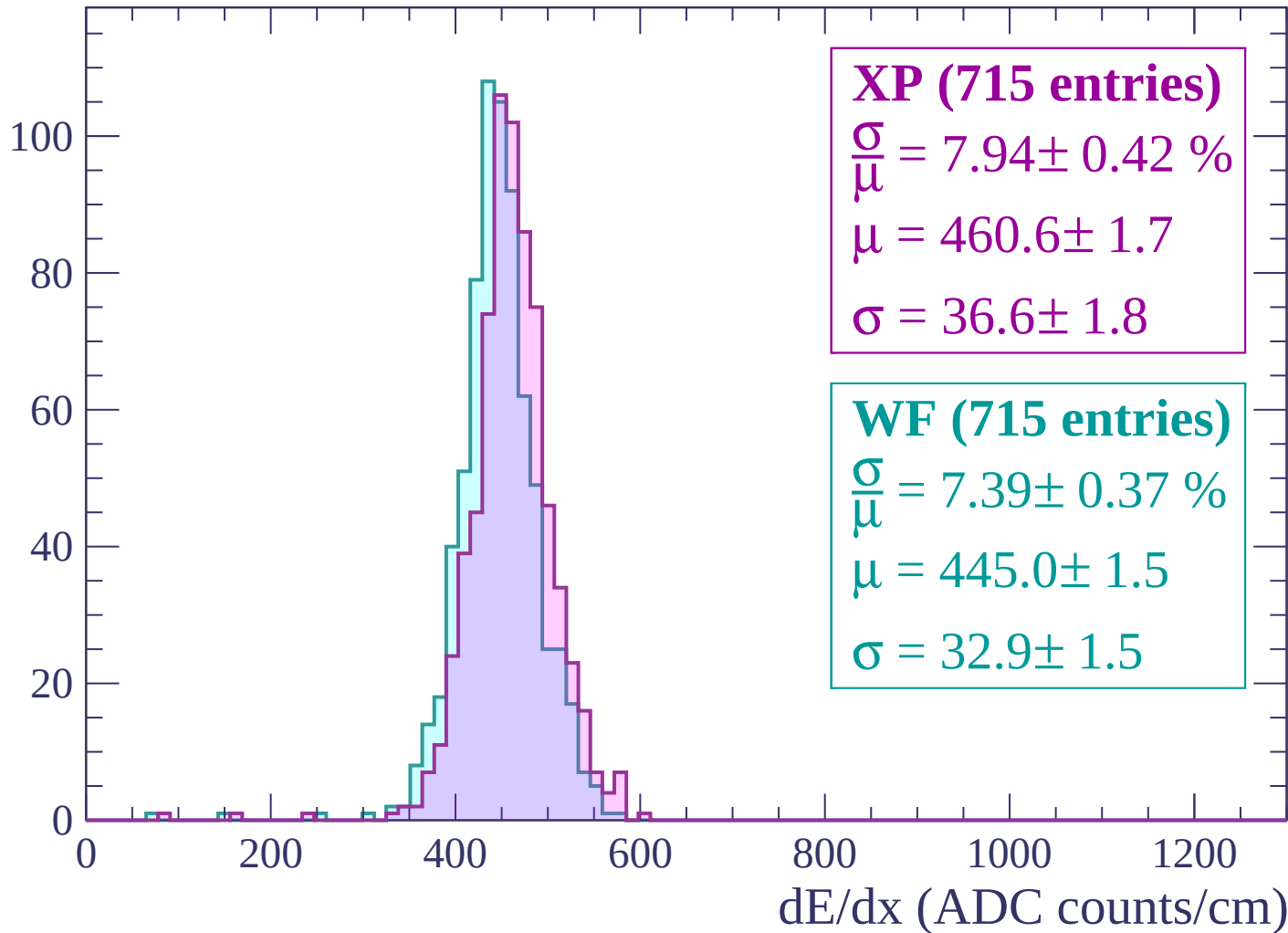
$\mu = 446.3 \pm 1.5$

$\sigma = 34.9 \pm 1.5$

dE/dx (ADC counts/cm)

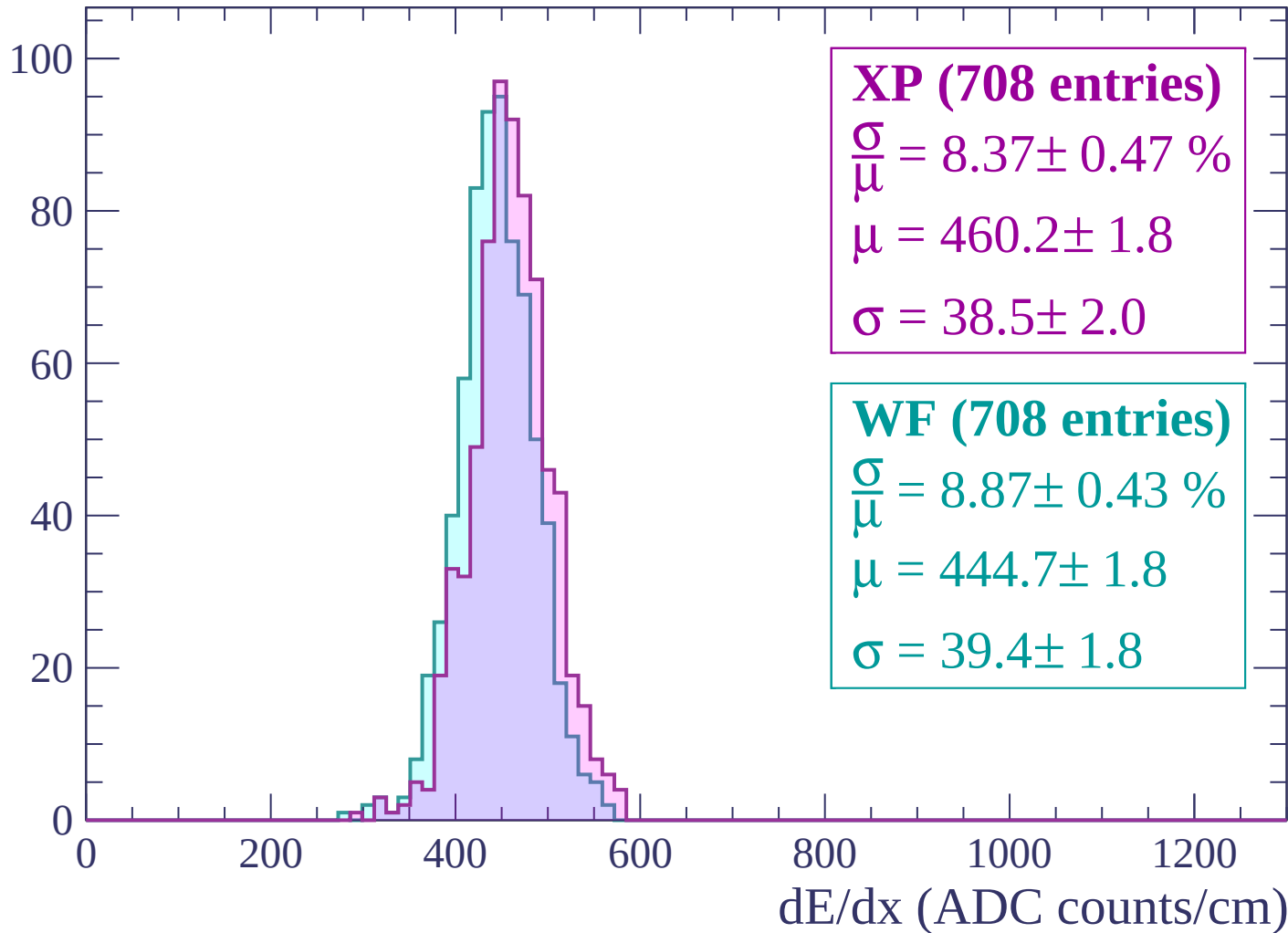
Energy loss | -1200 < p < -1140

Count



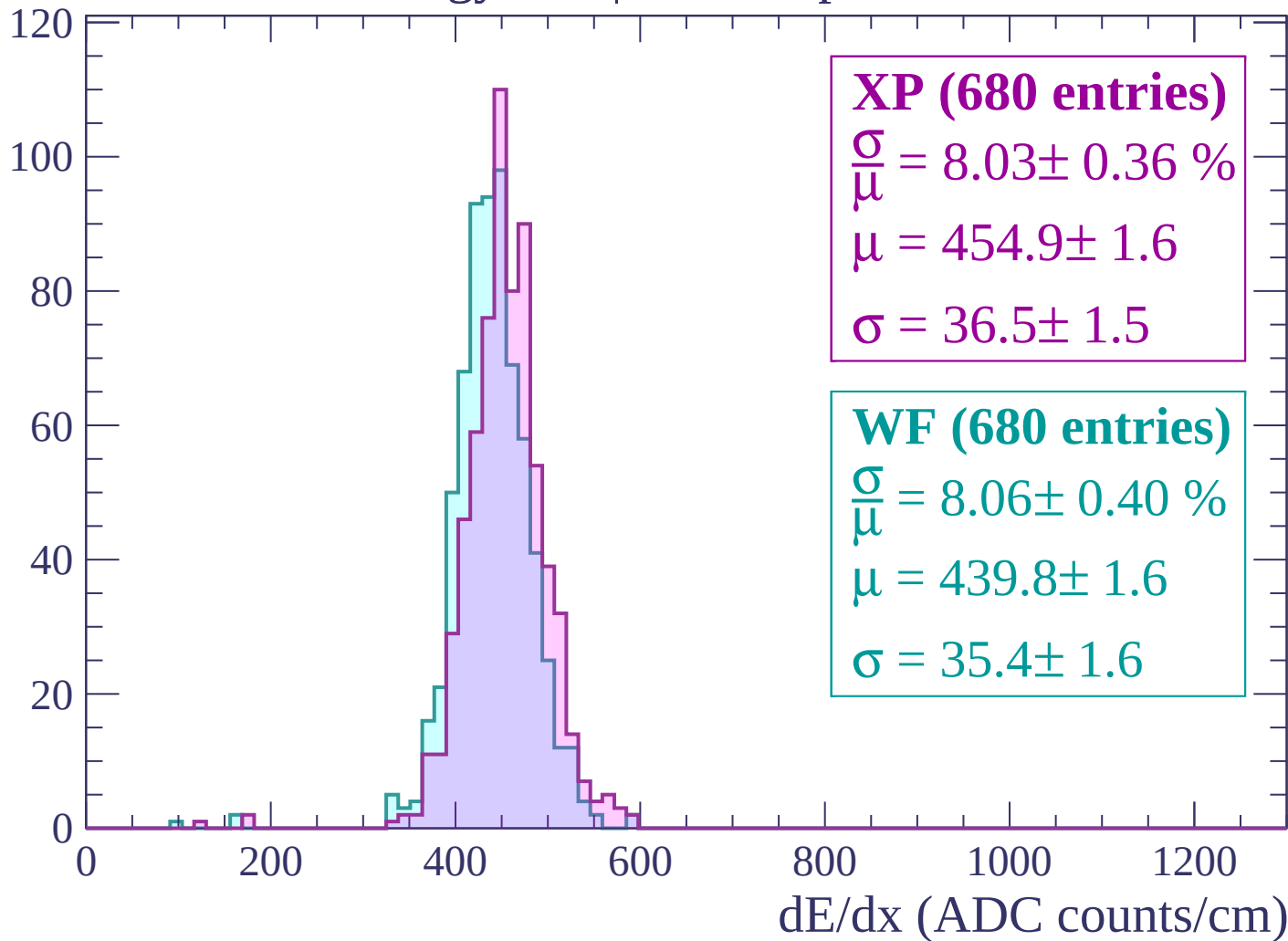
Energy loss | -1140 < p < -1080

Count



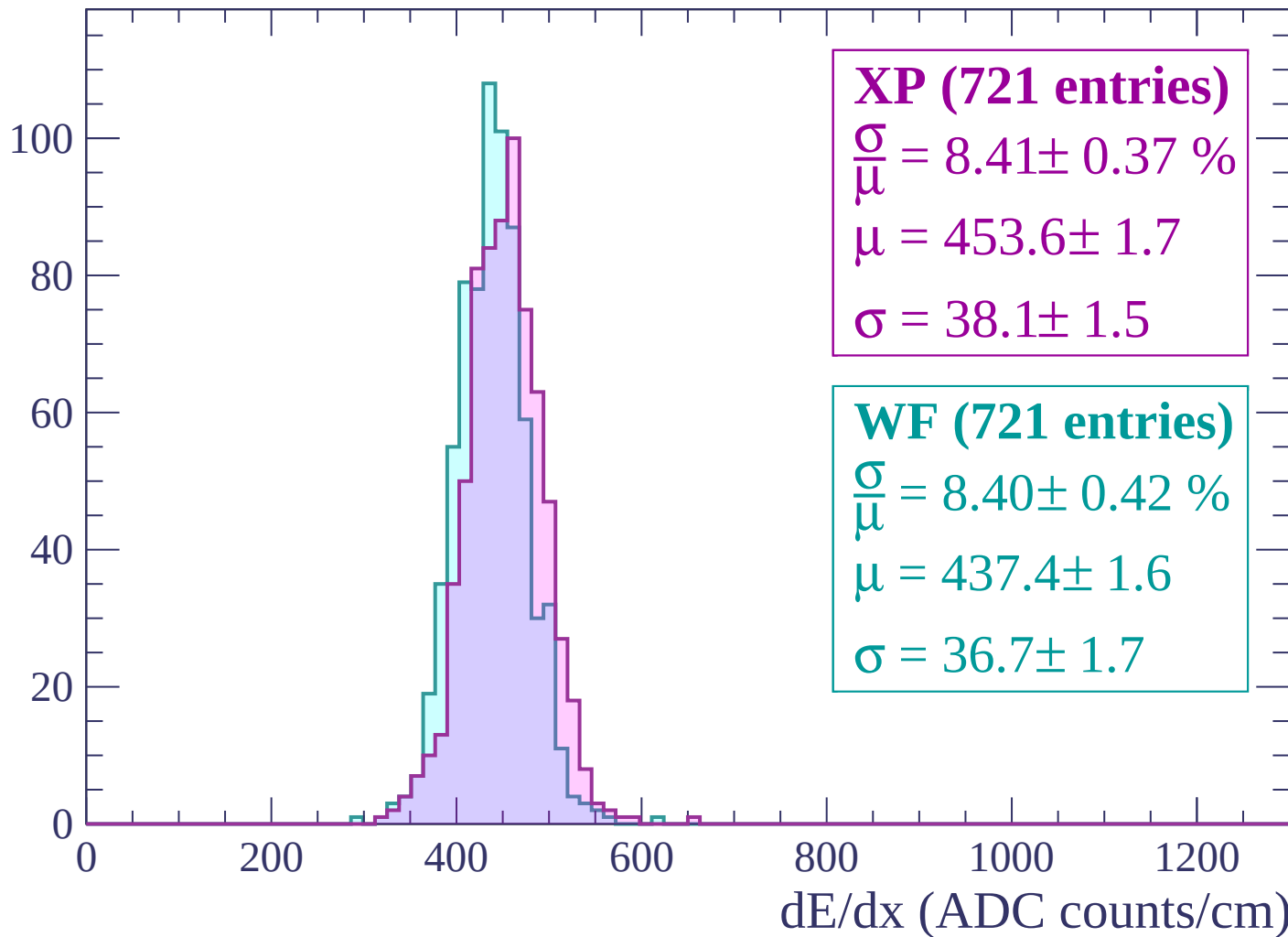
Energy loss | $-1080 < p < -1020$

Count



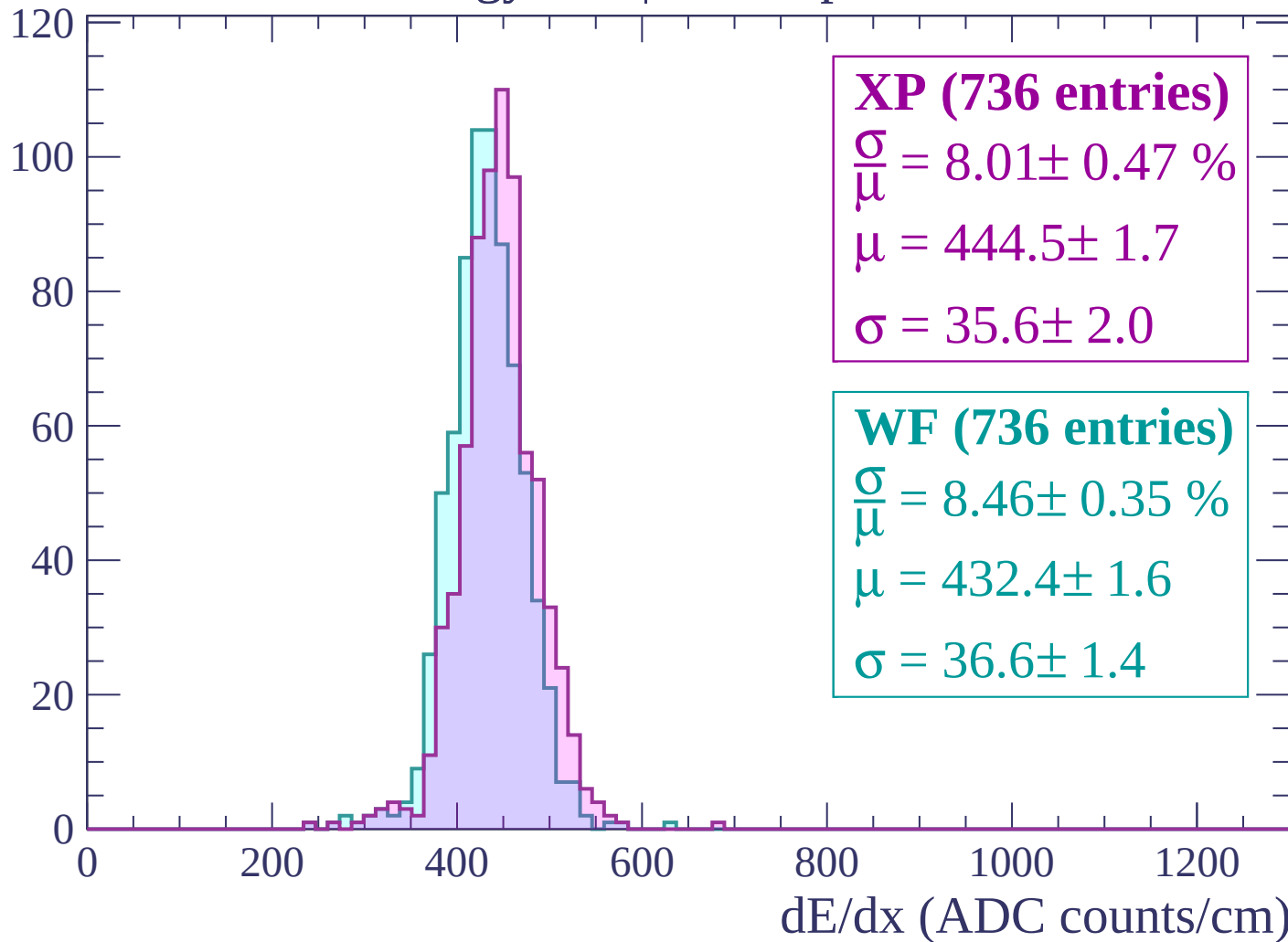
Energy loss | $-1020 < p < -960$

Count



Energy loss | $-960 < p < -900$

Count



Energy loss | $-900 < p < -840$

Count

100

80

60

40

20

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (682 entries)

$\frac{\sigma}{\mu} = 9.24 \pm 0.49 \%$

$\mu = 442.1 \pm 1.9$

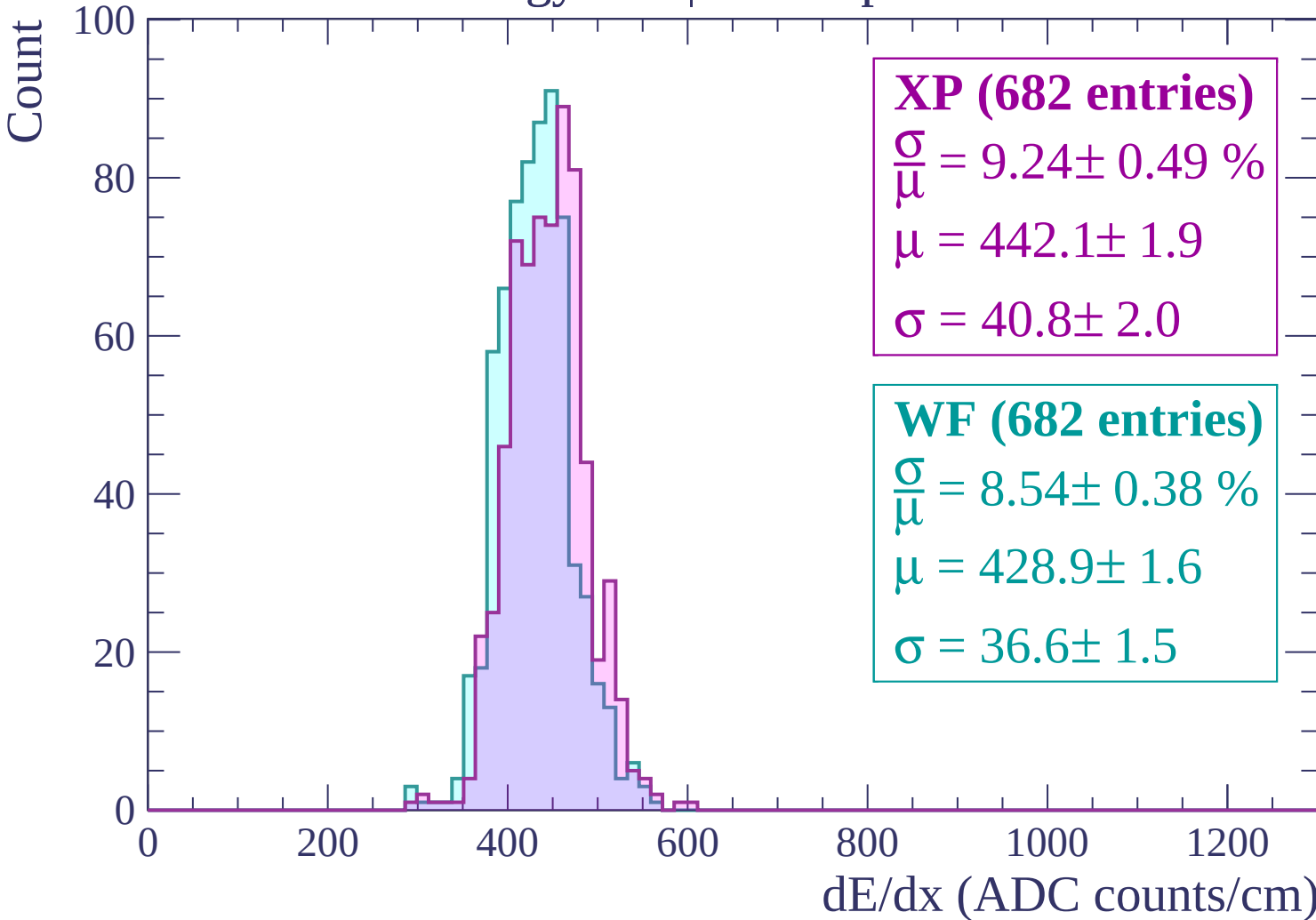
$\sigma = 40.8 \pm 2.0$

WF (682 entries)

$\frac{\sigma}{\mu} = 8.54 \pm 0.38 \%$

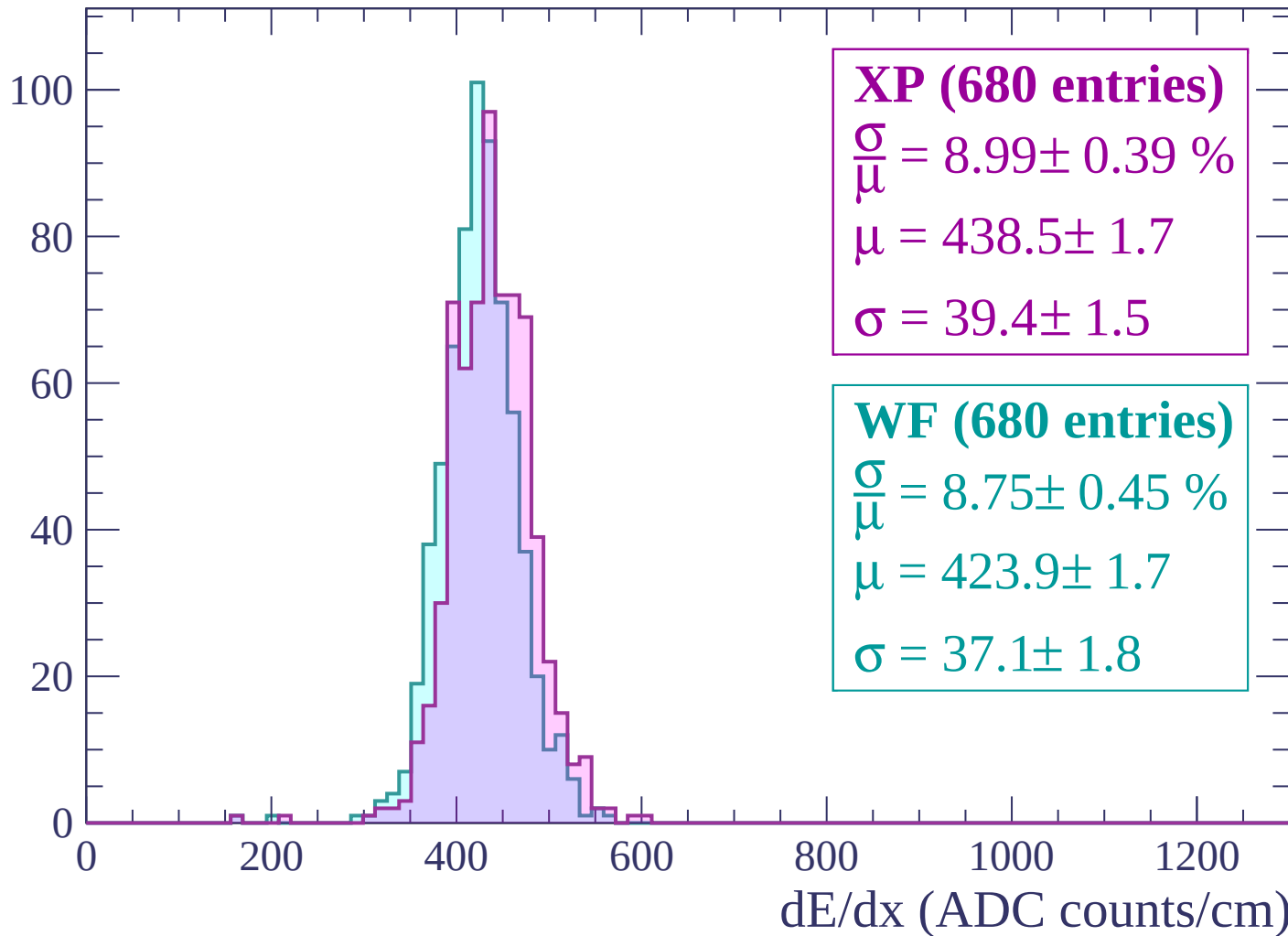
$\mu = 428.9 \pm 1.6$

$\sigma = 36.6 \pm 1.5$



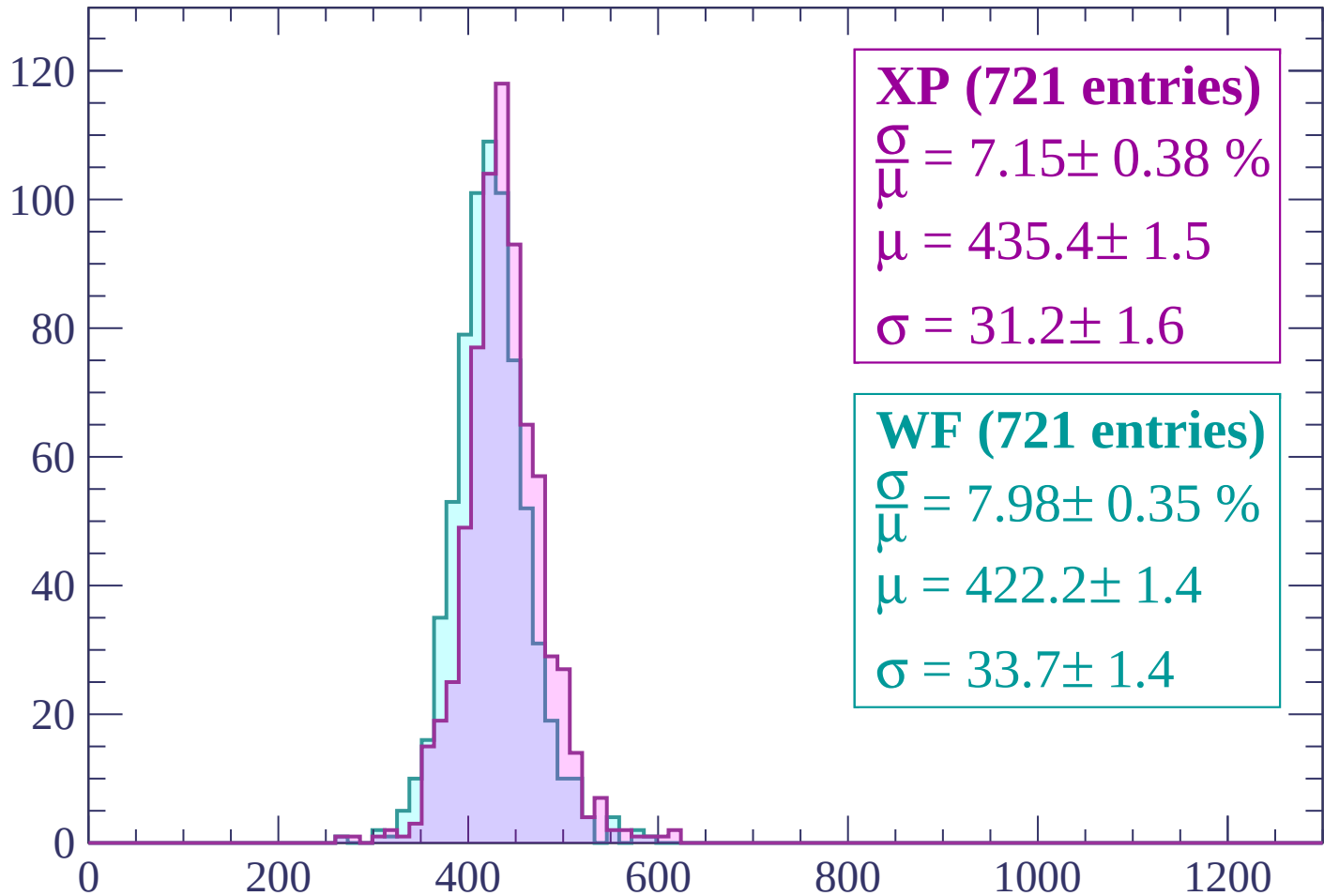
Energy loss | $-840 < p < -780$

Count



Energy loss | $-780 < p < -720$

Count



XP (721 entries)

$\frac{\sigma}{\bar{\mu}} = 7.15 \pm 0.38 \%$

$\mu = 435.4 \pm 1.5$

$\sigma = 31.2 \pm 1.6$

WF (721 entries)

$\frac{\sigma}{\bar{\mu}} = 7.98 \pm 0.35 \%$

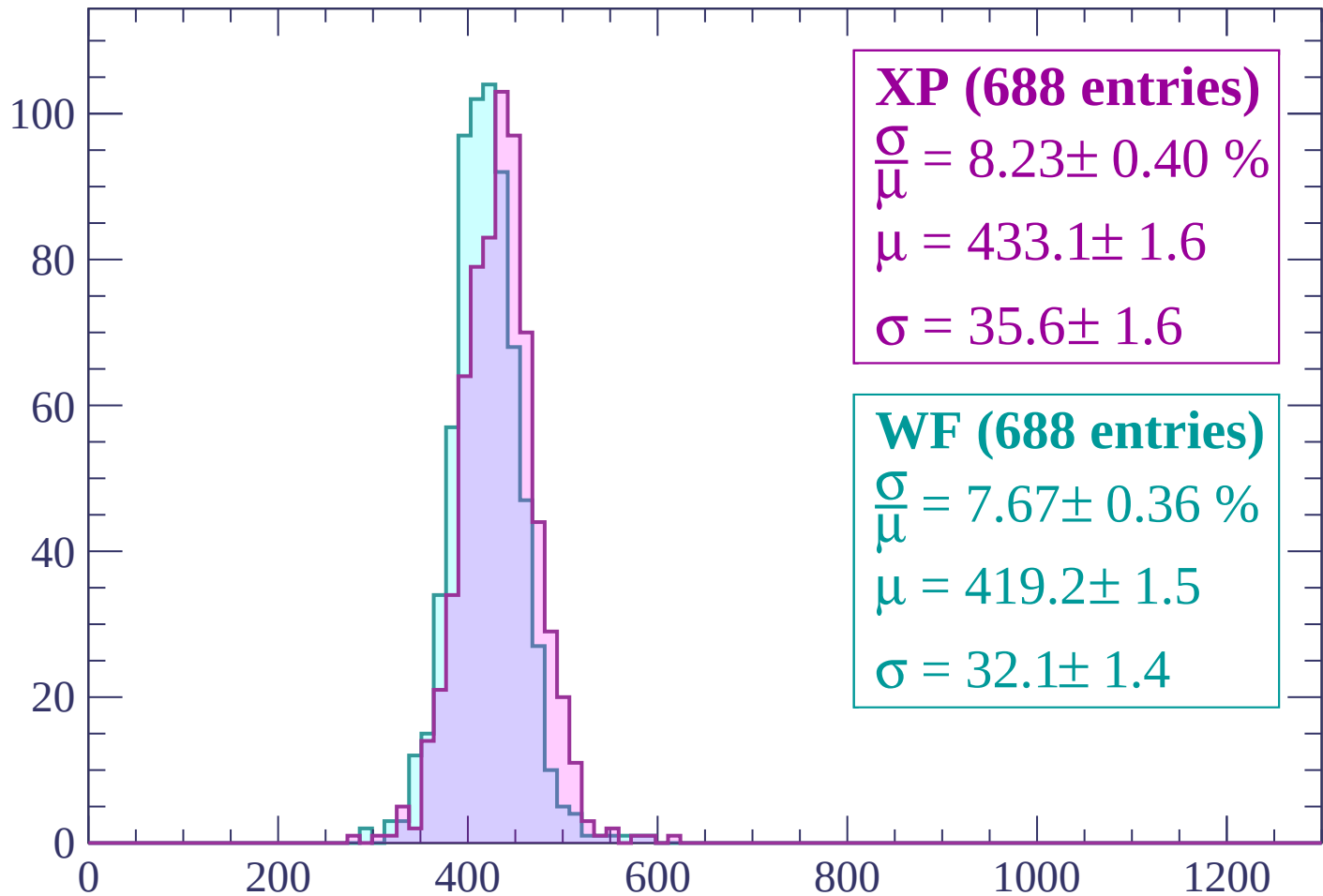
$\mu = 422.2 \pm 1.4$

$\sigma = 33.7 \pm 1.4$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -660$

Count



XP (688 entries)

$$\frac{\sigma}{\mu} = 8.23 \pm 0.40 \%$$

$$\mu = 433.1 \pm 1.6$$

$$\sigma = 35.6 \pm 1.6$$

WF (688 entries)

$$\frac{\sigma}{\mu} = 7.67 \pm 0.36 \%$$

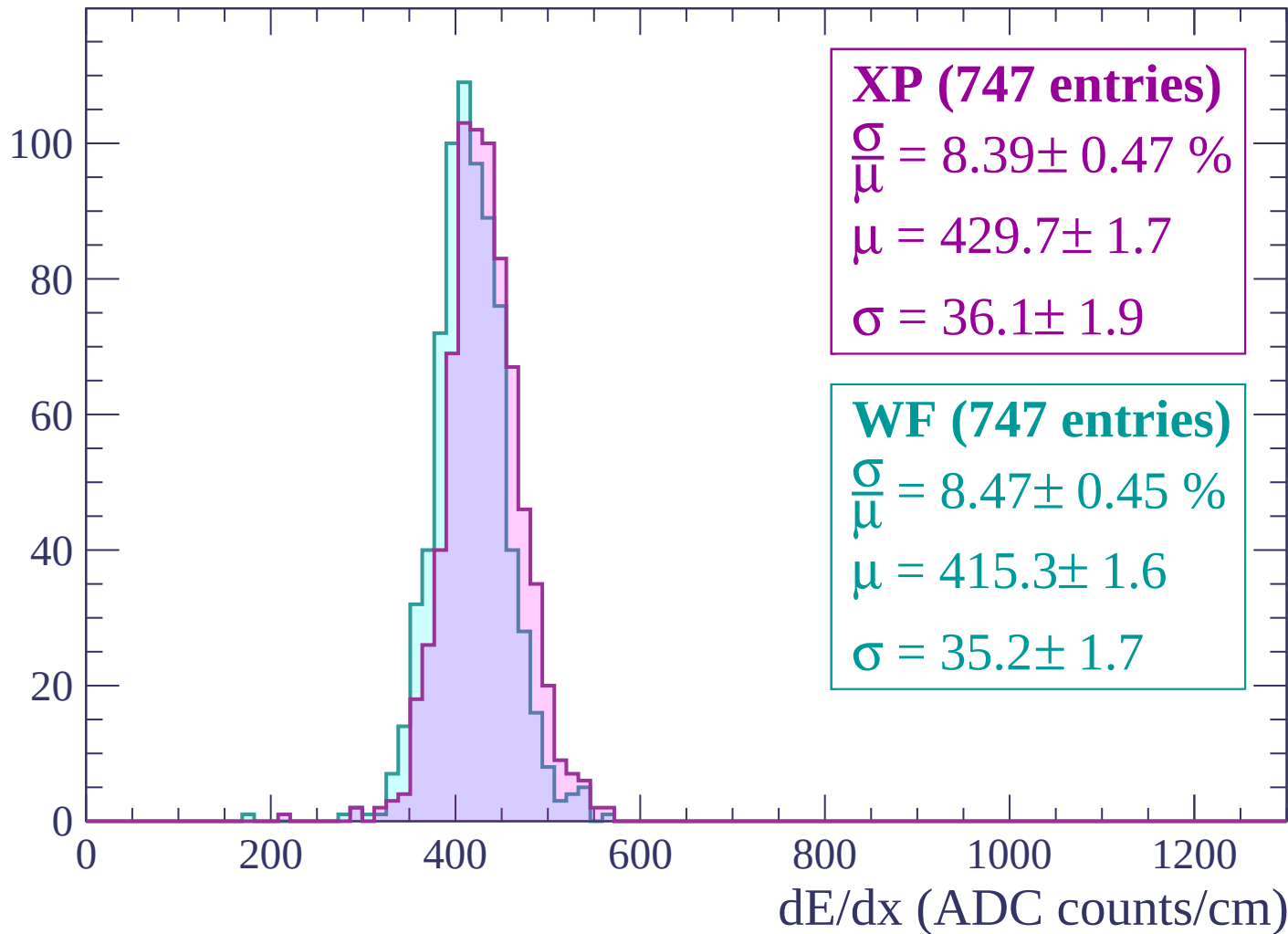
$$\mu = 419.2 \pm 1.5$$

$$\sigma = 32.1 \pm 1.4$$

dE/dx (ADC counts/cm)

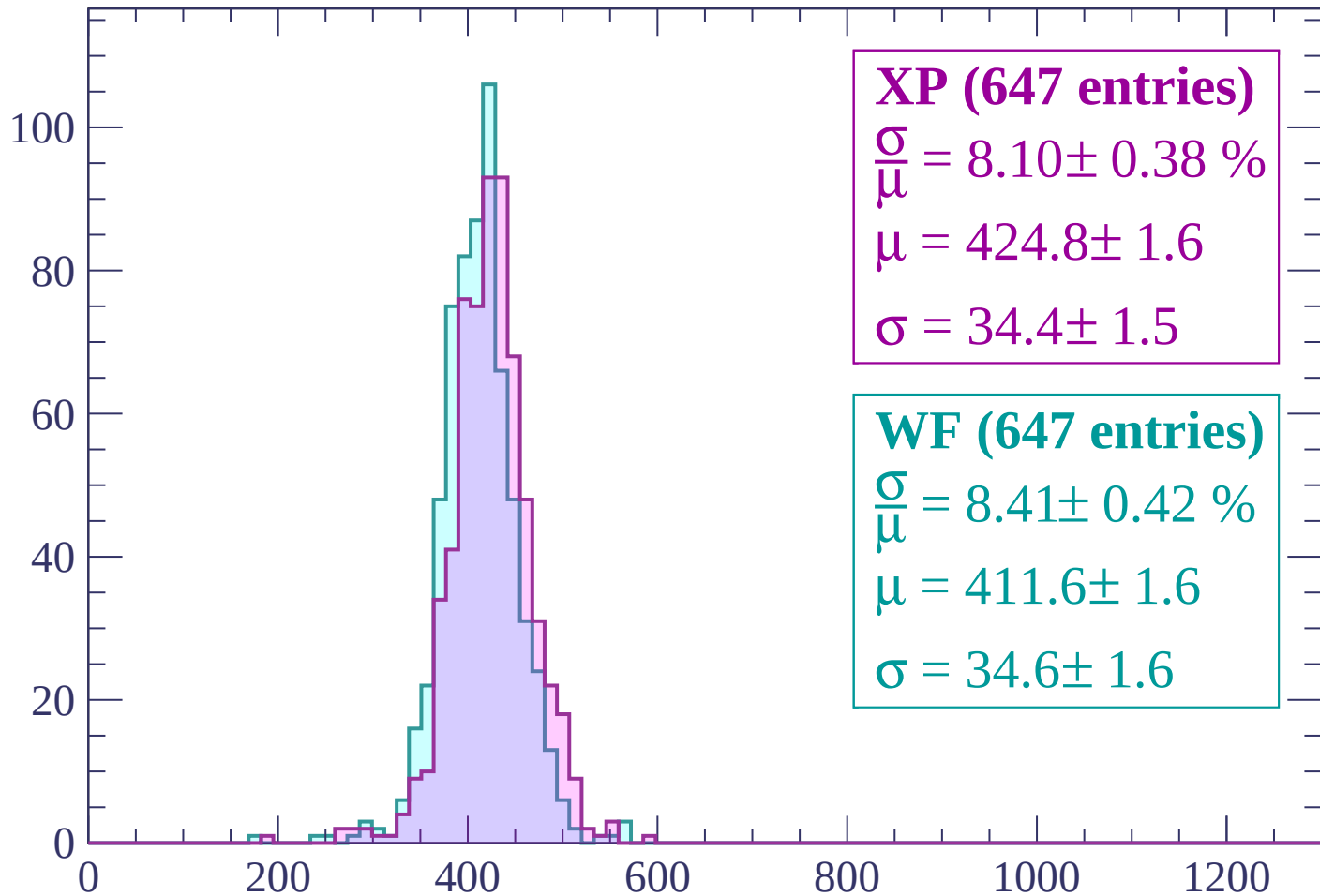
Energy loss | $-660 < p < -600$

Count



Energy loss | $-600 < p < -540$

Count



XP (647 entries)

$\frac{\sigma}{\mu} = 8.10 \pm 0.38 \%$

$\mu = 424.8 \pm 1.6$

$\sigma = 34.4 \pm 1.5$

WF (647 entries)

$\frac{\sigma}{\mu} = 8.41 \pm 0.42 \%$

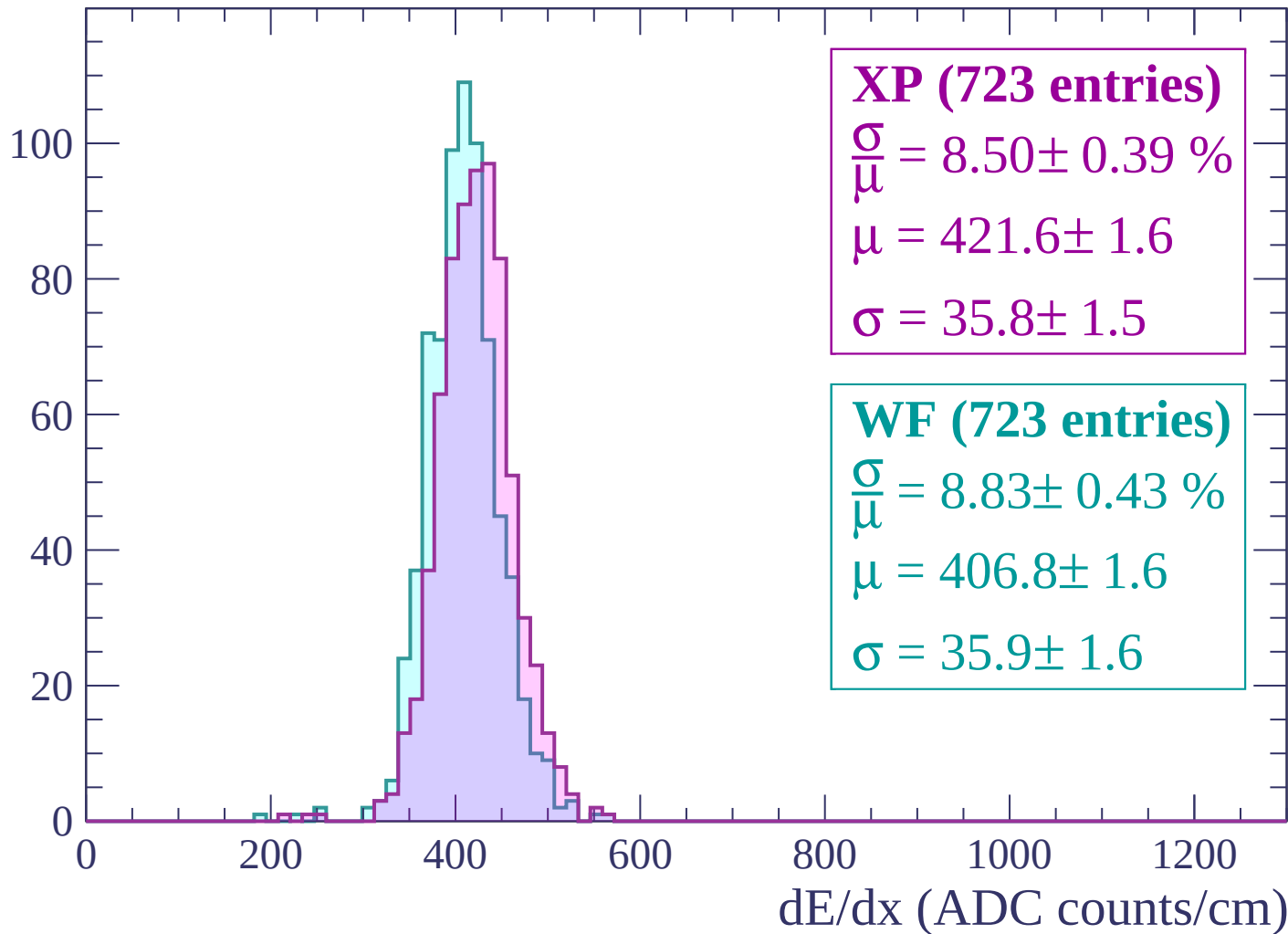
$\mu = 411.6 \pm 1.6$

$\sigma = 34.6 \pm 1.6$

dE/dx (ADC counts/cm)

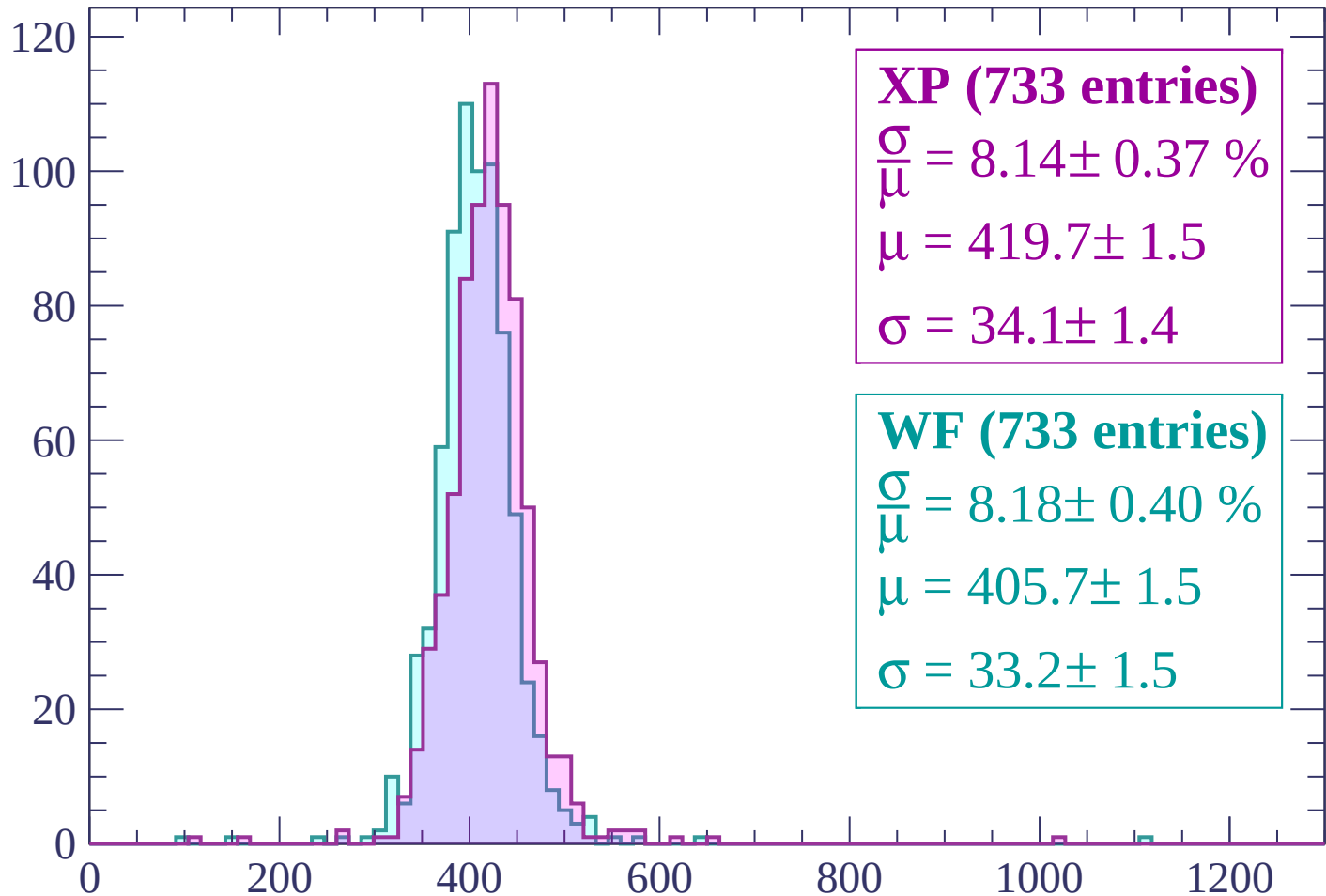
Energy loss | $-540 < p < -480$

Count



Energy loss | $-480 < p < -420$

Count



XP (733 entries)

$\frac{\sigma}{\mu} = 8.14 \pm 0.37 \%$

$\mu = 419.7 \pm 1.5$

$\sigma = 34.1 \pm 1.4$

WF (733 entries)

$\frac{\sigma}{\mu} = 8.18 \pm 0.40 \%$

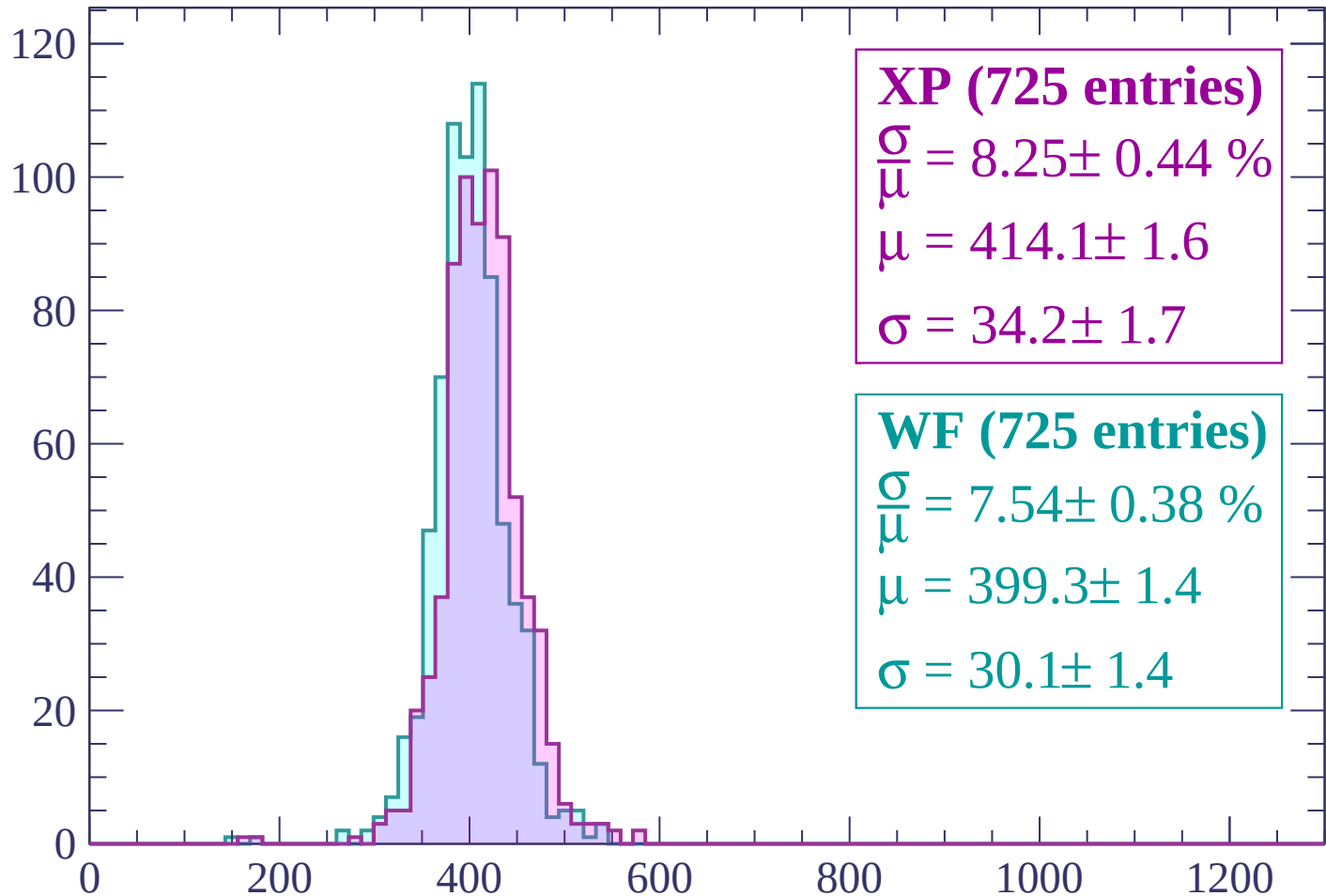
$\mu = 405.7 \pm 1.5$

$\sigma = 33.2 \pm 1.5$

dE/dx (ADC counts/cm)

Energy loss | $-420 < p < -360$

Count



XP (725 entries)

$\frac{\sigma}{\mu} = 8.25 \pm 0.44 \%$

$\mu = 414.1 \pm 1.6$

$\sigma = 34.2 \pm 1.7$

WF (725 entries)

$\frac{\sigma}{\mu} = 7.54 \pm 0.38 \%$

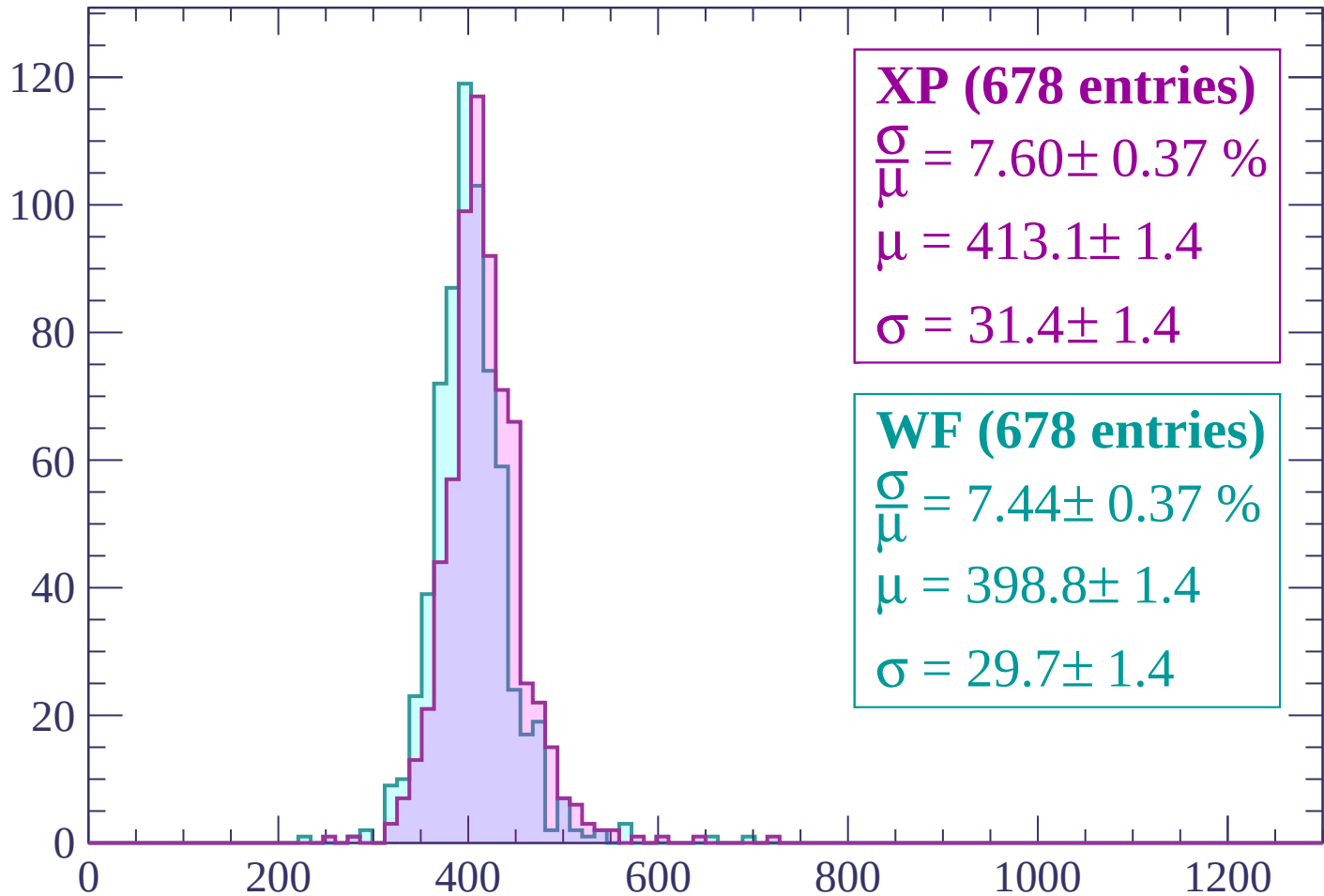
$\mu = 399.3 \pm 1.4$

$\sigma = 30.1 \pm 1.4$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -300$

Count



XP (678 entries)

$\frac{\sigma}{\mu} = 7.60 \pm 0.37 \%$

$\mu = 413.1 \pm 1.4$

$\sigma = 31.4 \pm 1.4$

WF (678 entries)

$\frac{\sigma}{\mu} = 7.44 \pm 0.37 \%$

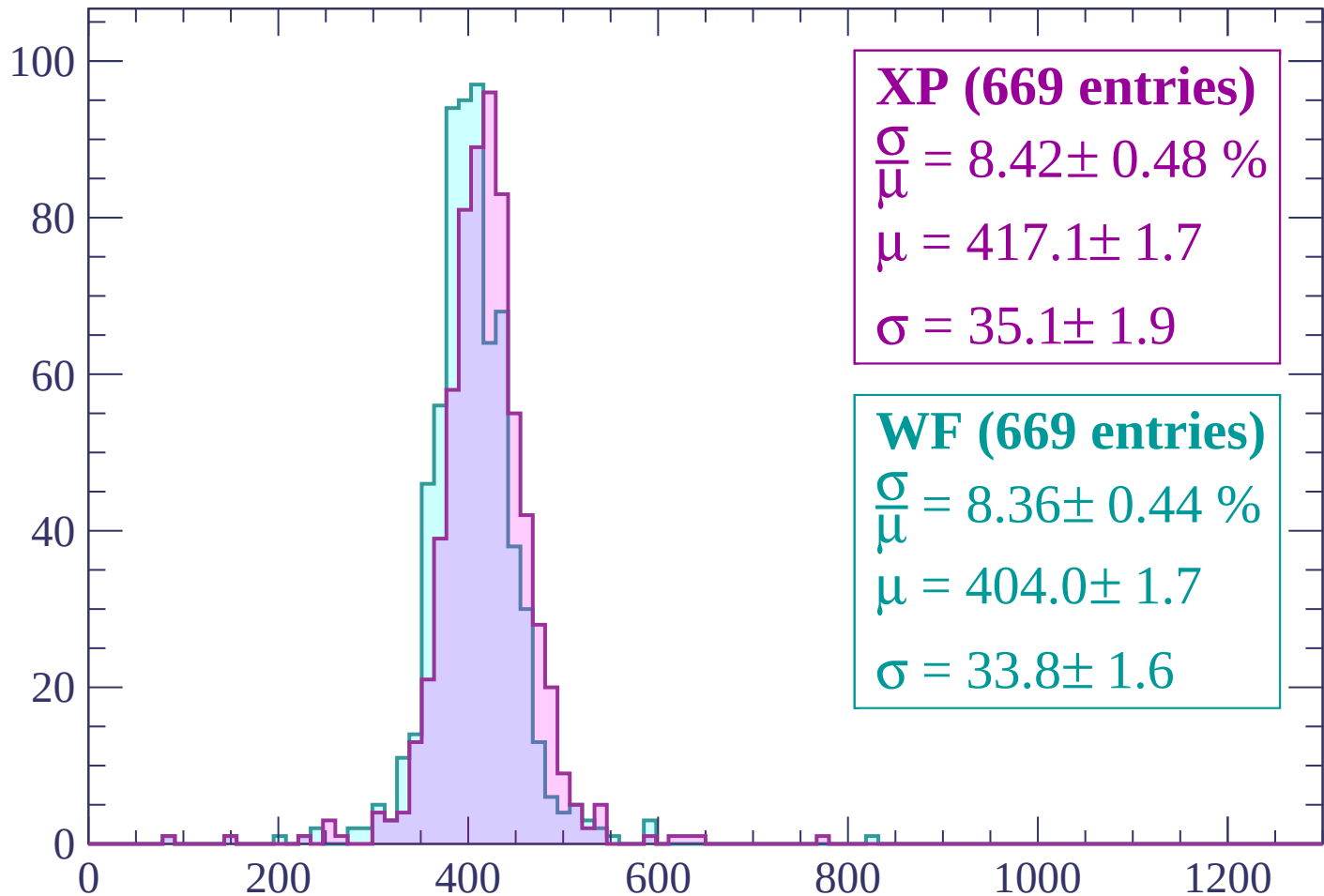
$\mu = 398.8 \pm 1.4$

$\sigma = 29.7 \pm 1.4$

dE/dx (ADC counts/cm)

Energy loss | $-300 < p < -240$

Count



XP (669 entries)

$$\frac{\sigma}{\mu} = 8.42 \pm 0.48 \%$$

$$\mu = 417.1 \pm 1.7$$

$$\sigma = 35.1 \pm 1.9$$

WF (669 entries)

$$\frac{\sigma}{\mu} = 8.36 \pm 0.44 \%$$

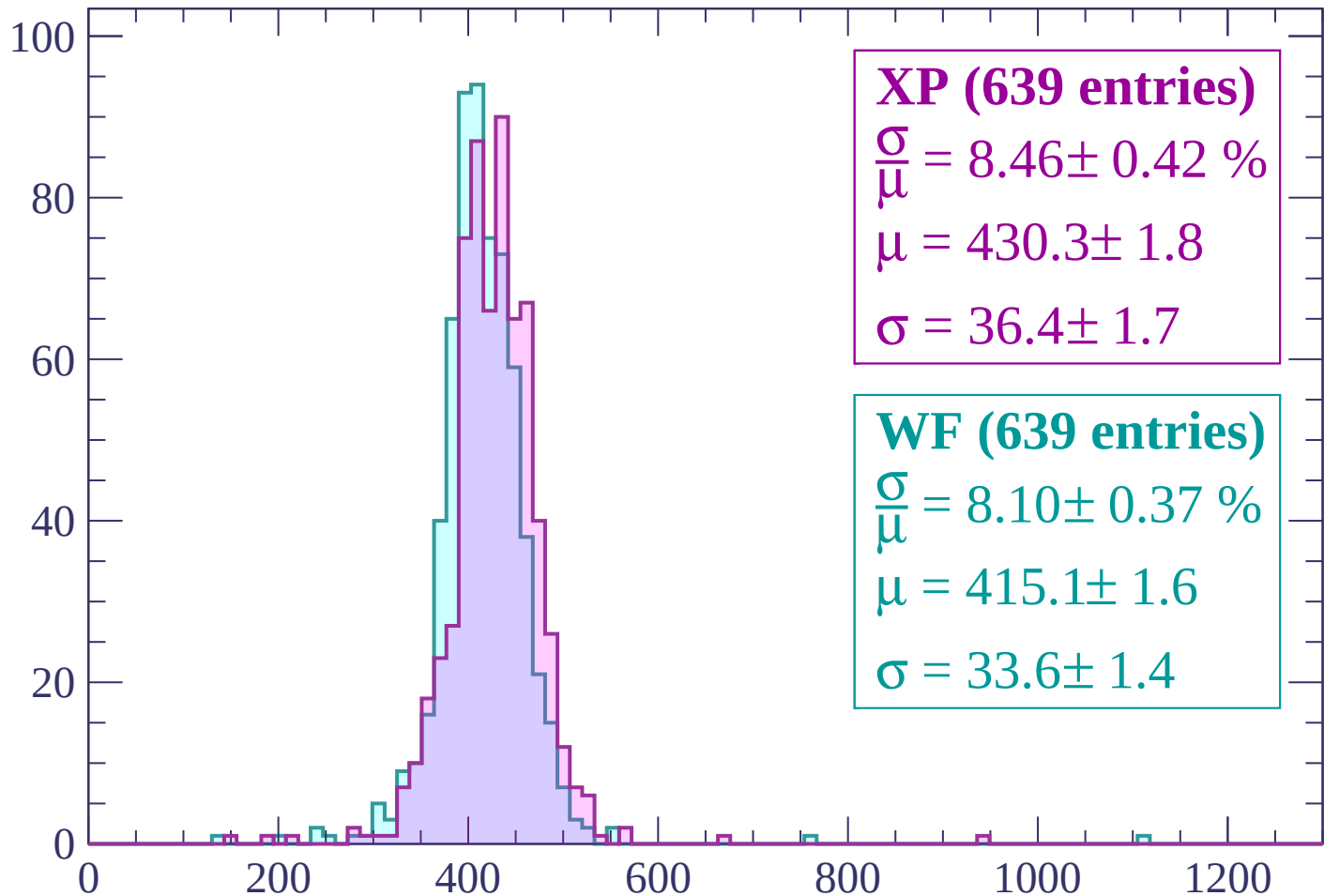
$$\mu = 404.0 \pm 1.7$$

$$\sigma = 33.8 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -180$

Count



XP (639 entries)

$$\frac{\sigma}{\mu} = 8.46 \pm 0.42 \%$$

$$\mu = 430.3 \pm 1.8$$

$$\sigma = 36.4 \pm 1.7$$

WF (639 entries)

$$\frac{\sigma}{\mu} = 8.10 \pm 0.37 \%$$

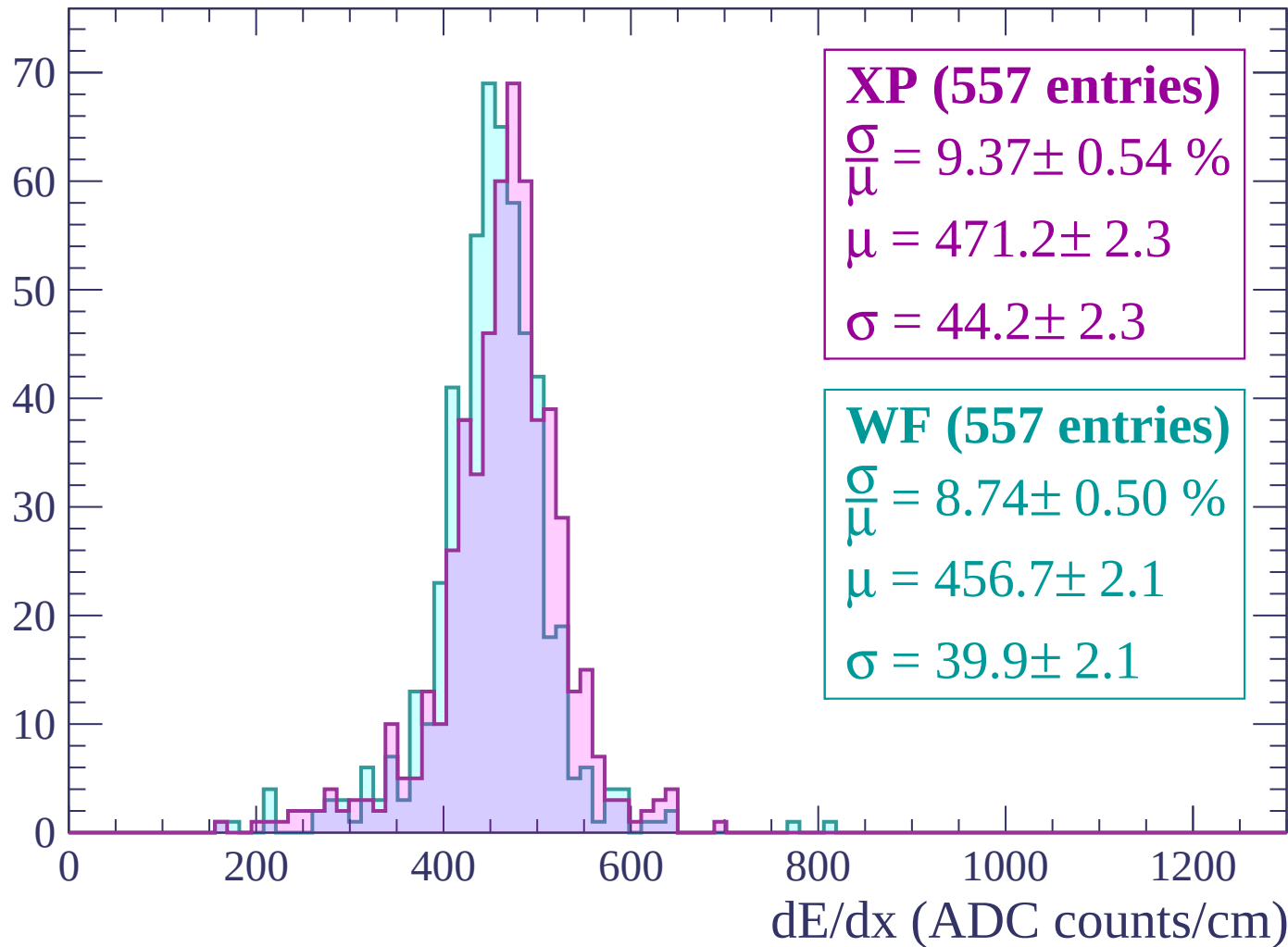
$$\mu = 415.1 \pm 1.6$$

$$\sigma = 33.6 \pm 1.4$$

dE/dx (ADC counts/cm)

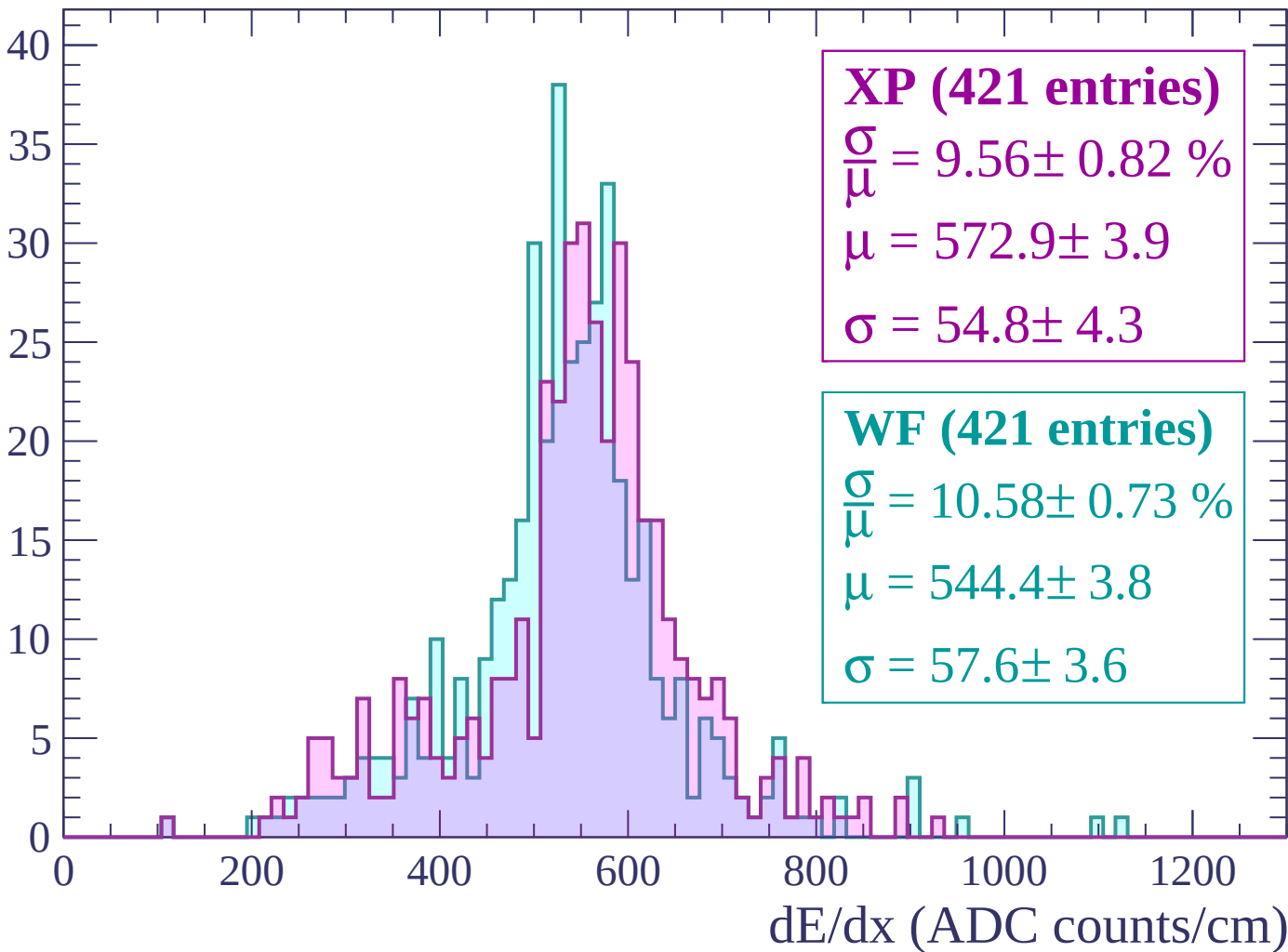
Energy loss | $-180 < p < -120$

Count



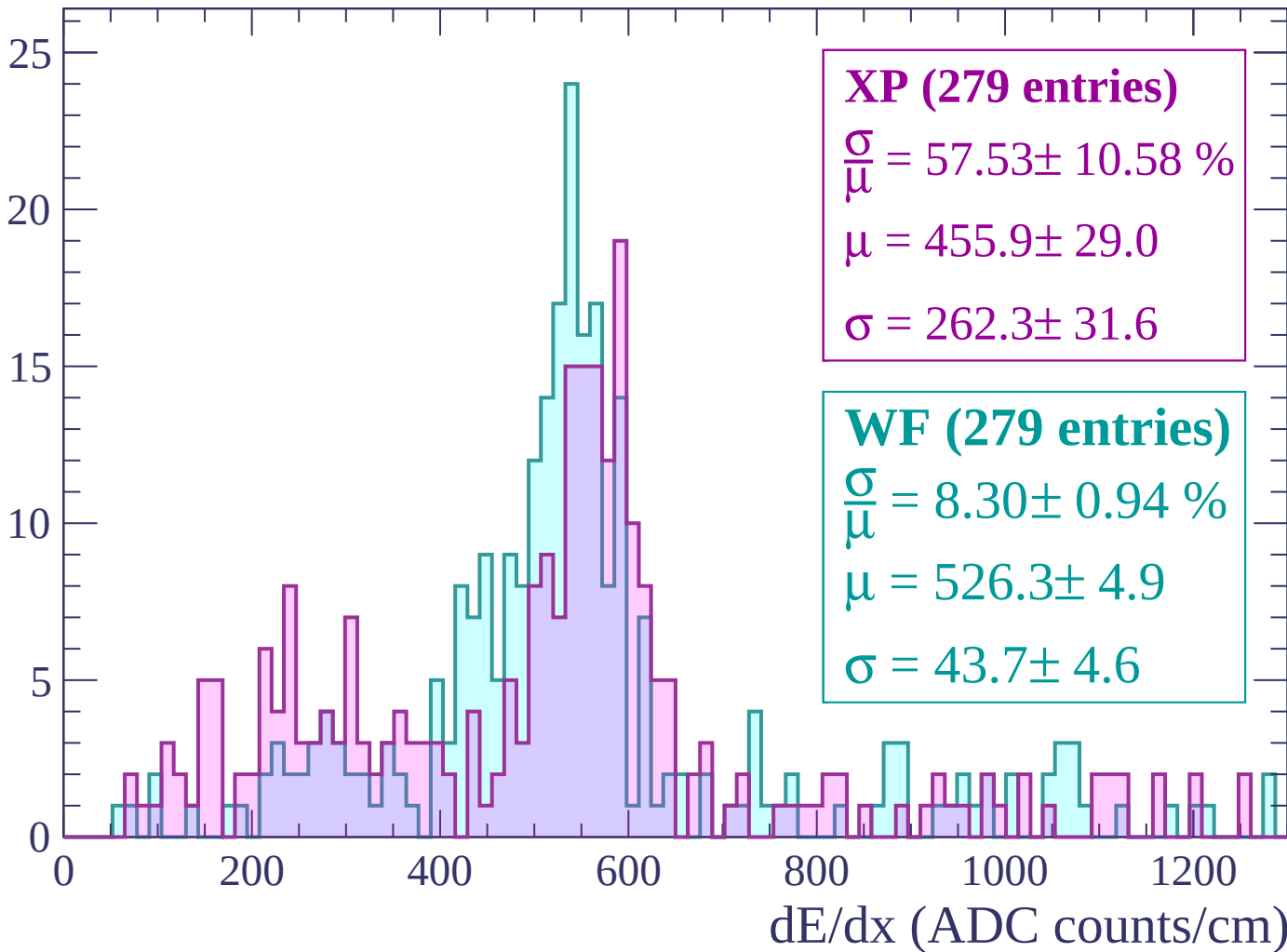
Energy loss | $-120 < p < -60$

Count



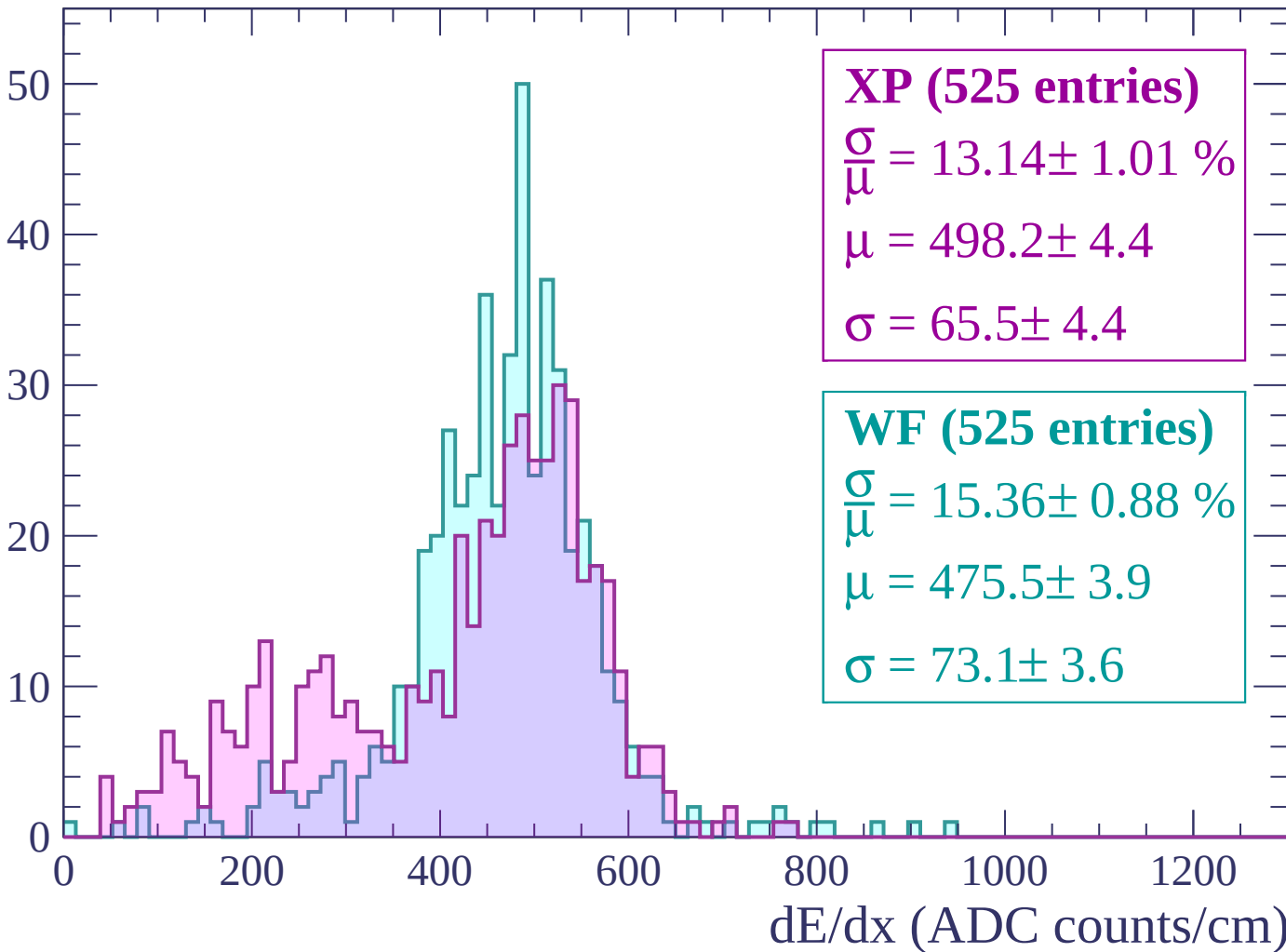
Energy loss | $-60 < p < 0$

Count



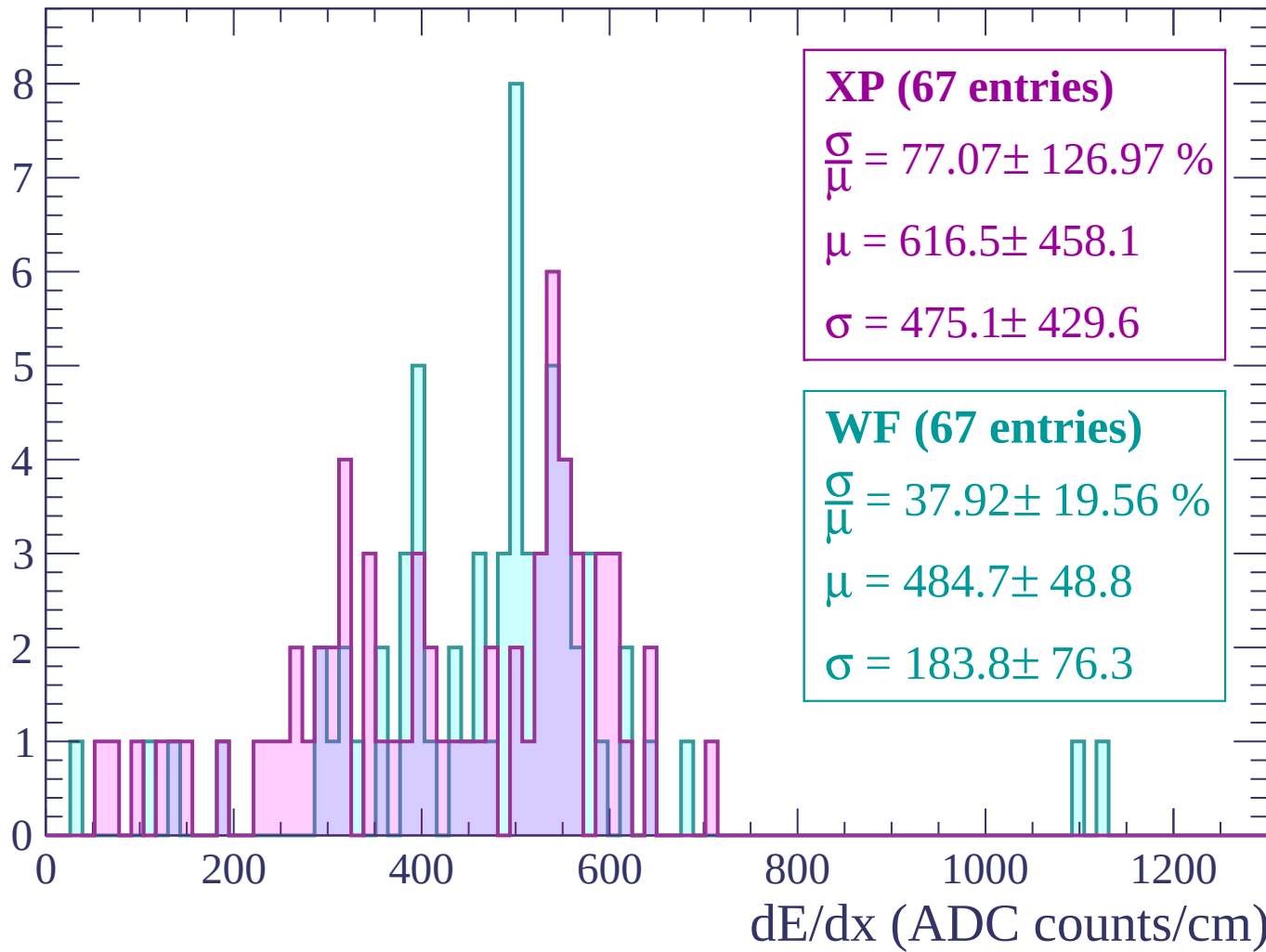
Energy loss | $0 < p < 60$

Count



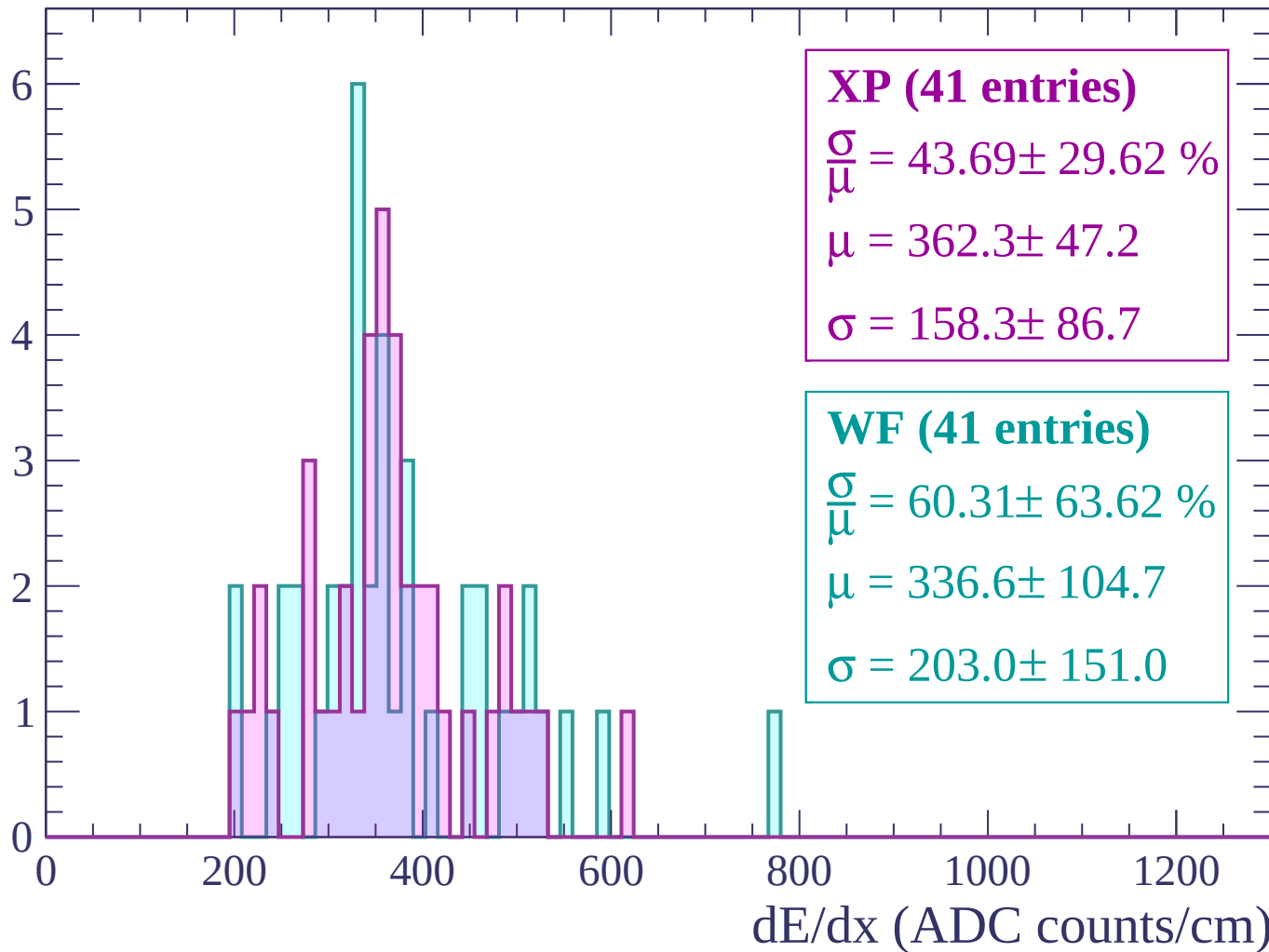
Energy loss | $60 < p < 120$

Count



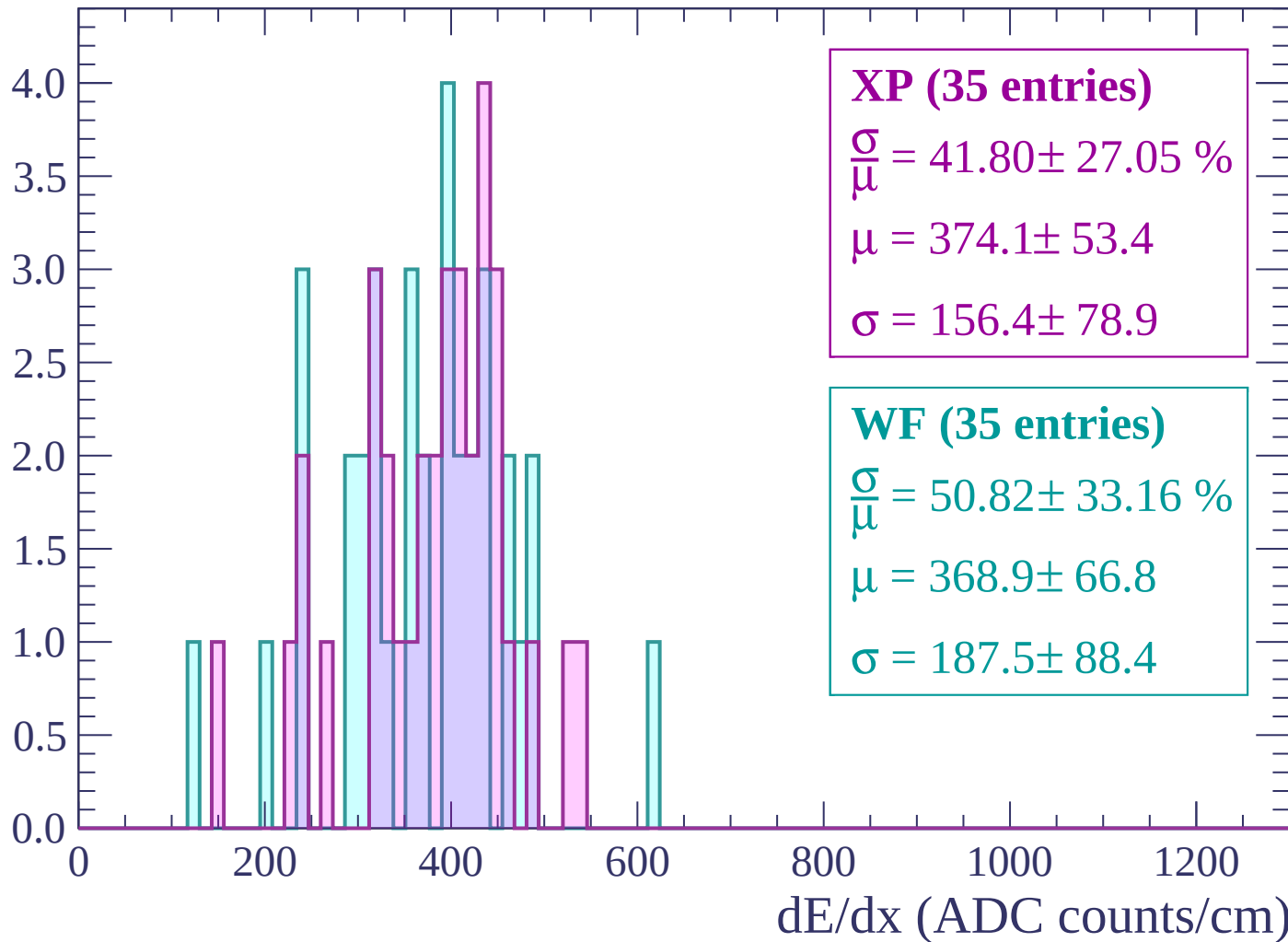
Energy loss | $120 < p < 180$

Count



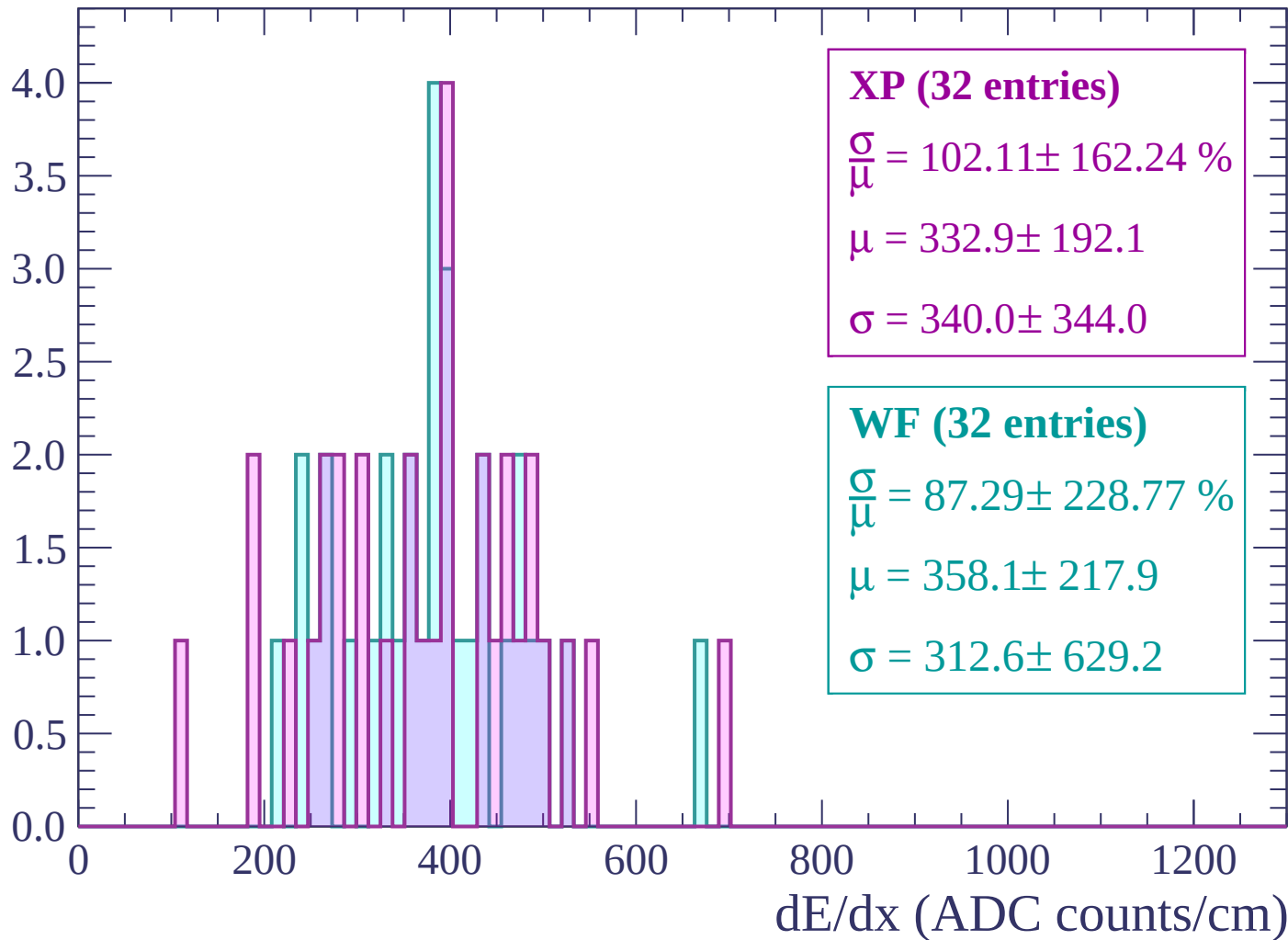
Energy loss | $180 < p < 240$

Count



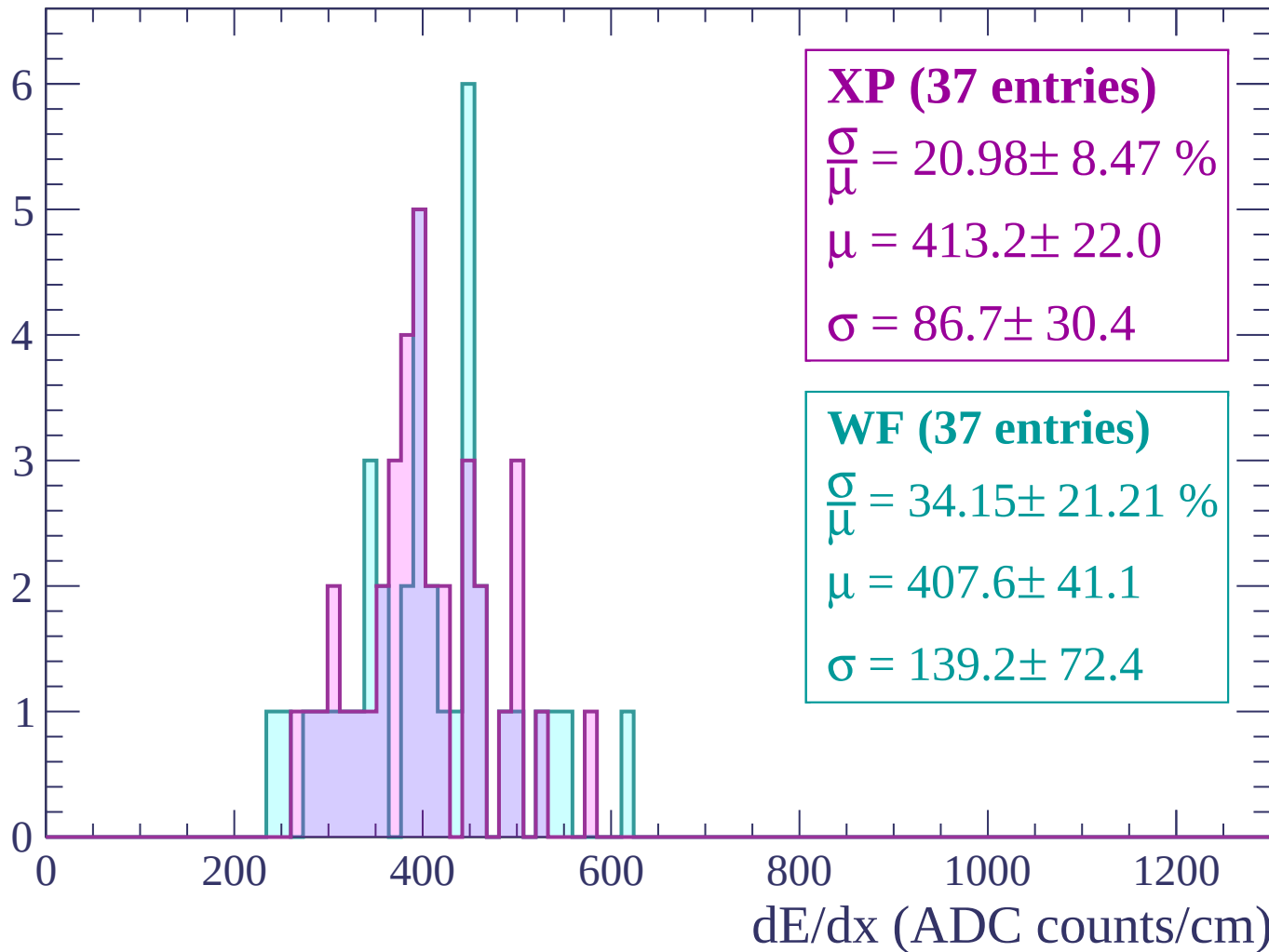
Energy loss | $240 < p < 300$

Count



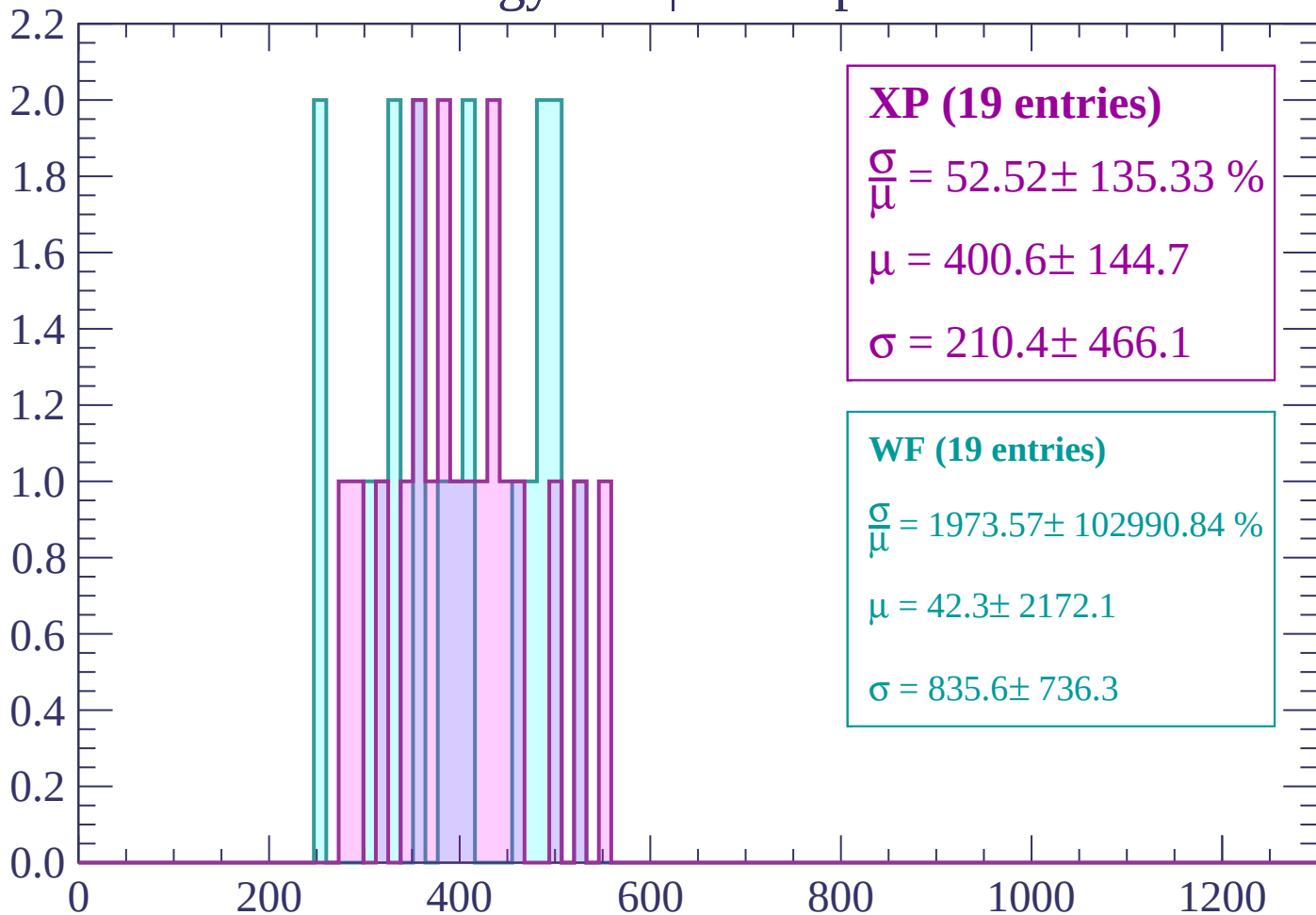
Energy loss | $300 < p < 360$

Count



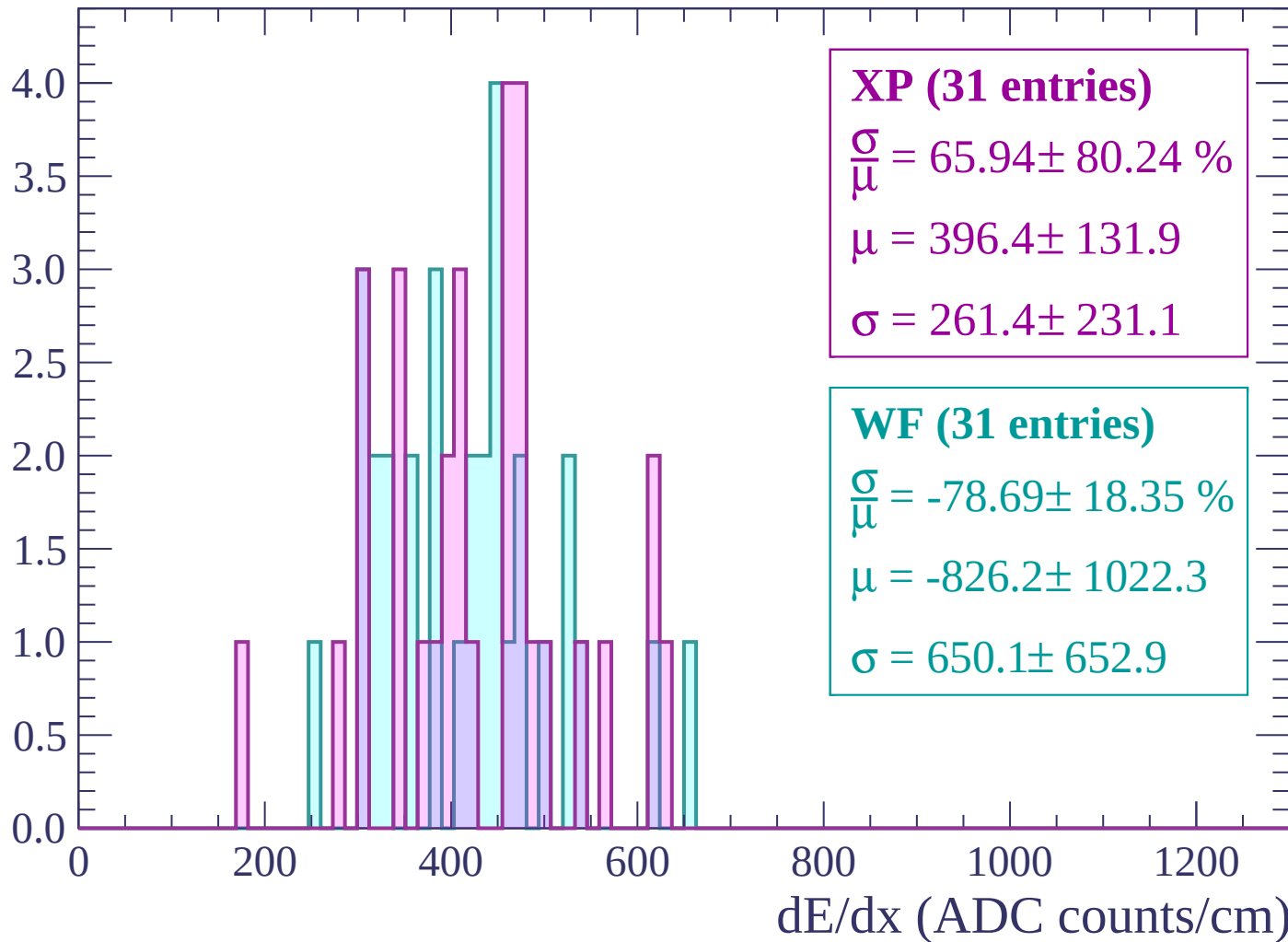
Energy loss | $360 < p < 420$

Count



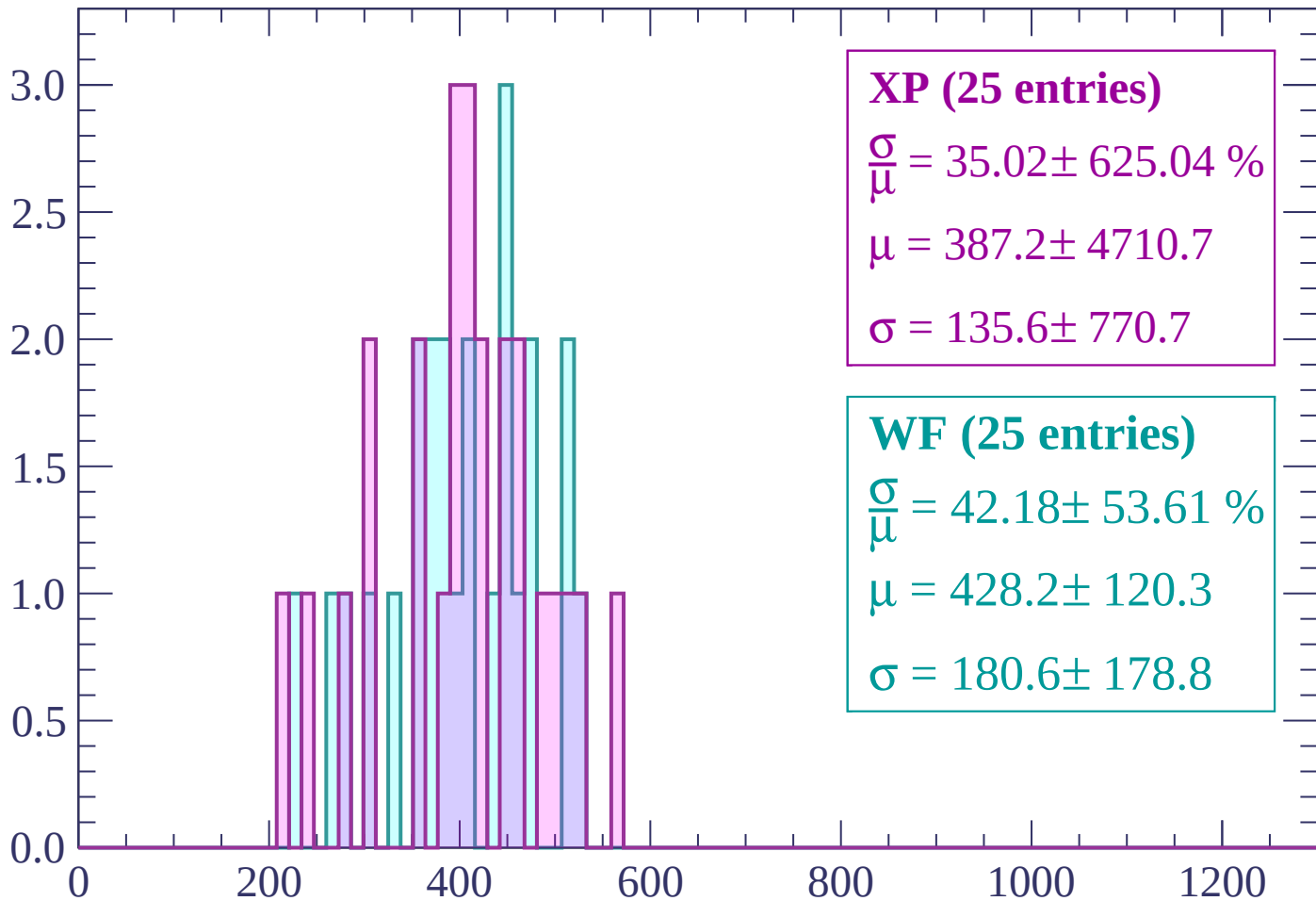
Energy loss | $420 < p < 480$

Count



Energy loss | $480 < p < 540$

Count



XP (25 entries)

$$\frac{\sigma}{\mu} = 35.02 \pm 625.04 \%$$

$$\mu = 387.2 \pm 4710.7$$

$$\sigma = 135.6 \pm 770.7$$

WF (25 entries)

$$\frac{\sigma}{\mu} = 42.18 \pm 53.61 \%$$

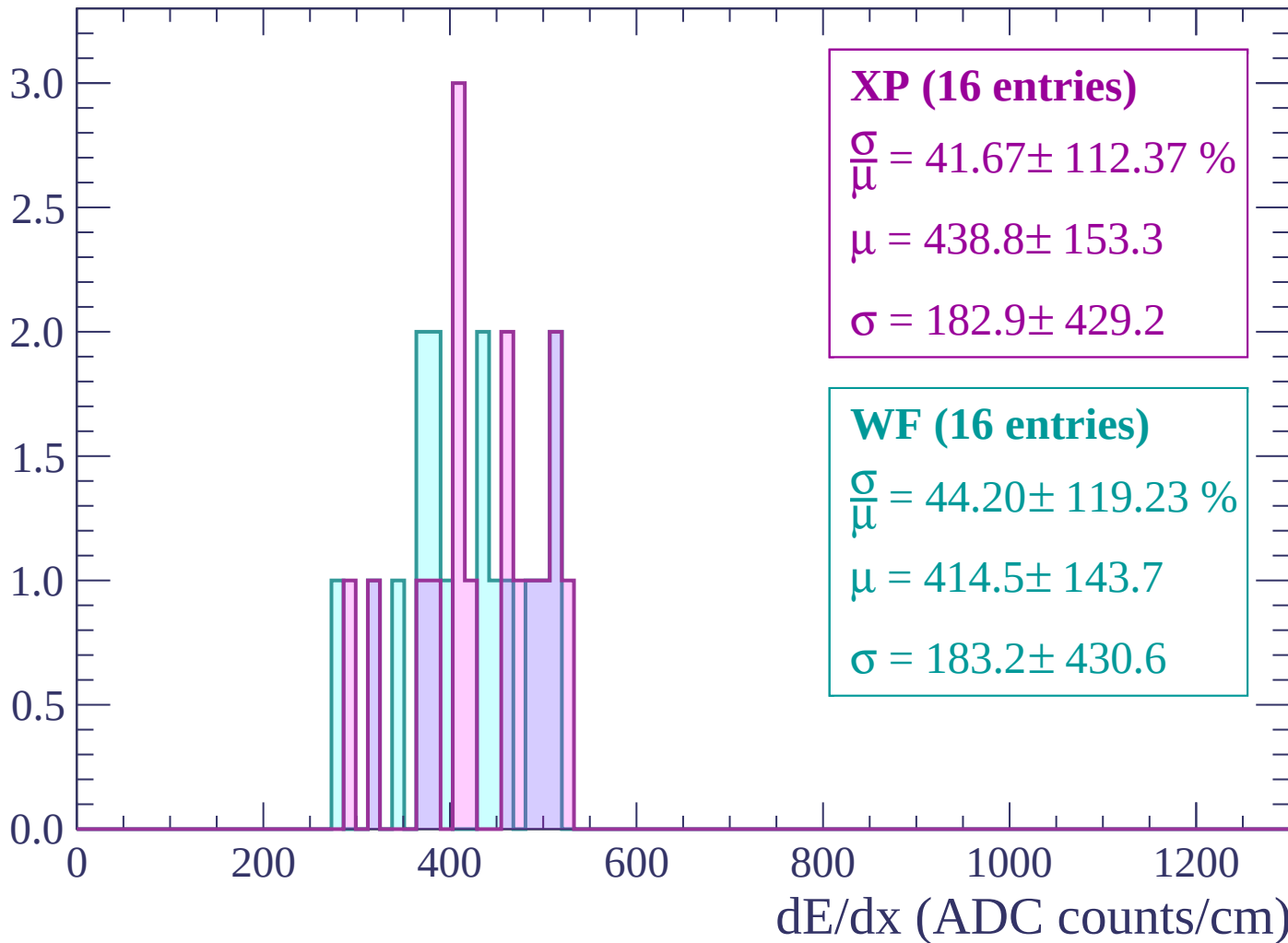
$$\mu = 428.2 \pm 120.3$$

$$\sigma = 180.6 \pm 178.8$$

dE/dx (ADC counts/cm)

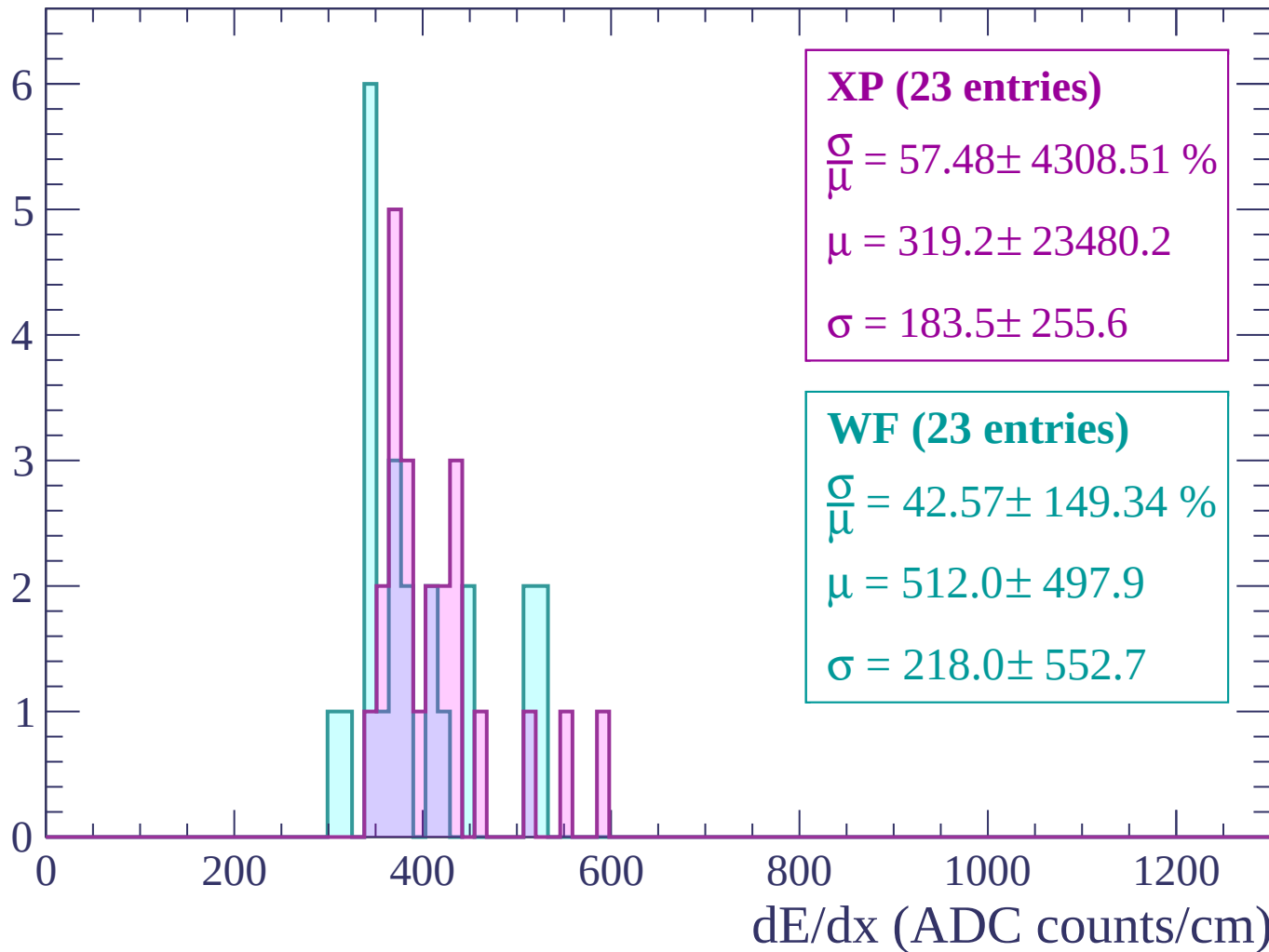
Energy loss | $540 < p < 600$

Count



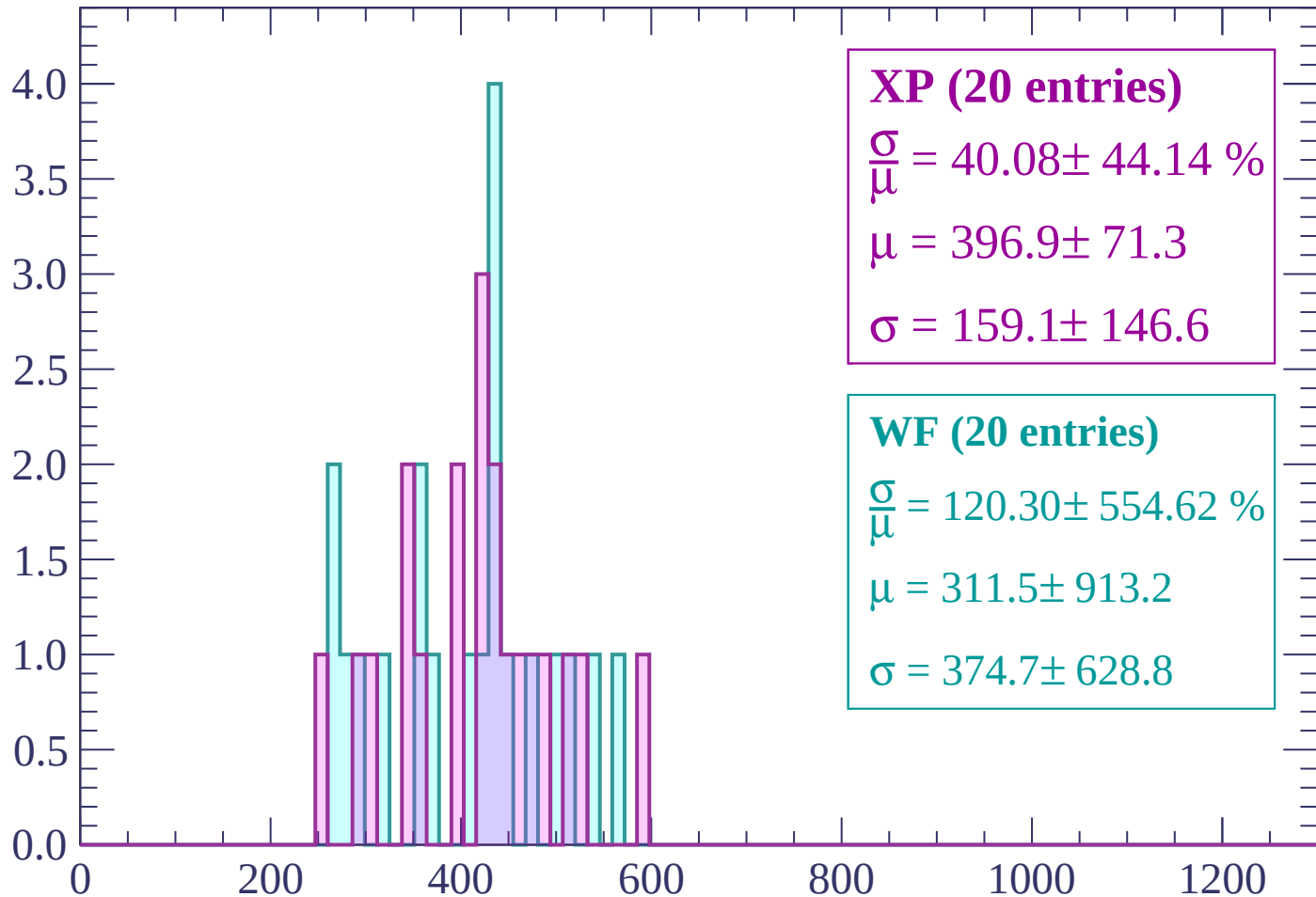
Energy loss | $600 < p < 660$

Count



Energy loss | $660 < p < 720$

Count



XP (20 entries)

$$\frac{\sigma}{\mu} = 40.08 \pm 44.14 \%$$

$$\mu = 396.9 \pm 71.3$$

$$\sigma = 159.1 \pm 146.6$$

WF (20 entries)

$$\frac{\sigma}{\mu} = 120.30 \pm 554.62 \%$$

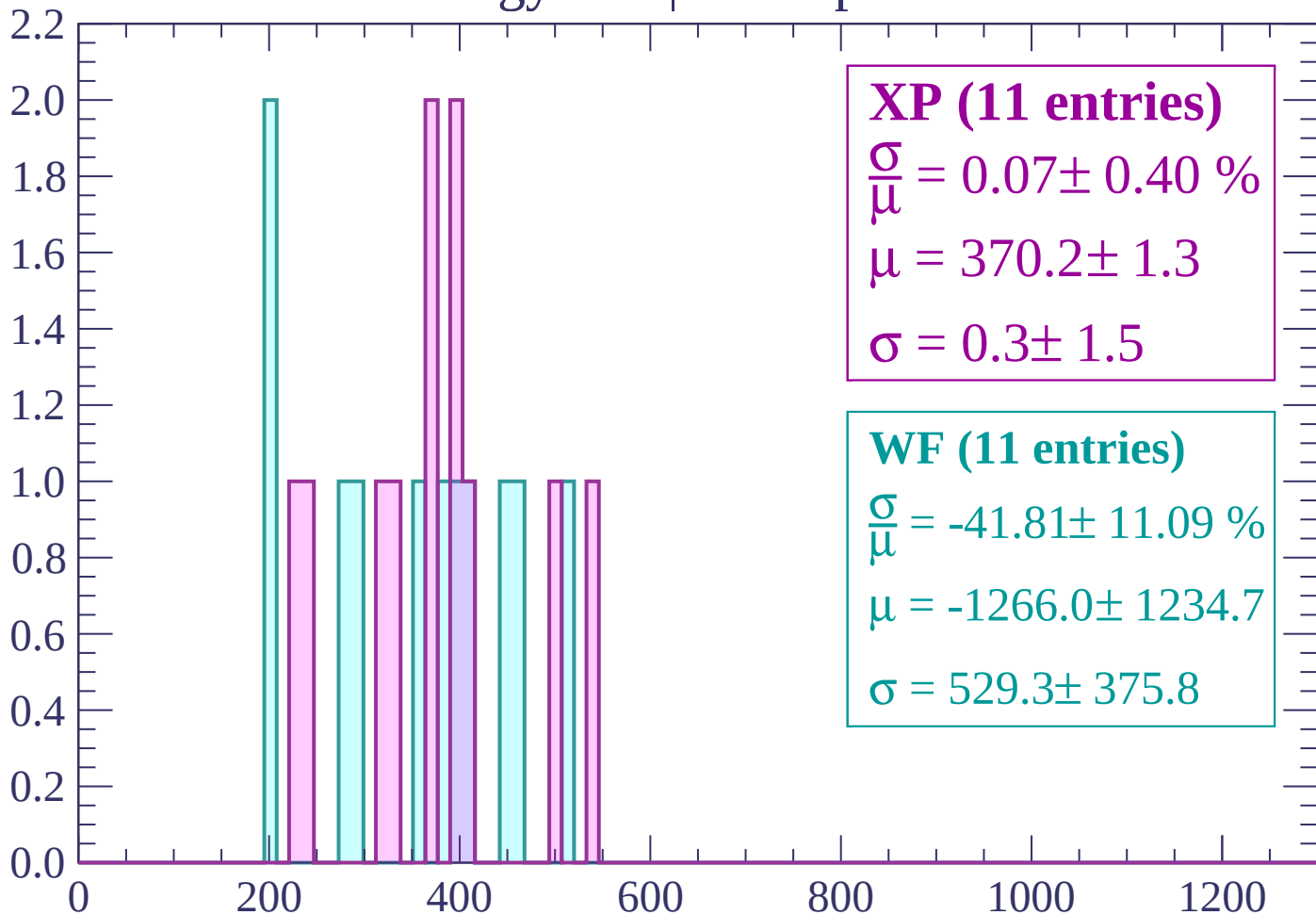
$$\mu = 311.5 \pm 913.2$$

$$\sigma = 374.7 \pm 628.8$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count



XP (11 entries)

$\frac{\sigma}{\mu} = 0.07 \pm 0.40 \%$

$\mu = 370.2 \pm 1.3$

$\sigma = 0.3 \pm 1.5$

WF (11 entries)

$\frac{\sigma}{\mu} = -41.81 \pm 11.09 \%$

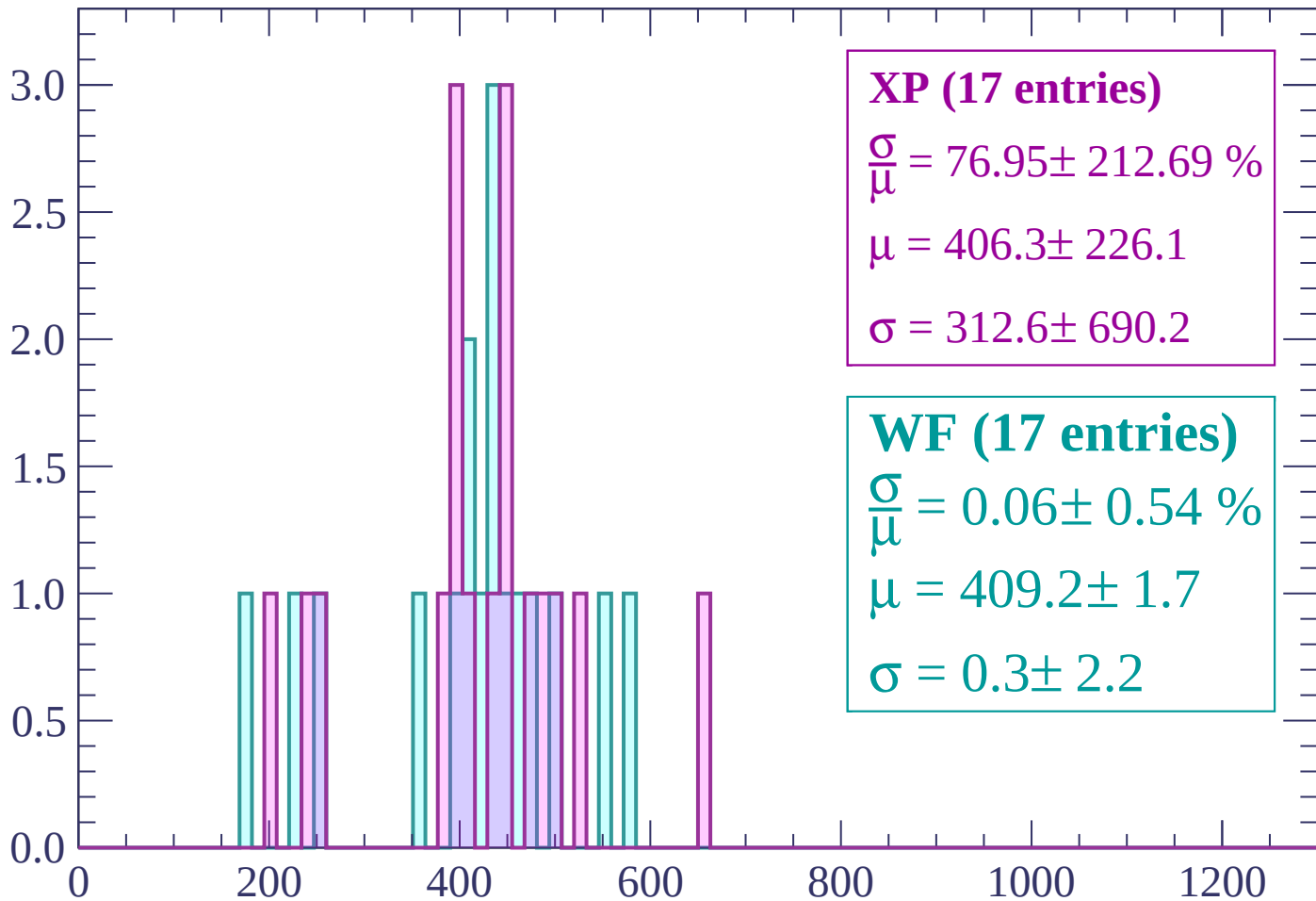
$\mu = -1266.0 \pm 1234.7$

$\sigma = 529.3 \pm 375.8$

dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$

Count



XP (17 entries)

$$\frac{\sigma}{\mu} = 76.95 \pm 212.69 \%$$

$$\mu = 406.3 \pm 226.1$$

$$\sigma = 312.6 \pm 690.2$$

WF (17 entries)

$$\frac{\sigma}{\mu} = 0.06 \pm 0.54 \%$$

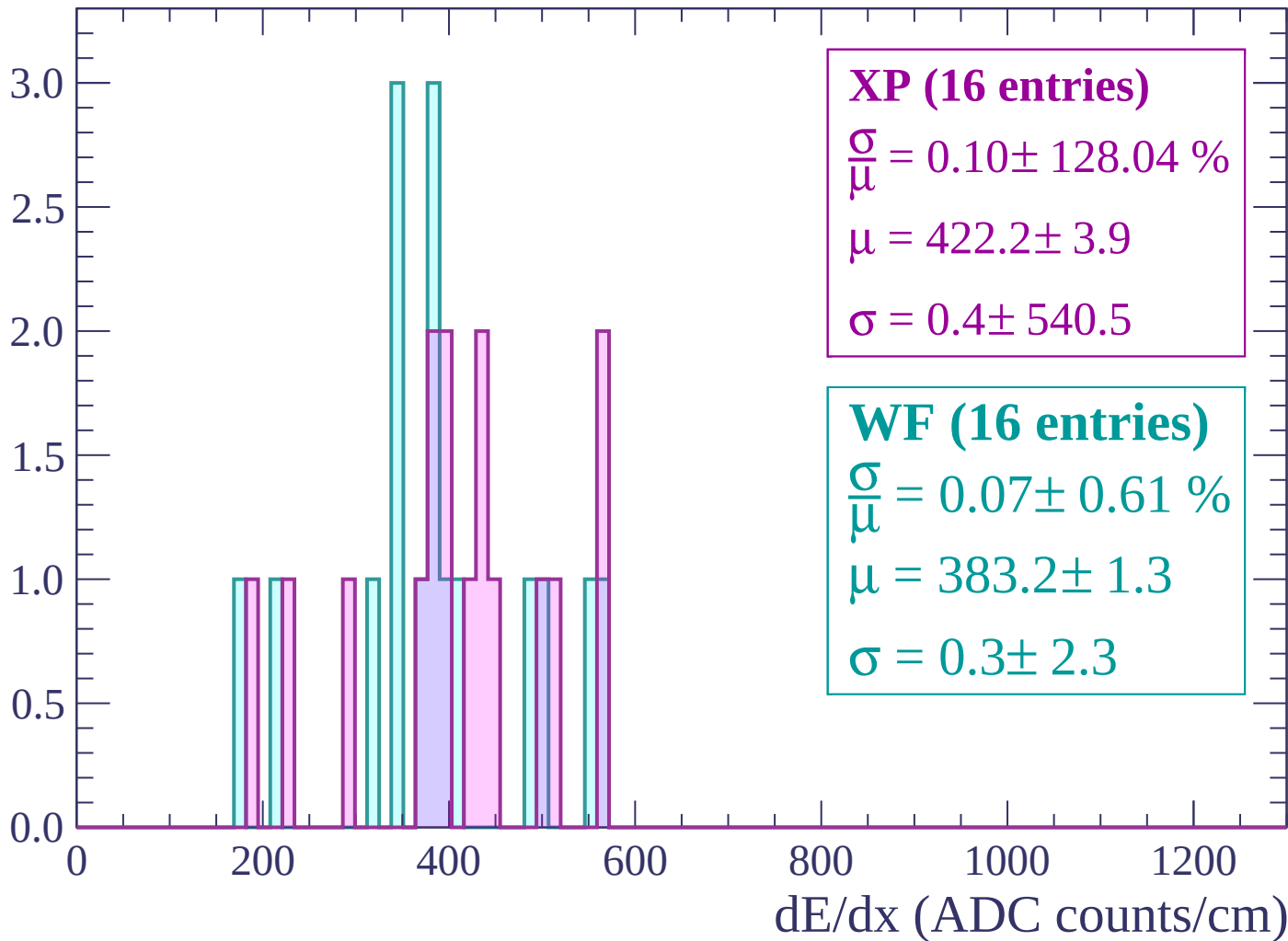
$$\mu = 409.2 \pm 1.7$$

$$\sigma = 0.3 \pm 2.2$$

dE/dx (ADC counts/cm)

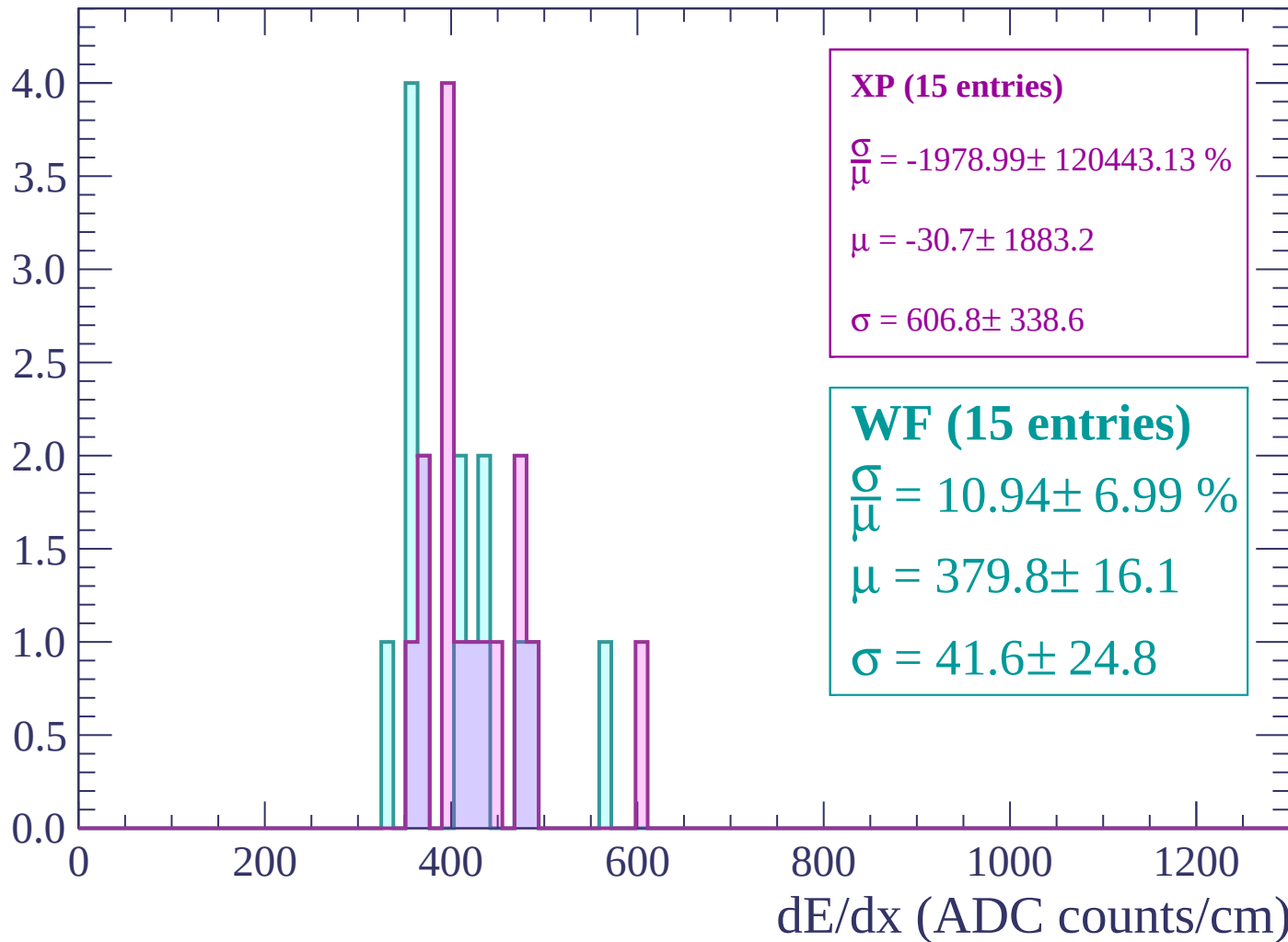
Energy loss | $840 < p < 900$

Count



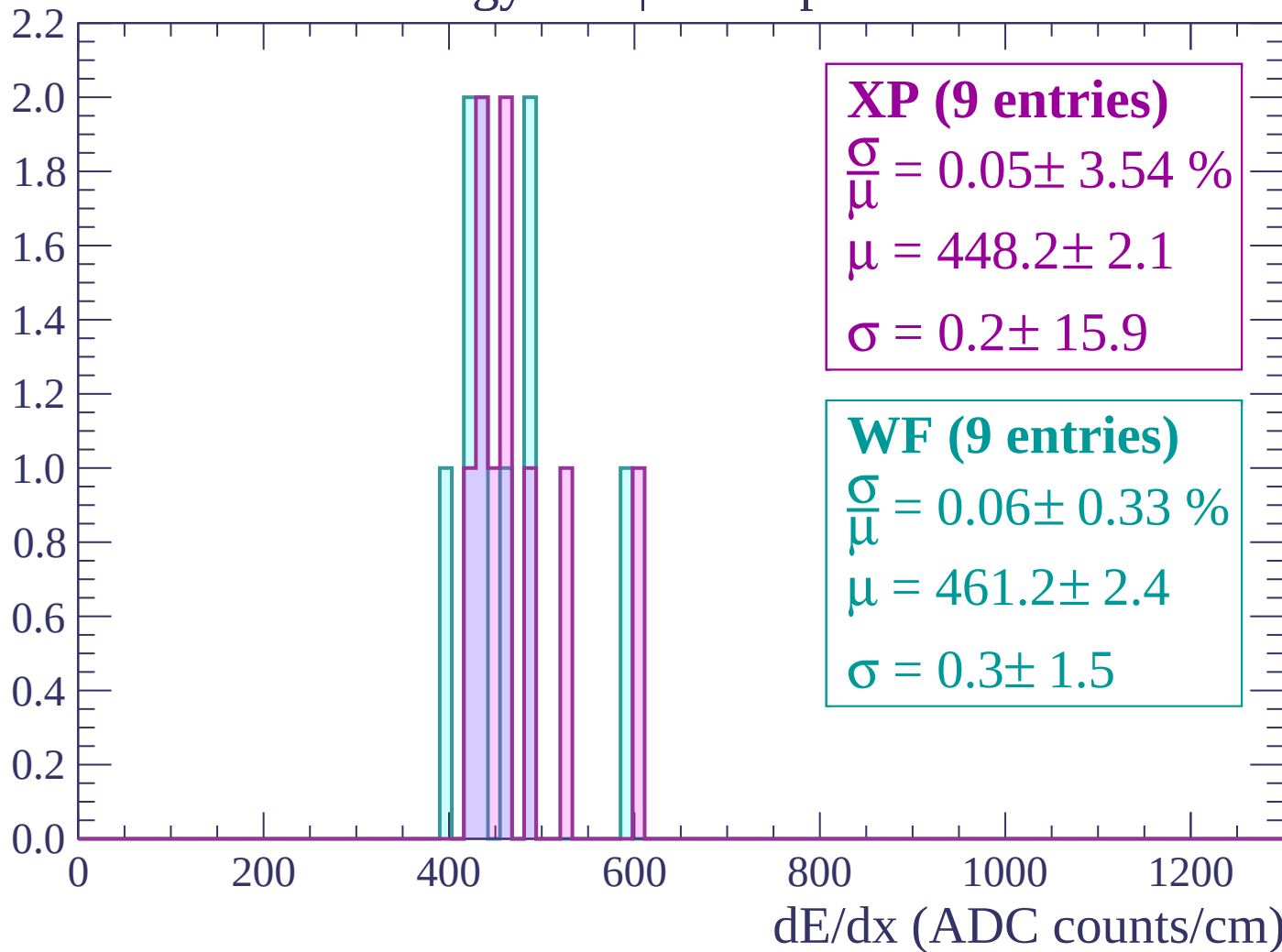
Energy loss | $900 < p < 960$

Count



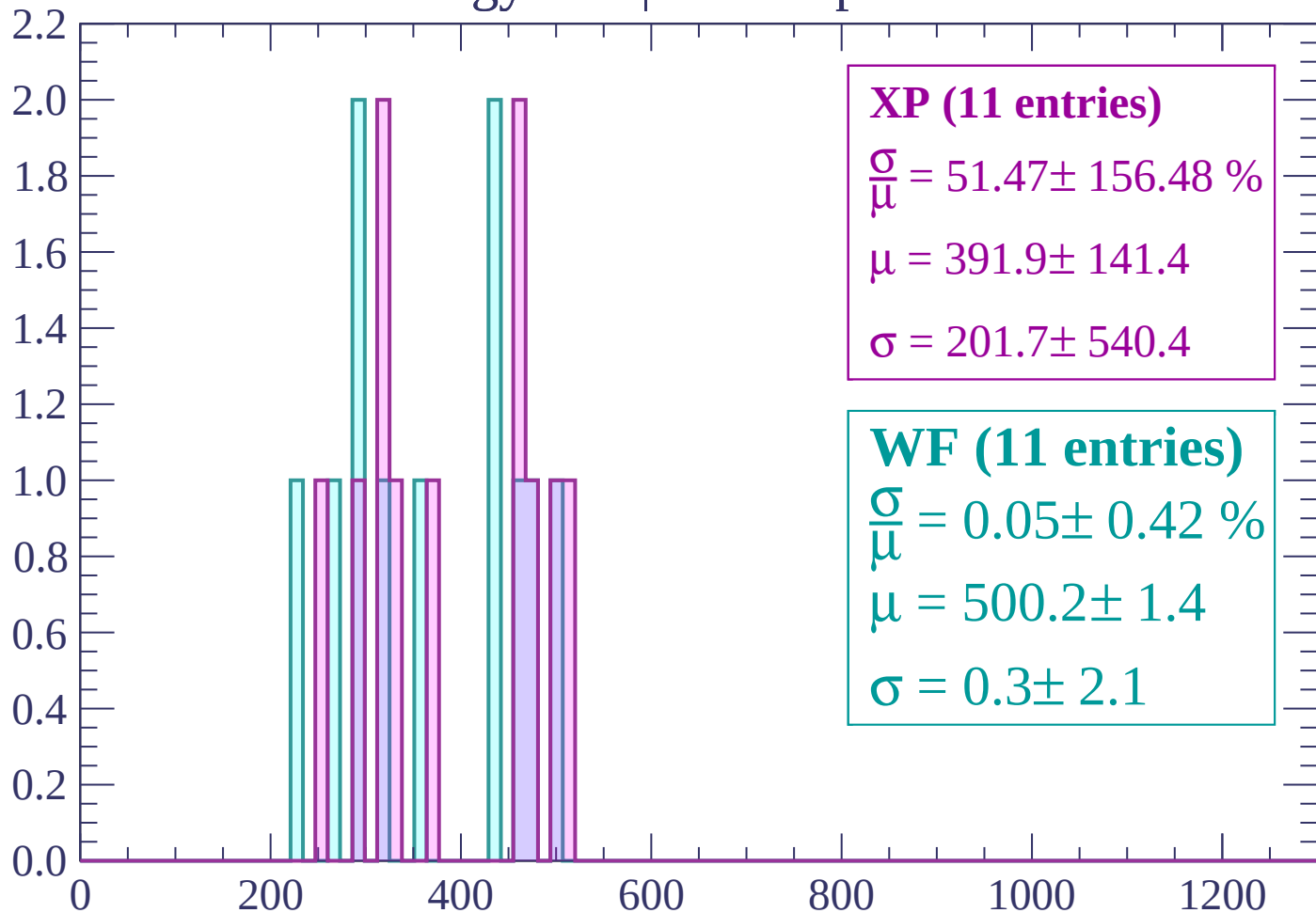
Energy loss | $960 < p < 1020$

Count



Energy loss | $1020 < p < 1080$

Count



XP (11 entries)

$$\frac{\sigma}{\mu} = 51.47 \pm 156.48 \%$$

$$\mu = 391.9 \pm 141.4$$

$$\sigma = 201.7 \pm 540.4$$

WF (11 entries)

$$\frac{\sigma}{\mu} = 0.05 \pm 0.42 \%$$

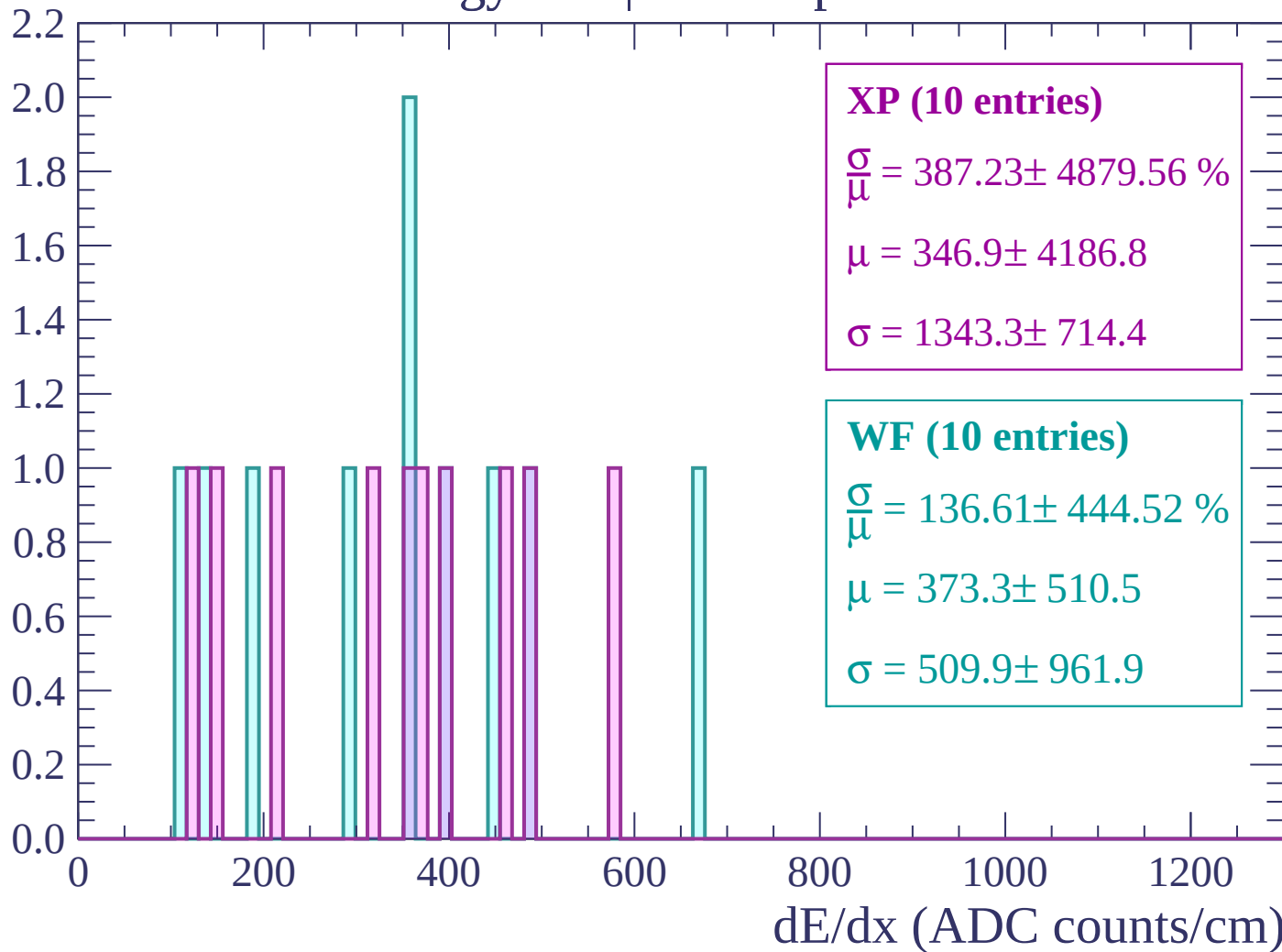
$$\mu = 500.2 \pm 1.4$$

$$\sigma = 0.3 \pm 2.1$$

dE/dx (ADC counts/cm)

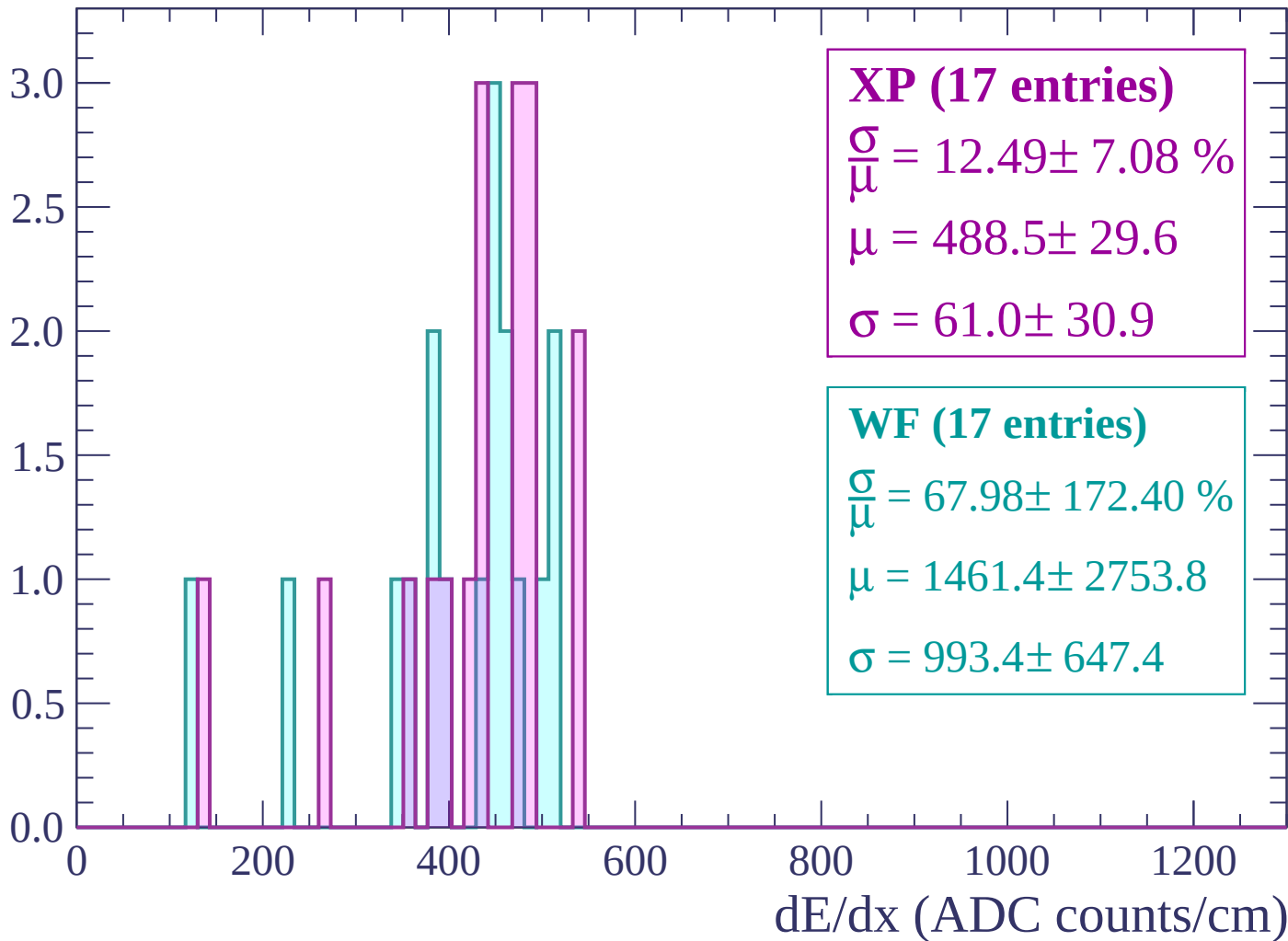
Energy loss | 1080 < p < 1140

Count



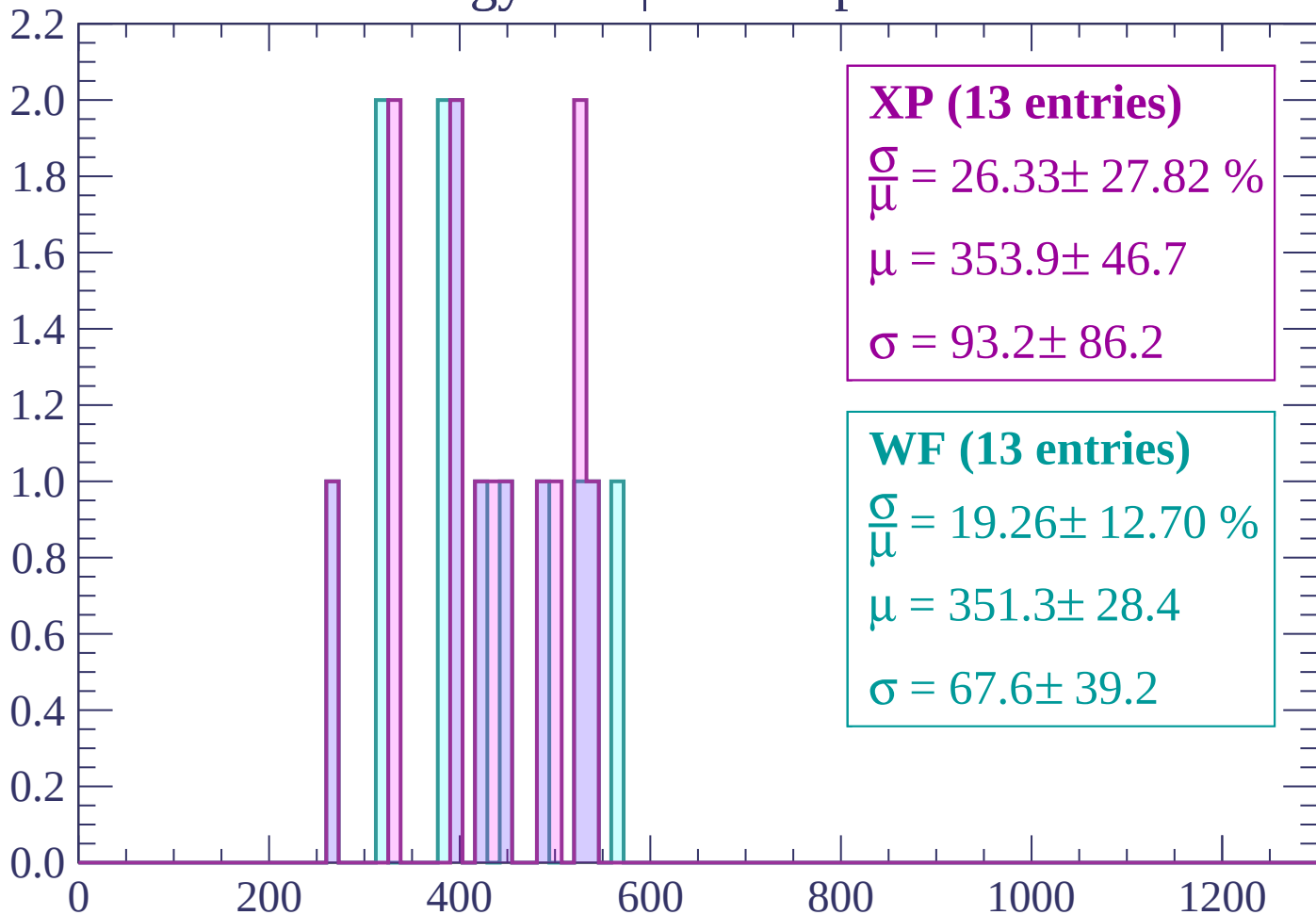
Energy loss | $1140 < p < 1200$

Count



Energy loss | $1200 < p < 1260$

Count



XP (13 entries)

$$\frac{\sigma}{\mu} = 26.33 \pm 27.82 \%$$

$$\mu = 353.9 \pm 46.7$$

$$\sigma = 93.2 \pm 86.2$$

WF (13 entries)

$$\frac{\sigma}{\mu} = 19.26 \pm 12.70 \%$$

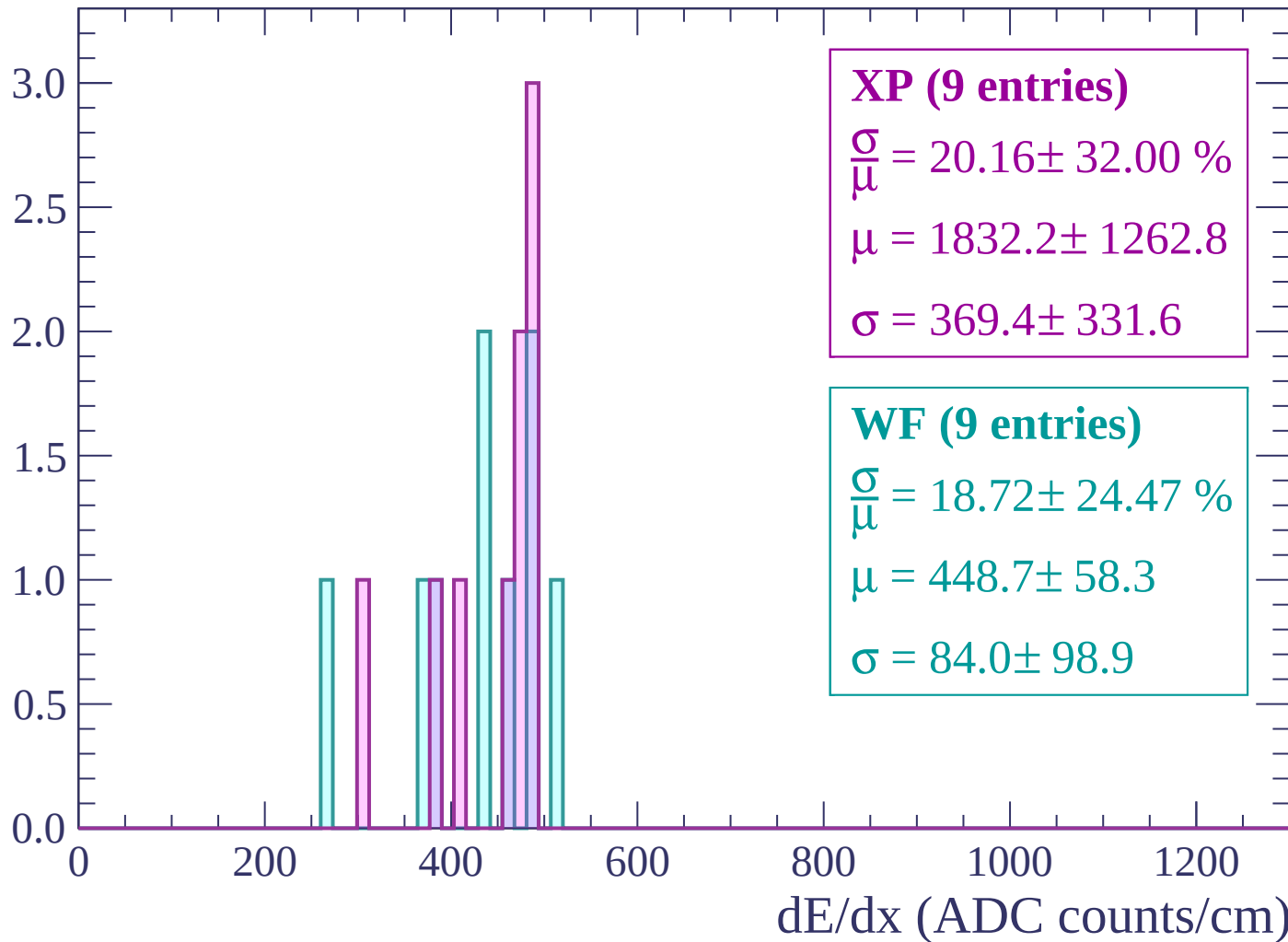
$$\mu = 351.3 \pm 28.4$$

$$\sigma = 67.6 \pm 39.2$$

dE/dx (ADC counts/cm)

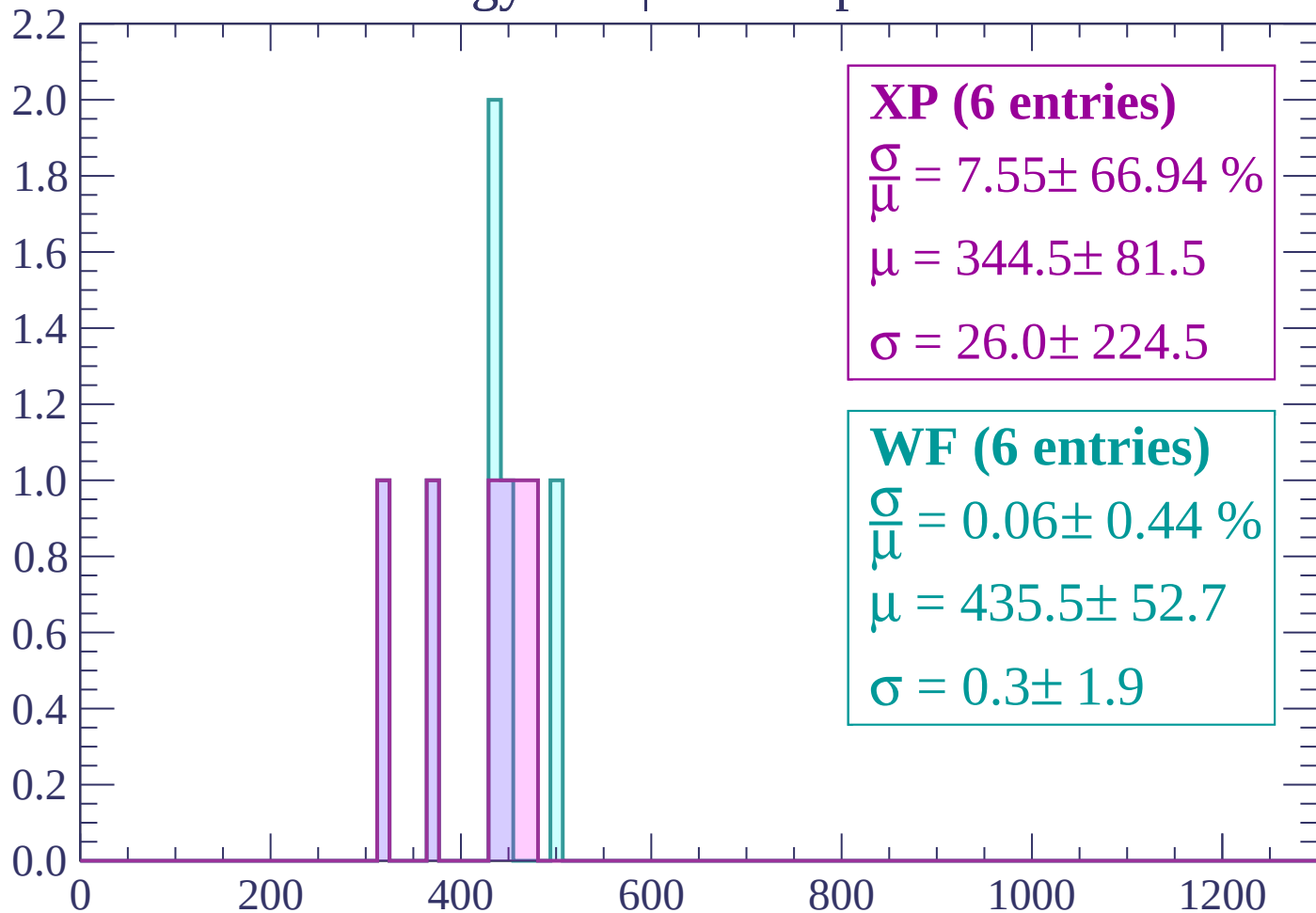
Energy loss | $1260 < p < 1320$

Count



Energy loss | $1320 < p < 1380$

Count



XP (6 entries)

$$\frac{\sigma}{\mu} = 7.55 \pm 66.94 \%$$

$$\mu = 344.5 \pm 81.5$$

$$\sigma = 26.0 \pm 224.5$$

WF (6 entries)

$$\frac{\sigma}{\mu} = 0.06 \pm 0.44 \%$$

$$\mu = 435.5 \pm 52.7$$

$$\sigma = 0.3 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | $1380 < p < 1440$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (6 entries)

$\frac{\sigma}{\mu} = 102.75 \pm 458.62 \%$

$\mu = 458.8 \pm 1758.4$

$\sigma = 471.4 \pm 297.3$

WF (6 entries)

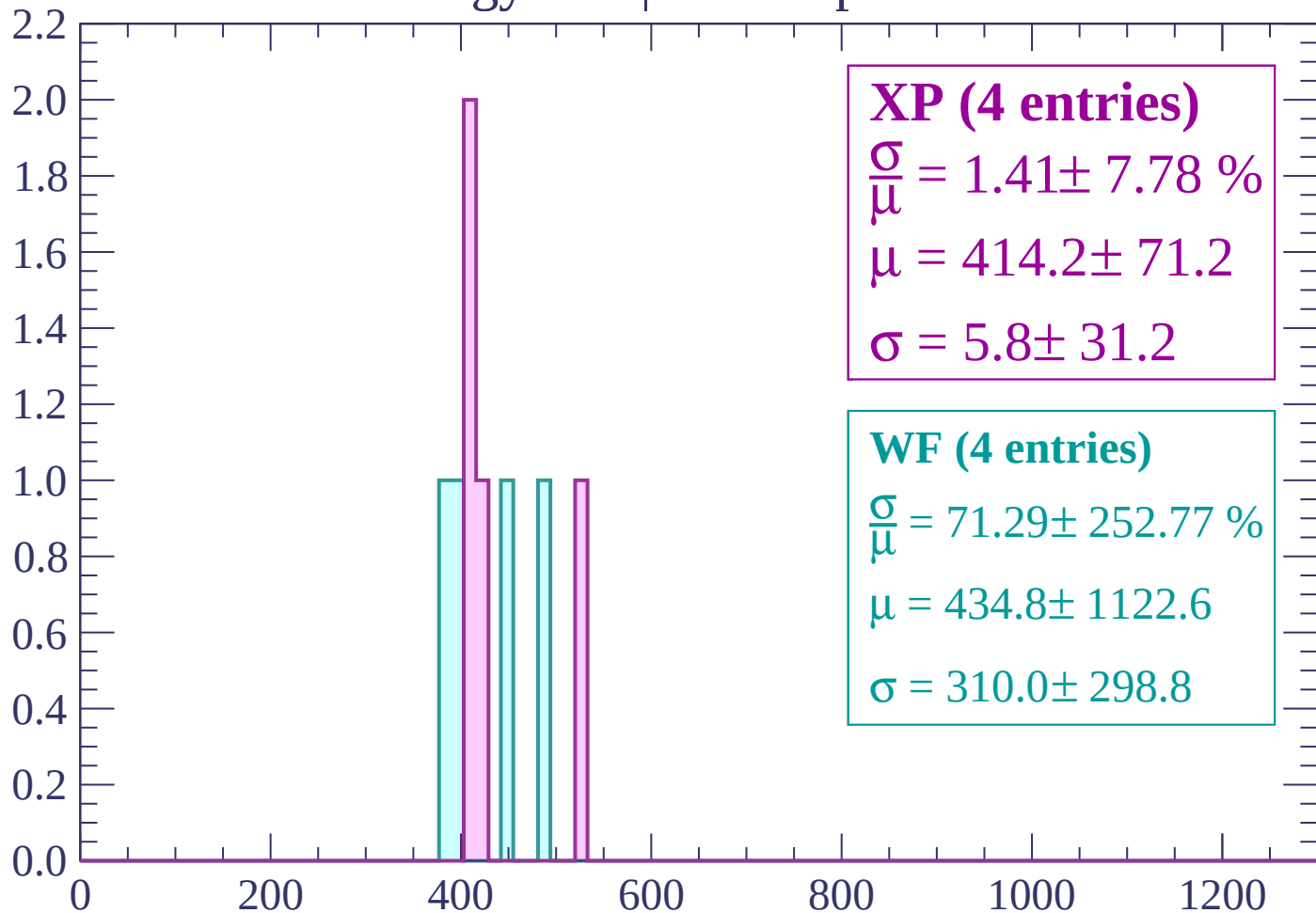
$\frac{\sigma}{\mu} = 3.51 \pm 28.94 \%$

$\mu = 370.5 \pm 47.2$

$\sigma = 13.0 \pm 105.6$

Energy loss | $1440 < p < 1500$

Count



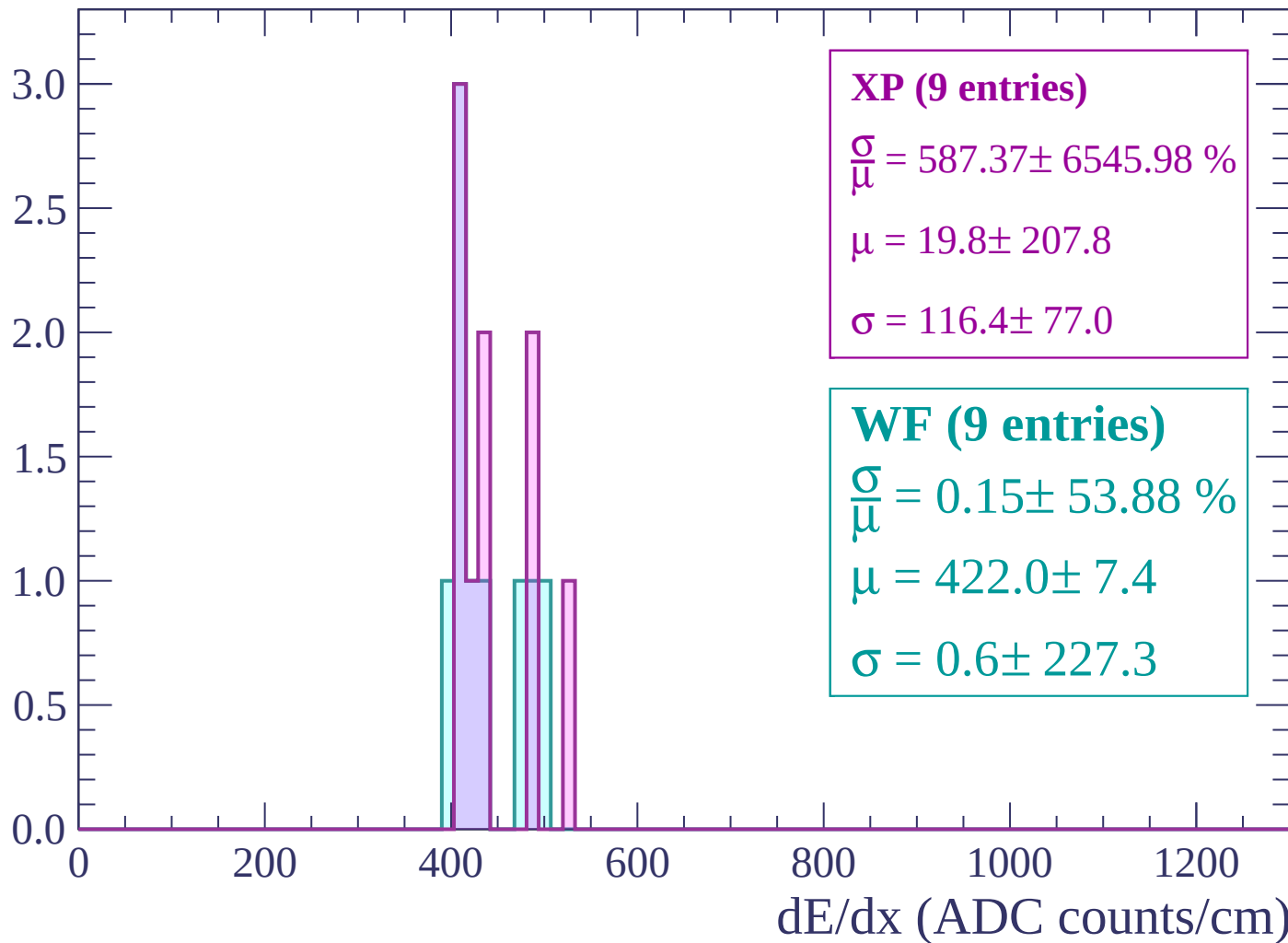
XP (4 entries)
 $\frac{\sigma}{\mu} = 1.41 \pm 7.78 \%$
 $\mu = 414.2 \pm 71.2$
 $\sigma = 5.8 \pm 31.2$

WF (4 entries)
 $\frac{\sigma}{\mu} = 71.29 \pm 252.77 \%$
 $\mu = 434.8 \pm 1122.6$
 $\sigma = 310.0 \pm 298.8$

dE/dx (ADC counts/cm)

Energy loss | $1500 < p < 1560$

Count



Energy loss | $1560 < p < 1620$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

XP (4 entries)

$$\frac{\sigma}{\mu} = 4.41 \pm 32.71 \%$$

$$\mu = 442.0 \pm 64.7$$

$$\sigma = 19.5 \pm 141.7$$

WF (4 entries)

$$\frac{\sigma}{\mu} = 2.99 \pm 24.50 \%$$

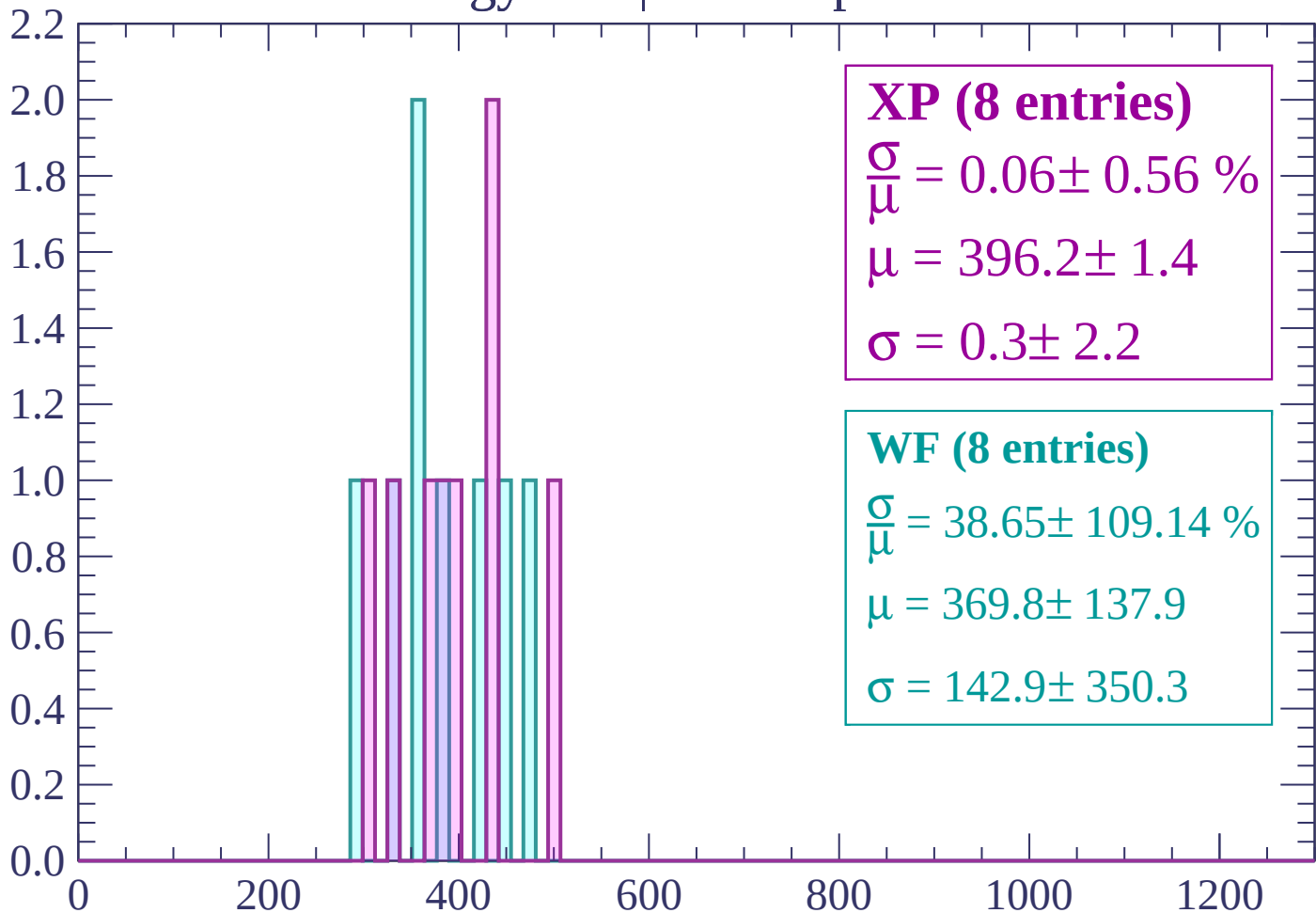
$$\mu = 435.5 \pm 47.2$$

$$\sigma = 13.0 \pm 105.3$$

dE/dx (ADC counts/cm)

Energy loss | $1620 < p < 1680$

Count



XP (8 entries)

$\frac{\sigma}{\mu} = 0.06 \pm 0.56 \%$

$\mu = 396.2 \pm 1.4$

$\sigma = 0.3 \pm 2.2$

WF (8 entries)

$\frac{\sigma}{\mu} = 38.65 \pm 109.14 \%$

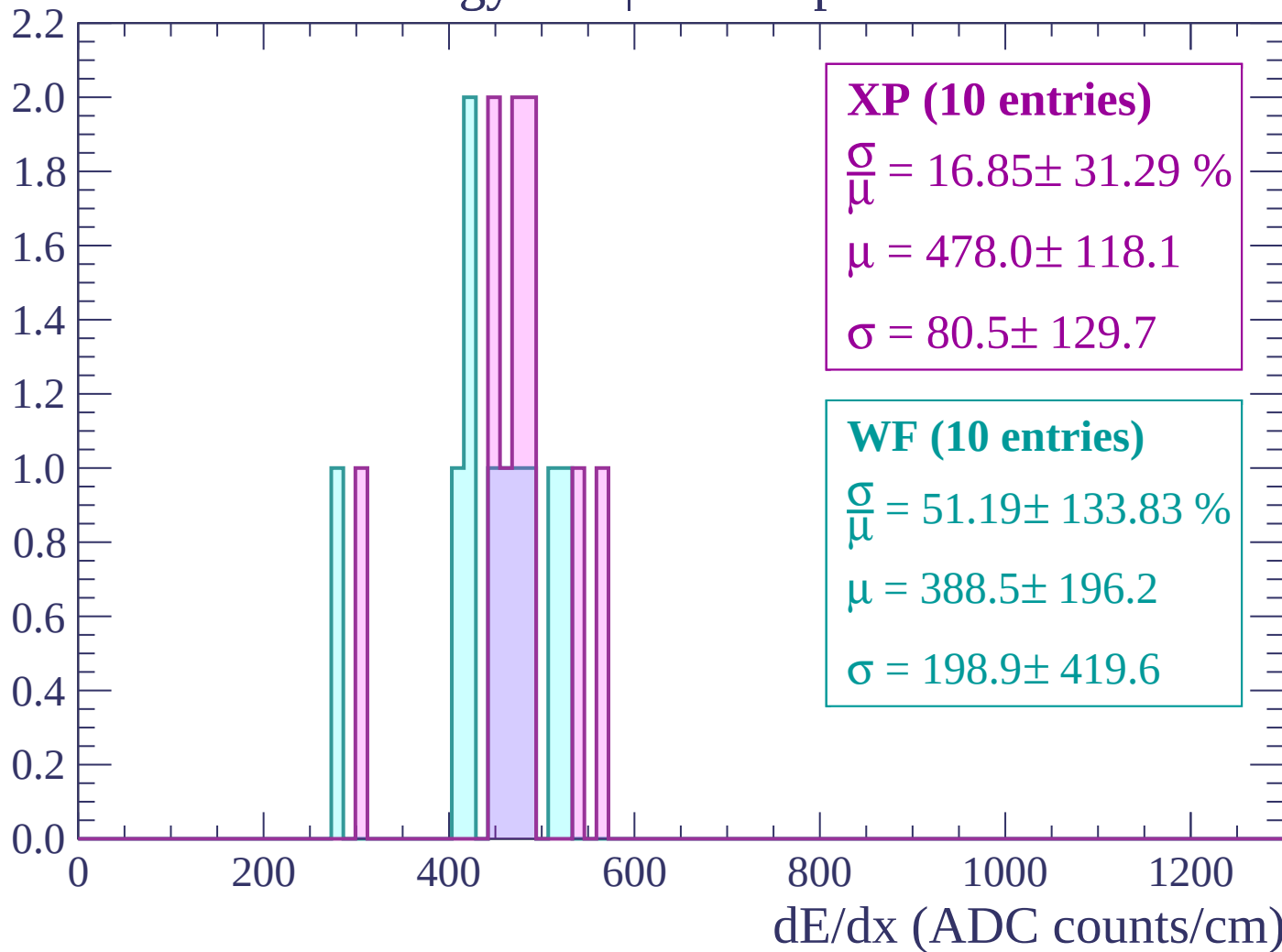
$\mu = 369.8 \pm 137.9$

$\sigma = 142.9 \pm 350.3$

dE/dx (ADC counts/cm)

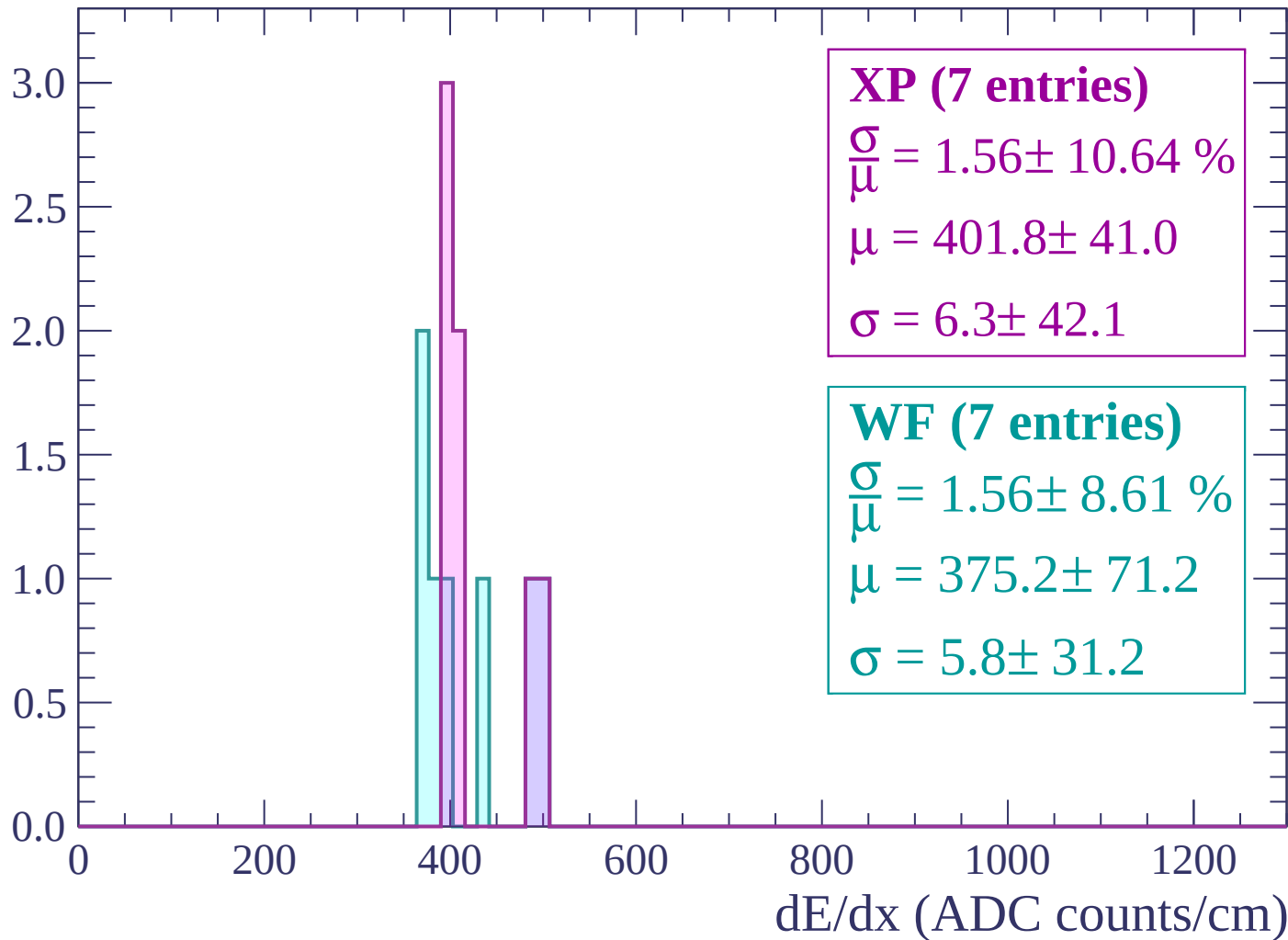
Energy loss | $1680 < p < 1740$

Count



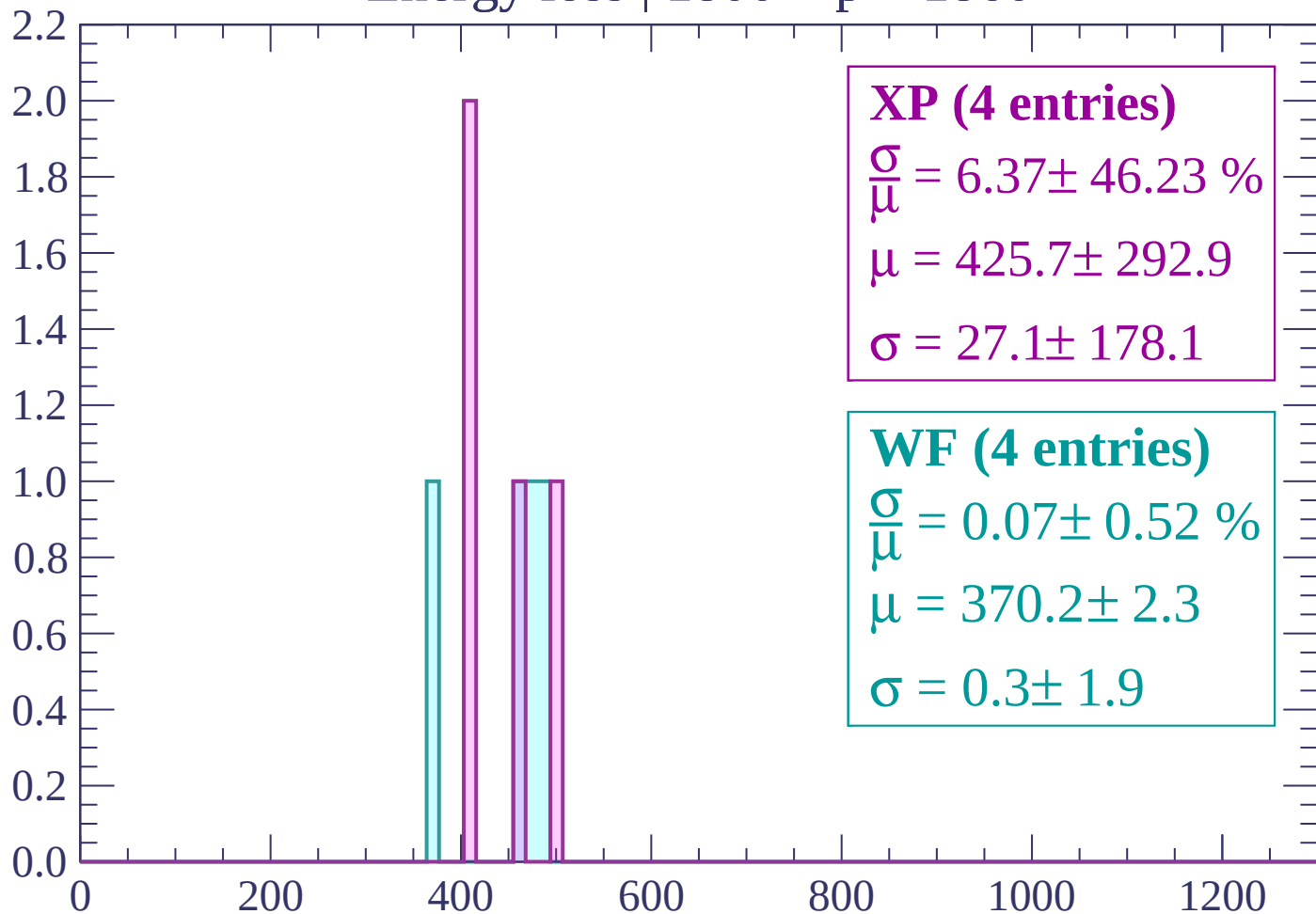
Energy loss | $1740 < p < 1800$

Count



Energy loss | $1800 < p < 1860$

Count



XP (4 entries)

$$\frac{\sigma}{\mu} = 6.37 \pm 46.23 \%$$

$$\mu = 425.7 \pm 292.9$$

$$\sigma = 27.1 \pm 178.1$$

WF (4 entries)

$$\frac{\sigma}{\mu} = 0.07 \pm 0.52 \%$$

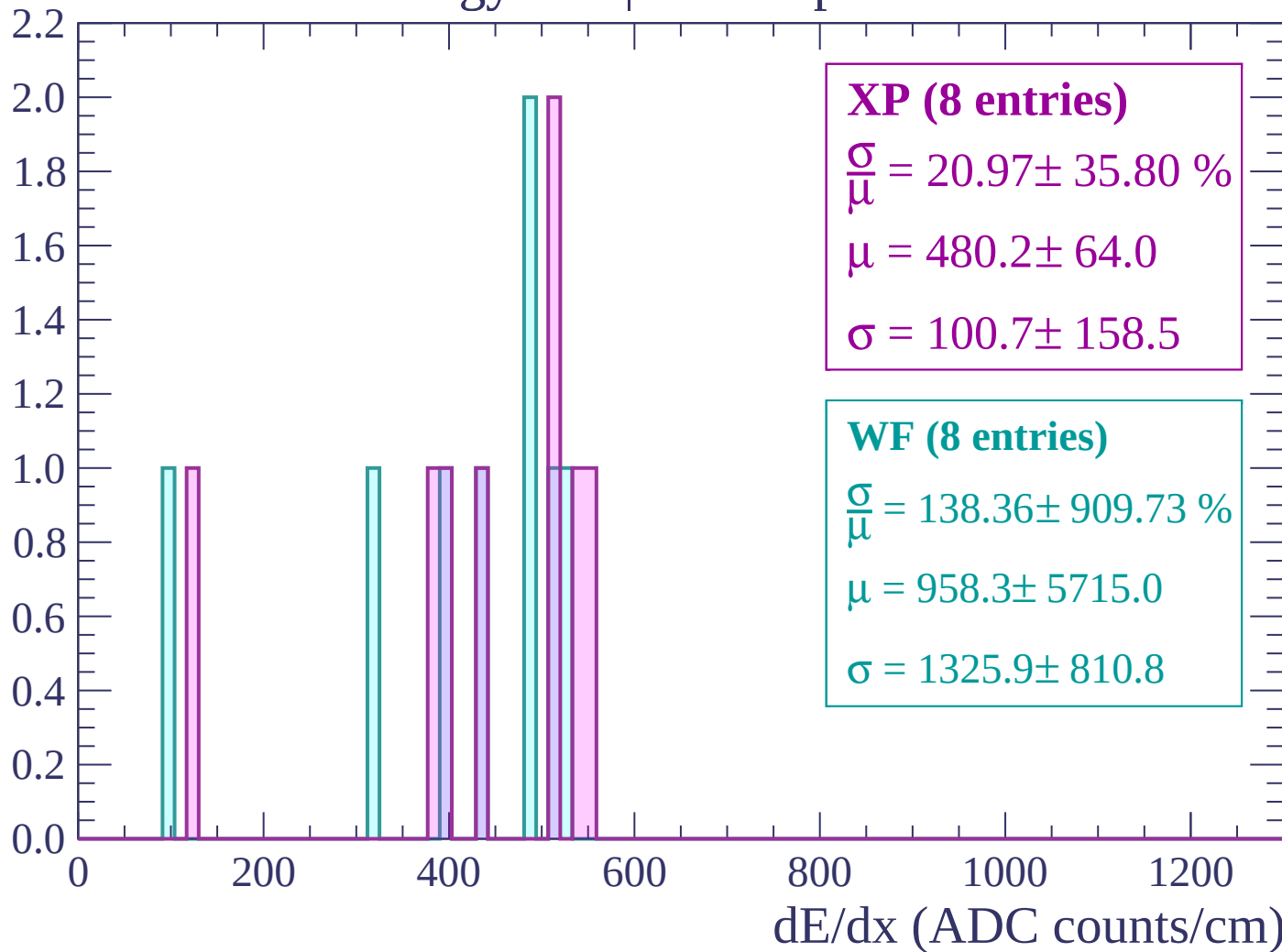
$$\mu = 370.2 \pm 2.3$$

$$\sigma = 0.3 \pm 1.9$$

dE/dx (ADC counts/cm)

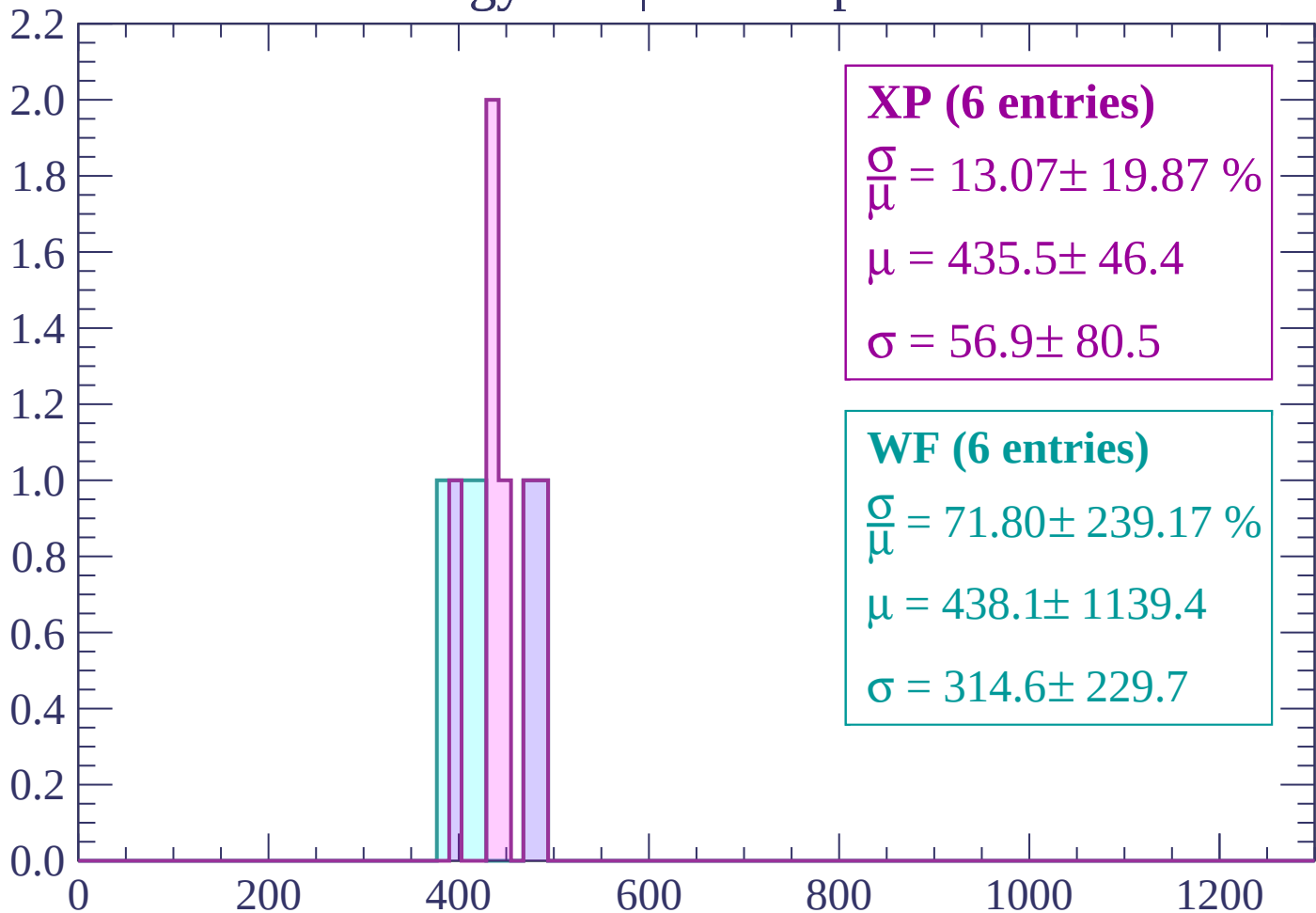
Energy loss | $1860 < p < 1920$

Count



Energy loss | $1920 < p < 1980$

Count



dE/dx (ADC counts/cm)

Energy loss | $1980 < p < 2040$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (5 entries)

$\frac{\sigma}{\mu} = 98.35 \pm 454.64 \%$

$\mu = 483.4 \pm 1898.4$

$\sigma = 475.4 \pm 330.8$

WF (5 entries)

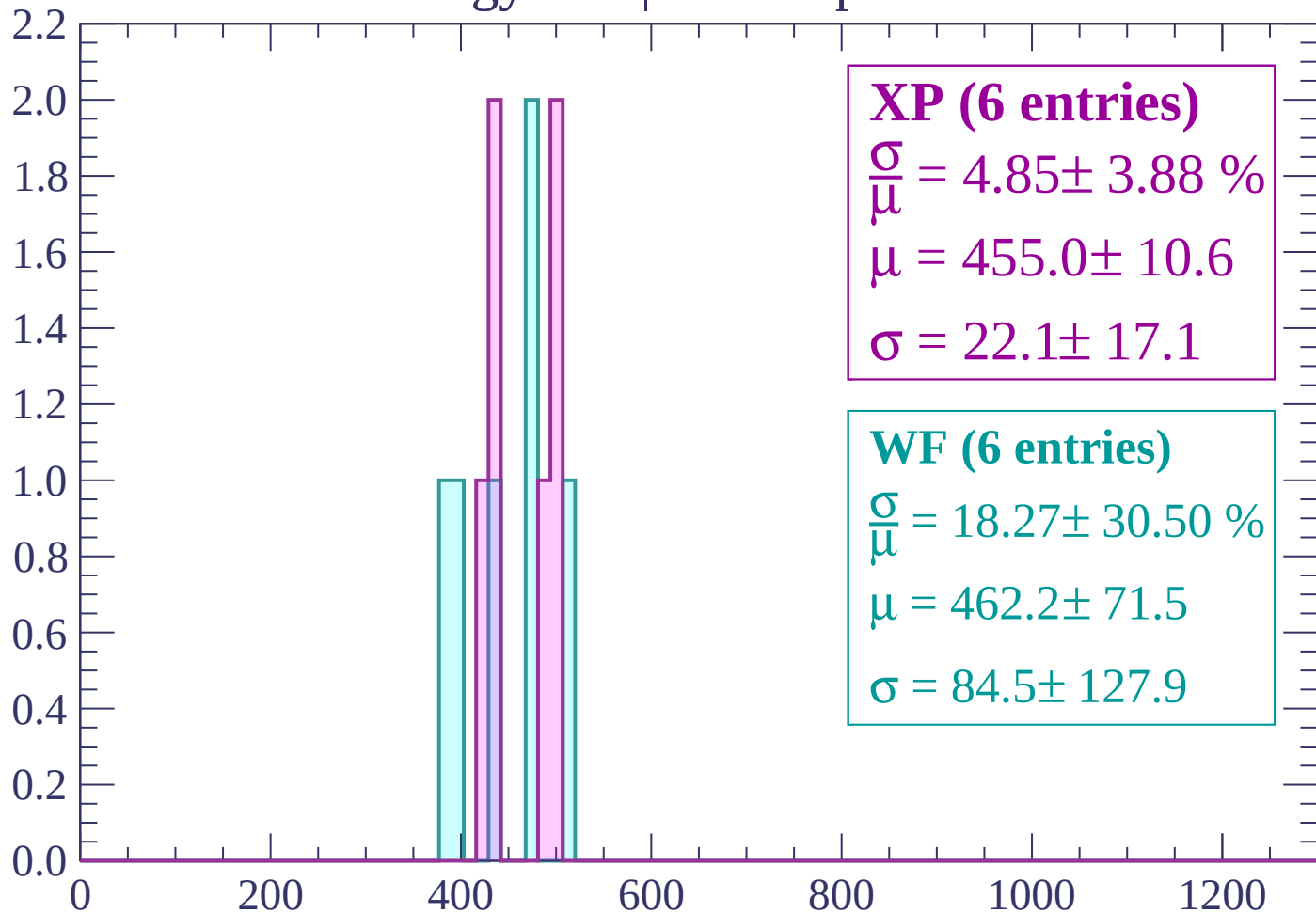
$\frac{\sigma}{\mu} = 1.67 \pm 9.28 \%$

$\mu = 390.0 \pm 27.8$

$\sigma = 6.5 \pm 35.7$

Energy loss | $2040 < p < 2100$

Count



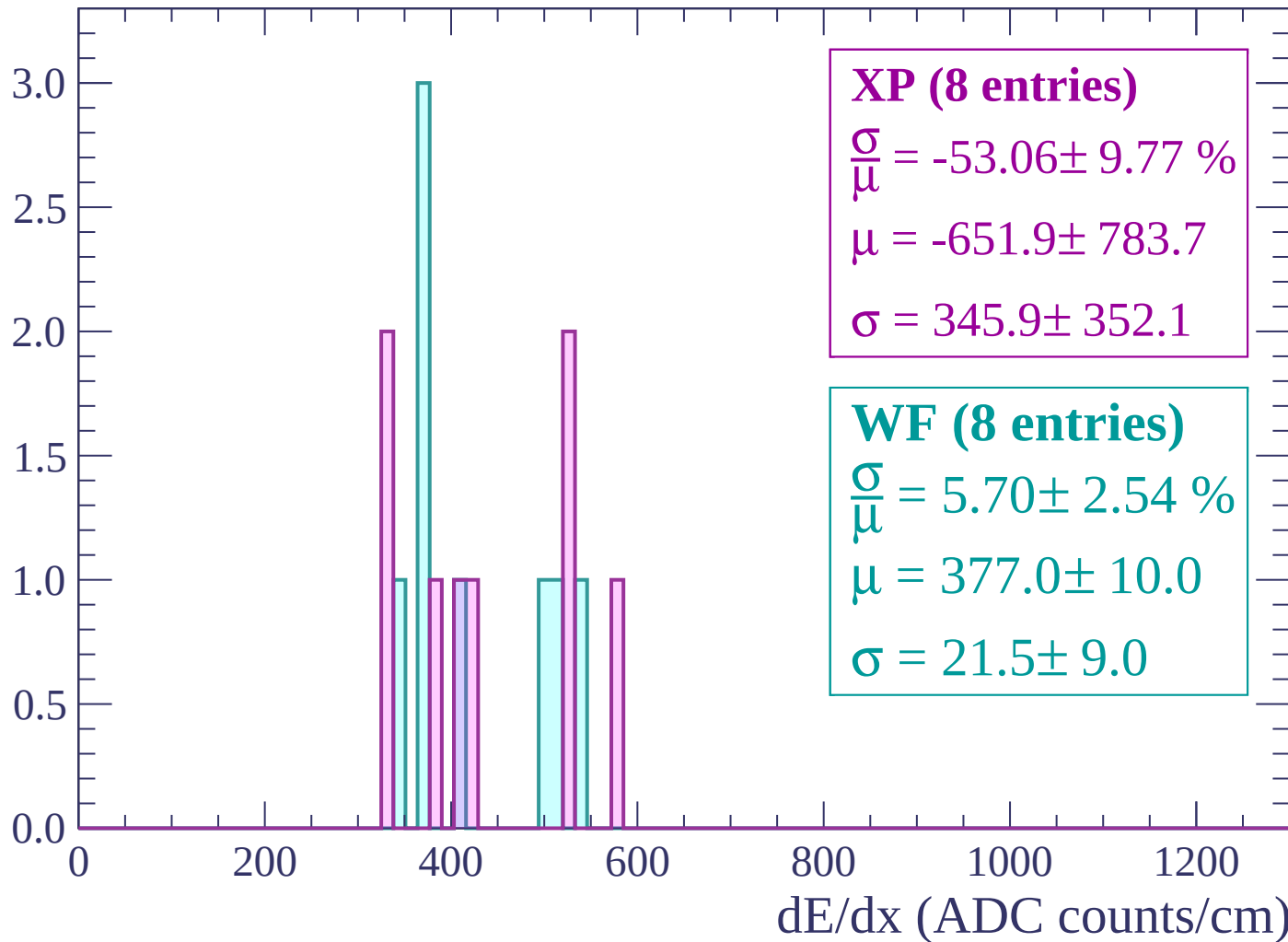
XP (6 entries)
 $\frac{\sigma}{\mu} = 4.85 \pm 3.88 \%$
 $\mu = 455.0 \pm 10.6$
 $\sigma = 22.1 \pm 17.1$

WF (6 entries)
 $\frac{\sigma}{\mu} = 18.27 \pm 30.50 \%$
 $\mu = 462.2 \pm 71.5$
 $\sigma = 84.5 \pm 127.9$

dE/dx (ADC counts/cm)

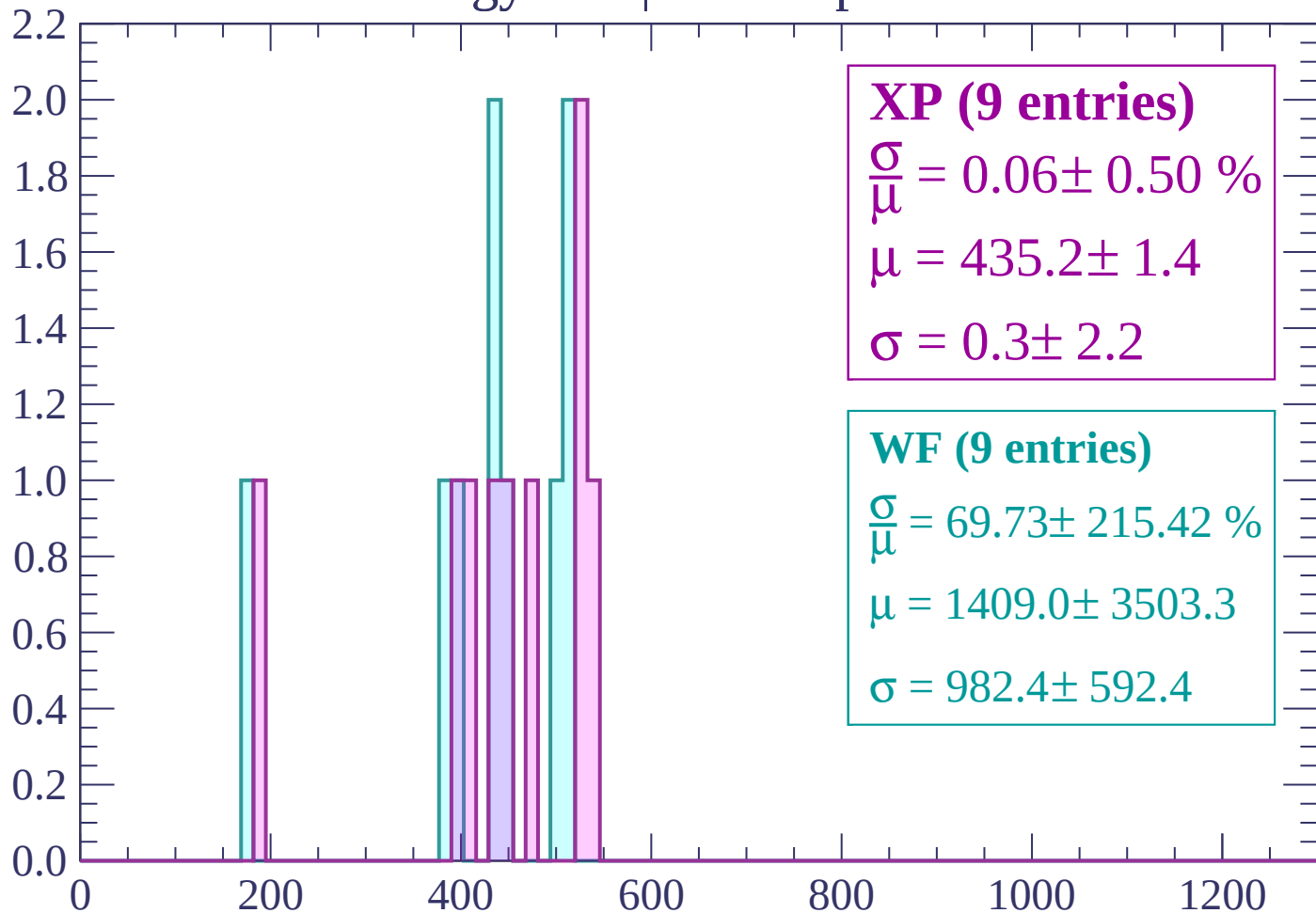
Energy loss | $2100 < p < 2160$

Count



Energy loss | $2160 < p < 2220$

Count



XP (9 entries)

$\frac{\sigma}{\mu} = 0.06 \pm 0.50 \%$

$\mu = 435.2 \pm 1.4$

$\sigma = 0.3 \pm 2.2$

WF (9 entries)

$\frac{\sigma}{\mu} = 69.73 \pm 215.42 \%$

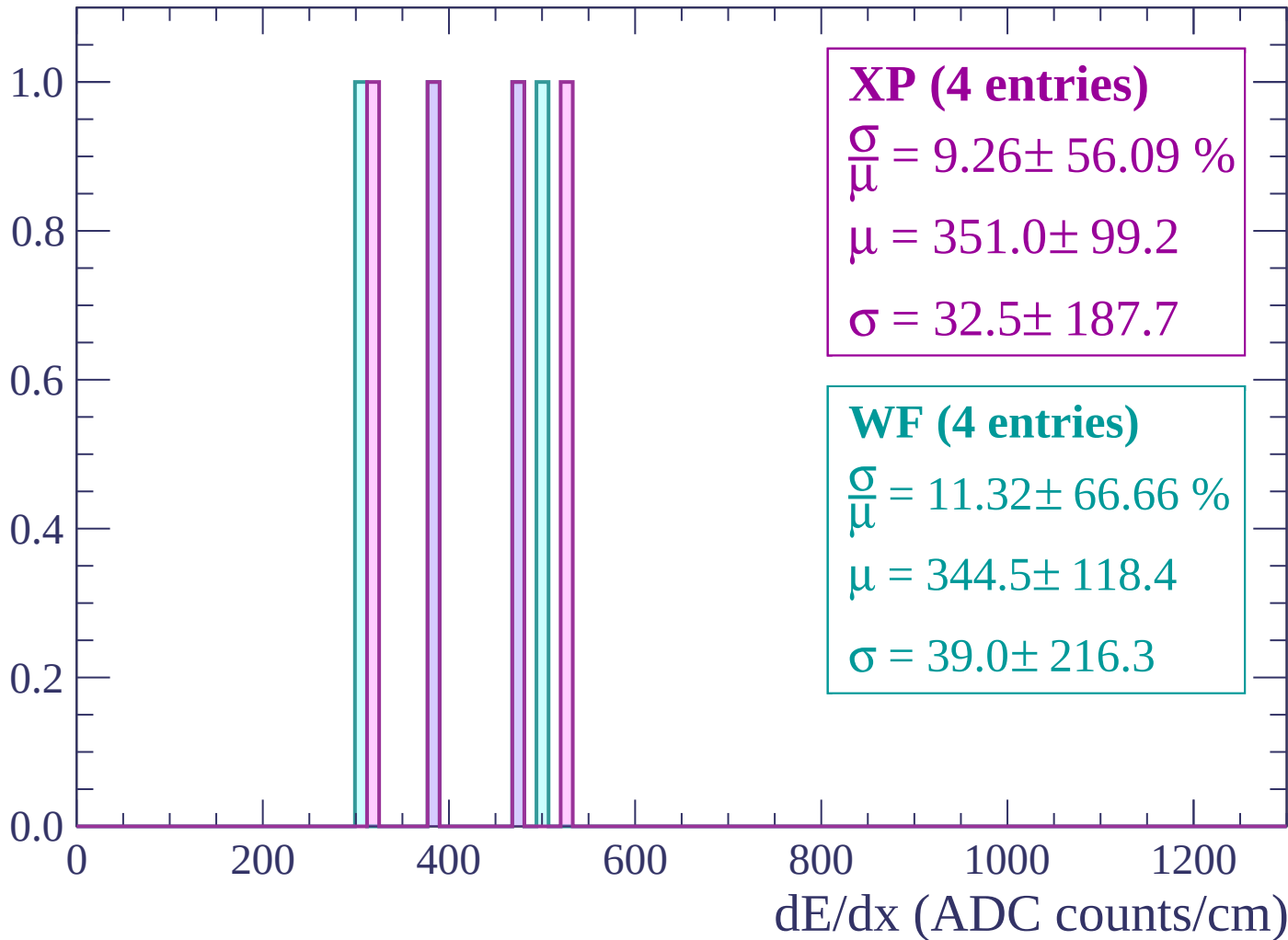
$\mu = 1409.0 \pm 3503.3$

$\sigma = 982.4 \pm 592.4$

dE/dx (ADC counts/cm)

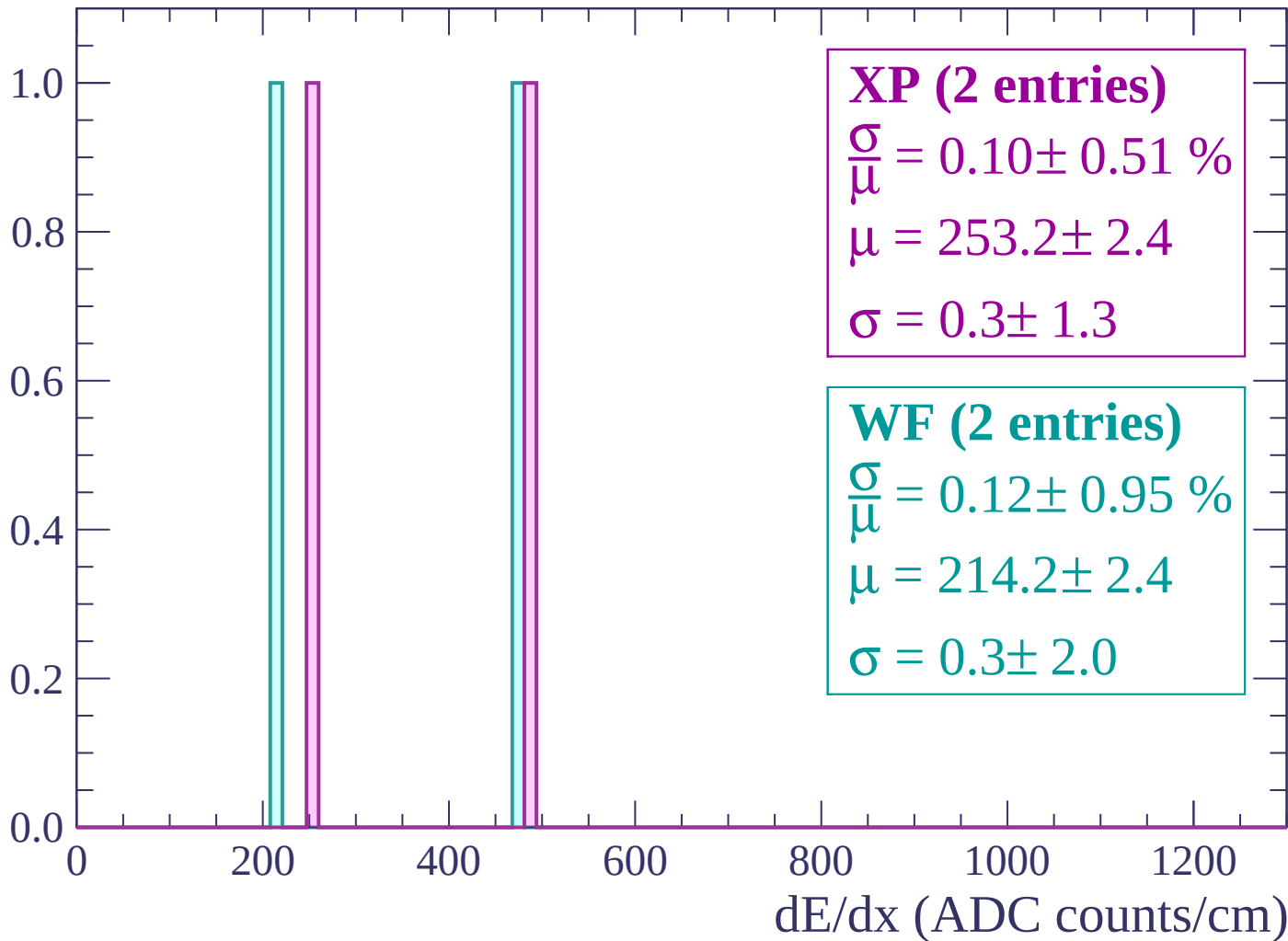
Energy loss | $2220 < p < 2280$

Count



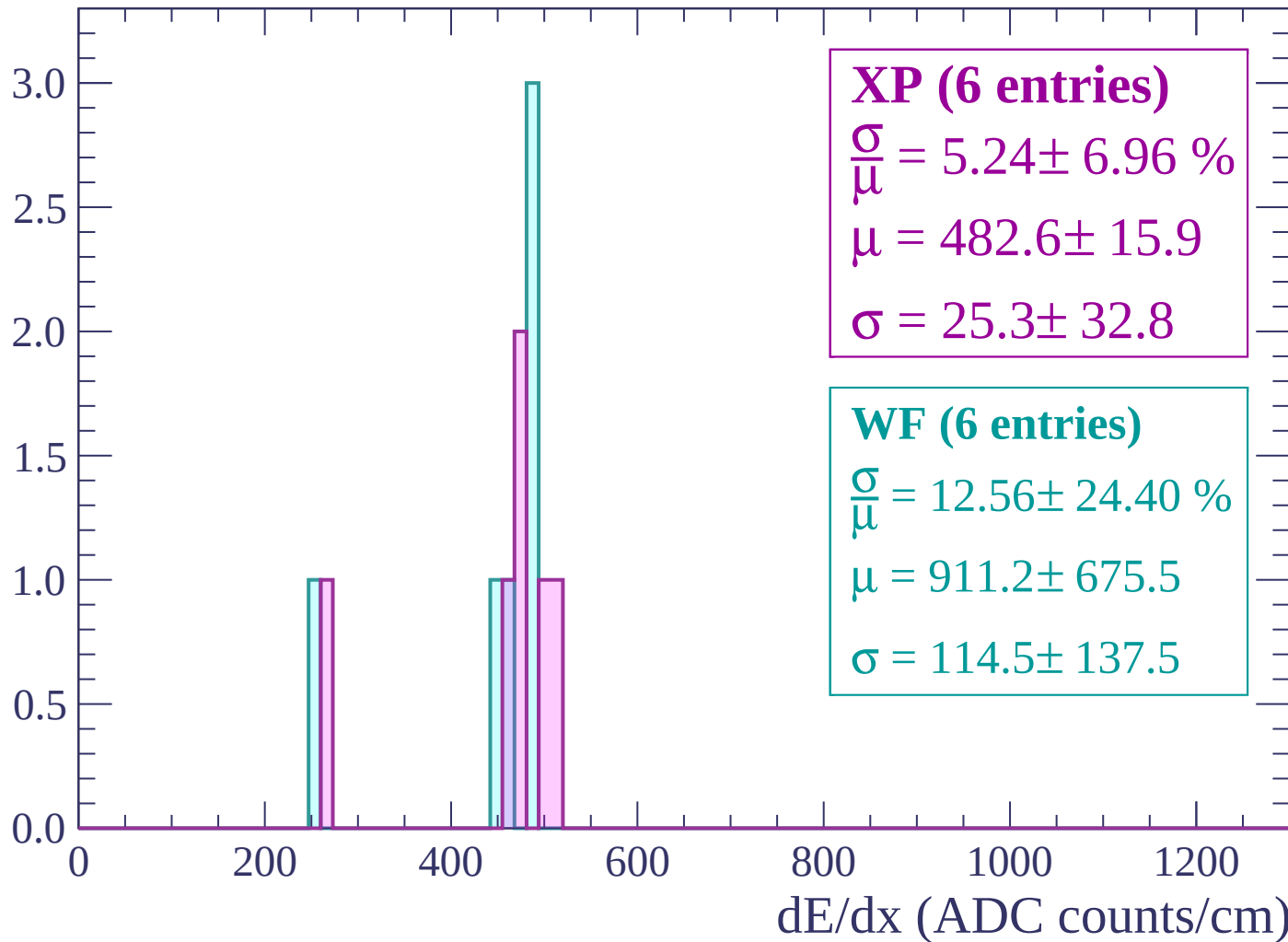
Energy loss | $2280 < p < 2340$

Count



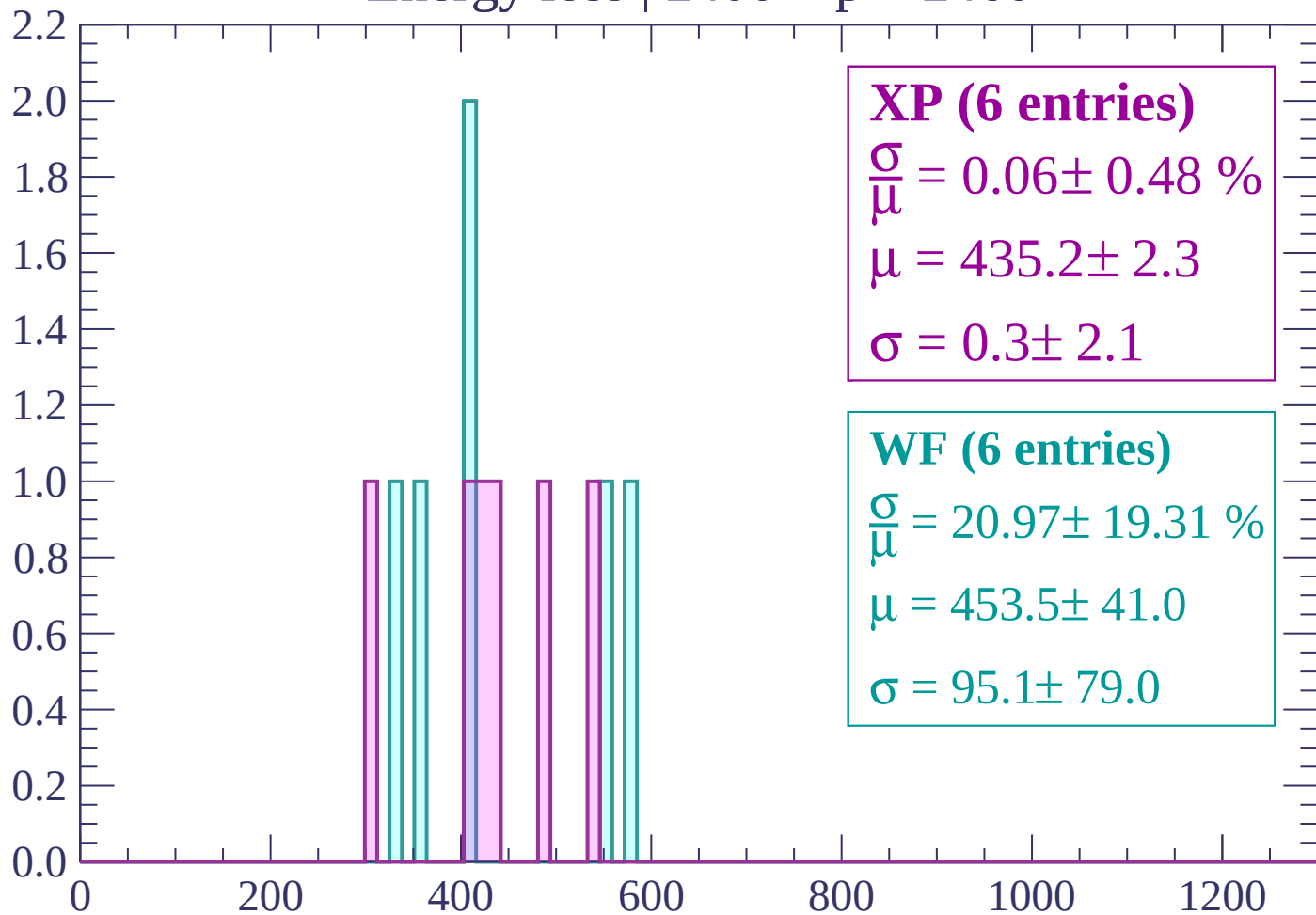
Energy loss | $2340 < p < 2400$

Count



Energy loss | $2400 < p < 2460$

Count



XP (6 entries)

$\frac{\sigma}{\mu} = 0.06 \pm 0.48 \%$

$\mu = 435.2 \pm 2.3$

$\sigma = 0.3 \pm 2.1$

WF (6 entries)

$\frac{\sigma}{\mu} = 20.97 \pm 19.31 \%$

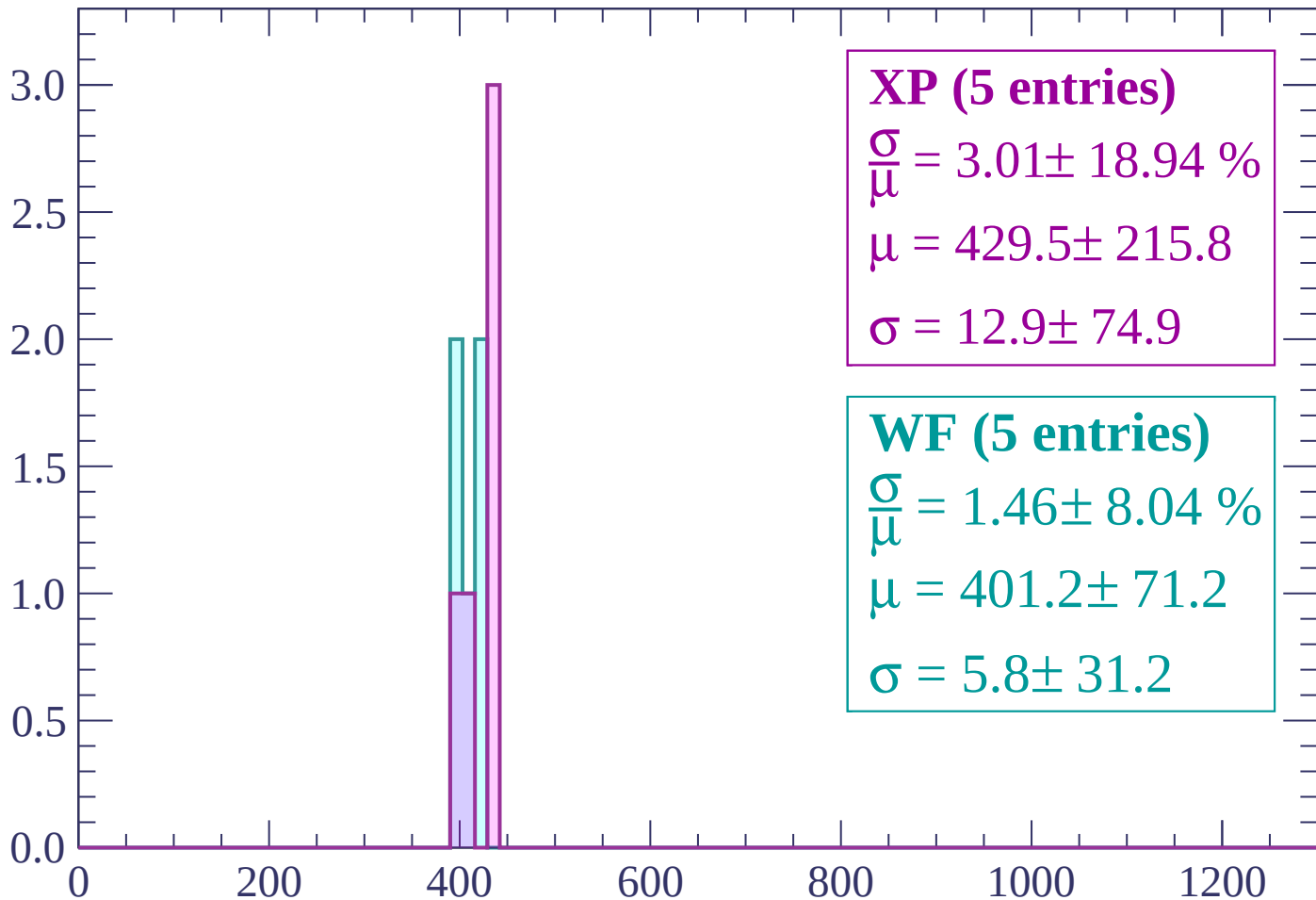
$\mu = 453.5 \pm 41.0$

$\sigma = 95.1 \pm 79.0$

dE/dx (ADC counts/cm)

Energy loss | $2460 < p < 2520$

Count



XP (5 entries)

$$\frac{\sigma}{\mu} = 3.01 \pm 18.94 \%$$

$$\mu = 429.5 \pm 215.8$$

$$\sigma = 12.9 \pm 74.9$$

WF (5 entries)

$$\frac{\sigma}{\mu} = 1.46 \pm 8.04 \%$$

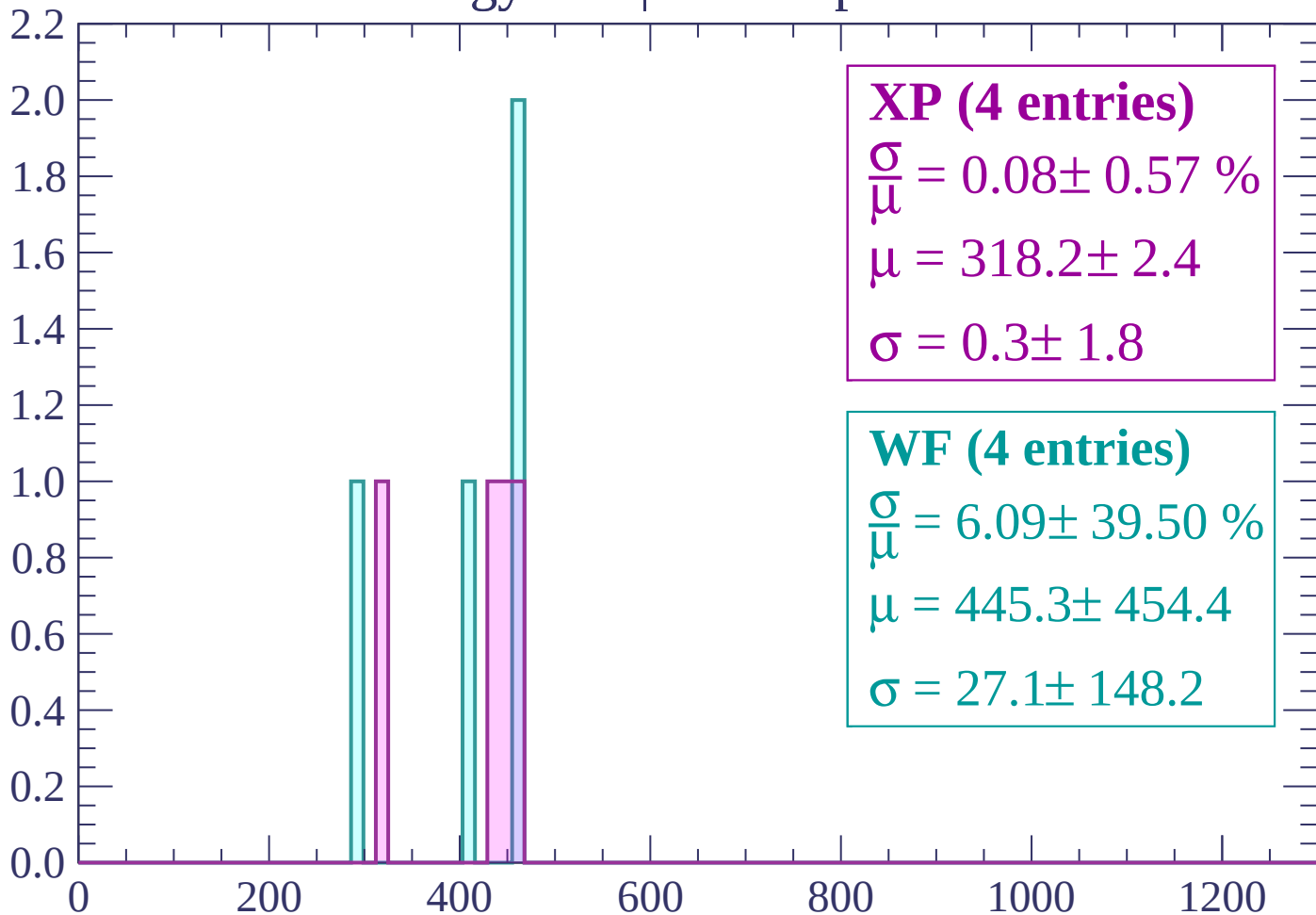
$$\mu = 401.2 \pm 71.2$$

$$\sigma = 5.8 \pm 31.2$$

dE/dx (ADC counts/cm)

Energy loss | $2520 < p < 2580$

Count



XP (4 entries)

$\frac{\sigma}{\mu} = 0.08 \pm 0.57 \%$

$\mu = 318.2 \pm 2.4$

$\sigma = 0.3 \pm 1.8$

WF (4 entries)

$\frac{\sigma}{\mu} = 6.09 \pm 39.50 \%$

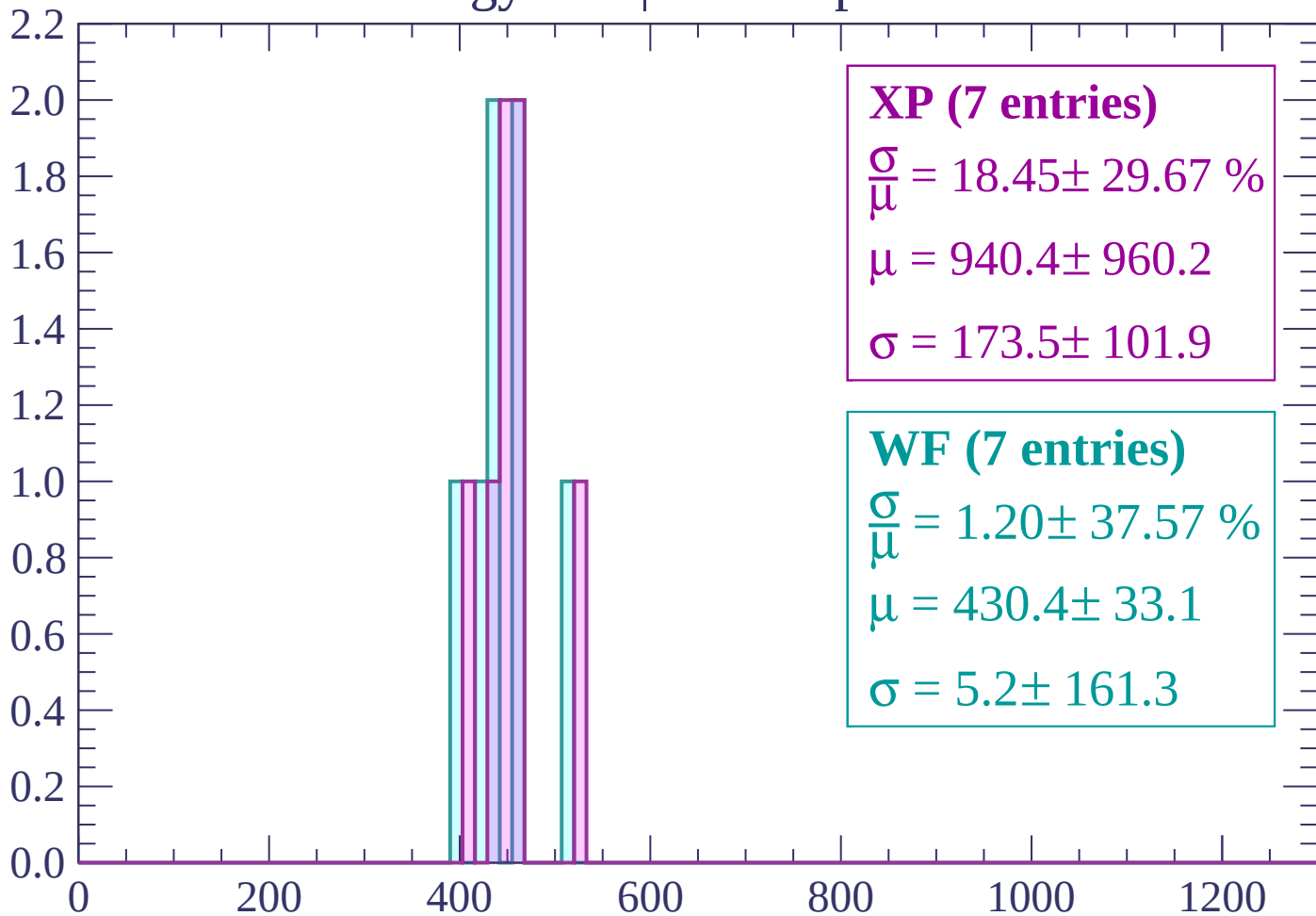
$\mu = 445.3 \pm 454.4$

$\sigma = 27.1 \pm 148.2$

dE/dx (ADC counts/cm)

Energy loss | $2580 < p < 2640$

Count



XP (7 entries)

$\frac{\sigma}{\mu} = 18.45 \pm 29.67 \%$

$\mu = 940.4 \pm 960.2$

$\sigma = 173.5 \pm 101.9$

WF (7 entries)

$\frac{\sigma}{\mu} = 1.20 \pm 37.57 \%$

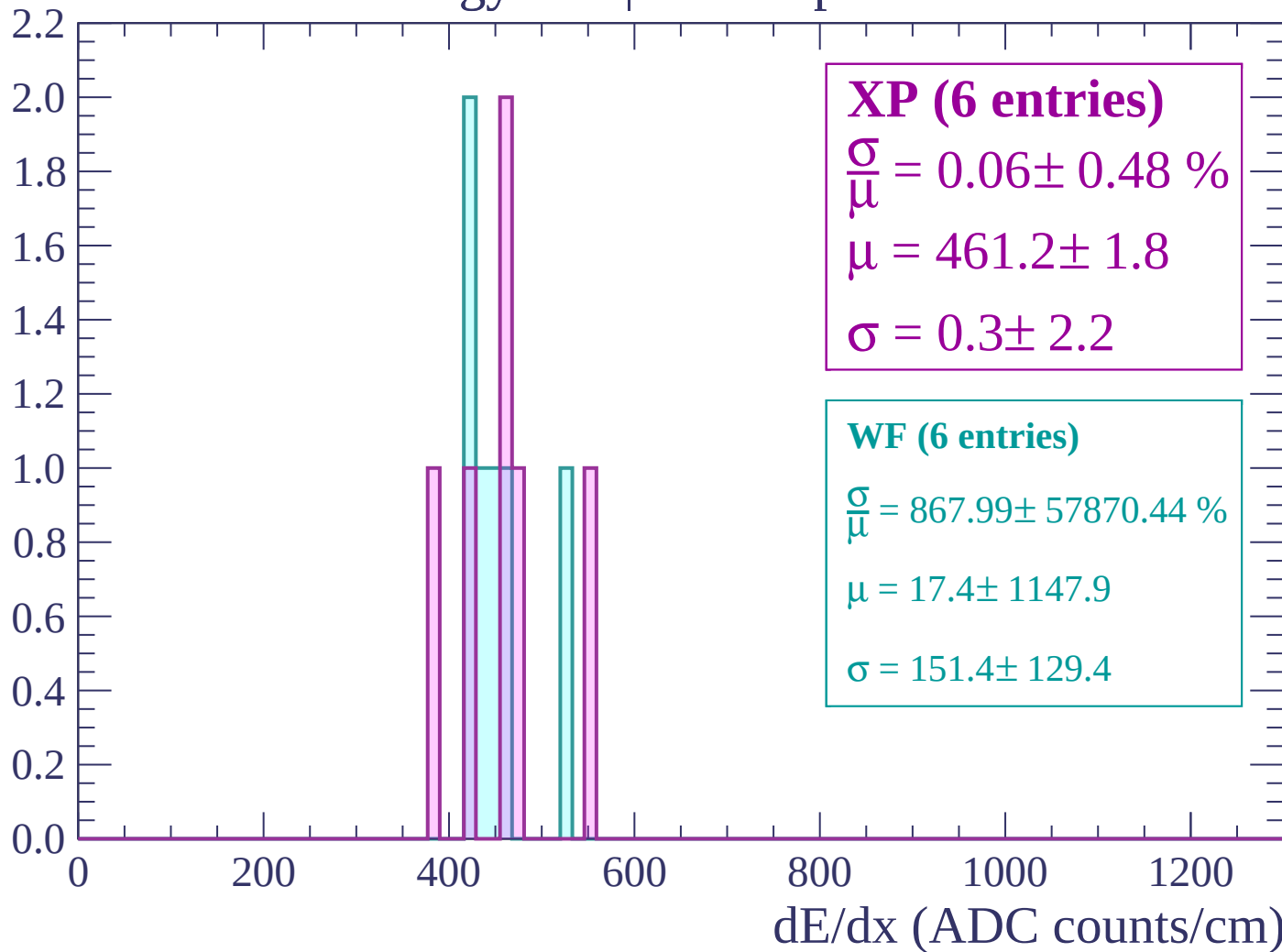
$\mu = 430.4 \pm 33.1$

$\sigma = 5.2 \pm 161.3$

dE/dx (ADC counts/cm)

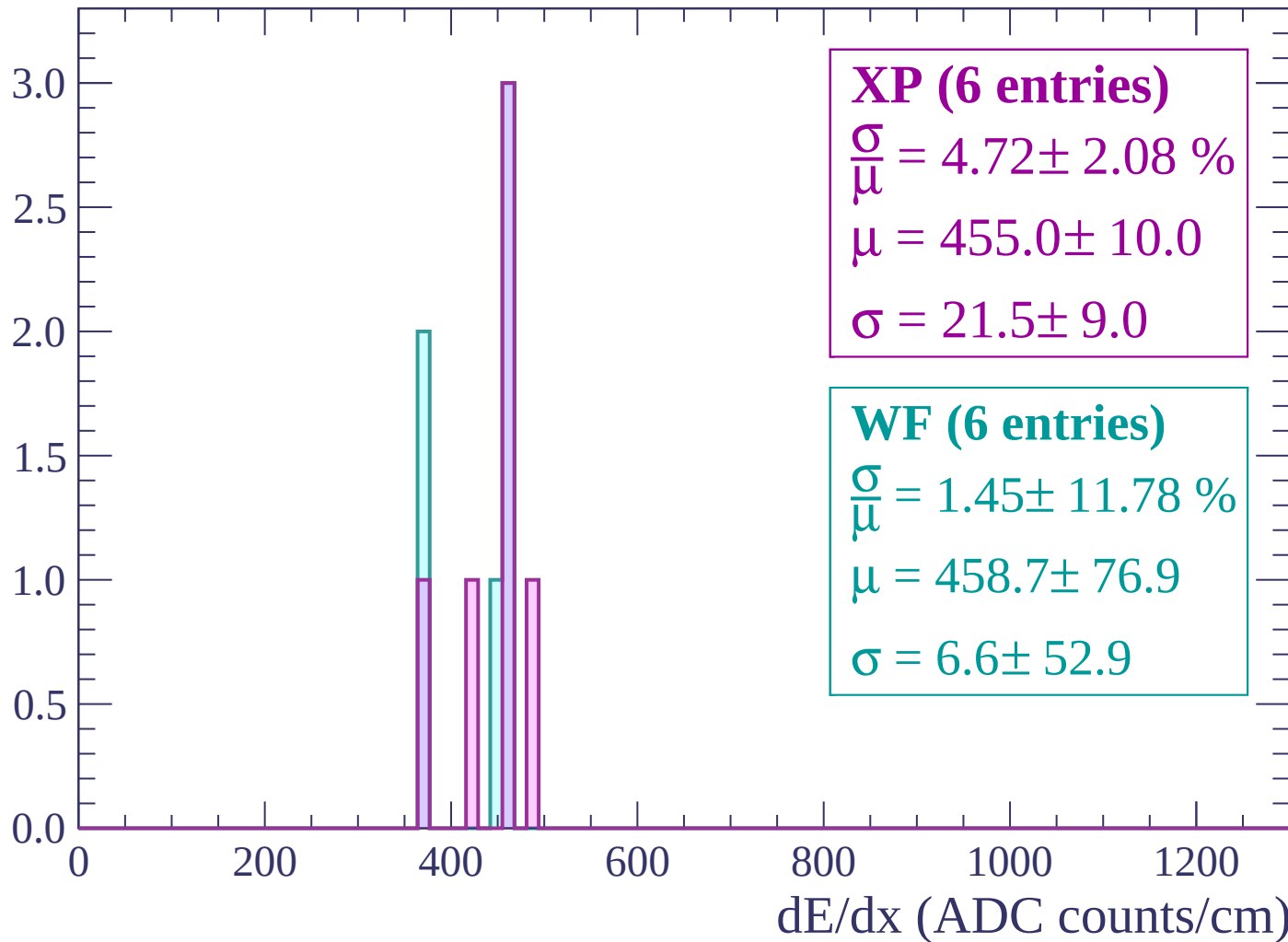
Energy loss | $2640 < p < 2700$

Count



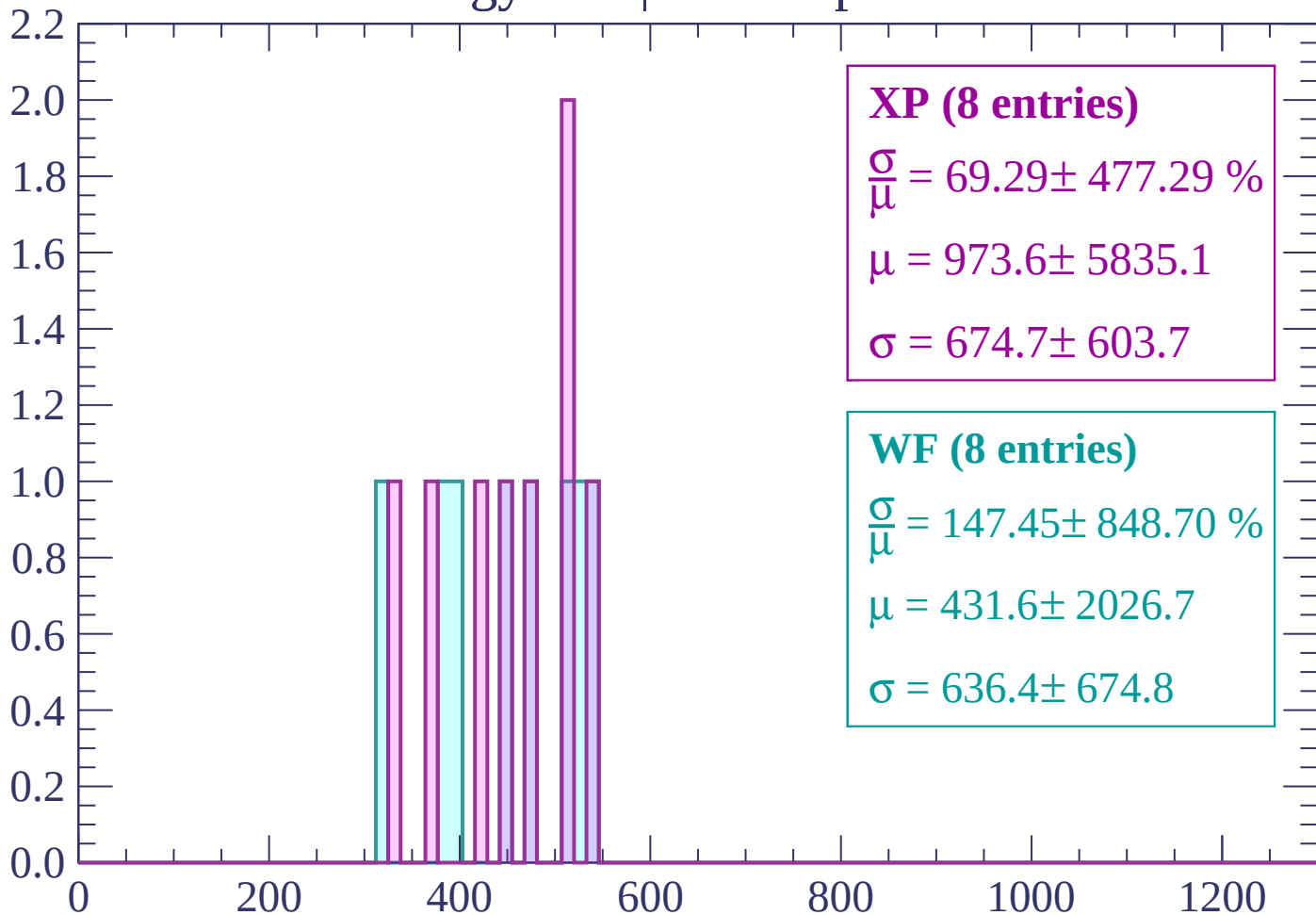
Energy loss | $2700 < p < 2760$

Count



Energy loss | $2760 < p < 2820$

Count



XP (8 entries)

$\frac{\sigma}{\mu} = 69.29 \pm 477.29 \%$

$\mu = 973.6 \pm 5835.1$

$\sigma = 674.7 \pm 603.7$

WF (8 entries)

$\frac{\sigma}{\mu} = 147.45 \pm 848.70 \%$

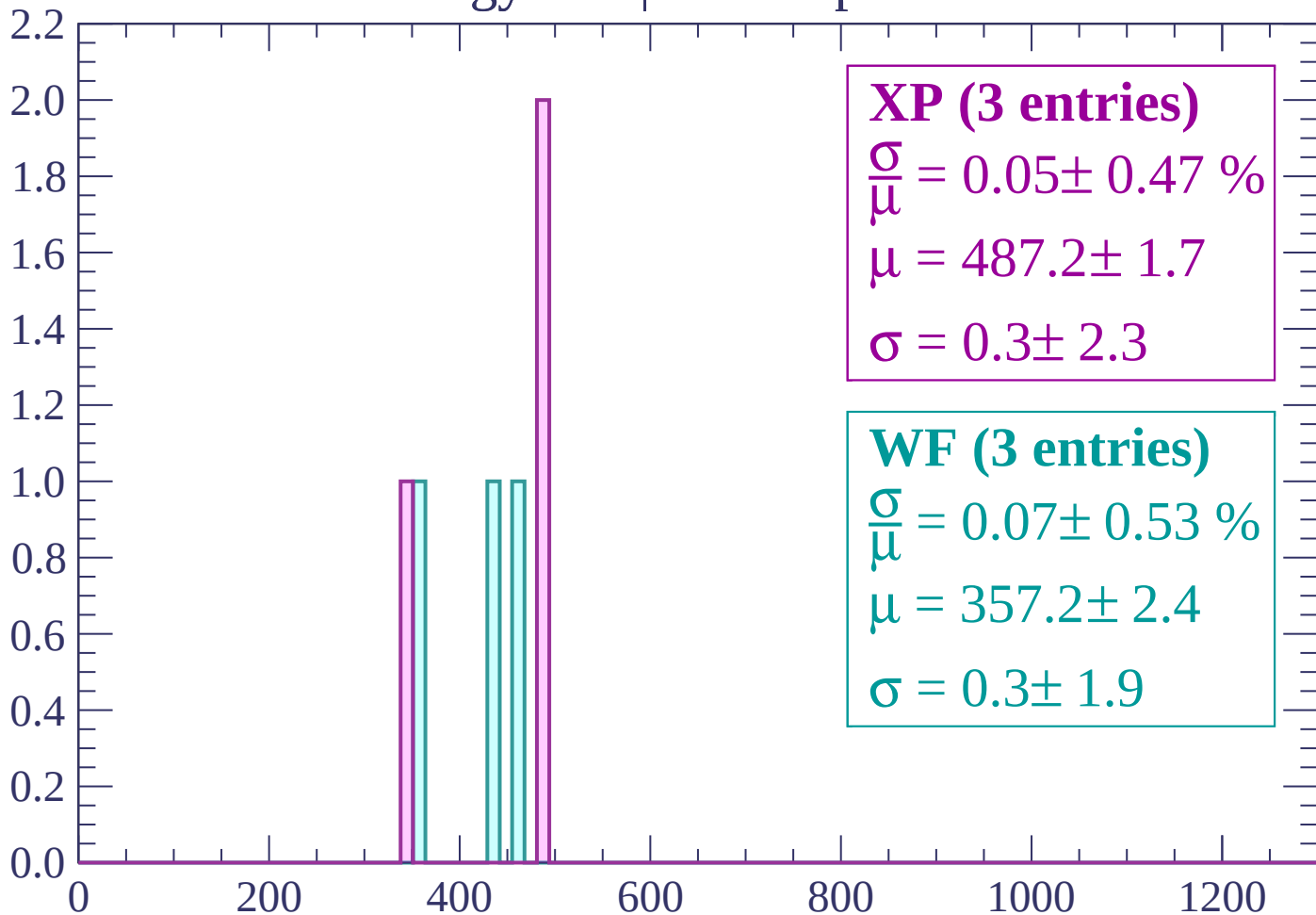
$\mu = 431.6 \pm 2026.7$

$\sigma = 636.4 \pm 674.8$

dE/dx (ADC counts/cm)

Energy loss | $2820 < p < 2880$

Count



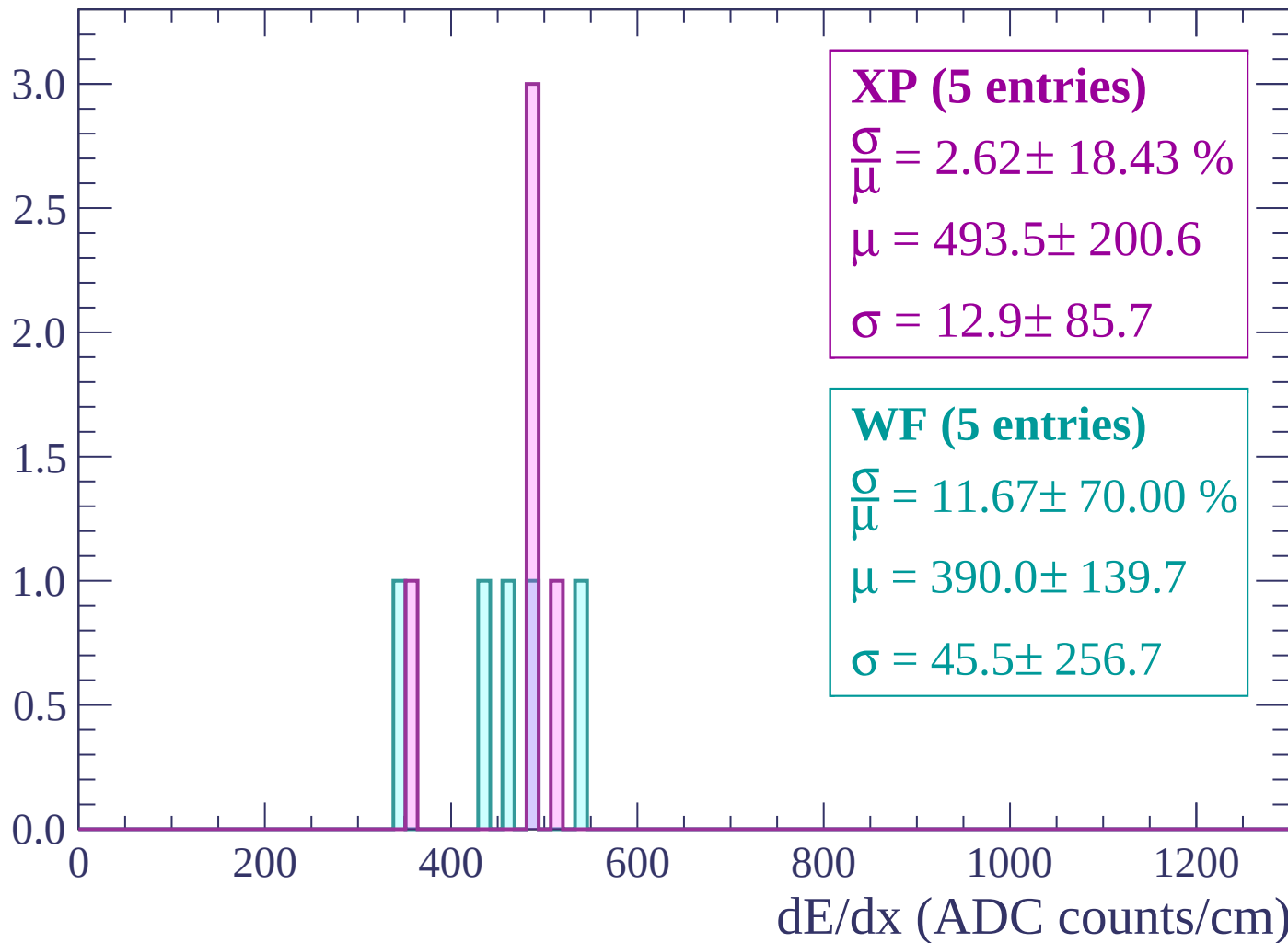
XP (3 entries)
 $\frac{\sigma}{\mu} = 0.05 \pm 0.47 \%$
 $\mu = 487.2 \pm 1.7$
 $\sigma = 0.3 \pm 2.3$

WF (3 entries)
 $\frac{\sigma}{\mu} = 0.07 \pm 0.53 \%$
 $\mu = 357.2 \pm 2.4$
 $\sigma = 0.3 \pm 1.9$

dE/dx (ADC counts/cm)

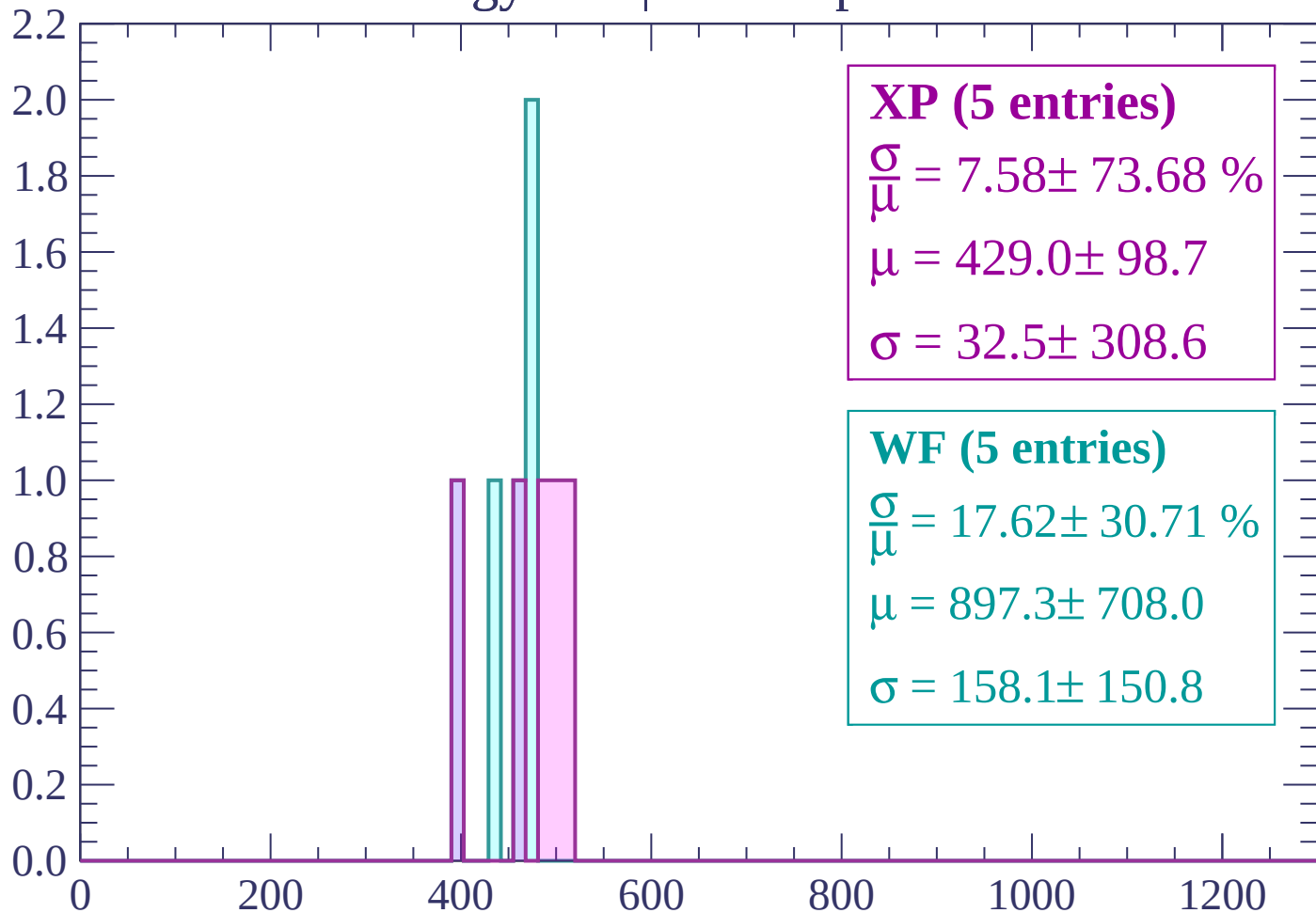
Energy loss | $2880 < p < 2940$

Count



Energy loss | $2940 < p < 3000$

Count



XP (5 entries)

$$\frac{\sigma}{\mu} = 7.58 \pm 73.68 \%$$

$$\mu = 429.0 \pm 98.7$$

$$\sigma = 32.5 \pm 308.6$$

WF (5 entries)

$$\frac{\sigma}{\mu} = 17.62 \pm 30.71 \%$$

$$\mu = 897.3 \pm 708.0$$

$$\sigma = 158.1 \pm 150.8$$

dE/dx (ADC counts/cm)

Energy loss | $3000 < p < 3060$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (4 entries)

$\frac{\sigma}{\mu} = 51.81 \pm 157.68 \%$

$\mu = 437.6 \pm 869.1$

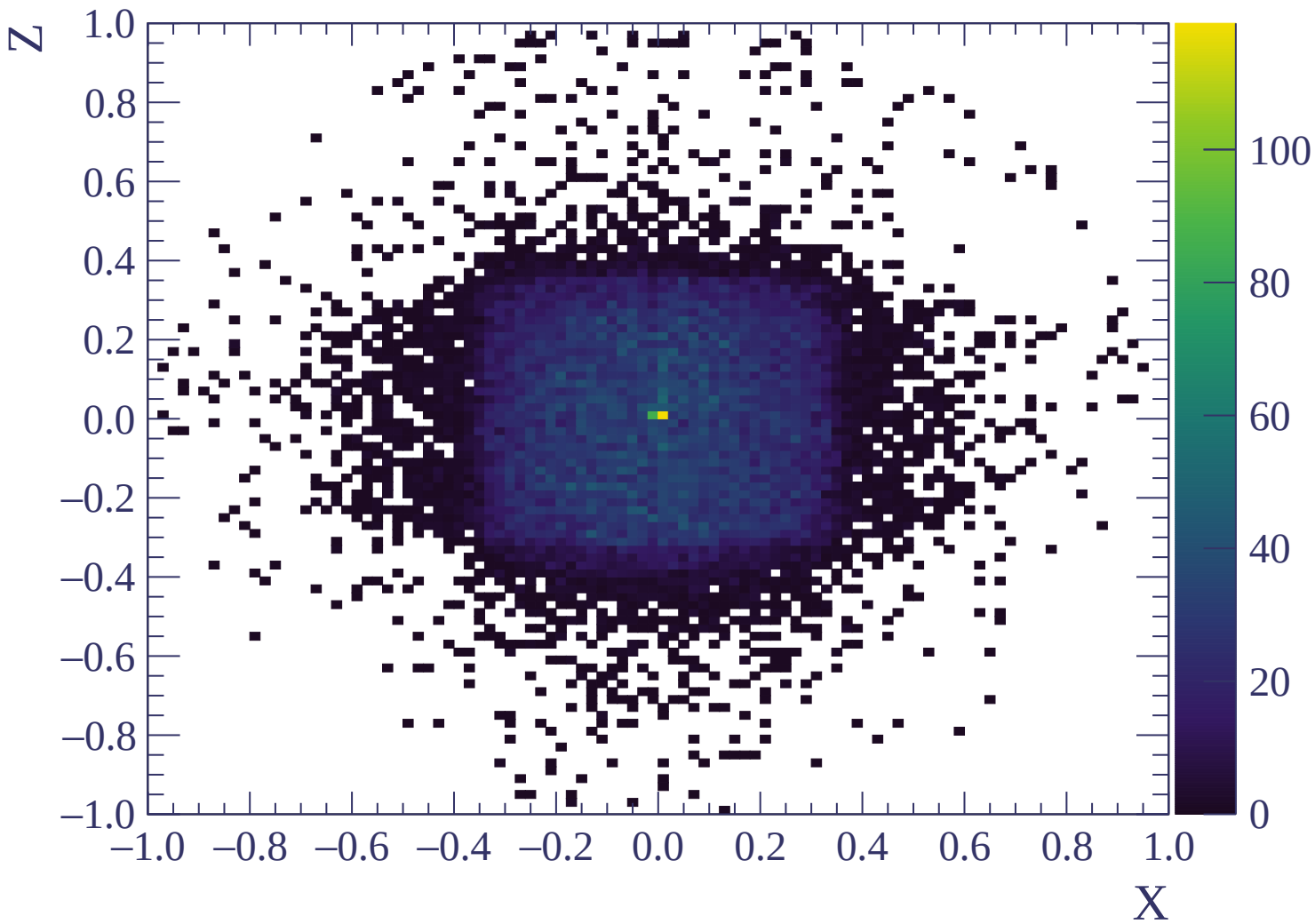
$\sigma = 226.7 \pm 239.8$

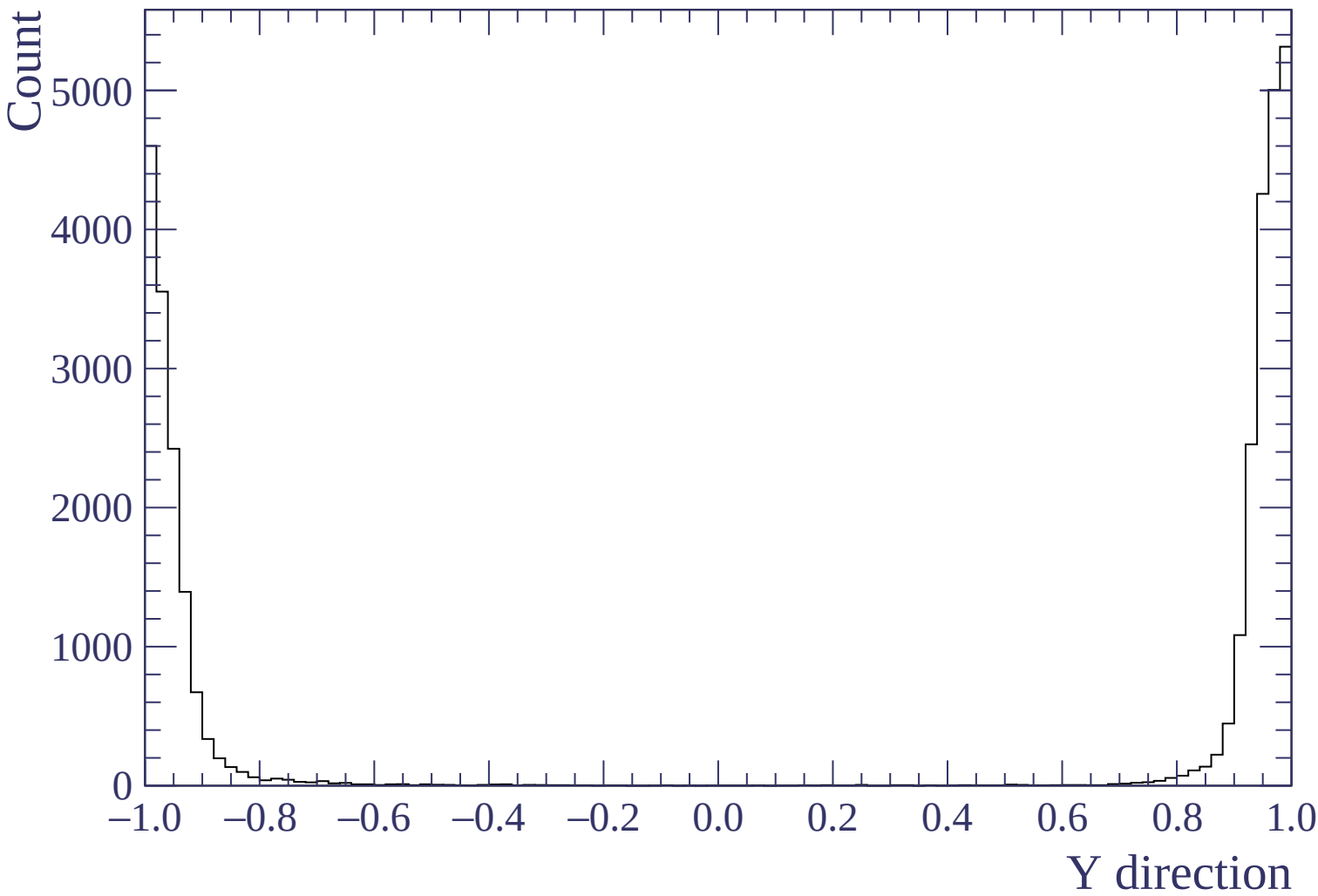
WF (4 entries)

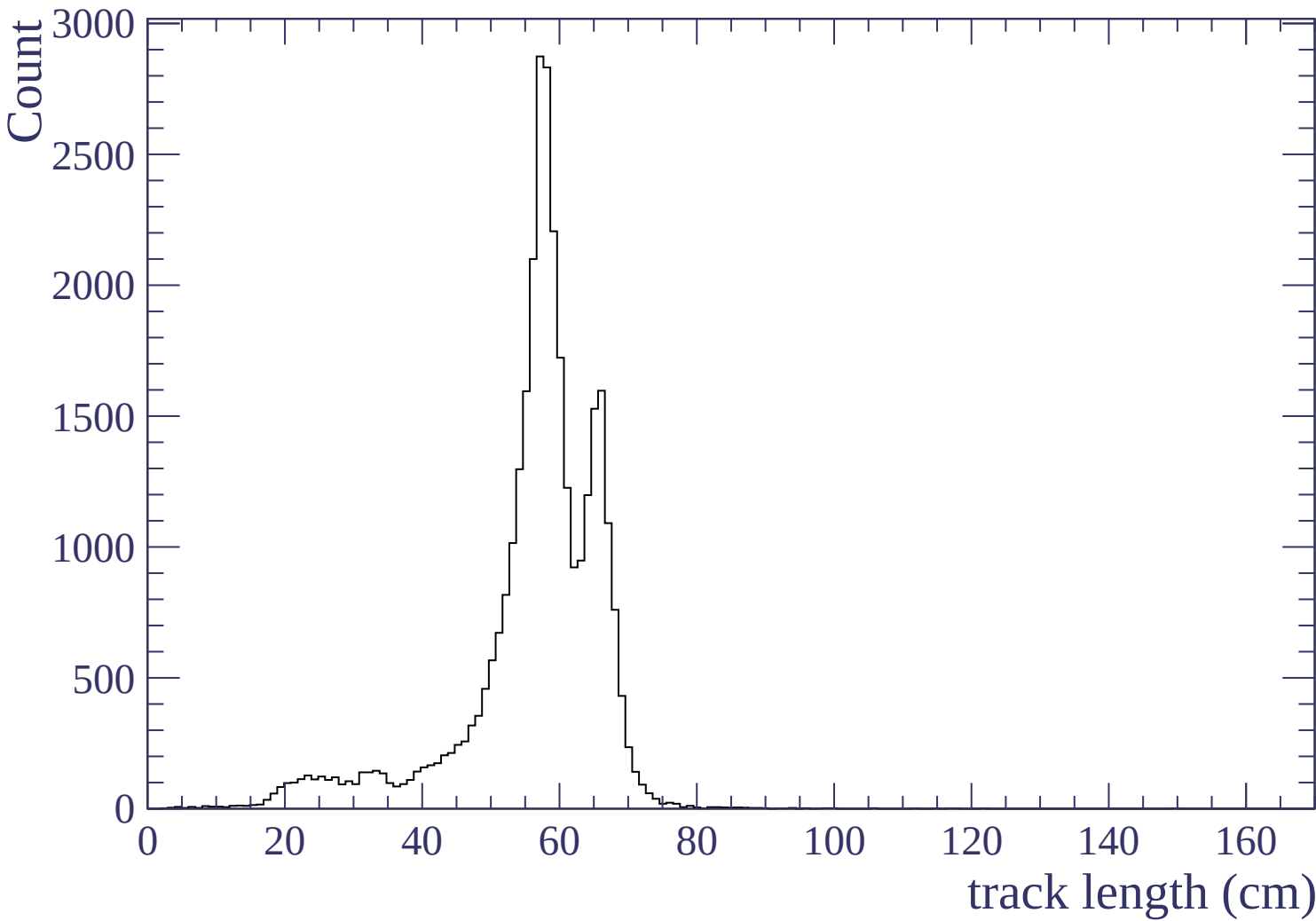
$\frac{\sigma}{\mu} = 59.37 \pm 218.38 \%$

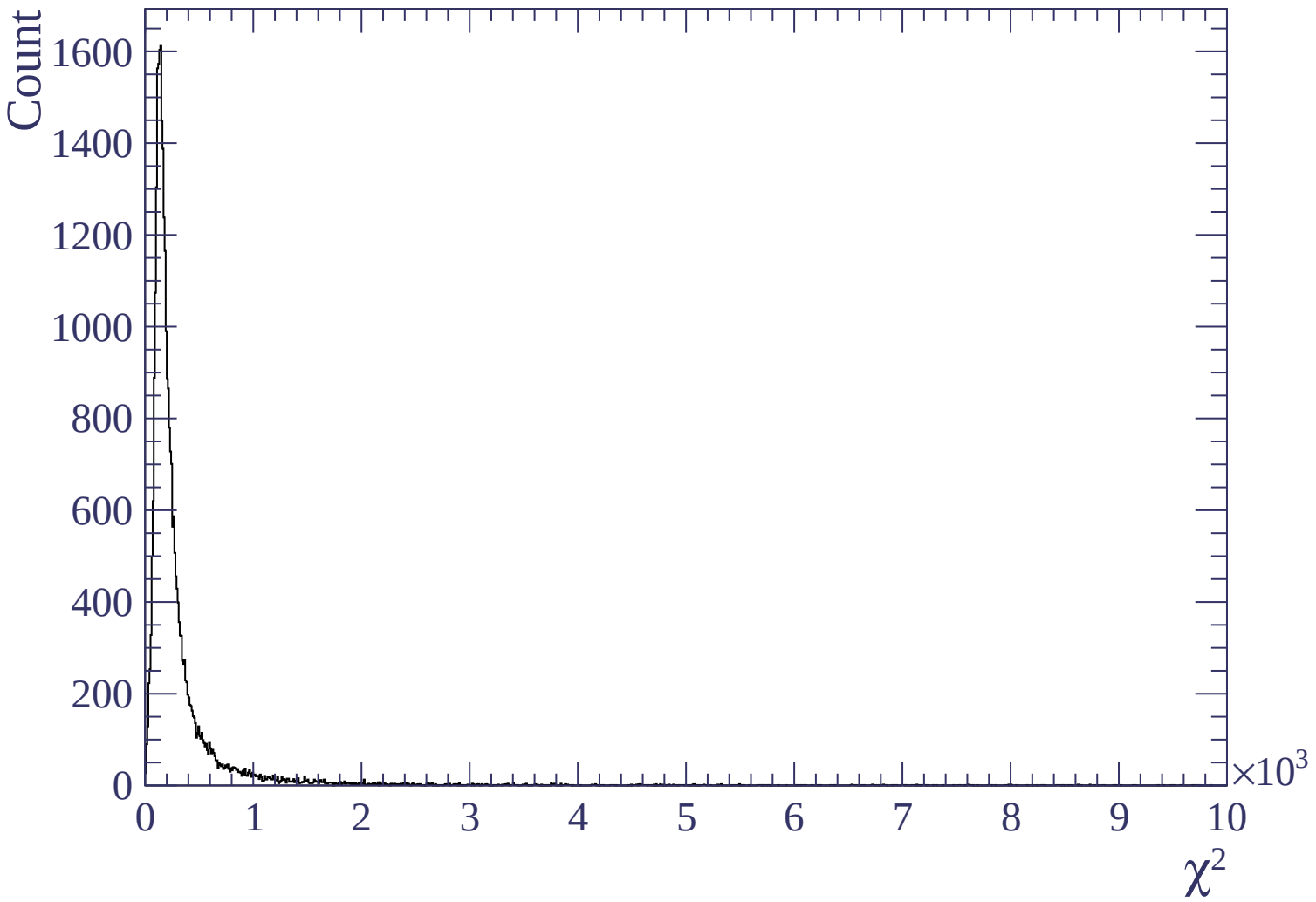
$\mu = 424.2 \pm 1144.9$

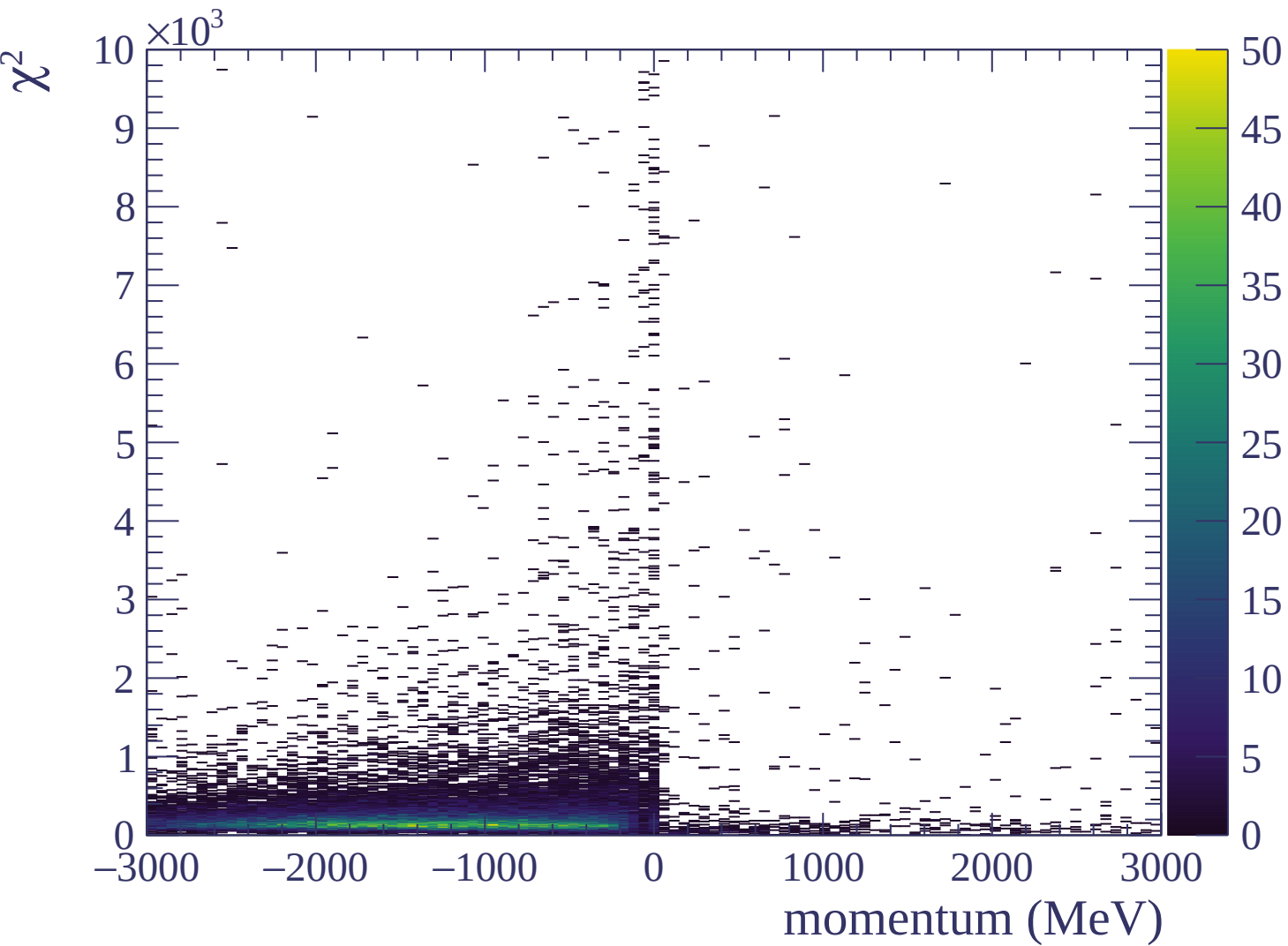
$\sigma = 251.9 \pm 246.7$





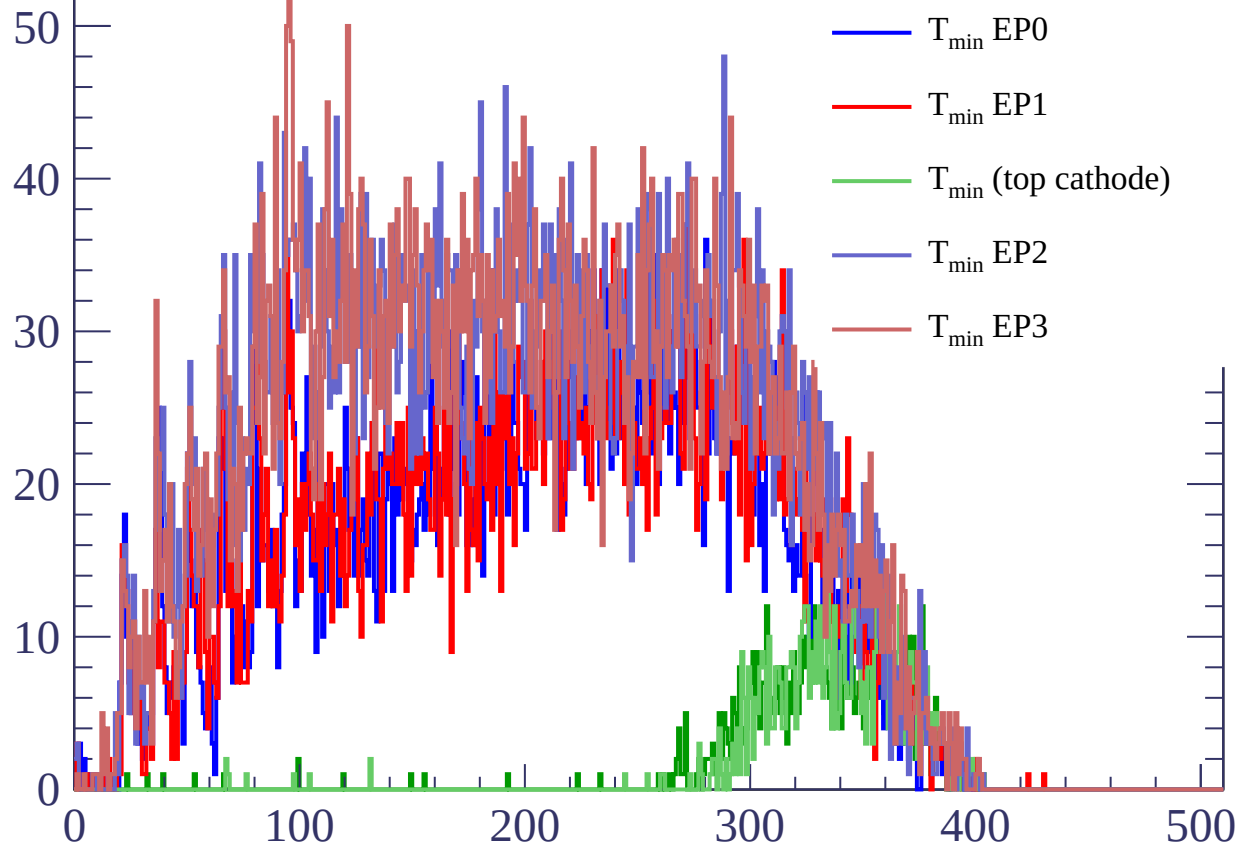






Count

- $T_{\min}$ (bottom cathode)
- $T_{\min}$ EP0
- $T_{\min}$ EP1
- $T_{\min}$ (top cathode)
- $T_{\min}$ EP2
- $T_{\min}$ EP3



time bin

Count

